What the Neutrino State Must Keep: Hidden Angular Structure, the Flux–Shear Link and the Threshold for Primordial Black Hole Formation

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AI-assisted. The ideas and research direction are the author's; AI tools (Claude, Anthropic) were used for literature search, computation, drafting and typesetting. The numbers come from the scripts, run logs and outputs printed in Appendix B. The simulations are runs of S. McCormick's public code on a cloud computer, not experiments.

Abstract

McCormick has shown that neutrinos which stream freely out of a collapsing overdensity raise the threshold for primordial black hole formation, by about $0.235\,R_\nu$ in the linear compaction, and that the effect fades when the neutrinos decouple gradually. This paper asks what sets that fading. First, the neutrino collision operator is read exactly as a cut between what is kept and what streams. All of its non-normality, $\frac{1}{8}\|[L, L^*]\|^2 = \kappa^2 k^2/3$, lies on the single link between density and flux. With separate rates for flux and shear it is $k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$, and the weight across the cut equals the sound speed, $\sqrt{E_B} = k/\sqrt{3}$. The kept sector is the charge-diffusion channel of relaxation kinetic theory, whose mode exists only for $\kappa > 2k/\pi$ (Romatschke); its non-orthogonality is $4\phi^2/\sin^2 2\phi$, $\phi = k/\kappa$, which diverges there. Second, McCormick's own code is run on cases outside his paper, with every prediction and kill line written down before the runs. A broad profile that collapses near $\epsilon \approx 7$–9 keeps the same share of the neutrino effect as a Gaussian that collapses near $\epsilon \approx 4$, or less, so the collapse time alone does not set the clock. A steeper decoupling history, $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^6$, gives $G = 0.648$, 0.317 and 0.154 at $\epsilon_{\rm dec} = 1$, 2 and 3, against 0.683, 0.431 and about 0.30 for the standard history: how the neutrinos leave matters, not only when. A kept-sector memory form, the population the density keeps under the decoupling operator, fitted once to McCormick's curve and then frozen, predicted those three histories within its kill line of 0.08 while a history-blind rule failed. Third, the memory has a location. Two neutrino states with the same density, flux and anisotropic stress, differing only in finer angular structure, give later density, flux and stress that differ by several per cent when the difference is set at or after decoupling; set deep in the coupled era it is erased, by the factor $\exp[-\epsilon_{\rm dec}^n/(n\tau_s^n)]$. An exact block elimination puts the whole effect in one hidden-source term, and the kept population obeys $(\dot{A})^2 \le 4E_B A(1-A)$. No instantaneous closure on density, flux and stress alone can carry this memory. Fourth, the anisotropic stress sets the threshold. With the flux relaxing at κ_1 and the shear and every higher moment at $\kappa_2 = r\kappa_1$, eleven thresholds from $r = 1/16$ to 16 show that the surviving share is not set by the connection energy, which is stationary at $r = 1$, but by the direction of the flux–shear mismatch: G runs from 0.776 to 0.129, and its offset $\ln[(1-G)/G]$ moves close to a straight line in the link rapidity $\frac{1}{2}\ln r$, at a rate of 1.11 ± 0.01 inside the window where the split operator is a Markov generator; with the collision rate evaluated at every Runge–Kutta stage the two inner pairs agree to 0.004, and the ± 0.01 is now set by grid resolution; a third grid level shrinks the resolution shift of the $r = 1.25$ threshold to 0.42 of the previous step. Damping the shear alone, with every higher moment left at the flux rate, gives a larger rate, 1.19, where any rule built

from link-weight ratios predicts less than 0.1: the threshold reads the shear itself. The frozen memory form misses this effect entirely, and so does the intrinsic displacement of the linear kept-sector state at readout, which the two collision models move by almost the same angle: the law lives in the operator's link rapidity, and the difference between the models is written in the nonlinear collapse. Halving the neutrino fraction to $R_\nu = 0.2$ leaves G at 0.409, against 0.425 at $R_\nu = 0.409$ measured with the same code. Fifth, the hidden sector is weighed. Its readout Gramian has no kernel, so the predictive quotient is the whole hidden space and the Gramian's spectrum ranks what a reduced state must keep: 10^{-7} for structure planted in the coupled era, 0.37 at decoupling, always led by $\ell = 3$. Two collision histories with equal optical depth and equal integrated connection energy, the rate reversed in time, differ in the memory form by up to 0.16: the order of the history is part of the record. Sixth, the hidden structure moves the threshold itself. In McCormick's full general-relativistic collapse, two neutrino states identical in density, flux, stress, lapse, constraints and every fluid and metric array, differing only in hidden angular structure planted at decoupling, have black-hole thresholds 3.7×10^{-3} apart for $\ell = 3$, 2.1×10^{-3} with the opposite sign for $\ell = 4$, and 5.5×10^{-3} along the Gramian's leading direction, two hundred bracket widths or more; planted in the coupled era they are 2×10^{-5} apart. The threshold reads angular information that no closure on density, flux and stress holds. Seventh, McCormick's multigroup collision term, with his archived per-group target, is exactly the split link, with $r = 1 + \kappa_n/\kappa_e$: neutrino self-scattering is the flux–shear split. Removing the split while keeping the isotropy history moves his threshold by +0.0003, a third of what the frozen link law predicts; most of what self-scattering does is timing. The algebraic identities, the counterexample and every verdict's arithmetic are machine-checked in Lean (Appendix C). Two further tests are stated for the $e^{\pm}$ annihilation epoch and for the mean of the rates.

1 The question

Primordial black holes heavier than about $10^6\,M_\odot$ form while the neutrinos, which carry 41–49% of the radiation, stop scattering and stream freely. McCormick evolved these neutrinos with their angle-resolved transport equation inside spherical general relativity and found that free streaming makes collapse harder [1]. The threshold on the linear compaction C_l, which sets abundance, rises by $\Delta C_{l,c}/C_{l,c} \simeq 0.235\,R_\nu$, and the threshold δ_c on the compaction peak rises by $\simeq 0.15\,R_\nu$. For Gaussian fluctuations a perfect-fluid collapse probability β becomes about $\beta^{1.18}$ above $6 \times 10^5\,M_\odot$.

He also ran gradual decoupling, with collision rate $\kappa = H(\epsilon_{\rm dec}/\epsilon)^3$, where $\epsilon = 1/(aHr_m)$ is the Hubble radius in units of the perturbation size. Write G for the share of the instantaneous free-streaming shift that survives:

$$G = \frac{\delta_c - \delta_c^{\rm fluid}}{\delta_c^{\rm inst} - \delta_c^{\rm fluid}}. \tag{1}$$

His Table II gives $G = 0.959$, 0.867, 0.683, 0.431, 0.190 and 0.063 at $\epsilon_{\rm dec} = 0.25$, 0.5, 1, 2, 4 and 8. He notes that linear theory gives estimates of opposite sign and "does not determine which diagnostic, if any, predicts the nonlinear threshold" [1, Sec. IV C].

Two questions follow and are answered here. What in the neutrino transport operator carries the effect (Section 2)? And which clock, and which part of the decoupling history, sets G (Sections 3–5)? Version 6 adds three more. Which hidden directions does the future read, and must a reduced state keep them (Section 8)? Does hidden angular structure move the threshold itself, in the full collapse (Section 9)? And what is McCormick's physical collision model in the language of the flux–shear link (Section 10)?

2 The collision operator as a cut

For one Fourier mode k, the linearised neutrino intensity in McCormick's relaxation model evolves under

$$\partial_t f = Lf, \qquad L = A + D, \qquad A = -ik\mu, \qquad D = -\kappa(I - P), \tag{2}$$

where μ is the cosine of the angle to the wave vector and P projects onto the isotropic part. Free streaming A is anti-Hermitian. Collisions D are Hermitian, conserve the isotropic part (the neutrino energy in the plasma frame) and remove the flux, which goes to the plasma. In the normalised Legendre basis $|\ell\rangle$, streaming couples only neighbours, $\langle \ell+1|\mu|\ell\rangle = c_\ell = (\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}$, so the operator lives on a chain $\ell = 0, 1, 2, \ldots$ (Figure 1).

Non-normality is connection energy. For Hermitian D and anti-Hermitian A, $[L, L^*] = 2[A, D]$, and for diagonal D, $[A, D]_{ij} = A_{ij}(D_j - D_i)$. Hence

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \sum_{\text{links } \ell\to\ell+1} k^2 c_\ell^2 \, (D_{\ell+1} - D_\ell)^2. \tag{3}$$

This is the scalar case of the identity in [8]: the self-commutator energy of a transport operator is the energy of its Hermitian field failing to be parallel along the links [derived; Appendix A.1]. With one rate, only the link $0 \to 1$ has a jump, so

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \frac{\kappa^2 k^2}{3} \qquad \text{[derived; checked to } 10^{-12}\text{, I1]}. \tag{4}$$

All the non-normality of the single-rate model lies on the link between density and flux, the same channel as the momentum drag κq in McCormick's plasma equation [1, Eq. (6)].

Beyond relaxation time. Let the flux relax at κ_1 (momentum transfer) and shear and higher moments at κ_2. Then two links carry jumps:

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \frac{k^2\kappa_1^2}{3} + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2 \qquad \text{[derived; checked, I2]}. \tag{5}$$

The energy is stationary in κ_2 at $\kappa_2 = \kappa_1$. If neutrino–neutrino scattering damps shear but not flux, with rates relative to $e^\pm$ scattering of 1.5 for ν_e and 7 for $\nu_{\mu,\tau}$, the shear link carries 64% and 97.5% of the non-normality [computed, under those assumed ratios].

The weight across the cut is the sound speed. With $H = k\mu$ the weight between kept and streaming parts is $E_B = \|PH(I-P)\|^2 = k^2 c_0^2 = k^2/3$, so $\sqrt{E_B} = k/\sqrt{3} = c_s k$, the radiation sound speed [9] [derived; I3]. A rate bound built on E_B therefore sees the root-mean-square streaming speed $1/\sqrt{3}$, not the light-speed front that empties the overdensity [10].

The kept sector, exactly. The isotropic part has the closed propagator

$$\langle 0|(s - L)^{-1}|0\rangle = \frac{g}{1 - \kappa g}, \qquad g = \frac{1}{k}\arctan\frac{k}{s + \kappa} \qquad \text{[derived; checked, I4]}. \tag{6}$$

The two-moment truncation, keeping $\ell = 0, 1$, has eigenvalues $-\kappa/2 \pm \sqrt{\kappa^2/4 - k^2/3}$, which merge at $\kappa = 2k/\sqrt{3}$ [derived; I5]. This exceptional point belongs to the truncation. The full chain has its own wall. The propagator is the charge-diffusion channel of the relaxation-time (BGK) equation,

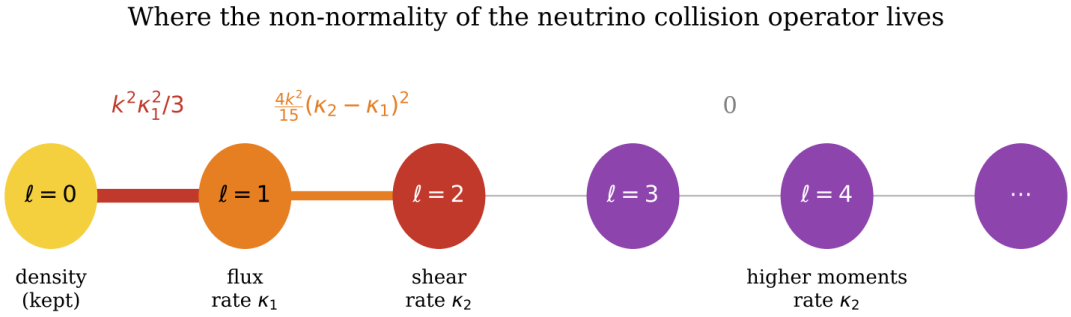


Figure 1: The neutrino moments form a chain. Streaming links each moment to its neighbours; collisions put a rate on each. Non-normality sits only where neighbouring rates differ: on the density–flux link ($k^2\kappa_1^2/3$) and, if shear relaxes at its own rate, on the flux–shear link. Every other link carries none.

and Romatschke found its hydrodynamic pole and the wall [13, Eq. (37)]; in the present variables it reads as follows. A mode is a pole, $\kappa g = 1$, so $\arctan[k/(s+\kappa)] = k/\kappa$ and

$$s = k \cot \frac{k}{\kappa} - \kappa, \tag{7}$$

and this pole exists only for $\kappa > \kappa_c = 2k/\pi \approx 0.6366\,k$ [derived; Appendix A.3a]. The mode is real all the way down; at κ_c it reaches the branch cut of g, the segment $\mathrm{Re}\, s = -\kappa$, $|\mathrm{Im}\, s| \le k$, at its centre $s = -\kappa$, and leaves the spectrum: Romatschke's hydrodynamic onset transition, $\tau_D = \pi/(2k)$ [13, 14]. Read here, it is the edge of the neutrino kept sector. With $\phi = k/\kappa$ the mode's non-orthogonality factor is

$$K = \frac{4\phi^2}{\sin^2 2\phi}, \tag{8}$$

which is 1 for a fully coupled gas and diverges at the onset $\phi = \pi/2$ [derived; Appendix A.3a]: the kept mode becomes infinitely non-orthogonal to its own left eigenvector before it leaves. A truncated chain approaches the onset from above: its least-damped mode turns real at $\kappa/k = 0.665$, 0.645 and 0.639 with 50, 200 and 800 moments, and at 400 moments its pole and K match the closed forms to ten and six digits (Figure 2) [computed; Appendix B.26].

3 New runs of McCormick's code

The code [2] was run unchanged except for one copied file in which the decoupling exponent n in $\kappa = H(\epsilon_{\mathrm{dec}}/\epsilon)^n$ is an input. Pure radiation, $R_\nu = 0.409$, $N = 400$ cells, $N_\mu = 32$ angle bins, $\epsilon_0 = 0.1$, as in his Table II. A reproduction run at $\delta = 0.5109$ with $n = 3$, $\epsilon_{\mathrm{dec}} = 2$ forms a black hole, inside his published bracket $(0.51074, 0.51109)$ [measured; Appendix B.4]. For the steep history, κ/H was capped at 5×10^5, the stiffness his own $\epsilon_{\mathrm{dec}} = 8$ runs reach at ϵ_0. At that cap deviations from isotropy are of relative order $H/\kappa \le 2 \times 10^{-6}$, the same estimate McCormick uses for his switch to transport [1, Sec. III].

For every test below each prediction and its kill condition were written before the runs (Appendix B.3). The threshold search runs two points per round; a run that neither collapses nor

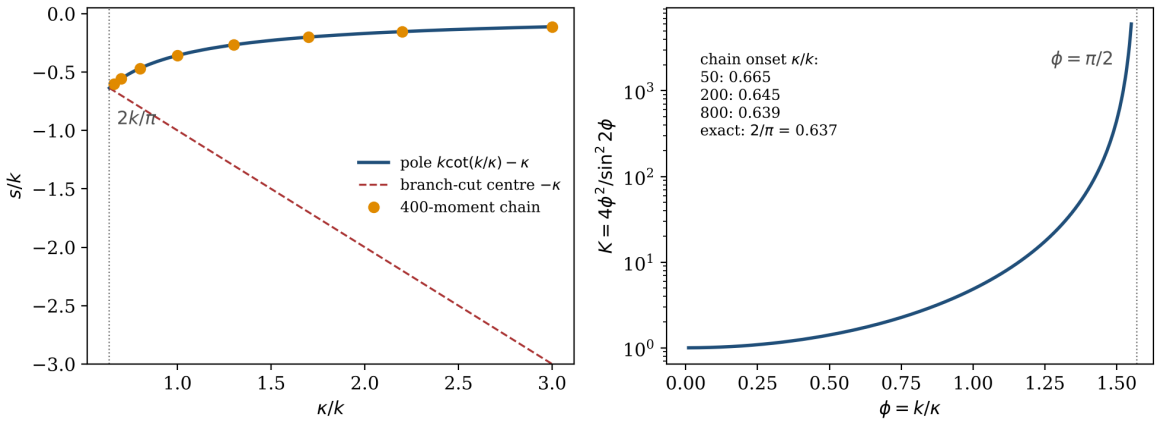


Figure 2: The wall of the full chain. Left: the kept-sector pole $s = k\cot(k/\kappa) - \kappa$ (line) and the least-damped eigenvalue of a 400-moment chain (dots); the pole meets the branch cut at its centre $s = -\kappa$ (dashed) at $\kappa = 2k/\pi$. Right: its non-orthogonality $K = 4\phi^2/\sin^2 2\phi$, with $\phi = k/\kappa$, diverging at the onset. Inset numbers: the truncated chain's onset at 50, 200 and 800 moments, closing on $2/\pi$ from above.

disperses by the end time, or that stops on an error, never moves the bracket, is retried once with a longer end time and is reported if still undecided. No point in any run below was undecided. These runs use McCormick's own end-state rule, in which a numerical breakdown after $2M/R > 0.95$ counts as collapse; Section 7 tightens that rule to a detected horizon and audits every bracket edge under it.

Test A: a profile that collapses late. The broad profile $\alpha = 0.3$ is decided near $\epsilon \approx 7$–9, against $\epsilon \approx 4$ for the Gaussian. Its thresholds are $\delta_c^{\rm fluid} = 0.44805$ and $\delta_c^{\rm inst} = 0.47747$ [2]. With $n = 3$:

	δ_c	G (broad)	G (Gaussian, [1])
$\epsilon_{\rm dec} = 4$	0.45337 ± 0.00013	0.181 ± 0.004	0.190
$\epsilon_{\rm dec} = 8$	0.44938 ± 0.00012	0.045 ± 0.004	0.063

The broad profile keeps the same share of the neutrino effect as the Gaussian or less (29% less at $\epsilon_{\rm dec} = 8$), although it collapses about twice as late (Figure 3) [measured]. A clock tied to the collapse time would have raised G; it fell. Because the test changes the profile shape together with the collapse time, it shows that the collapse time alone does not set the clock; it does not by itself separate the two, and it does not fix which earlier epoch does.

Tests B, C and D: a steeper decoupling history. Gaussian profile, $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^6$:

	δ_c	G ($n = 6$)	G ($n = 3$, [1])
C: $\epsilon_{\rm dec} = 1$	0.51756 ± 0.00019	0.648 ± 0.006	0.683
B: $\epsilon_{\rm dec} = 2$	0.50743 ± 0.00011	0.317 ± 0.004	0.431
D: $\epsilon_{\rm dec} = 3$	0.50245 ± 0.00019	0.154 ± 0.006	≈ 0.30

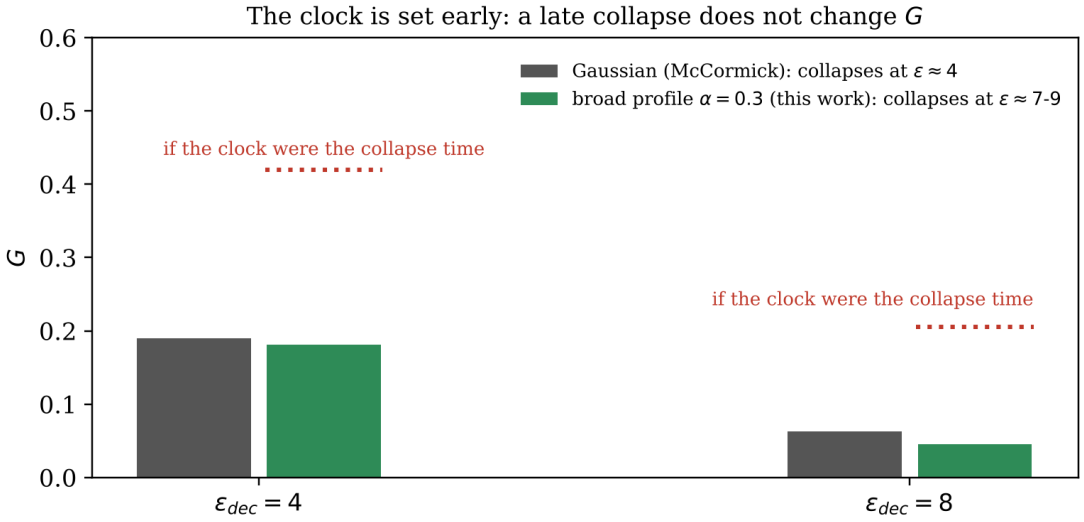


Figure 3: Test A. The broad profile collapses at $\epsilon \approx 7$–9, yet keeps the same share G of the free-streaming shift as the Gaussian. A clock tied to the collapse time would have given the dotted values.

[measured]. The $n = 3$ value at $\epsilon_{\rm dec} = 3$ lies between McCormick's 0.431 and 0.190 at 2 and 4. A neutrino population that decouples more abruptly, but by the same nominal time, keeps less of its effect, and the gap widens with $\epsilon_{\rm dec}$. The history matters, not only its end point.

Numerical controls. Test B was repeated with the rate cap raised tenfold, to 5×10^6, and on a finer grid, $N = 600$ cells and $N_\mu = 48$ angle bins. The thresholds are $\delta_c = 0.50740 \pm 0.00011$ and 0.50751 ± 0.00006, shifts of -0.00003 and $+0.00008$ against the pre-registered tolerance of 0.0005 [measured]. The cap acts only while the neutrinos are tightly coupled: the uncapped rate starts at $(\epsilon_{\rm dec}/\epsilon_0)^6$ and falls below the cap at $\epsilon \approx 0.11$, 0.22 and 0.34 for $\epsilon_{\rm dec} = 1$, 2 and 3. The cap control was run at $\epsilon_{\rm dec} = 2$ only; at $\epsilon_{\rm dec} = 1$ and 3 it is untested.

4 What does not set the threshold

Each reading below was tested against numbers. Only the last two were tested against runs set up for them in advance; the first three were tested against McCormick's published curve.

1. **The first maximum of linear theory.** The linear density contrast at its first maximum ($\epsilon \approx 1.7$) reproduces the instantaneous shift (centre factor 0.944, as McCormick finds). Under gradual decoupling the share it keeps is 0.86, 0.59, 0.19, -0.05, -0.02 and 0.00, a largest error of 0.50. Decoupling after $\epsilon \approx 1.7$ is nearly invisible to it, yet still moves the nonlinear threshold [computed].

2. **A sharp wall, and rate ratios.** Treat the two-moment exceptional point, crossed at $\epsilon_{\rm EP} = 0.811\,\epsilon_{\rm dec}^{3/4}$ for $kr_m = 2$, as a wall, or use the parameter-free escape fractions $k/(k+\kappa)$, $e^{-\kappa/k}$ and $1 - e^{-k/\kappa}$. Averaging each over two windows, with two weightings and two scales, gives 32 combinations. The wall variants miss by 0.14 or more. With the window starting at horizon entry, every combination misses by 0.17 or more. One combination, $e^{-\kappa/k}$ averaged

from ϵ_0, comes within 0.077; selected from 32, it is not counted. On Test B, which it never saw, it predicts 0.513 against 0.317 [computed; measured].

3. **Diffusive escape.** Diffusion with $D = 1/(3\sigma)$ from the conformal rate $\sigma = \epsilon_{\rm dec}^3/\tau^4$, combined in series with streaming at the collapse time ϵ^*, gives $G = 1/[1+\sqrt{15}(\epsilon_{\rm dec}/\epsilon^*)^{3/2}]$. With $\epsilon^* = 4$ it matches McCormick's curve to 0.02, but it was built after seeing that curve, and a free fit of its exponent gives 1.61 ± 0.03 rather than 3/2. Test A with $\epsilon^* = 7$–9 predicted 0.37–0.47 and 0.17–0.24, and Test B predicted 0.62. Both are excluded [measured].

4. **History-blind.** If only $\epsilon_{\rm dec}$ mattered, Tests B and D would give 0.431 and about 0.30. They give 0.317 and 0.154; Test D misses by 0.146 against a kill line of 0.08 [measured].

5 What survives: the memory of the kept sector

The reading that survives treats the isotropic part as a kept sector and asks how much of it the decoupling operator keeps [9]. Start a mode in $|0\rangle$ at $\tau_0 = 0.02$ and evolve it under $\partial_\tau v = -(ik\mu + a\kappa(\tau)(I-P))v$, with $a\kappa = \epsilon_{\rm dec}^n/\tau^{n+1}$. Let $A(\tau)$ be the population left in $|0\rangle$, averaged over the Gaussian profile with weights $k^2 e^{-k^2/2}$. Define

$$g(\epsilon^*) = \frac{1 - A_{\epsilon_{\rm dec}}(\epsilon^*)}{1 - A_{\rm inst}(\epsilon^*)}, \tag{9}$$

the share of the kept sector's loss that has happened by the readout time ϵ^*.

With no fitted number, read at the first compaction maximum $\epsilon^* = 1.7$, it gives 0.941, 0.860, 0.671, 0.344, 0.087 and 0.013 against McCormick's G. That is within 0.02 for $\epsilon_{\rm dec} \le 1$, with a largest error of 0.103, just outside the 0.10 bar set for it in advance [computed]. McCormick's own curve passes $G = \frac{1}{2}$ at $\epsilon_{\rm dec} \approx 1.67$, within 3% of the first maximum: the neutrinos must be decoupled by then for half the effect to survive.

The late shortfall says the collapse is still listening after the first maximum. Mixing the two epochs McCormick's numbers single out, $g_2 = w\, g(1.7) + (1-w)\, g(4.0)$, with one weight fitted to his curve, $w = 0.89$, gives a largest error of 0.022 [fit]. The form is not unique: with the scale fixed at $kr_m = 2$ the single-epoch reading misses by 0.093, and the best single readout epoch, $\epsilon^* = 1.87$, by 0.065. Its weight does not carry over between weightings: the same w with the scale fixed at $kr_m = 2$ misses by 0.115, outside the 0.10 bar declared for that check [computed]. The form is a fitted shape with a stated sensitivity, not a derived law.

The test is what that fitted form says about histories it never saw. All its $n = 6$ values were computed in one file dated 13:47 on 30 October, before any $n = 6$ run, and were not refitted afterwards:

	predicted	measured	miss	history-blind	miss
C: $\epsilon_{\rm dec} = 1$	0.663	0.648	0.014	0.683	0.035
B: $\epsilon_{\rm dec} = 2$	0.246	0.317	0.071	0.431	0.114
D: $\epsilon_{\rm dec} = 3$	0.118	0.154	0.037	≈ 0.30	0.146

Every miss of the memory form lies inside the kill line of 0.08 set before its run; the history-blind rule fails at $\epsilon_{\rm dec} = 3$ (Figures 4 and 5) [measured]. At $\epsilon_{\rm dec} = 2$ and 3 the form falls below the measurement: it counts slightly too much of the early erasure. Test C was set where the two rules nearly agree, as a survival check; Test D was set where they separate. A readout window set by the neutrino flux and the enclosed mass between first maximum and apparent horizon, as McCormick suggests [1, Sec. IV C], is the natural next object. It is untested.

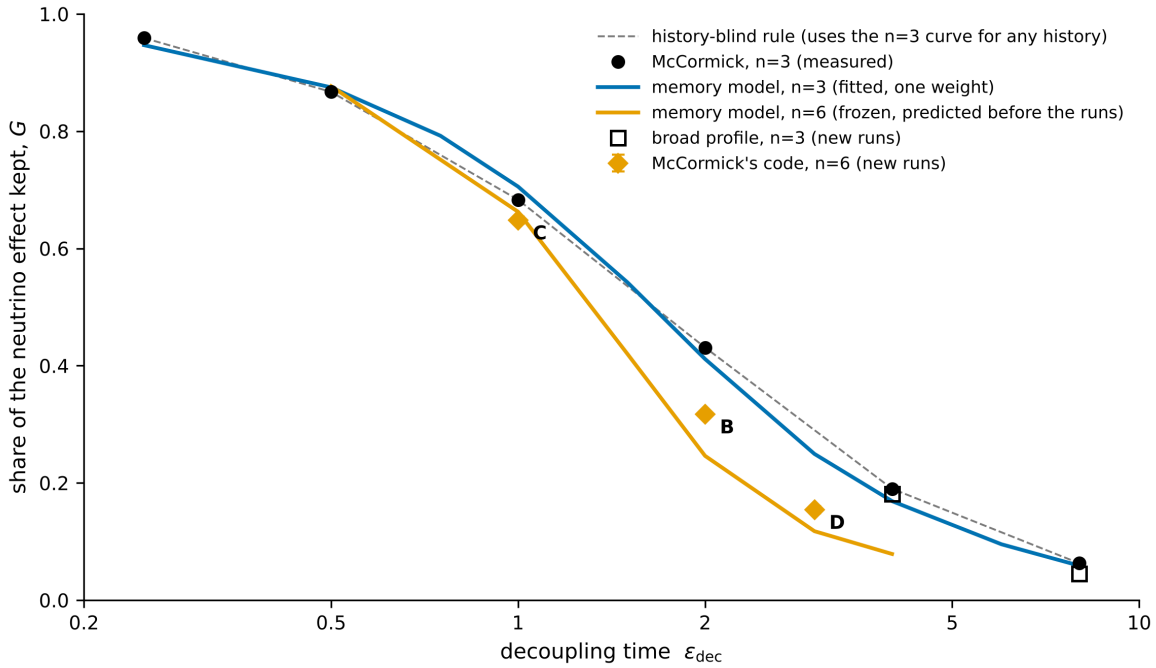


Figure 4: The surviving share G of the free-streaming shift against decoupling time. Black dots: McCormick's Gaussian runs with $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^3$; the grey dashed line through them is what a history-blind rule gives for any history. Blue: the kept-sector memory with one weight fitted to the black dots. Orange line: the same form for $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^6$, computed before any such run and not refitted. Orange diamonds: the new $n = 6$ runs C, B and D. Open squares: the broad profile (Test A).

6 Where the memory sits

A fluid closure carries the density, the flux and the anisotropic stress, the moments $\ell = 0, 1, 2$. The memory form of Section 5 says the threshold depends on how the kept sector was emptied. The question here is whether that history can sit in moments a closure does not carry.

Exact elimination of the hidden moments. Split $v = (a, h)$ with $a = v_0$ and $h = (v_1, v_2, \dots)$. In the relaxation model every hidden moment is damped at the same rate, so the collision term on the hidden block is $\sigma(\tau)$ times the identity and commutes with streaming. The hidden propagator is therefore exact in closed form,

$$U_Q(t, r) = e^{-[\Sigma(t)-\Sigma(r)]}\, e^{-ik(t-r)M_{QQ}}, \qquad \Sigma(\tau) = -\frac{\epsilon_{\rm dec}^n}{n\tau^n}, \quad \Sigma' = \sigma, \tag{10}$$

and eliminating h gives one equation for the kept amplitude,

$$\dot a(t) = S(t) - k^2 \int_s^t K(t, r)\, a(r)\, dr, \qquad K = c_0^2\, [U_Q(t, r)]_{11}, \qquad S = -ikc_0\, [U_Q(t, s)h(s)]_1 \tag{11}$$

[derived; Appendix A.5]. The kernel K is the memory; the source S is everything the hidden moments held at time s.

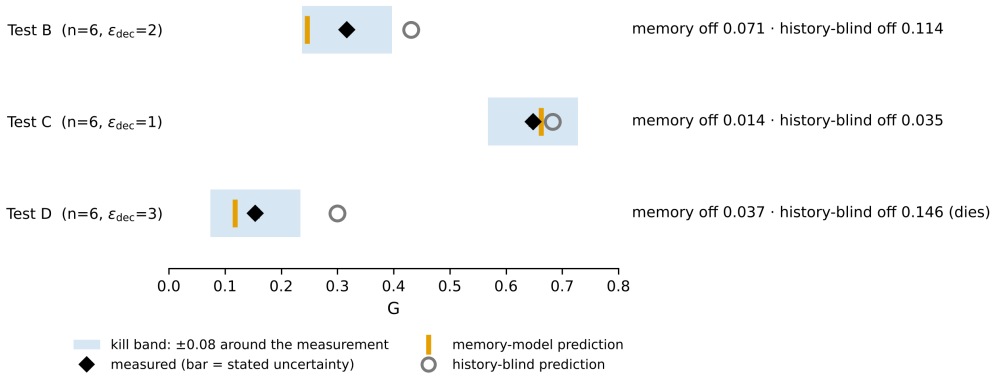


Figure 5: The three steep-history tests against their kill lines. The shaded band is ±0.08 around each measurement. Orange: the memory form's prediction, made before the run. Grey circle: the history-blind rule. At $\epsilon_{\mathrm{dec}} = 3$ the history-blind rule falls outside the band.

The matched pair. Take two states with the same $(v_0, v_1, v_2) = (1, 0.2, 0.1)$, differing only by ± 0.05 in v_3 (or v_4), scaled by one constant to unit norm. Both are positive angular distributions ($f(\mu) \geq 0.50$). Their kept amplitudes start equal, so their difference obeys the equation above driven only by S of the hidden difference. Both are evolved under the same history, $\epsilon_{\mathrm{dec}} = 0.5$, with the planting time τ_s scanned across decoupling. The readout is the profile-weighted change in (v_0, v_1, v_2), relative to its size, R_{ret}; the witness is on if $R_{\mathrm{ret}} > 10^{-3}$. Every on/off call was written down before the run, from the survival factor of the hidden block, $\exp[-\epsilon_{\mathrm{dec}}^n/(n\tau_s^n)]$, and the leakage $(k/\sigma)^\ell$ through damped rungs (Appendix B.14).

$\tau_s/\epsilon_{\mathrm{dec}}$	$n=3,\ \ell=3$	$n=3,\ \ell=4$	$n=6,\ \ell=3$	$n=6,\ \ell=4$
0.1	7×10^{-15}	5×10^{-19}	8×10^{-24}	5×10^{-31}
0.25	6×10^{-10}	6×10^{-11}	2×10^{-15}	7×10^{-20}
0.5	$\mathbf{8.6\times10^{-3}}$	$\mathbf{7.9\times10^{-3}}$	3×10^{-6}	3×10^{-6}
1	**0.081**	**0.074**	**0.094**	**0.086**
2	**0.095**	**0.085**	**0.098**	**0.088**

R_{ret} at $\tau = 4$; bold entries are on [computed]. Every call came out as predicted. The survival law is quantitative: for $n = 3$ the ratio of the entries at 0.5 and 1 is 0.106 against $e^{-8/3}/e^{-1/3} = 0.097$, and for $n = 6$ at 0.5 the law gives 1.4×10^{-6} against 1.6×10^{-6} at $\tau = 1.7$. Planted in the retained moments instead, deep in coupling the order is $\ell = 1 > 2 > 3 > 4$, each rung costing about $10^{-4} \approx k/\sigma$; after decoupling all four are of the same size. A hidden difference set at decoupling changes the later density, flux and stress as much as a visible one (Figure 6).

Checks. Doubling the number of moments, $60 \to 120$, and tightening the solver tolerances change every entry by at most 1.4×10^{-5} of its value. Solving the elimination equation from the closed-form U_Q alone, with no solve of the full system, reproduces the full difference to 3×10^{-7}–7×10^{-7}, falling fourfold with each halving of the step; leaving out the kernel misses by 19–234% [computed; Appendix B.14].

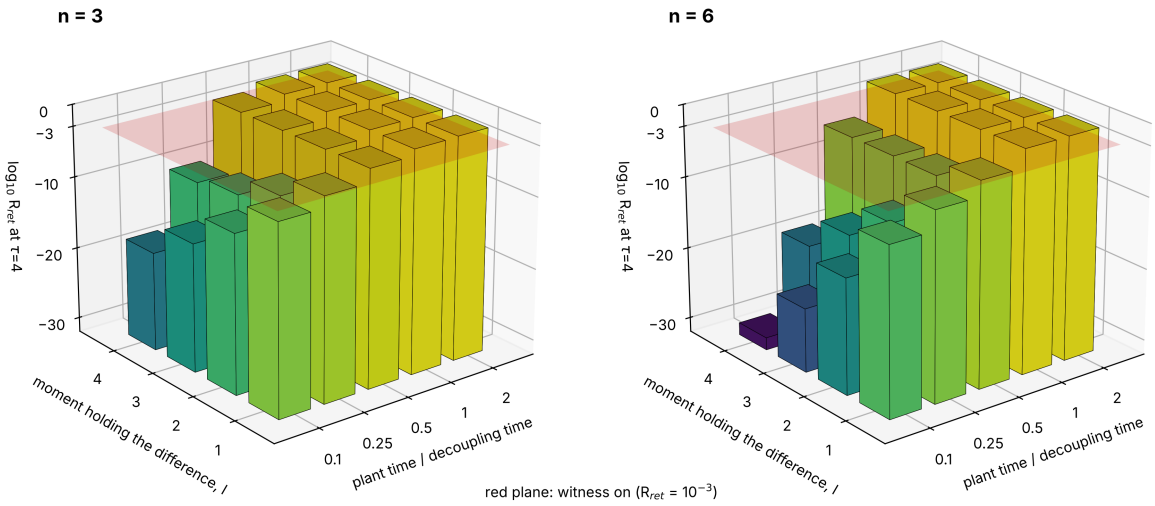


Figure 6: The door. Strength of the matched-pair witness, $\log_{10} R_{\mathrm{ret}}$ at $\tau = 4$, against the time the hidden difference was planted (in units of ϵ_{dec}) and the moment that held it. Planted deep in the coupled era the difference is erased, down to 10^{-30}; planted at or after decoupling it returns at the 10^{-1} level. The red plane is the witness threshold 10^{-3}. The steeper history ($n = 6$) closes the door more sharply.

A speed limit on the kept population. Collisions conserve $|0\rangle$ and remove the rest, so $\|v\|$ never grows, and streaming links $|0\rangle$ only to $|1\rangle$, with weight $kc_0 = \sqrt{E_B}$. Hence for $A = |v_0|^2$

$$\dot{A}^2 \le 4E_B\, A(1 - A), \qquad E_B = k^2/3 \tag{12}$$

[derived; Appendix A.6; Lean]. On every trajectory computed here the largest value of the ratio is 0.87 [computed]. The bound limits how fast the kept sector can be emptied; it does not set the threshold.

Reading. A closure on density, flux and stress is exact only for structure set while the neutrinos are tightly coupled. Structure set at or after decoupling survives in the finer angular moments and streams back into the visible ones by several per cent. The memory McCormick's runs record therefore has a place to live that no fluid closure contains. Whether realistic initial data carry such structure, and how far it moves the nonlinear threshold, are untested: the memory curves of Section 5 start with the hidden moments empty.

7 The shear sets the threshold

Section 2 put all the non-normality of a single-rate collision term on one link. A collision term with two rates has two links that carry it: density to flux, with jump κ_1, and flux to shear, with jump $\kappa_2 - \kappa_1$. The connection energy of the second link, $\frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$, is stationary at $\kappa_2 = \kappa_1$, while the exact kept response is not (Appendix A.7). Which does the nonlinear threshold follow, and which moment carries what it follows? This section answers both with McCormick's code, changed in one place, and then asks in which coordinate the answer is a law.

The change. In a copy of `ms_nuc.py` the collision term $\kappa(\varepsilon/4\pi - I)$ becomes $\kappa_2(\varepsilon/4\pi - I) + (\kappa_2 - \kappa_1)d_\mu$, where d_μ is the dipole part of the intensity, projected on the same angle bins that define the

flux (Appendix A.12). The dipole then relaxes at κ_1 and every moment $\ell \geq 2$ at κ_2; the neutrino energy is untouched, and the momentum handed to the plasma is still $\kappa_1 q$. The stiff time-step cap uses the larger rate. Write $r = \kappa_2/\kappa_1$, and keep $\kappa_1 = \kappa = H(\epsilon_{\rm dec}/\epsilon)^3$ with $\epsilon_{\rm dec} = 2$, the Gaussian profile, $R_\nu = 0.409$, $N = 400$, $N_\mu = 32$. At $r = 1$ the changed code reproduces the unchanged one: $\delta = 0.5109$ forms a black hole at $\epsilon = 6.61$ in both [measured].

Admissible ratios. On the angle bins the split collision operator is a Markov generator only for $\frac{3}{4} \leq r \leq \frac{3}{2}$ (all moments $\ell \geq 2$ at κ_2) and $0.6 \leq r \leq 1.2$ (shear alone at κ_2); outside those windows it has a negative off-diagonal entry and can, in principle, drive a nonnegative intensity negative. The pairs $r = 1.2$, $1/1.2$, 1.25 and 0.8 are inside the first window. The pairs $r = 1/2$ and 2 and the extreme pairs ($r = 1/16, 1/4, 4, 16$) are formal interventions on the solver: they are kept because they fix the sign and the size of the effect, and the minimum intensity was logged through every run of this section; it stayed positive in all of them except one run at $r = 16$, where it reached -2.4×10^{-10} [measured].

How the searches are run. Each threshold is found by trisection: two trial amplitudes per round, the bracket shrinking to its middle third. A run moves the bracket only if it ends in a detected horizon ($2M/R \geq 1$ with $U < 0$) or disperses. A run that stops on a numerical error, or that breaks down after $2M/R > 0.95$ without a detected horizon, is repeated once with the Courant numbers lowered from 0.4 to 0.3; if it fails again it is reported and never moves the bracket. Every decisive bracket edge of every threshold below was rerun under this rule: each upper edge is a detected horizon and each lower edge a dispersal [measured; Appendix B.15].

The link as a reflecting step. In the author's coordinate a link between rates d_i and d_j is one reflecting step, with Smith coordinate $\Gamma_{ij} = (d_i - d_j)/(d_i + d_j) = \tanh(\frac{1}{2} \ln d_i/d_j)$ and rapidity $\frac{1}{2} \ln(d_i/d_j)$; exchanging the ends negates Γ (Appendix A.8), and a matched link carries no connection energy [8]. For the flux–shear link, $\Gamma_{12} = (r-1)/(r+1)$. The surviving share G is itself a point on the disk, $\Gamma_G = 1 - 2G$, with offset $\tau_G = \ln[(1-G)/G]$. The question becomes: how does τ_G move with the link's rapidity?

Three readings, written down before the runs. (a) The frozen memory form of Section 5, with the collision diagonal $(0, 1, r, r, \dots)$, predicts almost no effect: $\Delta G = -0.004$ at $r = 1/4$ and $+0.003$ at $r = 4$. (b) If the connection energy controlled the threshold, the two shifts would share a sign, since the energy is even about $r = 1$. (c) If the threshold read the link's reflection coefficient, the shifts would be odd in Γ_{12}: equal and opposite.

Eleven thresholds. The reference thresholds are McCormick's, $\delta_c^{\rm fluid} = 0.49773$ and $\delta_c^{\rm inst} = 0.52832$, and $G = (\delta_c - \delta_c^{\rm fluid})/(\delta_c^{\rm inst} - \delta_c^{\rm fluid})$; the point $r = 1$ is measured here with the changed code. These eleven runs hold the collision rate fixed through each time step, as in version 5; the stage-correct reruns of the admissible points are given below:

r	Γ_{12}	$\frac{1}{2}\ln r$	δ_c	G	$\tau_G - \tau_G(1)$
1/16	-0.882	-1.386	0.521485 ± 0.000038	0.777	-1.547
1/4	-0.600	-0.693	0.516498 ± 0.000038	0.614	-0.763
1/2	-0.333	-0.347	0.513656 ± 0.000010	0.521	-0.384
1/1.2	-0.091	-0.091	0.5115145 ± 0.0000003	0.451	-0.103
0.8	-0.111	-0.112	0.5116957 ± 0.0000004	0.457	-0.127
1	0	0	0.510741 ± 0.000019	0.425	0
1.2	$+0.091$	$+0.091$	0.5100149 ± 0.0000005	0.402	$+0.098$
1.25	$+0.111$	$+0.112$	0.5098407 ± 0.0000002	0.396	$+0.122$
2	$+0.333$	$+0.347$	0.507973 ± 0.000010	0.335	$+0.385$
4	$+0.600$	$+0.693$	0.505442 ± 0.000038	0.252	$+0.786$
16	$+0.882$	$+1.386$	0.501673 ± 0.000025	0.129	$+1.610$

[measured]. At $r = 1/4$ and 4 the shifts from the measured $G(1) = 0.425$ are $+0.188$ and -0.173, so reading (a) fails by a factor of forty and with the wrong sign at $r = 4$, and reading (b) fails on the sign. Reading (c) holds in the offset, where the law is stated: -0.763 and $+0.786$, equal and opposite to 3% (Figure 7).

One line through the matched link. Read in the offset, the eleven points lie close to one line through $r = 1$ (residual 0.014 in τ_G). Each pair $r, 1/r$ splits into an odd part, $[\tau_G(r) - \tau_G(1/r)]/\ln r$, the rate at which the offset moves per unit of link rapidity, and an even part. The anchor $\tau_G(1)$ cancels from the odd part; the two reference thresholds do not, because each enters through $\ln(\delta_c - \delta_c^{\text{fluid}})$ at two different δ_c (Appendix A.9). The ranges below are hard bounds from the four corners of each pair of brackets:

$\lvert\frac{1}{2}\ln r\rvert$	0.091	0.112	0.347	0.693	1.386
odd rate	1.1006	1.1128	1.1093	1.1177	1.1384
bracket range	$[1.1000, 1.1011]$	$[1.1124, 1.1131]$	$[1.1052, 1.1134]$	$[1.1092, 1.1262]$	$[1.1332, 1.1436]$
even part	-0.003	-0.003	$+0.001$	$+0.012$	$+0.032$

[measured]. The rate is not one: a rule in which the offset of G moves by exactly the link rapidity is excluded at every pair. Inside the admissible window it lies at 1.10–1.11; it rises to 1.138 at the outermost, formal pair, where a small even part appears.

Stage-correct histories. In the runs above the collision rate $\kappa(\tau)$ was evaluated once per time step and held through the three stages of the Runge–Kutta step. Evaluating it at each stage time is the only change of Appendix B.19; it moves black-hole formation at $r = 1$, $\delta = 0.5109$, from $\epsilon = 6.6147$ to 6.6176. The admissible thresholds and the shear-only pair at $r = 1.2$ were rerun with it, each by bisection to a bracket width of at most 10^{-6} under the same endpoint rule, with predictions and kill lines written first:

r	δ_c, stage-correct	G	change
1	0.51077853	0.4266	$+3.8 \times 10^{-5}$
1.2	0.51000812	0.4014	-6.7×10^{-6}
1/1.2	0.51151594	0.4507	$+1.4 \times 10^{-6}$
1.25	0.50984158	0.3959	$+8.8 \times 10^{-7}$
0.8	0.51169233	0.4564	-3.4×10^{-6}
shear only, 1.2	0.50995463	0.3996	$+4.7 \times 10^{-7}$
shear only, 1/1.2	0.51157978	0.4528	$+5.0 \times 10^{-6}$

with half-widths 3–5 $\times 10^{-7}$ [measured; Appendix B.19]. The changes are a few 10^{-6}, but at $|\frac{1}{2}\ln r| \approx 0.1$ they move the rates. The inner odd rates become 1.1066, range $[1.1061, 1.1072]$, for $r = 1.2$ and $1/1.2$, and 1.1102, range $[1.1098, 1.1106]$, for $r = 1.25$ and 0.8: the two pairs now differ by 0.0036, against 0.012 before, so most of the earlier spread between them was the step-held collision rate and not a per-point systematic. The shear-only rate at $r = 1.2$ becomes 1.1928, range $[1.1922, 1.1933]$, still 0.086 above the all-$\ell \geq 2$ rate at the same pair. Of the move at $r = 1$, 2.2×10^{-5} is the stage fix and 1.6×10^{-5} is the distance from the earlier midpoint at tolerance 10^{-4} to the threshold; it moves $G(1)$ to 0.4266 and the even parts by about 0.003, and enters no odd rate.

The error the brackets do not show. The odd rate divides by $\ln r$, so at the inner pairs a shift of 10^{-4} in one threshold moves it by 0.06. Three systematics are larger than the brackets. The fluid reference carries $\pm 7.8 \times 10^{-5}$ in McCormick's archive, which moves every rate by ± 0.0038 together. Refining the $r = 1.25$ pair in radius, angle and time together ($N = 600$, $N_\mu = 48$, the Courant numbers and the Hubble step factor halved) raises both thresholds, by 7.5 and 6.4×10^{-5}, to 0.50991687 and 0.51175643, and lowers the rate to 1.1020, range $[1.1016, 1.1024]$: a shift of -0.0082, larger than the line of 0.005 set before the run. The angle alone moved the rate by less than 0.001 at $r = 4$ (below), so the shift comes from the radial grid or the time step; which one is untested. With the pair spread of 0.0036 the budget is 0.0097 in quadrature, 0.016 added linearly. Every central inner value, 1.1020 to 1.1102, lies in

$$\text{odd rate} = 1.11 \pm 0.01, \tag{13}$$

but the ± 0.01 is now the error budget itself, dominated by resolution, and not a margin around a converged number. A third level ($N = 900$, $N_\mu = 72$, Courant numbers and Hubble step halved again) was run with its predictions and kill line written first. At $r = 1.25$ it puts the threshold at 0.5099481 ± 0.0000063, a further rise of 3.1×10^{-5} against 7.5×10^{-5} for the previous step. The shift shrank to 0.42 of the previous one (range 0.33–0.50). Cells and angle bins shrink by 1.5 per level and the Courant-limited time step by 3. First order in the cell size predicted 0.50996706, outside the bracket; second order in the cell size predicted 0.50995033 (a factor 0.44) and first order in the time step 0.50994197 (a factor 0.33), both inside, the second at its lower edge. So the bracket excludes first order in space but does not separate second order in space from first order in time [measured; Appendix B.27]. Under either reading, if the $r = 0.8$ threshold shrinks by the same factor, the third-level rate is 1.099 and the extrapolated rate 1.094–1.098, just below the band; the $r = 0.8$ third-level run was still in progress when this version was closed, so the extrapolation is untested. Every threshold here starts at $\epsilon_0 = 0.1$, as McCormick's Table II; his runs started at $\epsilon_0 = 0.05$ and 0.025 move the reference thresholds by about 4×10^{-4} [2], and how the odd rate depends on the start time is untested. The rate is a measured number with no closed form attached [measured].

Which moment carries it. Damp only the shear, $\ell = 2$, at κ_2, and keep the flux and every $\ell \geq 3$ at κ_1: the collision term becomes $\kappa_1(\varepsilon/4\pi - I) + (\kappa_1 - \kappa_2)s_\mu$, with s_μ the $\ell = 2$ projection on the same angle bins (Appendix A.12). Now two links jump, flux to shear by $+\frac{1}{2}\ln r$ and shear to the next rung by $-\frac{1}{2}\ln r$. A rule in which the offset moves by a sum of link-weight ratios times link rapidities then nearly cancels, since c_1 and c_2 differ by 1.8%, and predicts $|\text{rate}| < 0.1$; a threshold that reads the shear damping itself predicts a rate near the one above. Both predictions and their kill lines were written before the runs, first at the extreme pair and then, after the admissibility window was found, inside it. At $r = 1$ the changed code again reproduces the unchanged one.

shear only	δ_c	G	odd rate	all $\ell \geq 2$, same pair
$r = 1.2$	0.5099542 ± 0.0000002	0.400	1.1895, range $[1.1892, 1.1898]$	1.1006
$r = 1/1.2$	0.5115748 ± 0.0000002	0.453		
$r = 4$	0.5052350 ± 0.0000037	0.245	1.2038, range $[1.2030, 1.2046]$	1.1177
$r = 1/4$	0.5170942 ± 0.0000037	0.633		

[measured]. The link-weight reading fails by an order of magnitude at both pairs; the shear reading holds at both. Damping the shear alone moves the threshold more than damping the shear and every higher moment together, by 0.09 in the rate at either pair: the moments above the shear, relaxed with it, pull the other way. With stage-correct histories the $r = 1.2$ pair gives 1.1928 against 1.1066 (above).

Angular resolution. Repeating the $r = 4$ and $1/4$ pair with 48 angle bins instead of 32, set before the run to pass if the odd rate moved by less than 0.01 and each threshold by less than 10^{-4}, gives $\delta_c = 0.5054959 \pm 0.0000082$ and 0.5165576 ± 0.0000041: both thresholds rise, by 5.3 and 6.0×10^{-5}, and the odd rate moves from 1.1177 to 1.1170 (range $[1.1156, 1.1184]$). The resolution shift is common to r and $1/r$ and cancels from the odd rate to below 0.001; it moves G itself by about 0.002 [measured].

Two small factors, checked. The reference point for $r = 1$ is measured here with the changed code at tolerance 10^{-4}: $G = 0.4253 \pm 0.0006$, at the lower edge of McCormick's bracket $[0.4253, 0.4367]$, whose midpoint 0.431 would shift every prediction that uses it by about 0.006. Only the $r > 1$ runs shrink the stiff time step, by the factor r; rerunning $r = 1/4$ with its step shrunk fourfold leaves δ_c unchanged to 3×10^{-6}, so the slight asymmetry between the two sides is not numerical [measured].

The neutrino fraction. The kept-sector operator contains no R_ν. With $R_\nu = 0.2$, $n = 3$, $\epsilon_{\rm dec} = 2$ and McCormick's instantaneous threshold at that fraction, 0.51297, the threshold is 0.50396 ± 0.00003, so $G = 0.409 \pm 0.005$. The memory form had predicted 0.411; against 0.425 measured here at $R_\nu = 0.409$ the difference is 0.016 [measured]. The share is nearly, not exactly, independent of how much of the radiation is neutrinos.

Which coordinate the law lives in. The rapidity $\frac{1}{2}\ln(\kappa_2/\kappa_1)$ is a coordinate on the parameters of the collision operator. In the author's framework [21] a parameter chart does not generate the geometry of a kept subspace; the intrinsic measure of how far the kept sector moved when r changed is the distance of its state in the moment chain. That distance was computed for the same histories, as the principal angle at the first compaction maximum between the kept-sector trajectory at ratio r and at $r = 1$ (Section 5's chain, profile-averaged), with the readings and kill lines written first: if the threshold read that distance, the all-$\ell \geq 2$ and shear-only families would fall on one line in it, and the rise of the odd rate with $|\frac{1}{2}\ln r|$ would flatten. Neither happens: the two families' slopes in the state angle differ by 5% (they differ by 6% in the rapidity), the odd rates spread by 36% against 3.4%, and the eleven offsets are far less linear in the angle (residual 0.13) than in the rapidity (0.014). The two collision models move the linear kept-sector state at readout by almost the same angle (0.0475 against 0.0467 at $r = 4$) while their thresholds differ by 8%. The eigenprojector of the least-damped mode is not available as an alternative: at the readout rate $\kappa = 0.958$ that mode exists only for $k < \kappa\pi/2 = 1.50$ (Section 2), which carries 51% of the profile

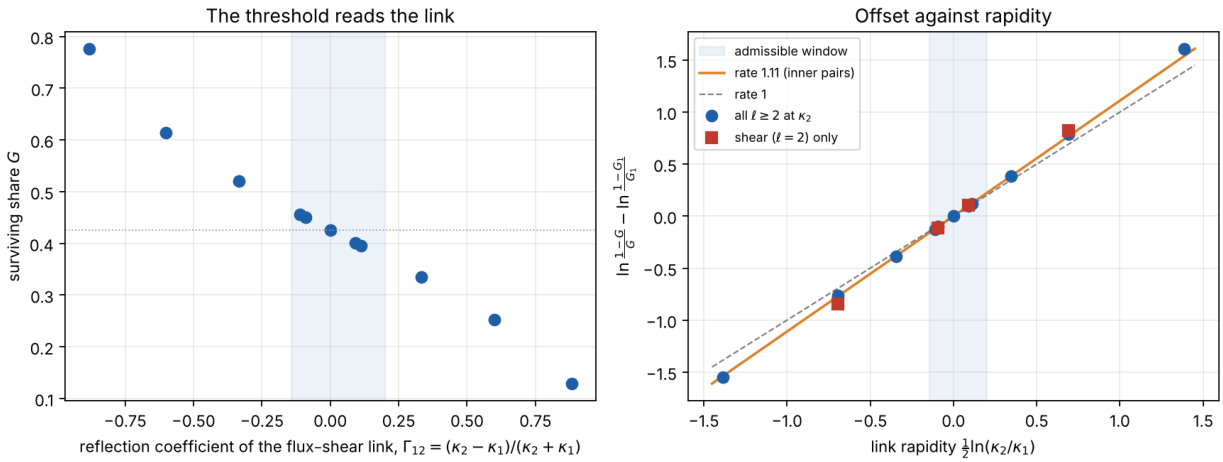


Figure 7: The flux–shear link. Left: the surviving share G against the link's reflection coefficient Γ_{12}, eleven thresholds from McCormick's code with the shear and higher moments relaxing at r times the flux rate. Right: the offset of G against the link rapidity $\frac{1}{2}\ln r$, with the shear-only pairs (squares); the shaded band is the admissible window $\frac{3}{4} \le r \le \frac{3}{2}$. Lines: rate one (dashed) and the inner-pair rate 1.11.

weight on the 36 nodes used here; for the other half the pole has met the branch cut and there is no isolated kept eigenvalue. So the law is a law in the operator's own link rapidity, and not in the displacement of the linear kept state: the difference between the two collision models is written after the readout epoch, in the nonlinear collapse [measured; Appendix B.17].

Reading. The anisotropic stress carries the threshold. Damping the shear faster than the flux keeps the stress from building and makes the collapse more fluid-like; damping it slower lets the stress the free-streaming effect is made of survive. The threshold reads the direction of that mismatch, in the rapidity of the collision operator's flux–shear link, and not how much energy sits on the link. That is the place Section 6 found the memory: in the angular structure a fluid closure drops. The kept-sector memory form, which counts only what the density keeps, cannot see it, and neither can the linear kept state at the first maximum [H].

An independent check of a 3 November review. An outside review of the companion paper reported two further thresholds on McCormick's code: a step in which collisions stop for good at $\epsilon_b = 1$, at $\delta_c = 0.513512$, and $n = 3$, $\epsilon_{\rm dec} = 1.20$, at 0.516797. Here 0.5132 and 0.5138 bracket the first, and 0.5165 and 0.5171 the second, each dispersing below and forming a black hole above [measured].

8 Which hidden directions the future reads

Section 6 showed that some hidden angular structure returns into density, flux and stress. This section asks which, and how much, in the same linear chain, and what that means for any reduced description that keeps only the low moments.

The predictive quotient and its weight. For a linear evolution $\dot{v} = L(\tau)v$ with readout R, the part of the state no readout can ever see is $N_\infty = \{v : RU(\tau, \tau_s)v = 0 \text{ for all } \tau\}$, and the minimal summary that predicts the readout is the quotient V/N_∞ (the predictive quotient of VOTKP, Theorem 3.1 [6], and of Dirac Time, Definition 1.1 [7]). Give the hidden sector an embedding Q (here the moments $\ell \geq 3$), the readout R onto $\ell \leq 2$, and a horizon T. The readout Gramian of the hidden sector,

$$\mathcal{G}_T = \int_{\tau_s}^{T} Q^* U(\tau, \tau_s)^* R^* R U(\tau, \tau_s) Q \, d\tau, \tag{14}$$

weighs each hidden direction h by its integrated squared output, $h^* \mathcal{G}_T h$. Its kernel, as $T \to \infty$, is $N_\infty \cap \operatorname{Ran} Q$. For the neutrino chain that kernel is zero: the entry $(RB^j)_{2,2+j} = k^j c_2 \cdots c_{j+1}$ of the observability matrix is nonzero for every j, so every hidden direction is eventually seen (Appendix A.14). The predictive quotient of this chain is therefore the whole hidden space, and what carries information is the spectrum of $\mathcal{G}_T$: the directions the future reads strongly and the ones it barely reads [derived]. The Gramian is the standard observability Gramian of control theory, restricted to the hidden sector; what is used here is its meaning for the predictive quotient and its use in choosing what a reduced state must keep.

The door, weighed. For McCormick's history $n = 3$, $\epsilon_{\text{dec}} = 2$ and the readout to $\epsilon = 4$, profile-summed over k and averaged over $\pm k$ (which makes its eigenvectors pure-parity), the top three eigenvalues are 1.1×10^{-7}, 9×10^{-12}, 10^{-15} for structure planted in the coupled era ($\epsilon_s = 0.5$), 0.367, 0.148, 0.040 at decoupling ($\epsilon_s = 2$), and 0.147, 0.024, 0.0016 in free streaming ($\epsilon_s = 3$), the last on a shorter horizon [measured; Figure 8]. The leading direction is always odd and led by $\ell = 3$ (weight 0.92–0.99, the rest mostly $\ell = 5$); the second is always even and led by $\ell = 4$. Halving the time step changes the leading eigenvalues by at most 3×10^{-6}; raising the chain to $L = 90$ changes nothing at double precision. The coupled-era suppression is not the survival factor alone: for this history it is $1.1 \times 10^{-7}/0.367 = 3 \times 10^{-7}$ against the squared survival factor e^{-42} (and 2.5×10^{-9} against the same e^{-42} for $\epsilon_{\text{dec}} = 0.5$), because the damped rungs still leak through the conserved $\ell = 0$ at order $(k/\sigma)^\ell$; the snapshot Gramian at the end of the horizon, which is the witness of Section 6, follows the survival factor to within a factor of ten for the $\epsilon_{\text{dec}} = 0.5$ history, where it was measured.

The direction to plant. Among hidden perturbations that keep the intensity positive on McCormick's 32 angle bins, the one with the largest readout at the positivity edge is odd, with weights $\ell = 3$: 0.851, $\ell = 5$: 0.117, $\ell = 7$: 0.010, and gives 2.47 times the readout of pure p_3 on the decoupling surface; pure p_4 gives 0.40 [measured]. Its shape is set by positivity; its parity and its $\ell = 3$ lead are set by the Gramian. Section 9 plants it in the full collapse.

Order, with the opacity held fixed. Take two collision histories with the same integrated opacity $\int \sigma \, d\tau$ over a window and the rate reversed in time inside it. The optical depth is equal by construction; the time-integrated connection energy $\int \frac{k^2}{3} \sigma^2 \, d\tau$ is equal, because it does not see order; and if the window closes before the readout, the operator at readout is equal too. The exact propagated readout is not. For six such pairs (single-rate, $n = 3$ and 6, and the split operator at $r = 4$ and 1/4), the density, flux and stress at readout differ by 33–72%, and the memory form g_2 by 0.06–0.16, at or past the 0.08 kill line of Section 5 [measured; Figure 8]. The rate falling late keeps more of the kept population than the rate falling early, in every pair whose window closes before the readout. Matching optical depth between an $n = 3$ and an $n = 6$ history over $[0.5, 4]$ still leaves g differing by up to 0.74. No scalar summary of the history, whether opacity, integrated

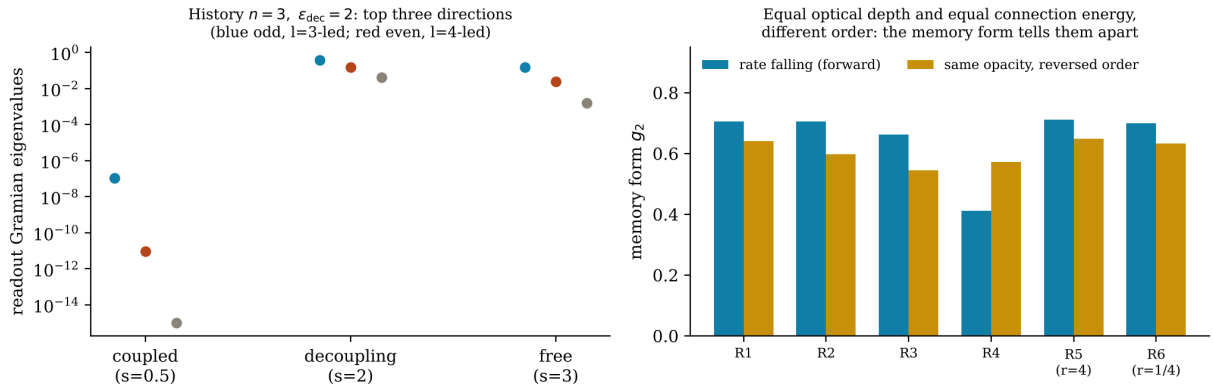


Figure 8: Left: eigenvalues of the readout Gramian of the hidden moments $\ell \geq 3$, for structure planted in the coupled era, at decoupling and in free streaming ($n = 3$, $\epsilon_{\rm dec} = 2$, readout to $\epsilon = 4$). Right: the memory form g_2 for pairs of collision histories with equal optical depth and equal integrated connection energy, the rate reversed in time inside a window; R4 is the one pair whose window contains the readout.

connection energy or the operator at readout, can carry what the threshold reads; the order is part of the record. Whether the threshold itself separates such a pair is untested.

Closures, at linear order. Against the exact chain, over the nine histories of Section 3, the profile-weighted relative error of the density, flux and stress trajectories to $\epsilon = 4$ falls from P_1 to P_5 in every history, and is worst for the earliest decoupling ($\epsilon_{\rm dec} = 0.25$): 0.79 for P_1, 0.50 for P_2, 0.28 for P_3, 0.071 for P_5. Entropy closures add nothing at this order: linearised M_1 and M_2 are exactly P_1 and P_2. The viscous (Chapman–Enskog) closure is worst everywhere (1.39 at the earliest history). A short-memory correction of the optimal-prediction type [5] beats P_2 in all nine. The exact elimination of the moments $\ell \geq 3$ (Appendix A.5 with the cut at $\ell = 2|3$) is exact to 4.6×10^{-6}; freezing its kernel to one history and reusing it costs up to 0.13 [measured]. At the first compaction maximum ($\epsilon^* = 1.7$) every truncation from P_2 upward gives the memory form to within 10^{-3}; by $\epsilon^* = 4$ they miss it by up to 0.26. The memory is in the history of the hidden block, and an instantaneous closure on the kept moments cannot hold it.

9 The door in the full collapse

The witness of Section 6 lives in the linear chain. The question it leaves is whether hidden angular structure that density, flux and stress do not carry can move the black-hole threshold itself, in the full general-relativistic collapse. This section answers it with McCormick's code.

The construction. Run the gradual-decoupling Gaussian case ($R_\nu = 0.409$, $n = 3$, $\epsilon_{\rm dec} = 2$, $N = 400$, $N_\mu = 32$, one collision rate evaluated at every Runge–Kutta stage) to a restart surface ϵ_s, landing on it exactly, and save the whole state. In every cell change only the hidden part of the angular intensity:

$$\delta I_{ik} = f\, b(r_i)\, m_i\, \frac{\varepsilon_i}{4\pi}\, p_\ell(\mu_k), \tag{15}$$

where p_ℓ is the discrete orthogonal polynomial on the 32 angle bins (orthogonal to $1, \mu, \mu^2$ to 2×10^{-17}), b a smooth bump that vanishes identically for $r \leq 0.3 r_m$ and $r \geq 2 r_m$, m_i the cell's

positivity margin, and $f = \pm\frac{1}{2}$. The bump keeps the perturbation regular at the centre, where a regular ℓ-th moment must vanish like r^ℓ; a perturbation uniform in radius is irregular there, and its first test moved the central neutrino density by 91%, so it was dropped as a verdict tool. Between the $+f$ and $-f$ states, at all three restart surfaces, the discrete $\ell = 0, 1, 2$ components agree to 4.4×10^{-16} of c_0; the energy, radial pressure and mass to 9×10^{-16}; the flux to 10^{-16} of the energy; the lapse to 2×10^{-15}; the constraint $\Gamma^2 - (1 + U^2 - 2M/R)$ to 7×10^{-15}; the fluid and metric arrays are bit-identical. The hidden component differs by 0.08–0.17 of c_0 [measured; Appendix A.15]. To any fluid description, and to every quantity that enters Einstein's equations at the restart, the two states are the same state.

The readings, written down first. The door of Section 6 predicts that structure planted at decoupling ($\epsilon_s = 2$) moves the threshold and structure planted deep in the coupled era ($\epsilon_s = 0.5$) does not. The opposite reading is that the nonlinear collapse washes out $\ell \geq 3$ whatever the timing. A third restart surface, $\epsilon_s = 4$, in free streaming, was added before any threshold run. A difference counts as real only if the two brackets, each widened by twice a measured resolution drift, stay apart; the drift, measured by rerunning the decoupling $\ell = 3$ pair with 48 angle bins, is $d_K = 1.65 \times 10^{-4}$. No sign was predicted.

Eight thresholds. The unperturbed restart at $\epsilon_s = 2$ gives $\delta_c = 0.5107500 \pm 0.0000093$, within one bracket of the $r = 1$ threshold of Section 7. Each threshold below is found by trisection under the endpoint rule of Section 7:

ϵ_s	hidden direction	$\delta_c(+f)$	$\delta_c(-f)$	difference
0.5	$\ell = 3$	0.5107685	0.5107510	$+1.75 \times 10^{-5}$
0.5	$\ell = 4$	0.5107685	0.5107500	$+1.85 \times 10^{-5}$
2	$\ell = 3$	0.5125833	0.5088395	$\mathbf{+3.744 \times 10^{-3}}$
2	$\ell = 3$, $N_\mu = 48$	0.5125134	0.5090041	$\mathbf{+3.509 \times 10^{-3}}$
2	$\ell = 4$	0.5096749	0.5117726	$\mathbf{-2.098 \times 10^{-3}}$
2	Gramian direction	0.5134352	0.5079300	$\mathbf{+5.505 \times 10^{-3}}$
4	$\ell = 3$	0.5108796	0.5106358	$+2.44 \times 10^{-4}$
4	$\ell = 4$	0.5106759	0.5108241	-1.48×10^{-4}

Bracket widths are 0.8–1.85×10^{-5} [measured; Figure 9]. At decoupling all three directions are real: the differences are 202, 204 and 297 bracket widths and survive the drift allowance with room. In the coupled era both agree within two bracket widths, as predicted. In free streaming the raw differences are 13 and 8 bracket widths with the decoupling signs, but they fall inside the drift allowance; by the rule written first they are not called real. The opposite reading is excluded by the decoupling pairs.

What the collapse reads. Odd hidden structure oriented like $+p_3$ raises the threshold, by $+1.83 \times 10^{-3}$ from the unperturbed restart, and $-p_3$ lowers it by 1.91×10^{-3}: the response is close to odd in the amplitude. Even structure works the other way: $+p_4$ lowers the threshold, at 0.56 of the p_3 size (the linear Gramian gave 0.40). The Gramian direction of Section 8 moves the threshold 1.47 times as much as pure p_3 at the same positivity margin, in the same direction: the predictive-quotient ranking of hidden directions carries into the nonlinear collapse, at about 60% of its linear size. For scale, the decoupling difference of 3.5×10^{-3} is 11% of McCormick's whole free-streaming shift, $0.49773 \to 0.52832$, and ten times the flux–shear link effect in his physical model (Section 10).

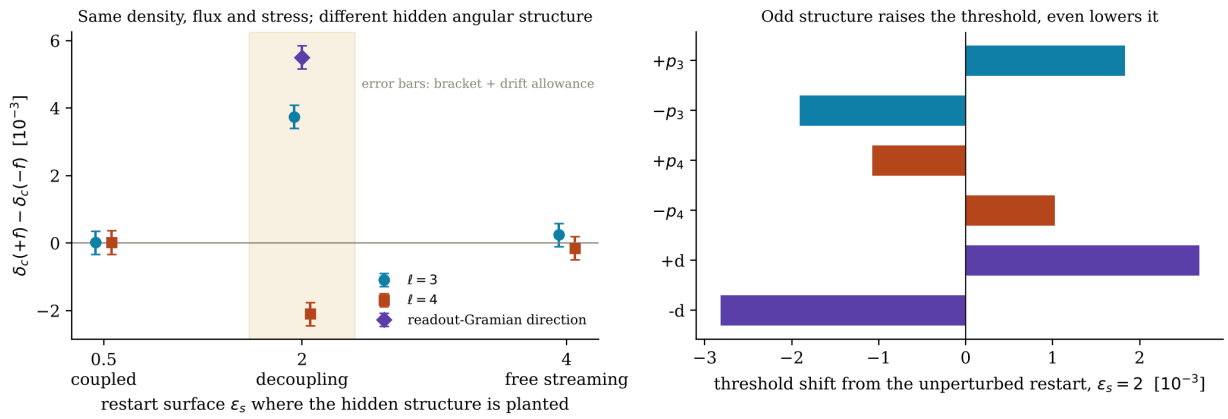


Figure 9: The door in the full collapse. Left: the difference in black-hole threshold between two neutrino states with identical density, flux, stress, energy, radial pressure, lapse and constraints, differing only in hidden angular structure, against the epoch at which the structure is planted; error bars are bracket half-widths plus twice the measured angular drift. Right: the threshold shifts from the unperturbed restart at decoupling.

The coupled era is not perfectly shut. At $\epsilon_s = 0.5$ the paired runs still separate in compaction by 10^5 times their noise floor, and the separation does not fall with angular refinement (48 against 32 bins: ratio 0.86–0.89, where an angular-grid leak would fall to about 0.44). It is a channel of the general-relativistic solver, and it carries about 0.3–0.5% of the decoupling effect, where the linear survival factor gives 5×10^{-10}. The witness chain has no such channel; a candidate, untested, is the fluid-frame shear and acceleration terms, which couple $\ell = 3$ to $\ell = 2$ at a rate set by the shear of the flow rather than by k, so that the coupled-era suppression is about S/κ per rung. Its dependence on radial resolution is untested.

Reading. The door is a window centred on decoupling: shut in the coupled era to within a fraction of a per cent, open at decoupling, narrowing again in free streaming because structure planted late has too little time to stream back into density and stress before the collapse is decided, near $\epsilon \approx 4$–6. The black-hole threshold reads angular information that no closure on density, flux and stress holds. This is an engineered sensitivity, at half each cell's positivity margin, on one profile: whether cosmological initial data carry such structure at decoupling is a separate question, not answered here [H].

10 In McCormick's physical model

McCormick's current calculation [3] replaces the single collision rate by six energy groups per flavour, with weak rates for plasma scattering and for neutrino self-scattering. This section reads that model in the two-rate language of Section 7, and runs the one simulation that decides what self-scattering does.

His collision term is the flux–shear link, exactly, with his archived target. For one energy group the collision term of his Eq. (C3), with his default target, isotropy in each group's own frame to first order, is $A\kappa_e(\varepsilon/4\pi - I) + A\kappa_n(\varepsilon/4\pi + c\mu - I)$, with c the dipole coefficient fixed

by the group's flux. On his 32 angle bins, with P_0 and P_1 the projections onto 1 and μ (orthogonal because the bins are symmetric), it is

$$C = -\kappa_e(I - P_0) - \kappa_n(I - P_0 - P_1) = -\operatorname{diag}(0,\ \kappa_e,\ \kappa_e + \kappa_n,\ \kappa_e + \kappa_n,\ \dots) \tag{16}$$

in the discrete orthonormal Legendre basis: plasma drag on the dipole, self-scattering on the shear and above. This is the split link with $\kappa_1 = \kappa_e$, $\kappa_2 = \kappa_e + \kappa_n$, $r = 1 + \kappa_n/\kappa_e$, and its connection energy is $k^2(\kappa_e^2/3 + 4\kappa_n^2/15)$ in the continuum (Appendix A.13). Taken straight from his compiled right-hand side, the operator matches the diagonal to 3.6×10^{-15}, its κ_n part to 4.9×10^{-15}; on the bins the two weights are $c_0^2 = \frac{1}{3} - \Delta\mu^2/12$ and $c_1^2 = \frac{4}{15}(1 - \Delta\mu^2)$, verified to machine precision at $N_\mu = 32$, 64, 128 and 1024 [derived; measured]. Only the plasma term hands momentum to the plasma. The identity is for this default target, the one behind his archived thresholds, which keeps each group's own flux. His boosted option, an isotropic distribution in one neutrino rest frame shared by all groups, $I \propto (1 - v\mu)^{-4}$ [2], ties every group's dipole to a common velocity and is outside the identity; it is the positive common-frame target used in the recorder runs below.

Where his rates sit. Along his background, $r = 1 + \kappa_n/\kappa_e$ is nearly constant in time and energy: 2.43–2.45 for ν_e and 7.69–7.77 for $\nu_{\mu,\tau}$ between $T_b = 0.3$ and 5 MeV. Inside the collapse, recorded at $r = r_m$ through the readout epoch $1.7 \le \epsilon \le 4$, the dominant group has $r = 7.9$–8.6 at $T_H = 1\,\mathrm{MeV}$ and 8.2–12.2 at $T_H = 0.51\,\mathrm{MeV}$, link rapidity 1.03–1.25, against the admissible window's upper edge 0.20 [measured]. His physical model lives far outside the window where the rate 1.11 was measured, and at readout both rates are below H ($\kappa_e/H \approx 0.02$–0.04, $\kappa_n/H \approx 0.16$–0.27 at $T_H = 1\,\mathrm{MeV}$): a large link rapidity on a weak link.

What self-scattering does. His archived thresholds at $T_H = 1\,\mathrm{MeV}$ are 0.519406 ± 0.000156 with self-scattering and 0.526391 ± 0.000109 without; the perfect fluid gives 0.49742. The instantaneous free-streaming threshold at that mass with the full equation of state was not in his archive; with his code it is $\delta_c^{\mathrm{inst}} = 0.53313 \pm 0.00012$ [measured]. Self-scattering does two things at once: it adds κ_n to the total isotropy rate, so the neutrinos stay isotropic longer, and it splits the flux from the shear. Two readings were written before the deciding run. In the timing reading only the isotropy history matters; with the flux–shear split removed and the isotropy history unchanged the threshold stays at 0.5194. In the link reading the frozen law of Section 7, with the energy-weighted link rapidity $u_{\mathrm{eff}} = 0.831$, places the split-free threshold near 0.5204 ± 0.0003.

The deciding run. In a copy of his solver the self-scattering term targets isotropy in the plasma frame, $\kappa_n(\varepsilon/4\pi - I)$, and its momentum $\kappa_n q$ is handed to the plasma: the flux and every higher moment then relax at $\kappa_e + \kappa_n$ ($r = 1$), and the total isotropy history is unchanged. On one frozen group the operator is $-(\kappa_e + \kappa_n)\operatorname{diag}(0, 1, 1, \dots)$ to 5.3×10^{-15}, energy is conserved to 1.5×10^{-16}, and the momentum lost by the neutrinos equals the momentum the plasma receives to 1.5×10^{-14}; with $\kappa_n = 0$ it reproduces his code bit for bit to $\epsilon = 0.6$. The bisection ($T_0 = 10\,\mathrm{MeV}$, six groups, $N = 300$, $N_\mu = 32$) gives

$$\delta_c(r = 1) = 0.51972 \pm 0.00009, \tag{17}$$

above his self-scattering threshold by $+0.00031 \pm 0.00018$, and strictly above it, since his 0.5195625 forms a black hole and here 0.519625 disperses [measured; Figure 10]. Against the lines written first this is neither reading: below the link line, just above the timing band. Most of what self-scattering does to the threshold is timing; the flux–shear split carries a real remainder, $+0.0003$ in δ_c, about a third of what the frozen law predicted and 4.5% of the whole self-scattering shift in G. The law

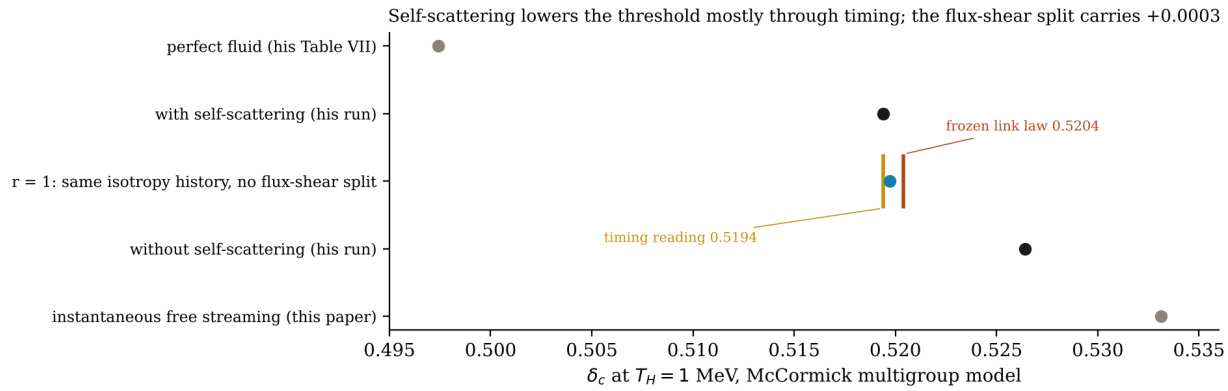


Figure 10: Thresholds at $T_H = 1\,\mathrm{MeV}$ in McCormick's multigroup model. Removing the flux–shear split while keeping the isotropy history ($r = 1$) moves the threshold by +0.0003 from his self-scattering run; the timing reading predicted no move, the frozen link law +0.0010.

measured inside $\frac{3}{4} \le r \le \frac{3}{2}$ does not carry linearly to $r \approx 8$; there the link's effect on the threshold is weaker than the straight line. The size of the shift alone could not have told the readings apart: with the measured δ_c^{inst} the frozen law at rate 1.11 and the timing reading both land inside their kill lines on his self-scattering threshold.

The kept share leads. With a read-only recorder in his multigroup solver (the intensities identical bit for bit to an unchanged run), his archived threshold edges were run with the positive common-frame target, at $T_H = 1$ and $0.51\,\mathrm{MeV}$. Both pairs still split, so with that target the thresholds lie in $[0.51925, 0.51956]$ and $[0.52475, 0.52513]$. At $r = r_m$ the kept-share anisotropy $1 - A$, with $A = I_0^2/\|I\|^2$ on the bins, separates the collapsing run from the dispersing run first, before $\epsilon = 3$ at both masses: about 0.85 in ϵ before the peak compaction does and 1.2–1.3 before the energy-frame velocity of the neutrino gas, $v_L = 2q/[e + P_r + \sqrt{(e + P_r)^2 - 4q^2}]$ [measured; Figure 11]. The gas frame peaks at $v_L \approx 0.11$ ($T_H = 1\,\mathrm{MeV}$) and 0.16 ($0.51\,\mathrm{MeV}$) near $\epsilon \approx 3.5$ and turns to infall after $\epsilon \approx 5.3$. Part of the kept share's lead may be that it is second order in the anisotropy; whether the lead is causal is what Section 9 tests, in the single-rate model.

11 Tests for McCormick's code

1. **The annihilation control.** During $e^{\pm}$ annihilation the plasma sound speed falls to $c_s = 0.519$ at $T = 0.131\,\mathrm{MeV}$, while the neutrinos' root-mean-square speed stays $1/\sqrt{3}$. The contest between front and sound, $f = (1 - c_s)/(1 - 1/\sqrt{3})$, reaches 1.135 at $1.6 \times 10^7\,M_\odot$ ($c_s = 0.521$ there). If the softer plasma enhanced the free-streaming effect, the shift would reach 7.5%, outside McCormick's envelope of 5.3–6.2%, so that reading already fails. If it reduces it, the shift stays at 5.5–6.3% [computed; reading]. At $1.6 \times 10^7\,M_\odot$ the two readings differ from the unmodified 6.31% by +13.5% and −11.9%. The run McCormick lists as missing, the same transport at fixed R_ν with $p = e/3$ against the full equation of state at 3×10^6–$1.6 \times 10^7\,M_\odot$, would confirm the sign.

2. **Which mean of the rates.** McCormick calibrates his single rate to an energy-group calculation. The arithmetic mean of the group rates reproduces his calibrated decoupling temperatures: $1.62\,\mathrm{MeV}$ against 1.6 ($e^{\pm}$ only), $1.02\,\mathrm{MeV}$ against 0.93–1.0 (with $\nu\nu$), and a mass

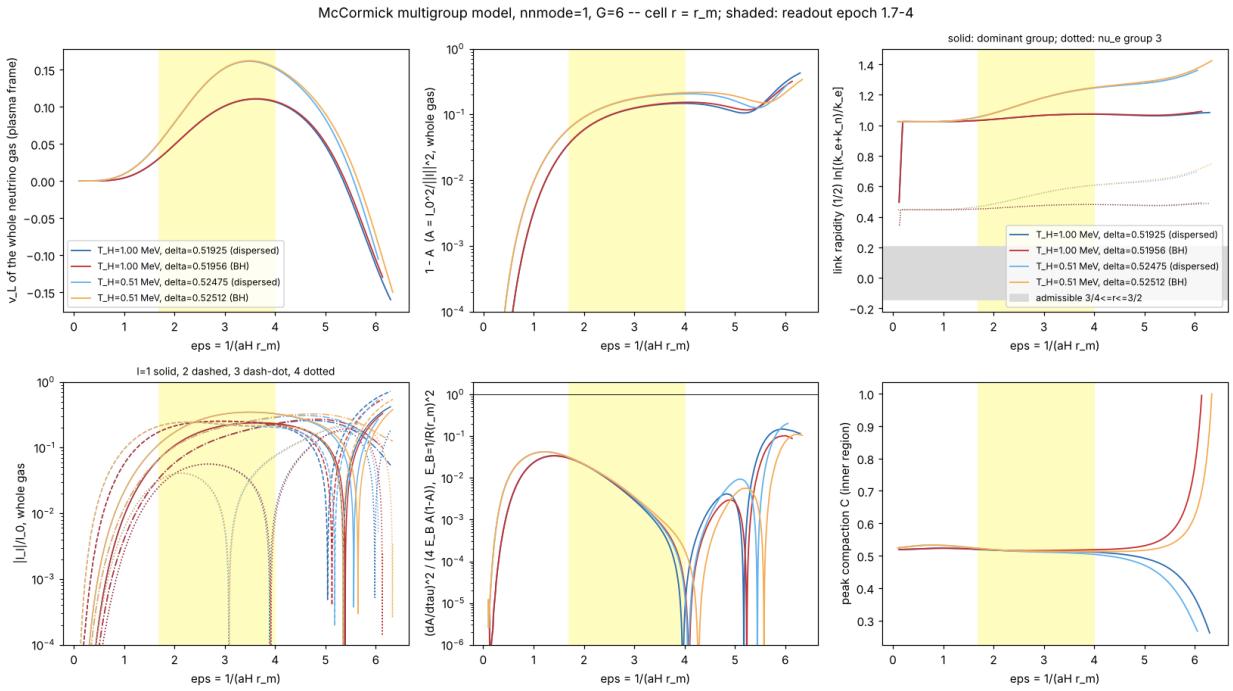


Figure 11: Inside McCormick's multigroup model at $r = r_m$, his archived threshold edges at $T_H = 1$ and 0.51 MeV (positive common-frame target): energy-frame velocity of the neutrino gas, kept-share anisotropy $1 - A$, link rapidity of the dominant and of a ν_e group against the admissible window (grey), moment ratios, the speed-limit ratio of the kept population, and the peak compaction. The readout epoch $1.7 \leq \epsilon \leq 4$ is shaded.

step of 2.53 against 2.5–3. The harmonic and geometric means fail; the mean of squares also passes, so the class of means is identified, not one mean [9] [computed; reading].

12 Relation to other work

The closest published analogue is bulk viscosity, which raises the threshold roughly in proportion to its strength in full numerical relativity [15]. Free streaming is a second dissipative channel with the same sign. Papanikolaou fixes thresholds at horizon crossing for a time-dependent equation of state, with δ_c following c_s^2 [16]. Escrivà ties them to the initial profile and the curvature of the compaction at its peak [17]. Test A is consistent with both: the neutrino clock is set early.

The 2026 thermal-history mass functions through neutrino decoupling and $e^{\pm}$ annihilation treat neutrinos as part of the fluid [18], so the step McCormick found is absent from them. For supermassive seeds, the μ-distortion bound leaves only heavy-tailed statistics [19], for which McCormick finds that the raised threshold costs a factor 1.2–8 in abundance in his examples [1].

The nearest analogue to the result of Section 7 is viscous damping. Bulk viscosity enters the collapse through the isotropic pressure [15]; the shear relaxation rate of a free-streaming component enters through the anisotropic stress, and no published threshold study varies it. In relaxation-time kinetic theory the kept sector's response is a continued fraction whose hydrodynamic pole meets a branch cut as the collision rate falls [13, 14]; Section 7 finds that at the readout scale the kept sector has no isolated eigenvalue for about half of the profile, which is why the threshold is a law

in the operator's link rapidity and not in the displacement of a kept state.

A search of the literature to October 2026 found no other work on thresholds with free-streaming components, on gradual kinetic decoupling in numerical relativity, on how the threshold depends on when a collision rate switches off, or on how it depends on the ratio of the shear and flux relaxation rates. The runs of Sections 3 and 7 are, as far as that search shows, the first such runs.

McCormick's revised calculation [3] replaces the single rate by energy groups with plasma and self-scattering rates; Section 10 shows that his collision term is exactly the split link of Section 7, so his model sits on the axis this paper varies, at $r \approx 2.4$ for ν_e and 7.7 for $\nu_{\mu,\tau}$. Froustey, Pitrou and Volpe solve neutrino decoupling in momentum space with full collision integrals and flavour mixing [4]; there the angular structure is that of a homogeneous background, while the hidden moments here are spatial. Seibold and Frank close moment models of radiative transfer by optimal prediction, with a memory term from the eliminated moments [5]; Section 8 finds that a short-memory correction of that type beats P_2 in every history tested, and that freezing the exact memory kernel to one history costs up to 0.13. The readout Gramian is the observability Gramian of linear control theory; what is new here is its use as the weight on the predictive quotient of VOTKP [6] and Dirac Time [7], and the test of its leading direction in the full collapse.

13 Summary

The neutrino collision operator is a cut whose whole non-normality sits on one link, density to flux, with weight equal to the sound speed; its kept-sector mode lives only above Romatschke's onset $\kappa = 2k/\pi$, where its non-orthogonality diverges. McCormick's code, run on cases outside his paper, shows that the collapse time alone does not set the neutrino clock, and that the decoupling history changes the threshold, not only its end point. The kept-sector memory, read mostly at the first compaction maximum and fitted once, predicted three histories it had not seen to within 0.071, 0.014 and 0.037, while a history-blind rule failed. The steep-history thresholds do not move with the rate cap or the grid. The memory sits in the angular structure a fluid closure drops: set at or after decoupling, it returns into density, flux and stress; set before, it is erased. And the anisotropic stress carries the threshold: with the shear relaxing at r times the flux rate, the offset of the surviving share moves close to a straight line in the link rapidity $\frac{1}{2}\ln r$, at a rate of 1.11 ± 0.01 inside the admissible window, and at 1.19–1.20 when the shear alone is damped. The direction of the flux–shear mismatch, not its connection energy, is what the collapse reads; the law lives in the collision operator's own link rapidity, not in the displacement of the linear kept state; and the rate is a measured number, 1.11 ± 0.01 with the error now set by grid resolution, with no closed form attached. The readout Gramian of the hidden moments has no kernel and ranks what must be kept, and collision histories that agree in every scalar summary still differ in the memory. In the full collapse, hidden angular structure planted at decoupling, invisible to density, flux, stress and Einstein's equations at the restart, moves the black-hole threshold by up to 5.5×10^{-3}, with the sign set by its parity and the size ranked by the Gramian; planted in the coupled era it moves it by less than one per cent of that. McCormick's physical collision term is the split link itself, and self-scattering acts on the threshold mostly through timing, with the flux–shear split carrying $+0.0003$. What the neutrino state must keep is not its density, flux and stress alone, but the hidden angular structure the future reads, and the order of the history that made it.

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A Proofs

A.1 Non-normality is connection energy, for the neutrino chain

Let $L = A + D$ with $A^* = -A$ and $D^* = D$. Then $L^* = D - A$, and

$$[L, L^*] = (A + D)(D - A) - (D - A)(A + D) = 2(AD - DA) = 2[A, D].$$

For diagonal $D = \mathrm{diag}(D_\ell)$, $[A, D]_{ij} = A_{ij}(D_j - D_i)$. So

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \tfrac{1}{2}\sum_{i\neq j} |A_{ij}|^2 (D_i - D_j)^2.$$

Streaming couples only ℓ and $\ell \pm 1$, with $|A_{\ell,\ell+1}| = kc_\ell$. Each link appears twice in the ordered sum, which gives $\sum_\ell k^2 c_\ell^2 (D_{\ell+1} - D_\ell)^2$. With $D = (0, -\kappa_1, -\kappa_2, -\kappa_2, \dots)$, $c_0^2 = 1/3$ and $c_1^2 = 4/15$, this is $k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$. With $\kappa_1 = \kappa_2 = \kappa$ it is $\kappa^2 k^2/3$. □

A.2 The weight across the cut

$PH(I - P)$ with $H = k\mu$ has the single entry $\langle 0|k\mu|1\rangle = kc_0$, so $E_B = k^2/3$. The variance of $k\mu$ in the isotropic state is $k^2\langle\mu^2\rangle = k^2/3$, the same number. □

A.3 The kept-sector propagator

Write $u = (s - L)^{-1}|0\rangle$, so $(s+\kappa+ik\mu)u = |0\rangle + \kappa Pu$. Projecting onto $|0\rangle$ with $\langle 0|(s+\kappa+ik\mu)^{-1}|0\rangle = \frac{1}{2}\int_{-1}^{1} d\mu/(s + \kappa + ik\mu) = k^{-1}\arctan[k/(s+\kappa)] = g$ gives $u_0 = g(1 + \kappa u_0)$, so $u_0 = g/(1 - \kappa g)$. □

A.4 The truncated exceptional point

On $\mathrm{span}\{|0\rangle, |1\rangle\}$, $L = \begin{pmatrix} 0 & -ik/\sqrt{3} \\ -ik/\sqrt{3} & -\kappa \end{pmatrix}$, with characteristic polynomial $\lambda^2 + \kappa\lambda + k^2/3$. Its discriminant $\kappa^2 - 4k^2/3$ vanishes at $\kappa = 2k/\sqrt{3}$, where the matrix is not diagonalisable. □

A.3a The onset of the kept-sector pole (Romatschke's onset [13, Eq. (37)], with its non-orthogonality)

A pole of $u_0 = g/(1 - \kappa g)$ needs $\kappa g = 1$, that is $\arctan[k/(s + \kappa)] = \phi$ with $\phi = k/\kappa$. For real $s > -\kappa$ the principal arctangent takes values in $(0, \pi/2)$, so a solution exists exactly when $\phi < \pi/2$, and then $s + \kappa = k\cot\phi$. Because the principal arctangent is one-to-one, there is no other solution on the physical sheet. As $\phi \to \pi/2$, $s \to -\kappa$, the centre of the cut of g, which joins the branch points $-\kappa \pm ik$. The eigenvector is $u(\mu) = 1/(s + \kappa + ik\mu)$. The operator is complex

symmetric in the Legendre basis, so its left eigenvector is u itself under the bilinear pairing, and $K = \big(\frac{1}{2}\int |u|^2\big)^2 / \big|\frac{1}{2}\int u^2\big|^2$. With $a = s + \kappa = k\cot\phi$,

$$\tfrac{1}{2}\int_{-1}^{1} \frac{d\mu}{a^2 + k^2\mu^2} = \frac{\arctan(k/a)}{ak} = \frac{\phi\tan\phi}{k^2}, \qquad \tfrac{1}{2}\int_{-1}^{1} \frac{d\mu}{(a+ik\mu)^2} = \frac{1}{a^2+k^2} = \frac{\sin^2\phi}{k^2},$$

so $K = \phi^2\tan^2\phi/\sin^4\phi = \phi^2/(\sin^2\phi\cos^2\phi) = 4\phi^2/\sin^2 2\phi$. □

A.5 Exact elimination of the hidden moments

Write $\partial_\tau v = -(ikM + \sigma(\tau)Q)v$ with $Q = I - P$, and split $v = (a, h)$ along P and Q. Then $\dot a = -ikc_0 h_1$ and $\dot h = -(ikM_{QQ} + \sigma)h - ikc_0\, a\, e_1$. On the hidden block $\sigma(\tau)I$ commutes with M_{QQ}, so the generators at different times commute and the propagator is the product $U_Q(t,r) = e^{-\int_r^t \sigma}\, e^{-ik(t-r)M_{QQ}}$, with $\int_r^t \sigma = \Sigma(t) - \Sigma(r)$, $\Sigma(\tau) = -\epsilon_{\mathrm{dec}}^n/(n\tau^n)$. Variation of constants gives $h(t) = U_Q(t,s)h(s) - ikc_0\int_s^t U_Q(t,r)e_1 a(r)\,dr$; substituting into $\dot a$ gives

$$\dot a(t) = -ikc_0[U_Q(t,s)h(s)]_1 - k^2c_0^2\int_s^t [U_Q(t,r)]_{11}a(r)\,dr.$$

For two states with equal $a(s)$ the difference Δa obeys the same equation with source $S[\Delta h(s)]$ and $\Delta a(s) = 0$; if $\Delta h(s) = 0$ it vanishes identically. The time-independent case $e^{t(X-\sigma I)} = e^{-\sigma t}e^{tX}$ is checked in Lean (Appendix C). □

A.6 The speed limit on the kept population

With $D = -\sigma Q \le 0$, $\frac{d}{d\tau}\|v\|^2 = -2\sigma\|Qv\|^2 \le 0$, so $\|v\| \le 1$ for a unit initial state. Since $\langle 0|D = 0$ and $\langle 0|M = c_0\langle 1|$, $\dot v_0 = -ikc_0 v_1$, and

$$\dot A = 2\,\mathrm{Re}\big(\bar v_0\,\dot v_0\big) = 2\,\mathrm{Re}\big(\bar v_0(-ikc_0)v_1\big), \qquad |\dot A| \le 2kc_0|v_0||v_1|.$$

With $|v_1|^2 \le \|v\|^2 - |v_0|^2 \le 1 - A$ and $k^2c_0^2 = E_B$, $\dot A^2 \le 4E_B A(1-A)$. The core inequality $|2\,\mathrm{Re}(\bar a(-ih)b)| \le 2|h||a||b|$ and the corollary under $|a|^2+|b|^2 \le 1$ are checked in Lean (Appendix C). □

A.7 Energy stationarity is not response stationarity

Keep three moments, $\ell = 0, 1, 2$, with streaming weights $c_0^2 = 1/3$, $c_1^2 = 4/15$, and $L = -ikM - D$ with $D = \mathrm{diag}(0, \kappa_1, \kappa_2)$, so that $s - L = s + ikM + D$ (the damping enters L with a minus sign, as in Section 2). The kept response $R_{00} = \langle 0|(s-L)^{-1}|0\rangle$ is the continued fraction

$$R_{00} = \cfrac{1}{s + \cfrac{k^2c_0^2}{s + \kappa_1 + \cfrac{k^2c_1^2}{s+\kappa_2}}}.$$

At $k = s = \kappa_1 = 1$ this is $R_{00}(\kappa_2) = \big(1 + \frac{1}{3}/(2 + \frac{4/15}{1+\kappa_2})\big)^{-1}$, so $R_{00}(1) = 32/37$ and $R_{00}'(1) = -5/1369$. The energy $E = k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$ has $\partial E/\partial\kappa_2 = \frac{8k^2}{15}(\kappa_2 - \kappa_1) = 0$ there. Checked by direct inversion of the 3×3 matrix; checked in Lean (Appendix C). □

A.8 The link coordinate is odd

For $r > 0$, $\Gamma(1/r) = (1/r - 1)/(1/r + 1) = (1 - r)/(1 + r) = -\Gamma(r)$, and $\frac{1}{2}\ln(1/r) = -\frac{1}{2}\ln r$. Hence $\Gamma(r) = \tanh(\frac{1}{2}\ln r)$, since $\tanh u = (e^{2u} - 1)/(e^{2u} + 1)$ with $e^{2u} = r$. □

A.9 What cancels from the odd part, and what does not

Write $\tau_G(r) = \ln[(1 - G(r))/G(r)]$ and let $\tau_1 = \tau_G(1)$. The odd part of the pair $r, 1/r$ is $[(\tau_G(r) - \tau_1) - (\tau_G(1/r) - \tau_1)]/\ln r = [\tau_G(r) - \tau_G(1/r)]/\ln r$: the anchor τ_1 cancels identically. With $G = (\delta - \delta_f)/(\delta_i - \delta_f)$, $\tau_G = \ln(\delta_i - \delta) - \ln(\delta - \delta_f)$, so

$$\frac{\partial}{\partial \delta_f}\big[\tau_G(r) - \tau_G(1/r)\big] = \frac{1}{\delta_r - \delta_f} - \frac{1}{\delta_{1/r} - \delta_f}, \qquad \frac{\partial}{\partial \delta_i}\big[\tau_G(r) - \tau_G(1/r)\big] = -\frac{1}{\delta_i - \delta_r} + \frac{1}{\delta_i - \delta_{1/r}},$$

which vanish only if $\delta_r = \delta_{1/r}$. The reference thresholds therefore enter the odd rate, with coefficients $\partial(\text{rate})/\partial\delta_f \approx +50$ and $\partial(\text{rate})/\partial\delta_i \approx -28$ per unit for the inner pairs, which is the ± 0.004 common-mode term of Section 7. □

A.10 The streaming weights

$c_\ell^2 = (\ell + 1)^2/((2\ell + 1)(2\ell + 3))$ gives $c_0^2 = 1/3$, $c_1^2 = 4/15$, $c_2^2 = 9/35$, so $c_0^2/c_1^2 = 5/4$ and $c_1^2/c_2^2 = 28/27$: the adjacent weights differ by 1.8% from the second link on, which is why a rule additive in link-weight ratios nearly cancels when two adjacent links jump by opposite rapidities (Section 7). Checked in Lean. □

A.11 Every verdict as a whole-number inequality

Let $O(\delta) = (\delta_i - \delta)/(\delta - \delta_f)$, the odds $(1 - G)/G$, so $\tau_G = \ln O$, and let $Q = O(\delta_r)/O(\delta_{1/r})$. The odd rate is $\ln Q/\ln r$, and for $r = u/v$ and a rational bound p/q,

$$\frac{\ln Q}{\ln r} \geq \frac{p}{q} \iff Q^q \geq r^p \iff \big[(\delta_i - \delta_r)(\delta_{1/r} - \delta_f)\big]^q v^p \geq \big[(\delta_r - \delta_f)(\delta_i - \delta_{1/r})\big]^q u^p,$$

with every threshold written as an integer on a common scale. O decreases in δ, so over a pair of brackets Q is smallest at the upper corner of the r bracket with the lower corner of the $1/r$ bracket, and largest at the opposite pair of corners; checking both corners bounds the rate over the whole bracket pair. Each such inequality is decided exactly by Lean's kernel (Appendix C). □

A.12 The discrete dipole and shear projections

On the midpoint bins μ_k with weight $\Delta\mu$, write $\langle f, g\rangle = \sum_k f_k g_k \Delta\mu$, $m_2 = \langle \mu, \mu\rangle$, and $P_2 = \mu^2 - m_2/2$. Then $\langle 1, \mu\rangle = 0$ by the symmetry of the bins, $\langle 1, P_2\rangle = m_2 - \frac{m_2}{2}\langle 1, 1\rangle = 0$ because $\langle 1, 1\rangle = 2$, and $\langle \mu, P_2\rangle = 0$ by symmetry. So $d = \langle \mu, I\rangle \mu/m_2$ and $s = \langle P_2, I\rangle P_2/\langle P_2, P_2\rangle$ are the orthogonal projections of the intensity onto the discrete $\ell = 1$ and $\ell = 2$ directions, orthogonal to each other and to the isotropic part. The collision terms of Section 7, $\kappa_2(\varepsilon/4\pi - I) + (\kappa_2 - \kappa_1)d$ and $\kappa_1(\varepsilon/4\pi - I) + (\kappa_1 - \kappa_2)s$, therefore conserve the bin-summed energy exactly, leave the dipole relaxing at κ_1 in both, and relax the $\ell = 2$ part at κ_2 in both (together with every $\ell \geq 3$ in the first, alone in the second); $m_2 = 2/3 - \Delta\mu^2/6$ differs from the continuum $2/3$ by 0.1% at $N_\mu = 32$. Checked numerically to 10^{-16} (Appendix B.15). □

A.13 McCormick's self-scattering term is the flux–shear link

On N_μ symmetric midpoint bins with weights $w_k = \Delta\mu/2$ and inner product $\langle f, g\rangle = \sum_k w_k f_k g_k$, let $p_0 = 1$, $p_1 = \mu/\|\mu\|$, and complete to an orthonormal basis $p_0, \dots, p_{N_\mu - 1}$ by Gram–Schmidt on $1, \mu, \mu^2, \dots$. Write $P_0 f = \langle p_0, f\rangle p_0$ and $P_1 f = \langle p_1, f\rangle p_1$. For one frozen group, $\varepsilon/4\pi$ is the bin mean of I, so $\varepsilon/4\pi - I = -(I - P_0)I$. The dipole coefficient of the common-frame target is $c = q/(2\pi \sum_k \mu_k^2 \Delta\mu)$, chosen so that $c\mu$ carries the group's flux; on the bins $c\mu = P_1 I$. Hence the collision term is

$$CI = -\kappa_e (I - P_0)I - \kappa_n (I - P_0 - P_1)I.$$

Since $\langle 1, \mu\rangle = 0$ on symmetric bins, P_0 and P_1 are orthogonal projections onto orthogonal lines, and in the basis $\{p_\ell\}$, $C = -\operatorname{diag}(0, \kappa_e, \kappa_e + \kappa_n, \dots, \kappa_e + \kappa_n)$. This is the damping matrix of Appendix A.1 with $\kappa_1 = \kappa_e$, $\kappa_2 = \kappa_e + \kappa_n$, so by the split-rate identity there

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = k^2\big(c_0^2\kappa_e^2 + c_1^2\kappa_n^2\big),$$

which in the continuum ($c_0^2 = \frac{1}{3}$, $c_1^2 = \frac{4}{15}$) is $k^2(\kappa_e^2/3 + 4\kappa_n^2/15)$. On the bins the streaming couplings are $c_\ell = \langle p_\ell, \mu p_{\ell+1}\rangle$; the values $c_0^2 = \frac{1}{3} - \Delta\mu^2/12$ and $c_1^2 = \frac{4}{15}(1 - \Delta\mu^2)$ hold to machine precision at $N_\mu = 32, 64, 128, 1024$ and are not proved here. The momentum exchanged with the plasma is $\kappa_e q$ only: the self-scattering term has $\langle p_1, (I - P_0 - P_1)I\rangle = 0$. □

A.14 The hidden sector of the chain is fully observable

Let $B = -ikM - D$ with M the tridiagonal streaming matrix of Section 2 (couplings $c_\ell \neq 0$) and D diagonal; let R project onto $\ell \leq 2$ and Q embed $\ell \geq 3$. The observability matrix of the pair (R, B) has rows RB^j. Because B is tridiagonal with nonzero off-diagonal entries $-ikc_\ell$, the entry $(RB^j)_{2,2+j}$ collects only the path that climbs one rung per step from $\ell = 2$ to $\ell = 2 + j$, so $(RB^j)_{2,2+j} = (-ik)^j c_2 c_3 \cdots c_{j+1} \neq 0$ for $k \neq 0$, whatever D is. The rows RB^j, $j = 0, \dots, L - 2$, therefore reach every moment, the pair is observable, and $N_\infty = \{0\}$ for the frozen-time operator. For a smooth time-dependent $D(\tau)$ the Gramian $\mathcal{G}_T$ is positive definite for every $T > \tau_s$. Suppose $h^* \mathcal{G}_T h = 0$; then $y(\tau) = RU(\tau, \tau_s)Qh$ vanishes on $[\tau_s, T]$, so all its derivatives vanish at τ_s. Let $\ell_0 \geq 3$ be the lowest moment with $h_{\ell_0} \neq 0$. Each derivative of U is a sum of products of B and of derivatives of D; the diagonal factors cannot change ℓ, and only the streaming part climbs one rung per factor. So the $\ell = 2$ component of the $(\ell_0 - 2)$-th derivative of y at τ_s is $(-ik)^{\ell_0 - 2} c_2 \cdots c_{\ell_0 - 1} h_{\ell_0} \neq 0$, a contradiction. Its eigenvalues then rank the hidden directions by the integrated squared readout they produce; $h^* \mathcal{G}_T h = \int_{\tau_s}^{T} \|RU(\tau, \tau_s)Qh\|^2 d\tau$ follows from the definition. Averaging over $\pm k$ commutes with the parity $\ell \mapsto (-1)^\ell$, so the averaged Gramian's eigenvectors can be chosen of pure parity. □

A.15 Matched states at a restart surface

Let p_ℓ ($\ell \geq 3$) be orthogonal on the 32 bins to 1, μ and μ^2 under the bin weights. Then $\delta I_{ik} = f\, b_i\, m_i\, (\varepsilon_i/4\pi)\, p_\ell(\mu_k)$ has zero discrete moments of orders 0, 1 and 2 in every cell, so the neutrino energy density, flux and radial pressure of the two states $I \pm \delta I$ are equal cell by cell. The stress–energy tensor of the neutrinos in the spherical solver is built from exactly these three moments, so the sources of the Hamiltonian and momentum constraints, the lapse equation and the fluid equations are equal; with the fluid and metric arrays copied unchanged, every field of the restart state except the hidden part of I is identical, and the constraint residual is identical. Positivity of $I \pm \delta I$ in every bin holds if $|f|\, b_i\, m_i\, |p_\ell(\mu_k)| < 4\pi I_{ik}/\varepsilon_i$ for all k; the margin $m_i = \min_k [4\pi I_{ik}/\varepsilon_i] / \max_k |p_\ell(\mu_k)|$ makes $|f| < 1$ sufficient, and $f = \pm\frac{1}{2}$ uses half of it. On an

isotropic cell the bound is $|a| < 1/\max_k |p_3(\mu_k)| = 1.21498$ for $\ell = 3$, and $+2.3509$, -1.3885 for $\ell = 4$. $\square$

B Evidence: scripts, run logs and outputs

B.1 Identity checks

```
# SCRIPT: PBH-NU-IDENTITIES  checks of every derived identity used in the paper
# Kill conditions (written first): each identity must hold to 1e-10 relative, else the statement is withdrawn.
import numpy as np
from scipy.integrate import quad
LM=80
c=lambda l:(l+1)/np.sqrt((2*l+1)*(2*l+3))          # <l+1|mu|l> in the normalised Legendre basis
Mu=np.zeros((LM+1,LM+1))
for l in range(LM): Mu[l,l+1]=Mu[l+1,l]=c(l)
def L_op(k,k1,k2):
    D=np.diag([0.0,-k1]+[-k2]*(LM-1)); A=-1j*k*Mu; return A,D,A+D
fro2=lambda X:np.sum(np.abs(X)**2)
print('I1  single rate: (1/8)|[L,L*]|^2 = kappa^2 k^2/3')
for k,kap in [(1.0,0.3),(2.0,1.5),(0.7,4.0)]:
    A,D,L=L_op(k,kap,kap); lhs=fro2(L@L.conj().T-L.conj().T@L)/8
    print('    k=%.1f kappa=%.1f  lhs=%.12f  rhs=%.12f'%(k,kap,lhs,kap**2*k**2/3))
print('I2  split rates: (1/8)|[L,L*]|^2 = k^2 k1^2/3 + (4k^2/15)(k2-k1)^2')
for k,k1,k2 in [(1.0,0.3,2.0),(2.0,1.5,0.4),(0.7,4.0,9.0)]:
    A,D,L=L_op(k,k1,k2); lhs=fro2(L@L.conj().T-L.conj().T@L)/8
    print('    k=%.1f k1=%.1f k2=%.1f lhs=%.12f rhs=%.12f'%(k,k1,k2,lhs,k**2*k1**2/3+4*k**2/15*(k2-k1)**2))
print('I3  weight across the cut: |P (k mu) (1-P)|^2 = k^2/3, so sqrt(E_B) = k/sqrt3 = c_s k')
P=np.zeros((LM+1,LM+1)); P[0,0]=1; Q=np.eye(LM+1)-P
for k in [1.0,2.5]: print('    k=%.1f  E_B=%.12f  k^2/3=%.12f'%(k,fro2(P@(k*Mu)@Q),k**2/3))
print('I4  monopole propagator <0|(s+i k mu+kappa(1-P))^-1|0> = g/(1-kappa g), g=arctan(k/(s+kappa))/k')
for k,kap,s in [(1.0,0.5,0.3),(3.0,2.0,0.1),(0.5,0.2,1.0)]:
    Lfull=s*np.eye(LM+1)+1j*k*Mu+kap*Q; num=np.linalg.solve(Lfull,np.eye(LM+1)[:,0])[0]
    g=np.arctan(k/(s+kap))/k; print('    k=%.1f kappa=%.1f s=%.1f  matrix=%.12f  closed=%.12f'%(k,kap,s,num.
     real,(g/(1-kap*g))))
print('I5  two-moment truncation: eigenvalues of [[0,-ik/sqrt3],[-ik/sqrt3,-kappa]] merge at kappa=2k/sqrt3')
k=1.0; kap=2*k/np.sqrt(3); ev=np.linalg.eigvals(np.array([[0,-1j*k/np.sqrt(3)],[-1j*k/np.sqrt(3),-kap]]));
     print('    eigenvalues',np.round(ev,10))
```

```
I1  single rate: (1/8)|[L,L*]|^2 = kappa^2 k^2/3
    k=1.0 kappa=0.3  lhs=0.030000000000  rhs=0.030000000000
    k=2.0 kappa=1.5  lhs=3.000000000000  rhs=3.000000000000
    k=0.7 kappa=4.0  lhs=2.613333333333  rhs=2.613333333333
I2  split rates: (1/8)|[L,L*]|^2 = k^2 k1^2/3 + (4k^2/15)(k2-k1)^2
    k=1.0 k1=0.3 k2=2.0 lhs=0.800666666667 rhs=0.800666666667
    k=2.0 k1=1.5 k2=0.4 lhs=4.290666666667 rhs=4.290666666667
    k=0.7 k1=4.0 k2=9.0 lhs=5.880000000000 rhs=5.880000000000
I3  weight across the cut: |P (k mu) (1-P)|^2 = k^2/3, so sqrt(E_B) = k/sqrt3 = c_s k
    k=1.0  E_B=0.333333333333  k^2/3=0.333333333333
    k=2.5  E_B=2.083333333333  k^2/3=2.083333333333
I4  monopole propagator <0|(s+i k mu+kappa(1-P))^-1|0> = g/(1-kappa g), g=arctan(k/(s+kappa))/k
    k=1.0 kappa=0.5 s=0.3  matrix=1.623370180076  closed=1.623370180076
    k=3.0 kappa=2.0 s=0.1  matrix=0.889069885739  closed=0.889069885739
    k=0.5 kappa=0.2 s=1.0  matrix=0.937653083725  closed=0.937653083725
I5  two-moment truncation: eigenvalues of [[0,-ik/sqrt3],[-ik/sqrt3,-kappa]] merge at kappa=2k/sqrt3
    eigenvalues [-0.57735027+9.2e-09j -0.57735027-9.2e-09j]
```

B.2 The one change to McCormick's code

In a copy of `run_nuc.py`, the collision rate line became
`kap = (1/(2*t))*min((eps_dec/(2*np.sqrt(t*t0)))**nexp, kapcap)`
with `nexp` (default 3) and `kapcap` $= 5 \times 10^5$ as inputs. Nothing else was changed.

```
# SCRIPT: BEYOND-STEVE-BISECT  threshold bisection with gradual decoupling, profile alpha and rate exponent as
      inputs
import sys, json, time, os
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_nuc_n import run
tag=sys.argv[1]; qshape=float(sys.argv[2]); epsd=float(sys.argv[3]); nexp=float(sys.argv[4])
lo,hi,tol=float(sys.argv[5]),float(sys.argv[6]),float(sys.argv[7]); emax=float(sys.argv[8])
log=[]; t0=time.time()
while hi-lo>tol:
    mid=0.5*(lo+hi); t1=time.time()
    out=run(mid,fnu=0.409,N=400,K=32,eps0=0.1,eps_dec=epsd,eps_max=emax,qshape=qshape,nexp=nexp)
    print(f'{tag} delta={mid:.5f}: {out["status"]} eps={out["eps"]:.2f} ({time.time()-t1:.0f}s)',flush=True)
    log.append((mid,out['status'],float(out['eps'])))
    if out['status']=='BH': hi=mid
    elif out['status']=='dispersed': lo=mid
    else: break
print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} ({(time.time()-t0)/60:.1f} min)',flush=
     True)
json.dump(dict(tag=tag,qshape=qshape,eps_dec=epsd,nexp=nexp,dc=0.5*(lo+hi),w=hi-lo,log=log),open(f'res_{tag}.
     json','w'))
```

```
# SCRIPT: BEYOND-STEVE-TRISECT  two trial amplitudes per round in parallel (thirds of the bracket)
import sys, json, time, os
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_nuc_n import run
def one(a):
    d,q,ed,n,em=a; o=run(d,fnu=0.409,N=400,K=32,eps0=0.1,eps_dec=ed,eps_max=em,qshape=q,nexp=n); return d,o['
     status'],float(o['eps'])
if __name__=='__main__':
    tag=sys.argv[1]; q=float(sys.argv[2]); ed=float(sys.argv[3]); n=float(sys.argv[4]); lo,hi,tol=float(sys.
     argv[5]),float(sys.argv[6]),float(sys.argv[7]); em=float(sys.argv[8])
    log=[]; t0=time.time()
    with Pool(2) as p:
        while hi-lo>tol:
            a=lo+(hi-lo)/3; b=lo+2*(hi-lo)/3
            res=p.map(one,[(a,q,ed,n,em),(b,q,ed,n,em)])
            for d,s,e in res: print(f'{tag} delta={d:.5f}: {s} eps={e:.2f}',flush=True); log.append((d,s,e))
            (da,sa,_),(db,sb,_)=res
            if 'BH' not in (sa,sb) and 'dispersed' not in (sa,sb): break
            if sa=='BH': hi=da
            elif sb=='BH': lo,hi=da,db
            else: lo=db
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} ({(time.time()-t0)/60:.1f} min)',
     flush=True)
    json.dump(dict(tag=tag,dc=0.5*(lo+hi),w=hi-lo,log=log),open(f'res_{tag}.json','w'))
```

B.3 Predictions written before the runs

```
BEYOND-STEVE pre-registration, written 2026-10-30 before any of these runs.
G = (delta_c - delta_c_fluid) / (delta_c_instant - delta_c_fluid), same profile.
Gaussian (alpha=1): fluid 0.49773, instant 0.52832 (Steve Table II).
alpha=0.3: fluid 0.44805, instant 0.47747 (Steve res_fl_q03 / res_nu409_q03).

TEST A  broad profile alpha=0.3, rate kappa=H(eps_dec/eps)^3, eps_dec = 4 and 8.
  Diffusive-escape law G=1/(1+sqrt15 (eps_dec/eps*)^1.5), eps* = 7-9 (alpha=0.3 decided late):
     G(4) = 0.38-0.46 ; G(8) = 0.17-0.24
  Null (curve does not move with eps*): G(4)=0.19, G(8)=0.063
  Kill: the law dies if G(4) < 0.30 or G(8) < 0.13.  Null dies if G(4) > 0.28.

TEST B  Gaussian, steeper rate kappa=H(eps_dec/eps)^6, eps_dec = 2.
  Diffusive-escape law with n=6: G = 1/(1+sqrt24 (eps_dec/eps*)^3), eps*=4  ->  G = 0.62
  Memory model (Weight Between agent): G = 0.25
  History-blind model: G = 0.43
```

```
  Kill: each model dies if measured G is more than 0.08 from its number.
```

B.4 Run logs

```
Reproduction check, 2026-10-30 ~14:25 CT, printed by the session (not logged to file at the time):
repro check n=3 eps_dec=2 at 0.5109 (Steve threshold 0.51092): BH 6.6146961105642985 90
(status BH, apparent horizon at eps 6.61, 90 s run time)
McCormick logG.txt bracket for the same case: 0.51109 BH, 0.51074 dispersed.
```

```
A_q03_ed4 delta=0.46300: BH eps=4.70 (116s)
A_q03_ed4 delta=0.45500: BH eps=6.24 (133s)
A_q03_ed4 delta=0.45100: dispersed eps=7.01 (145s)
A_q03_ed4 delta=0.45300: dispersed eps=7.97 (154s)
A_q03_ed4 delta=0.45400: BH eps=6.89 (141s)
A_q03_ed4 delta=0.45350: BH eps=7.63 (153s)
A_q03_ed4 delta=0.45325: dispersed eps=9.51 (176s)
RESULT A_q03_ed4: delta_c = 0.45337 +- 0.00013 (17.0 min)
```

```
A_q03_ed8 delta=0.46300: BH eps=4.41 (539s)
A_q03_ed8 delta=0.45500: BH eps=5.30 (545s)
```

```
A_q03_ed8 delta=0.45100: BH eps=6.43 (561s)
A_q03_ed8 delta=0.44900: dispersed eps=7.81 (576s)
A_q03_ed8 delta=0.45000: BH eps=7.23 (564s)
A_q03_ed8 delta=0.44950: BH eps=9.08 (598s)
A_q03_ed8 delta=0.44925: dispersed eps=8.15 (577s)
RESULT A_q03_ed8: delta_c = 0.44938 +- 0.00012 (47.9 min)
```

```
B_n6_ed2 delta=0.51350: BH eps=4.54 (1433s)
B_n6_ed2 delta=0.50525: dispersed eps=5.51 (1441s)
B_n6_ed2 delta=0.50938: BH eps=5.28 (1434s)
B_n6_ed2 delta=0.50731: dispersed eps=6.58 (1464s)
```

```
B_n6_ed2_fin delta=0.50800: BH eps=5.99
B_n6_ed2_fin delta=0.50869: BH eps=5.54
B_n6_ed2_fin delta=0.50754: BH eps=6.96
B_n6_ed2_fin delta=0.50777: BH eps=6.27
RESULT B_n6_ed2_fin: delta_c = 0.50743 +- 0.00011 (48.5 min)
```

B.5 Kept-sector memory curves

```
# SCRIPT: PBH-NU-MEMORY-CURVE  kept-sector survival (memory form) curves for the figures; same object as WB-
    STEVE-07
import numpy as np, json
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
kn,kw=roots_legendre(36); kmin,kmax=0.25,9.0; ks=0.5*(kmax-kmin)*(kn+1)+kmin; wk=0.5*(kmax-kmin)*kw
wE=wk*ks**2*np.exp(-ks**2/2); wE/=wE.sum()
LM=60; M=np.zeros((LM+1,LM+1))
for l in range(1,LM+1): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
tau0=0.02; eg=np.linspace(tau0,6.0,1200); n=LM+1; P=np.zeros(n); P[0]=1
def surv(k,ed,p):
    def rhs(t,v):
        K=0.0 if ed is None else ed**p/t**(p+1)
        x,y=v[:n],v[n:]
        return np.concatenate([-K*(1-P)*x+k*(M@y),-K*(1-P)*y-k*(M@x)])
    v0=np.zeros(2*n); v0[0]=1
    s=solve_ivp(rhs,(tau0,eg[-1]),v0,method='BDF',t_eval=eg,rtol=1e-9,atol=1e-12)
    return s.y[0]**2+s.y[n]**2
f=lambda ed,p: sum(w_*surv(kk,ed,p) for kk,w_ in zip(ks,wE))
```

```
i17,i40=np.searchsorted(eg,1.7),np.searchsorted(eg,4.0); w=0.890; Ai=f(None,3)
out={}
for p,eds in [(3,[0.25,0.5,0.75,1,1.5,2,3,4,6,8]),(6,[0.5,1,1.5,2,3,4])]:
    for ed in eds:
        g=(1-f(ed,p))/(1-Ai); out['%d_%g'%(p,ed)]=dict(first_max=float(g[i17]),two_epoch=float(w*g[i17]+(1-w)*
     g[i40]))
        print(p,ed,round(g[i17],3),round(w*g[i17]+(1-w)*g[i40],3),flush=True)
json.dump(out,open('memory_curve.json','w'))
```

```
3 0.25 0.941 0.947
3 0.5 0.86 0.875
3 0.75 0.768 0.792
3 1 0.671 0.705
3 1.5 0.489 0.542
3 2 0.344 0.411
3 3 0.168 0.249
3 4 0.087 0.169
3 6 0.03 0.095
3 8 0.013 0.059
6 0.5 0.862 0.877
6 1 0.623 0.662
6 1.5 0.352 0.42
6 2 0.159 0.246
6 3 0.026 0.118
6 4 0.005 0.079
```

B.6 The kept-sector readings against McCormick's curve and the history prediction

```
# SCRIPT: WB-STEVE-03-PAPER-OBJECTS
# D2 (memory survival of the kept sector under the time-dependent relaxing operator) and D3 (energy that has
     crossed into the
# hidden sector) as predictors of Steve's G(eps_dec).  Kill conditions are those stated in WB-STEVE-02-LINEAR
     (D2, D3, H1 and
# opposite runs); restated:
#  D2  g = (1-A)/(1-A_inst), A = population of the kept sector l=0 at readout epoch eps* in {1.7, 4.0} (the
     two epochs Steve's
#      own numbers give), no fitted parameter.  Dies if max_eps |g - G| >= 0.10 at BOTH epochs.
#  D3  g = X/X_inst, X = sum_{l>=2} |F_l|^2/(2l+1) summed over the Gaussian perturbation (weights k^2 zeta^2),
      same epochs, same kill.
#  Opposite runs: kept sector = l in {0,1} (flux kept), and k r_m fixed at 2 instead of the profile average.
#  Fitted-epoch scan is diagnostic only (a fitted epoch is not a pass).
import numpy as np
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
import wb_lin
Gs = {0.25: 0.960, 0.5: 0.868, 1.0: 0.684, 2.0: 0.431, 4.0: 0.190, 8.0: 0.063}
kn, kw = roots_legendre(36); kmin, kmax = 0.25, 9.0; ks = 0.5*(kmax-kmin)*(kn+1)+kmin; wk = 0.5*(kmax-kmin)*kw
wE = wk*ks**2*np.exp(-ks**2/2)          # energy weights k^2 zeta(k)^2 of the Gaussian perturbation
wE /= wE.sum()
LM = 60; a = lambda l: l/np.sqrt(4.0*l*l-1)
M = np.zeros((LM+1, LM+1))
for l in range(1, LM+1): M[l, l-1] = M[l-1, l] = a(l)
tau0 = 0.02; eg = np.linspace(tau0, 6.0, 1200)

def survival(k, ed, keep):
    """population of kept sector (keep=1: l=0; keep=2: l=0,1) vs tau, under dv/dtau = -(i k M + Kc (1-P)) v, v
     (tau0)=|0>."""
    n = LM+1; P = np.zeros(n); P[:keep] = 1
    def rhs(t, v):
        K = 0.0 if ed is None else ed**3/t**4
        x, y = v[:n], v[n:]
        return np.concatenate([-K*(1-P)*x+k*(M @ y), -K*(1-P)*y-k*(M @ x)])
    v0 = np.zeros(2*n); v0[0] = 1
```

```
    sol = solve_ivp(rhs, (tau0, eg[-1]), v0, method='BDF', t_eval=eg, rtol=1e-9, atol=1e-12)
    pop = sol.y[:keep]**2+sol.y[n:n+keep]**2
    return np.sum(pop, axis=0)

def avg_surv(ed, keep, single=None):
    if single is not None: return survival(single, ed, keep)
    out = np.zeros_like(eg)
    for kk, w_ in zip(ks, wE): out += w_*survival(kk, ed, keep)
    return out

res = {}
for keep, tag in [(1, 'kept=l0'), (2, 'kept=l0,1')]:
    for single, stag in [(None, 'profile-averaged'), (2.0, 'k r_m=2')]:
        Ai = avg_surv(None, keep, single); curves = {}
        for ed in Gs:
            A = avg_surv(ed, keep, single); curves[ed] = (1-A)/(1-Ai)
        res[(tag, stag)] = curves
        for es in [1.7, 4.0]:
            i = np.searchsorted(eg, es); err = {ed: curves[ed][i]-Gs[ed] for ed in Gs}
            print('D2 %-10s %-17s eps*=%.1f : g = %s  max|err| = %.3f' % (tag, stag, es, np.round([curves[ed][
     i] for ed in Gs], 3), max(abs(v) for v in err.values())))
        # fitted-epoch scan (diagnostic only)
        best = min(((max(abs(curves[ed][i]-Gs[ed]) for ed in Gs), eg[i]) for i in range(40, len(eg), 5)))
        print('   fitted-epoch scan (diagnostic only): best eps* = %.2f with max|err| = %.3f' % (best[1], best
     [0]))
print('Steve G :', [Gs[e] for e in Gs])

# D3 from the full linear solver
def hidden_energy(ed, Rn=0.409):
    X = np.zeros_like(eg)
    for kk, w_ in zip(ks, wE):
        _, Y = wb_lin.run_mode(kk, tau0, eg, Rn, ed)
        F = Y[4:]; ls = np.arange(2, 2+F.shape[0])
        X += w_*np.sum(F**2/(2*ls[:, None]+1), axis=0)
    return X
Xi = hidden_energy(None); D3 = {}
for ed in Gs:
    D3[ed] = hidden_energy(ed)/Xi
for es in [1.7, 4.0]:
    i = np.searchsorted(eg, es); print('D3 eps*=%.1f : g = %s  max|err| = %.3f' % (es, np.round([D3[ed][i] for
      ed in Gs], 3), max(abs(D3[ed][i]-Gs[ed]) for ed in Gs)))
best = min(((max(abs(D3[ed][i]-Gs[ed]) for ed in Gs), eg[i]) for i in range(40, len(eg), 5)))
print('D3 fitted-epoch scan (diagnostic only): best eps* = %.2f with max|err| = %.3f' % (best[1], best[0]))
```

```
D2 kept=l0    profile-averaged  eps*=1.7 : g = [0.941 0.86  0.671 0.344 0.087 0.013]  max|err| = 0.103
D2 kept=l0    profile-averaged  eps*=4.0 : g = [0.997 0.994 0.987 0.959 0.83  0.429]  max|err| = 0.640
   fitted-epoch scan (diagnostic only): best eps* = 1.87 with max|err| = 0.065
/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/a_wb/wb_steve_03_paper_objects.py
     :46: RuntimeWarning: invalid value encountered in divide
  A = avg_surv(ed, keep, single); curves[ed] = (1-A)/(1-Ai)
D2 kept=l0    k r_m=2           eps*=1.7 : g = [1.    0.955 0.772 0.393 0.097 0.014]  max|err| = 0.093
D2 kept=l0    k r_m=2           eps*=4.0 : g = [1.001 1.007 1.016 1.006 1.008 0.587]  max|err| = 0.818
   fitted-epoch scan (diagnostic only): best eps* = 1.69 with max|err| = 0.095
D2 kept=l0,1  profile-averaged  eps*=1.7 : g = [0.981 0.922 0.735 0.373 0.09  0.013]  max|err| = 0.100
D2 kept=l0,1  profile-averaged  eps*=4.0 : g = [0.995 0.983 0.941 0.817 0.544 0.217]  max|err| = 0.386
   fitted-epoch scan (diagnostic only): best eps* = 1.84 with max|err| = 0.082
D2 kept=l0,1  k r_m=2           eps*=1.7 : g = [0.986 0.933 0.752 0.385 0.094 0.014]  max|err| = 0.096
D2 kept=l0,1  k r_m=2           eps*=4.0 : g = [0.998 0.993 0.979 0.882 0.584 0.261]  max|err| = 0.451
   fitted-epoch scan (diagnostic only): best eps* = 1.77 with max|err| = 0.089
Steve G : [0.96, 0.868, 0.684, 0.431, 0.19, 0.063]
/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/a_wb/wb_steve_03_paper_objects.py
     :66: RuntimeWarning: invalid value encountered in divide
  D3[ed] = hidden_energy(ed)/Xi
D3 eps*=1.7 : g = [1.006e+00 9.420e-01 6.770e-01 2.620e-01 3.200e-02 1.000e-03]  max|err| = 0.169
D3 eps*=4.0 : g = [1.024 1.071 1.154 1.168 0.872 0.198]  max|err| = 0.737
D3 fitted-epoch scan (diagnostic only): best eps* = 1.92 with max|err| = 0.112
```

```
# SCRIPT: WB-STEVE-07-HISTORY
# Discriminating prediction for the next step.  Table II interpolation cannot predict a DIFFERENT collision history, the
# two-epoch memory-survival form can.  kappa = H (eps_dec/eps)^n  with n = 2 and 6 (Steve used n = 3), eps_dec = 1 and 2.
# KILL CONDITION (before the run): the test is only worth asking Steve for if the predicted G for n = 2 or n = 6 differs from the
# n = 3 value by at least 0.10 at some eps_dec; otherwise the memory form is insensitive to history and the step is dropped.
import numpy as np
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
kn, kw = roots_legendre(36); kmin, kmax = 0.25, 9.0; ks = 0.5*(kmax-kmin)*(kn+1)+kmin; wk = 0.5*(kmax-kmin)*kw
wE = wk*ks**2*np.exp(-ks**2/2); wE /= wE.sum()
LM = 60; M = np.zeros((LM+1, LM+1))
for l in range(1, LM+1): M[l, l-1] = M[l-1, l] = l/np.sqrt(4.0*l*l-1)
tau0 = 0.02; eg = np.linspace(tau0, 6.0, 1200); n = LM+1; P = np.zeros(n); P[0] = 1
def surv(k, ed, p):
    def rhs(t, v):
        K = 0.0 if ed is None else ed**p/t**(p+1)       # a*kappa = (1/tau)(eps_dec/tau)^p
        x, y = v[:n], v[n:]
        return np.concatenate([-K*(1-P)*x+k*(M @ y), -K*(1-P)*y-k*(M @ x)])
    v0 = np.zeros(2*n); v0[0] = 1
    sol = solve_ivp(rhs, (tau0, eg[-1]), v0, method='BDF', t_eval=eg, rtol=1e-9, atol=1e-12)
    return sol.y[0]**2+sol.y[n]**2
f = lambda ed, p: sum(w_*surv(kk, ed, p) for kk, w_ in zip(ks, wE))
i17, i40 = np.searchsorted(eg, 1.7), np.searchsorted(eg, 4.0); w = 0.890
Ai = f(None, 3)
res = {}
for p in [2, 3, 6]:
    for ed in [1.0, 2.0]:
        g = (1-f(ed, p))/(1-Ai); res[(p, ed)] = w*g[i17]+(1-w)*g[i40]
        print('n = %d, eps_dec = %.0f : G_pred(two-epoch, w = 0.89) = %.3f  (n=3 Table II value %.3f)' % (p, ed, res[(p, ed)], {1.0: 0.684, 2.0: 0.431}[ed]))
dmax = max(abs(res[(p, ed)]-res[(3, ed)]) for p in [2, 6] for ed in [1.0, 2.0])
print('largest change from n=3: %.3f -> %s' % (dmax, 'discriminating' if dmax >= 0.10 else 'DROPPED (insensitive)'))
```

```
n = 2, eps_dec = 1 : G_pred(two-epoch, w = 0.89) = 0.730  (n=3 Table II value 0.684)
n = 2, eps_dec = 2 : G_pred(two-epoch, w = 0.89) = 0.511  (n=3 Table II value 0.431)
n = 3, eps_dec = 1 : G_pred(two-epoch, w = 0.89) = 0.705  (n=3 Table II value 0.684)
n = 3, eps_dec = 2 : G_pred(two-epoch, w = 0.89) = 0.411  (n=3 Table II value 0.431)
n = 6, eps_dec = 1 : G_pred(two-epoch, w = 0.89) = 0.662  (n=3 Table II value 0.684)
n = 6, eps_dec = 2 : G_pred(two-epoch, w = 0.89) = 0.246  (n=3 Table II value 0.431)
largest change from n=3: 0.165 -> discriminating
```

```
original wb_steve_07_out.txt written 2026-10-30 13:47:53.348475647 -0500; Test B launched about 14:30 CT the same day
```

B.7 Shear and flux links, and which mean of the rates

```
# SCRIPT: WB-STEVE-01-ALGEBRA
# Tests T0-T4 mapping "The Weight Between" v1.1 onto McCormick's free-streaming PBH preprint (algebraic part).
# KILL CONDITIONS (written before any run):
#  T0  Steve-internal consistency (so the targets are trustworthy). Table II: recompute Delta C_l,c/C_l,c from the delta_c
#      columns through Eq.(15) delta = C - 3C^2/8; dies (flag only) if any row differs from the printed value by > 0.3 pt.
#      Also recompute G = shift/shift_inst for the gradual rows (flag if any differs from the brief's G by > 0.005).
#  T1  Prop 5 / Def 4 on Steve's operator L = -i k mu - kappa (I-P), P = projector on l=0 (orthonormal Legendre basis).
#      H = k mu, E_B = ||P H (1-P)||_F^2 must equal k^2/3 = (1/2)||[H,P]||_F^2 = Var_{|0>}(H) to 1e-12 relative. Dies otherwise.
```

```
#      Connection-energy identity (scalar form, H' = -L = D + iK, D=diag(kappa_l), K = k mu):
#      (1/4)||[H',H'^dag]||^2 = ||[D,K]||^2 = S_2. Dies if off by > 1e-10 relative.
#  T2  Hidden-walk (Schur) formula of the paper: G_B(z) = [z - E - W (z-F)^{-1} W^dag]^{-1} with E=0, W the l
    =0->l>=1 block,
#      F the hidden block of L, must equal the closed form G(z) = g(s)/(1 - kappa g(s)), g(s)=arctan(k/s)/k, s
    =z+kappa,
#      to 1e-8 relative at 50 random complex z (Re s>0). Dies otherwise.
#      Two-level EP: eigenvalues of [[0,-i k/sqrt3],[-i k/sqrt3,-kappa]] coalesce at kappa = 2 k/sqrt3 = 2
    sqrt(E_B). Dies otherwise.
#  T3  Ladder on Steve's single-rate model: all gaps equal kappa, so S_0^2 = S_-2 S_2 and S_-1^2 = S_-2 S_0
    must be EQUALITIES
#      (1e-12). Dies otherwise.  For the energy-dependent (multigroup) rates the same inequalities must be
    strict.
#  T4  Which rung is the effective single rate of a spread of rates?  Pre-registered claim (to be killed if
    wrong): the paper's
#      ladder says the effective rate is a power mean of the group rates, M_p, p in {-2,-1,0,+1,+2}. PASS only
     if ONE rung p
#      reproduces BOTH of Steve's calibrated decoupling temperatures, e+- only: 1.6 MeV and with nu-nu:
    0.93-1.0 MeV,
#      each within 15 percent, using only the rate numbers printed in his Appendix C. Opposite run: every
    other p tried.
import numpy as np
from scipy.special import erfc
from scipy.optimize import brentq
rng = np.random.default_rng(3010)
fro2 = lambda X: float(np.real(np.vdot(X, X)))

print('=== T0 Steve-internal consistency ===')
dc_f = 0.49773
rows = [('inst eps0=0.1 Rnu=.409', 0.52832, 9.6), ('dec 0.25', 0.52709, 9.2), ('dec 0.5', 0.52428, 8.3), ('dec
     1', 0.51865, 6.5),
       ('dec 2', 0.51092, 4.0), ('dec 4', 0.50354, 1.8), ('dec 8', 0.49967, 0.6),
       ('Rnu=.10', 0.50547, 2.3), ('Rnu=.20', 0.51297, 4.7), ('Rnu=.30', 0.52027, 7.0), ('Rnu=.488', 0.53379,
     11.5),
       ('alpha .3', 0.47747, 9.3), ('alpha .5', 0.49682, 9.4), ('alpha 2', 0.56342, 9.9), ('alpha 4',
    0.59734, 10.0)]
Cl = lambda d: (4/3)*(1-np.sqrt(1-1.5*d))
fl = {'alpha .3': 0.44805, 'alpha .5': 0.46687, 'alpha 2': 0.53274, 'alpha 4': 0.56819}
for nm, dn, printed in rows:
    df = fl.get(nm, dc_f); s = 100*(Cl(dn)/Cl(df)-1)
    print('%-24s Delta C_l,c = %5.2f %%  printed %5.1f  diff %+5.2f %s' % (nm, s, printed, s-printed, 'FLAG'
     if abs(s-printed) > 0.3 else ''))
sh = {0.25: 5.90, 0.5: 5.33, 1: 4.20, 2: 2.65, 4: 1.17, 8: 0.39}; inst = 6.15
Gbr = {0.25: .959, .5: .867, 1: .683, 2: .431, 4: .190, 8: .063}
G = {e: (v)/inst for e, v in sh.items()}
# use the delta_c columns directly (more digits than the percent column)
dcs = {0.25: 0.52709, .5: 0.52428, 1: 0.51865, 2: 0.51092, 4: 0.50354, 8: 0.49967}
G = {e: (dcs[e]-dc_f)/(0.52832-dc_f) for e in dcs}
for e in G: print('eps_dec %5.2f  G(from delta_c) = %.3f   brief %.3f   diff %+.3f %s' % (e, G[e], Gbr[e], G[e
    ]-Gbr[e], 'FLAG' if abs(G[e]-Gbr[e]) > .005 else ''))
Rv = np.array([.1, .2, .3, .409, .488]); dC = np.array([2.3, 4.7, 7.0, 9.6, 11.5]); dd = np.array([1.56, 3.06,
     4.53, 6.15, 7.24])
print('shift/R_nu  C_l: ', np.round(dC/Rv, 2), ' spread %.1f %%' % (100*np.std(dC/Rv)/np.mean(dC/Rv)))
print('shift/R_nu  delta_c: ', np.round(dd/Rv, 2), ' spread %.1f %%' % (100*np.std(dd/Rv)/np.mean(dd/Rv)))
# Table III slopes and ratios
x = np.array([1e-1, 3e-2, 1e-2, 3e-3, 1e-3, 3e-4, 1e-4]); pf = np.array([.679, .497, .390, .288, .215, .151,
    .105]); fs = np.array([.799, .590, .470, .343, .255, .178, .123])
print('Table III ratios', np.round(fs/pf, 3), ' slopes', np.polyfit(np.log(x), np.log(pf), 1)[0].round(3), np.
    polyfit(np.log(x), np.log(fs), 1)[0].round(3))
# beta^1.18 rule: Gaussian C_l, erfc(C/(sqrt2 sigma))
for b in [1e-9, 1e-12, 1e-15]:
    sig = 0.662/np.sqrt(2)/brentq(lambda y: np.log(erfc(y))-np.log(b), 0.5, 10)
    bn = erfc(0.726/(np.sqrt(2)*sig)); print('beta_pf %.0e -> beta_nu/beta_pf = %.4f (Steve:
     0.021/0.0052/0.0013), exponent %.3f' % (b, bn/b, np.log(bn)/np.log(b)))
print('coincidence list for the 1.19 mass ratio: (1-Rnu)^(-1/3) = %.4f ; (delta_c ratio)^3 = %.4f ; (C_l ratio
    )^2 = %.4f ; Table III 1.17-1.20' %
      ((1-.409)**(-1/3), (0.52832/0.49773)**3, (Cl(0.52832)/Cl(0.49773))**2))
```

```
print('Rnu-dependence of a power law (1-Rnu)^-x needed for the C_l shift: x = %.3f at .409, %.3f at .488, %.3f
      at .1 (not constant -> rejected)' %
      tuple(np.log(Cl(d)/Cl(dc_f))/(-np.log(1-r)) for d, r in [(0.52832, .409), (0.53379, .488), (0.50547, .1)
     ]))

print('\n=== T1 Prop 5 / connection energy on Steve operator ===')
lmax = 80; a = lambda l: l/np.sqrt(4.0*l*l-1)
M = np.zeros((lmax+1, lmax+1))
for l in range(1, lmax+1): M[l, l-1] = M[l-1, l] = a(l)
k, kap = 1.7, 0.9
P = np.zeros_like(M); P[0, 0] = 1; I = np.eye(lmax+1)
H = k*M
EB = fro2(P @ H @ (I-P)); comm = 0.5*fro2(H @ P - P @ H); var = (H @ H)[0, 0]-(H)[0, 0]**2
print('E_B = %.15f  k^2/3 = %.15f  (1/2)||[H,P]||^2 = %.15f  Var_0(H) = %.15f' % (EB, k*k/3, comm, var))
assert abs(EB/(k*k/3)-1) < 1e-12 and abs(comm/EB-1) < 1e-12 and abs(var/EB-1) < 1e-12
Dg = np.diag([0.0]+[kap]*lmax); Hp = Dg+1j*H
lhs = 0.25*fro2(Hp @ Hp.conj().T-Hp.conj().T @ Hp); rhs = fro2(Dg @ H-H @ Dg)
print('(1/4)||[H,H^dag]||^2 = %.12f   ||[D,K]||^2 = %.12f   kappa^2 k^2 * 2/3 = %.12f' % (lhs, rhs, kap**2*k*k
     *2/3))
assert abs(lhs/rhs-1) < 1e-10
print('T1 PASS: sqrt(E_B)/k = %.6f = c_s = 1/sqrt3 (= <mu^2>^(1/2), the radiation sound speed)' % (np.sqrt(EB)
     /k))

print('\n=== T2 hidden-walk (Schur) formula vs arctan closed form ===')
lm = 600; Mb = np.zeros((lm+1, lm+1))
for l in range(1, lm+1): Mb[l, l-1] = Mb[l-1, l] = a(l)
Lm = -1j*k*Mb-kap*(np.eye(lm+1)-np.diag([1.0]+[0]*lm))
W = Lm[0:1, 1:]; F = Lm[1:, 1:]; Wd = Lm[1:, 0:1]
worst = 0
for _ in range(50):
    z = complex(rng.uniform(0.2, 2), rng.uniform(-3, 3))
    GB = 1/(z-0-(W @ np.linalg.solve(z*np.eye(lm)-F, Wd))[0, 0])      # paper formula with E=0
    s = z+kap; g = np.arctan(k/s)/k; Gc = g/(1-kap*g)
    full = np.linalg.inv(z*np.eye(lm+1)-Lm)[0, 0]
    worst = max(worst, abs(GB-Gc)/abs(Gc), abs(full-Gc)/abs(Gc))
print('worst relative difference (Schur vs closed form vs full inverse): %.2e' % worst); assert worst < 1e-8
for kp in [0.5, 2*k/np.sqrt(3)-1e-3, 2*k/np.sqrt(3), 2*k/np.sqrt(3)+1e-3, 3.0]:
    ev = np.linalg.eigvals(np.array([[0, -1j*k/np.sqrt(3)], [-1j*k/np.sqrt(3), -kp]]))
    print('  kappa = %.4f  two-level eigenvalue separation %.2e' % (kp, abs(ev[0]-ev[1])))
print('T2 PASS. EP at kappa = 2 sqrt(E_B) = %.4f' % (2*np.sqrt(EB)))

print('\n=== T3 ladder on the single-rate model ===')
Kc = k*M; gaps = np.zeros((lmax+1, lmax+1)); gaps[0, 1:] = kap; gaps[1:, 0] = kap   # only the cut edge has a
     gap kappa; hidden-hidden gaps are 0
def Sk(kk, gp):
    m = (np.abs(Kc) > 0) & (gp > 0); return float(np.sum(np.abs(Kc[m])**2*gp[m]**kk))
S = {kk: Sk(kk, gaps) for kk in [-2, -1, 0, 1, 2]}
print('S_k:', {kk: round(v, 6) for kk, v in S.items()})
print('equality checks: S0^2/(S-2 S2) = %.12f ; S-1^2/(S-2 S0) = %.12f' % (S[0]**2/(S[-2]*S[2]), S[-1]**2/(S
     [-2]*S[0])))
assert abs(S[0]**2/(S[-2]*S[2])-1) < 1e-12 and abs(S[-1]**2/(S[-2]*S[0])-1) < 1e-12
print('rung +1 for this model: S_1 = E_B*2*kappa (counting both orderings) = %.6f' % S[1])

print('\n=== T4 which rung is the effective rate? (Steve Appendix C numbers only) ===')
# Gamma = c G_F^2 T^4 E.  Calibrate with the unblocked E=T coefficients: 0.506 -> 1.9 MeV (nu_e), 0.108 -> 3.1
      MeV (nu_mu,tau) : T_dec = 1.9 (0.506/C)^(1/3)
def Tdec(C): return 1.9*(0.506/C)**(1/3)
print('sanity: C=0.108 -> %.2f MeV (Steve 3.1)' % Tdec(0.108))
# energy groups: six groups x=0.4..20, edges uniform in ln x (Steve App. C), energy weight x^4/(e^x+1) dlnx, c
     (x) interpolated linearly in ln x between his endpoints
edges = np.linspace(np.log(0.4), np.log(20), 7); xs = np.exp(0.5*(edges[1:]+edges[:-1]))
w = xs**4/(np.exp(xs)+1)*np.diff(edges); w /= w.sum()
def cfun(x, c0, c1): return c0+(c1-c0)*np.log(x/0.5)/np.log(20/0.5)
flav = {'nu_e': dict(wt=1/3, ee=(0.34, 0.47)), 'nu_mutau': dict(wt=2/3, ee=(0.073, 0.101))}
cnn = (0.49, 0.68)
def rates(fl, withnn):
    r = xs*cfun(xs, *flav[fl]['ee']);
    if withnn: r = r+xs*cfun(xs, *cnn)       # anisotropy (l>=2) damped by e+- plus nu-nu
```

```
    return r
def pmean(r, wt, p):
    if p == 0: return np.exp(np.sum(wt*np.log(r)))
    return np.sum(wt*r**p)**(1/p)
print('group energy weights', np.round(w, 3))
print('model check: single-group at x=3.15 (e+- only) T_dec: nu_e %.2f MeV (Steve 1.3), nu_mutau %.2f MeV (
    Steve 2.3)' %
      (Tdec(3.15*cfun(3.15, *flav['nu_e']['ee'])), Tdec(3.15*cfun(3.15, *flav['nu_mutau']['ee']))))
res = {}
for withnn in [False, True]:
    for p in [-2, -1, 0, 1, 2]:
        # level 1: power mean over energy groups within a flavour; level 2: same-p mean over flavours (flavour
     weights 1/3, 2/3)
        Cf = [pmean(rates(f, withnn), w, p) for f in flav]; wf = np.array([flav[f]['wt'] for f in flav])
        Ceff = pmean(np.array(Cf), wf, p); res[(withnn, p)] = Tdec(Ceff)
for p in [-2, -1, 0, 1, 2]:
    t1, t2 = res[(False, p)], res[(True, p)]
    ok = abs(t1/1.6-1) < .15 and (0.93*0.85 <= t2 <= 1.0*1.15)
    print('rung p=%+d : T_dec(e+- only) = %.2f MeV (target 1.6)   T_dec(with nu-nu) = %.2f MeV (target
    0.93-1.0)   both within 15%%: %s' % (p, t1, t2, ok))
print('ratio of masses (T^-2) e+- only -> with nu-nu, rung +1: %.2f, rung -1: %.2f (Steve: 2.5-3)' % ((res[(
    False, 1)]/res[(True, 1)])**2, (res[(False, -1)]/res[(True, -1)])**2))
```

```
=== T0 Steve-internal consistency ===
inst eps0=0.1 Rnu=.409   Delta C_l,c =  9.64 %  printed   9.6  diff +0.04
dec 0.25                 Delta C_l,c =  9.23 %  printed   9.2  diff +0.03
dec 0.5                  Delta C_l,c =  8.31 %  printed   8.3  diff +0.01
dec 1                    Delta C_l,c =  6.48 %  printed   6.5  diff -0.02
dec 2                    Delta C_l,c =  4.04 %  printed   4.0  diff +0.04
dec 4                    Delta C_l,c =  1.76 %  printed   1.8  diff -0.04
dec 8                    Delta C_l,c =  0.58 %  printed   0.6  diff -0.02
Rnu=.10                  Delta C_l,c =  2.35 %  printed   2.3  diff +0.05
Rnu=.20                  Delta C_l,c =  4.68 %  printed   4.7  diff -0.02
Rnu=.30                  Delta C_l,c =  7.00 %  printed   7.0  diff +0.00
Rnu=.488                 Delta C_l,c = 11.47 %  printed  11.5  diff -0.03
alpha .3                 Delta C_l,c =  9.34 %  printed   9.3  diff +0.04
alpha .5                 Delta C_l,c =  9.43 %  printed   9.4  diff +0.03
alpha 2                  Delta C_l,c =  9.91 %  printed   9.9  diff +0.01
alpha 4                  Delta C_l,c = 10.05 %  printed  10.0  diff +0.05
eps_dec  0.25  G(from delta_c) = 0.960   brief 0.959   diff +0.001
eps_dec  0.50  G(from delta_c) = 0.868   brief 0.867   diff +0.001
eps_dec  1.00  G(from delta_c) = 0.684   brief 0.683   diff +0.001
eps_dec  2.00  G(from delta_c) = 0.431   brief 0.431   diff +0.000
eps_dec  4.00  G(from delta_c) = 0.190   brief 0.190   diff -0.000
eps_dec  8.00  G(from delta_c) = 0.063   brief 0.063   diff +0.000
shift/R_nu  C_l:  [23.   23.5  23.33 23.47 23.57]  spread 0.9 %
shift/R_nu  delta_c:  [15.6  15.3  15.1  15.04 14.84]  spread 1.7 %
Table III ratios [1.177 1.187 1.205 1.191 1.186 1.179 1.171]  slopes 0.266 0.267
beta_pf 1e-09 -> beta_nu/beta_pf = 0.0208 (Steve: 0.021/0.0052/0.0013), exponent 1.187
beta_pf 1e-12 -> beta_nu/beta_pf = 0.0053 (Steve: 0.021/0.0052/0.0013), exponent 1.190
beta_pf 1e-15 -> beta_nu/beta_pf = 0.0013 (Steve: 0.021/0.0052/0.0013), exponent 1.192
coincidence list for the 1.19 mass ratio: (1-Rnu)^(-1/3) = 1.1916 ; (delta_c ratio)^3 = 1.1959 ; (C_l ratio)^2
     = 1.2020 ; Table III 1.17-1.20
Rnu-dependence of a power law (1-Rnu)^-x needed for the C_l shift: x = 0.175 at .409, 0.162 at .488, 0.220 at
    .1 (not constant -> rejected)

=== T1 Prop 5 / connection energy on Steve operator ===
E_B = 0.963333333333333  k^2/3 = 0.963333333333333  (1/2)||[H,P]||^2 = 0.963333333333333  Var_0(H) =
    0.963333333333333
(1/4)||[H,H^dag]||^2 = 1.560600000000   ||[D,K]||^2 = 1.560600000000   kappa^2 k^2 * 2/3 = 1.560600000000
T1 PASS: sqrt(E_B)/k = 0.577350 = c_s = 1/sqrt3 (= <mu^2>^(1/2), the radiation sound speed)

=== T2 hidden-walk (Schur) formula vs arctan closed form ===
worst relative difference (Schur vs closed form vs full inverse): 5.29e-16
  kappa = 0.5000  two-level eigenvalue separation 1.90e+00
  kappa = 1.9620  two-level eigenvalue separation 6.26e-02
  kappa = 1.9630  two-level eigenvalue separation 4.18e-08
  kappa = 1.9640  two-level eigenvalue separation 6.27e-02
```

```
  kappa = 3.0000  two-level eigenvalue separation 2.27e+00
T2 PASS. EP at kappa = 2 sqrt(E_B) = 1.9630

=== T3 ladder on the single-rate model ===
S_k: {-2: 2.378601, -1: 2.140741, 0: 1.926667, 1: 1.734, 2: 1.5606}
equality checks: S0^2/(S-2 S2) = 1.000000000000 ; S-1^2/(S-2 S0) = 1.000000000000
rung +1 for this model: S_1 = E_B*2*kappa (counting both orderings) = 1.734000

=== T4 which rung is the effective rate? (Steve Appendix C numbers only) ===
sanity: C=0.108 -> 3.18 MeV (Steve 3.1)
group energy weights [0.004 0.038 0.229 0.528 0.199 0.003]
model check: single-group at x=3.15 (e+- only) T_dec: nu_e 1.40 MeV (Steve 1.3), nu_mutau 2.33 MeV (Steve 2.3)
rung p=-2 : T_dec(e+- only) = 2.35 MeV (target 1.6)   T_dec(with nu-nu) = 1.22 MeV (target 0.93-1.0)   both
    within 15%: False
rung p=-1 : T_dec(e+- only) = 2.12 MeV (target 1.6)   T_dec(with nu-nu) = 1.14 MeV (target 0.93-1.0)   both
    within 15%: False
rung p=+0 : T_dec(e+- only) = 1.86 MeV (target 1.6)   T_dec(with nu-nu) = 1.07 MeV (target 0.93-1.0)   both
    within 15%: False
rung p=+1 : T_dec(e+- only) = 1.62 MeV (target 1.6)   T_dec(with nu-nu) = 1.02 MeV (target 0.93-1.0)   both
    within 15%: True
rung p=+2 : T_dec(e+- only) = 1.44 MeV (target 1.6)   T_dec(with nu-nu) = 0.97 MeV (target 0.93-1.0)   both
    within 15%: True
ratio of masses (T^-2) e+- only -> with nu-nu, rung +1: 2.53, rung -1: 3.44 (Steve: 2.5-3)
```

```
# SCRIPT: WB-STEVE-04-CHANNELS
# (i) Non-normality channels of a collision operator with different rates per multipole (Steve open item (c)),
     via the paper's
#    connection-energy identity.  (ii) Item (e): neutrinos as a self-interacting fluid that exchanges NO
    momentum with the plasma.
# KILL CONDITIONS (written before the run):
#  C1  For L = -i k mu - diag(kappa_l), S_2 = ||[D,K]||^2 = (1/4)||[H',H'^dag]||^2 splits over Legendre edges
    l<->l+1 as
#      sum_l 2 k^2 a_{l+1}^2 (kappa_l - kappa_{l+1})^2, a_l^2 = l^2/(4l^2-1).  Dies if off by > 1e-10 relative
    .
#      Consequence (derived): the relaxation-time model (all kappa_l equal for l>=1) puts ALL non-normality on
     edge 0-1; any
#      model with kappa_1 != kappa_2 puts weight on edge 1-2.
#  E1  Two perfect fluids with equal sound speed and no momentum exchange are one fluid: in pure radiation (p
    = e/3) the shift of
#      the first maximum must vanish as the neutrino self-interaction rate K0 = kappa/H -> infinity.
    Prediction from the paper's
#      rung -1 (viscosity weight of the 1-2 edge, S_-1 = (4/15) k^2/kappa): the shift times K0 tends to a
    constant.
#      Dies if (a) shift does not decrease monotonically with K0, or (b) shift*K0 differs by more than 15
    percent between
#      K0 = 100 and K0 = 1000.  Opposite run: the same operator with the drag rate on (flux relaxed to the
    plasma) instead.
import numpy as np
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
fro2 = lambda X: float(np.real(np.vdot(X, X)))
lmax = 60; a2 = lambda l: l*l/(4.0*l*l-1)
M = np.zeros((lmax+1, lmax+1))
for l in range(1, lmax+1): M[l, l-1] = M[l-1, l] = np.sqrt(a2(l))
print('=== C1 non-normality channels ===')
k = 1.3
for name, kap in [('relaxation time (all l>=1 equal) kappa_e=1', [0]+[1.0]*lmax),
                  ('nu_e: flux damped by e+- (1), stress by e+- + nu-nu (1+1.5)', [0, 1.0]+[2.5]*(lmax-1)),
                  ('nu_mu,tau: flux (1), stress (1+7)', [0, 1.0]+[8.0]*(lmax-1))]:
    D = np.diag(kap); Hp = D+1j*k*M
    S2 = fro2(D @ (k*M)-(k*M) @ D); chk = 0.25*fro2(Hp @ Hp.conj().T-Hp.conj().T @ Hp)
    edges = [2*k*k*a2(l+1)*(kap[l]-kap[l+1])**2 for l in range(0, 6)]
    tot = sum(2*k*k*a2(l+1)*(kap[l]-kap[l+1])**2 for l in range(0, lmax))
    print('%-62s S_2 = %.6f (commutator form %.6f, edge sum %.6f)' % (name, S2, chk, tot))
    assert abs(S2/chk-1) < 1e-10 and abs(S2/tot-1) < 1e-10
    print('    edge energies 0-1, 1-2, 2-3, 3-4: %s ; share on edge 1-2: %.1f %%' % (np.round(edges[:4], 4),
     100*edges[1]/tot))
wt = np.array([1/3, 2/3]); e12 = np.array([2.25*(4/15), 49*(4/15)]); e01 = np.array([1/3, 1/3])
```

```
print('flavour-weighted ratio (stress edge)/(flux edge) in units of kappa_e^2 k^2 : %.2f' % (np.sum(wt*e12)/np
    .sum(wt*e01)))
print('E_B for the cut P0 (plasma-coupled monopole): k^2/3 ; for the cut P01 (flux kept, i.e. nu as own fluid)
    : 4 k^2/15 ; ratio %.4f' % ((4/15)/(1/3)))
E_B01 = fro2(np.diag([1, 1]+[0]*(lmax-1)) @ (k*M) @ np.diag([0, 0]+[1]*(lmax-1)))
print('numerical E_B(P01) = %.10f ; 4k^2/15 = %.10f' % (E_B01, 4*k*k/15))

print('\n=== E1 self-interacting neutrinos with no momentum exchange, pure radiation (first-max shift of Delta
    ) ===')
LM = 40
def run(k, Rn, K0, drag, tau0=0.02, tg=None):
    n = 4+(LM-1); LS = np.arange(3, LM+1)
    def rhs(t, y):
        dg, tg_, dn, tn = y[:4]; F = np.zeros(LM+2); F[2:LM+1] = y[4:]; F[LM+1] = (2*LM+1)/(k*t)*F[LM]-F[LM-1]
        th = (1-Rn)*tg_+Rn*tn; de = (1-Rn)*dg+Rn*dn; sg = F[2]/2; Dl = de+4*th/(k*k*t)
        phi = -1.5*Dl/(k*t)**2; psi = phi-6*Rn*sg/(k*t)**2; dphi = -psi/t+2*th/(k*k*t*t)
        K = K0/t; dy = np.empty_like(y)               # a*kappa = K0 * (a H) = K0 / tau
        Kd = K if drag else 0.0
        dy[0] = -4/3*tg_+4*dphi
        dy[1] = k*k*dg/4+k*k*psi-Kd*(Rn/(1-Rn))*(tg_-tn)
        dy[2] = -4/3*tn+4*dphi
        dy[3] = k*k*(dn/4-sg)+k*k*psi+Kd*(tg_-tn)
        dy[4] = 8/15*tn-0.6*k*F[3]-K*F[2]
        dy[5:] = k/(2*LS+1)*(LS*F[2:LM]-(LS+1)*F[4:LM+2])-K*F[3:LM+1]
        return dy
    psi0 = 2/3; d0 = -2*psi0-(2/3)*(k*tau0)**2*psi0; t0 = 0.5*k*k*tau0*psi0
    y0 = np.zeros(n); y0[:4] = [d0, t0, d0, t0]
    sol = solve_ivp(rhs, (tau0, tg[-1]), y0, method='BDF', t_eval=tg, rtol=1e-10, atol=1e-13)
    Y = sol.y; th = (1-Rn)*Y[1]+Rn*Y[3]; de = (1-Rn)*Y[0]+Rn*Y[2]
    return de+4*th/(k*k*tg)
def fmax(x, y):
    s = -y
    for i in range(1, len(s)-1):
        if s[i] > s[i-1] and s[i] >= s[i+1]:
            A, B, C = s[i-1], s[i], s[i+1]; off = 0.5*(A-C)/(A-2*B+C); return B-0.25*(A-C)*off
tg = np.linspace(0.02, 8.0, 3000); kk = 1.0
m_pf = fmax(tg, run(kk, 0.0, 0.0, False, tg=tg))
for drag, tag in [(False, 'self-interaction only (no momentum exchange with plasma)'), (True, 'drag on the
    plasma included (Steve relaxation model)')]:
    print(tag)
    prev = None; vals = {}
    for K0 in [3, 10, 30, 100, 300, 1000, 3000]:
        r = fmax(tg, run(kk, 0.409, K0, drag, tg=tg))/m_pf; vals[K0] = 1/r-1
        print('   K0 = %5d : first-max ratio %.5f ; implied threshold rise %.4f %% ; rise*K0 = %.3f' % (K0, r,
     100*vals[K0], 100*vals[K0]*K0))
    if not drag:
        mono = all(vals[a] > vals[b] for a, b in zip([3, 10, 30, 100, 300, 1000], [10, 30, 100, 300, 1000,
    3000]))
        const = abs(vals[1000]*1000/(vals[100]*100)-1)
        print('   E1 (a) monotone decrease: %s ; (b) rise*K0 changes by %.1f %% between K0=100 and 1000 -> %s'
     % (mono, 100*const, 'PASS' if mono and const < .15 else 'DIES'))
```

```
=== C1 non-normality channels ===
relaxation time (all l>=1 equal) kappa_e=1                    S_2 = 1.126667 (commutator form 1.126667, edge
    sum 1.126667)
    edge energies 0-1, 1-2, 2-3, 3-4: [1.1267 0.     0.     0.    ] ; share on edge 1-2: 0.0 %
nu_e: flux damped by e+- (1), stress by e+- + nu-nu (1+1.5)    S_2 = 3.154667 (commutator form 3.154667, edge
    sum 3.154667)
    edge energies 0-1, 1-2, 2-3, 3-4: [1.1267 2.028  0.     0.    ] ; share on edge 1-2: 64.3 %
nu_mu,tau: flux (1), stress (1+7)                             S_2 = 45.292000 (commutator form 45.292000,
    edge sum 45.292000)
    edge energies 0-1, 1-2, 2-3, 3-4: [ 1.1267 44.1653  0.      0.    ] ; share on edge 1-2: 97.5 %
flavour-weighted ratio (stress edge)/(flux edge) in units of kappa_e^2 k^2 : 26.73
E_B for the cut P0 (plasma-coupled monopole): k^2/3 ; for the cut P01 (flux kept, i.e. nu as own fluid): 4 k
    ^2/15 ; ratio 0.8000
numerical E_B(P01) = 0.4506666667 ; 4k^2/15 = 0.4506666667

=== E1 self-interacting neutrinos with no momentum exchange, pure radiation (first-max shift of Delta) ===
```

```
self-interaction only (no momentum exchange with plasma)
   K0 =     3 : first-max ratio 0.96022 ; implied threshold rise 4.1424 % ; rise*K0 = 12.427
   K0 =    10 : first-max ratio 0.98310 ; implied threshold rise 1.7192 % ; rise*K0 = 17.192
   K0 =    30 : first-max ratio 0.99370 ; implied threshold rise 0.6345 % ; rise*K0 = 19.035
   K0 =   100 : first-max ratio 0.99804 ; implied threshold rise 0.1969 % ; rise*K0 = 19.688
   K0 =   300 : first-max ratio 0.99934 ; implied threshold rise 0.0662 % ; rise*K0 = 19.849
   K0 =  1000 : first-max ratio 0.99980 ; implied threshold rise 0.0199 % ; rise*K0 = 19.918
   K0 =  3000 : first-max ratio 0.99993 ; implied threshold rise 0.0067 % ; rise*K0 = 20.088
   E1 (a) monotone decrease: True ; (b) rise*K0 changes by 1.2 % between K0=100 and 1000 -> PASS
drag on the plasma included (Steve relaxation model)
   K0 =     3 : first-max ratio 0.95931 ; implied threshold rise 4.2421 % ; rise*K0 = 12.726
   K0 =    10 : first-max ratio 0.98281 ; implied threshold rise 1.7488 % ; rise*K0 = 17.488
   K0 =    30 : first-max ratio 0.99364 ; implied threshold rise 0.6396 % ; rise*K0 = 19.187
   K0 =   100 : first-max ratio 0.99803 ; implied threshold rise 0.1974 % ; rise*K0 = 19.743
   K0 =   300 : first-max ratio 0.99934 ; implied threshold rise 0.0664 % ; rise*K0 = 19.910
   K0 =  1000 : first-max ratio 0.99980 ; implied threshold rise 0.0200 % ; rise*K0 = 20.042
   K0 =  3000 : first-max ratio 0.99993 ; implied threshold rise 0.0067 % ; rise*K0 = 20.210
```

B.8 The full operator and the truncated wall

```
# SCRIPT: OSL-STEVE-B-EP-FULL-OPERATOR
# Question: Steve's neutrino operator L = -i k mu - kappa (I - P0) (momentum given to plasma, so l=1 is damped
      too).
# Paper-derived claim to test: the two-moment truncation has its exceptional point at kappa_EP = 2 sqrt(E_B) =
      2k/sqrt3
# (critical damping of a pair coupled with weight E_B = k^2/3). Does that EP survive when all l are kept?
# KILL CONDITIONS (written before the run):
# KB1 (truncation control) N=2 EP must come out at 2/sqrt3 = 1.1547 (k=1) to 1e-6, else the code is wrong.
# KB2 the claim 'kappa_EP = 2 sqrt(E_B)' survives in the full operator iff the N=200 value is within 10% of
     1.1547.
# KB3 (opposite / inverse reading) the statement 'above kappa_EP the neutrinos are overdamped (no oscillating
     mode)' survives
#     iff for N=200 every eigenvalue with Re > -kappa/2... is real for kappa > 1.5*kappa_EP; we report the
     least-damped eigenvalue's Im.
import numpy as np
def L(N, kap, k=1.0):
    n = np.arange(1, N); a = n / np.sqrt(4.0 * n**2 - 1.0)       # mu P_n = a_{n+1} P_{n+1} + a_n P_{n-1},
     orthonormal Legendre
    J = np.diag(a, 1) + np.diag(a, -1)
    D = np.eye(N); D[0, 0] = 0.0
    return -1j * k * J - kap * D
def slowest(N, kap):
    ev = np.linalg.eigvals(L(N, kap)); i = np.argmax(ev.real); return ev[i], ev
def ep(N, lo=0.01, hi=6.0):
    # smallest kappa such that the least-damped eigenvalue is real (|Im| < 1e-9) and stays real above it
    ks = np.linspace(lo, hi, 5901); flag = np.array([abs(slowest(N, x)[0].imag) < 1e-7 for x in ks])
    # last kappa at which a non-real least-damped eigenvalue occurs
    bad = np.where(~flag)[0]
    return ks[bad.max() + 1] if len(bad) and bad.max() + 1 < len(ks) else (ks[0] if len(bad) == 0 else np.nan)
print('2/sqrt3 = %.6f' % (2 / 3**0.5))
res = {}
for N in [2, 3, 5, 10, 40, 200]:
    res[N] = ep(N); print('N = %3d : least-damped eigenvalue becomes real (and stays real) for kappa >= %.4f'
     % (N, res[N]))
print('KB1: |EP(N=2) - 2/sqrt3| = %.2e' % abs(res[2] - 2 / 3**0.5))
print('KB2: EP(N=200)/(2/sqrt3) = %.3f' % (res[200] / (2 / 3**0.5)))
for kap in [0.5, 1.0, 1.5, 2.0, 3.0, 6.0]:
    for N in [2, 200]:
        lam, ev = slowest(N, kap); print('kappa=%.1f N=%3d least-damped lambda = %+.4f %+.4fi ; diffusion
     estimate -k^2/(3 kappa) = %+.4f' % (kap, N, lam.real, lam.imag, -1 / (3 * kap)))
```

```
2/sqrt3 = 1.154701
N =   2 : least-damped eigenvalue becomes real (and stays real) for kappa >= 1.1552
N =   3 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.0100
N =   5 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.0100
```

```
N =  10 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.7552
N =  40 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.6719
N = 200 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.6455
KB1: |EP(N=2) - 2/sqrt3| = 5.06e-04
KB2: EP(N=200)/(2/sqrt3) = 0.559
kappa=0.5 N=  2 least-damped lambda = -0.2500 -0.5204i ; diffusion estimate -k^2/(3 kappa) = -0.6667
kappa=0.5 N=200 least-damped lambda = -0.4947 -0.0078i ; diffusion estimate -k^2/(3 kappa) = -0.6667
kappa=1.0 N=  2 least-damped lambda = -0.5000 +0.2887i ; diffusion estimate -k^2/(3 kappa) = -0.3333
kappa=1.0 N=200 least-damped lambda = -0.3579 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.3333
kappa=1.5 N=  2 least-damped lambda = -0.2713 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.2222
kappa=1.5 N=200 least-damped lambda = -0.2291 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.2222
kappa=2.0 N=  2 least-damped lambda = -0.1835 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1667
kappa=2.0 N=200 least-damped lambda = -0.1695 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1667
kappa=3.0 N=  2 least-damped lambda = -0.1156 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1111
kappa=3.0 N=200 least-damped lambda = -0.1119 -0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1111
kappa=6.0 N=  2 least-damped lambda = -0.0561 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.0556
kappa=6.0 N=200 least-damped lambda = -0.0557 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.0556

[exited with code 0]
```

B.9 The annihilation contest

```
# SCRIPT: OSL-STEVE-C-SPEED-CONTEST-EOS
# Mapping under test (from the paper + Check A): the neutrino rms streaming speed is sqrt(E_B)/k = sqrt(<mu
    ^2>) = 1/sqrt3 (exact) and the
# plasma sound speed is c_s = sqrt(dp/de). In pure radiation they coincide; during e+- annihilation c_s falls
    below 1/sqrt3, so a
# 'speed contest' reading predicts the free-streaming advantage GROWS there. Pre-registered quantitative form
    (no fit):
#   shift_pred(T_H) = 0.15 * R_nu(T_H) * G(eps_dec = T_H/T_dec) * f,  f = (1 - c_s(T_H)) / (1 - 1/sqrt3)   (
    front speed 1 against sound)
#   baseline        = 0.15 * R_nu(T_H) * G(...)                                                          (
    no contrast factor)
# Steve's published envelope: relative shift of delta_c is 5.3% to 6.2% at ALL sample masses 3e5 .. 7e7 Msun (
    Sec. IV D).
# VALIDATION (must pass first): my plasma p/e minimum must be 0.274 +- 0.003 near T = 0.16 MeV; R_nu = 0.488 (
    T>>m_e) and 0.405 (T<<m_e).
# KILL CONDITIONS (written before the run):
# KC0 validation fails -> stop, code not trusted.
# KC1 'contrast enhancement' mapping dies if shift_pred (with f) exceeds 6.5% at any sample mass between 3e6
    and 7e7 Msun (G ~ 1 there).
# KC2 opposite reading (the contrast SHRINKS the effect: f' = 1/f) dies if shift_pred(f') is below 5.3% at any
     sample mass 3e5..7e7.
import numpy as np
from scipy.integrate import quad
me = 0.511
def fd(T, n):  # fermi-dirac integrals for g=4 (e+-), zero chem. potential
    pass
def eP(T):
    f = lambda p, w: p**2 * w(p) / (np.exp(np.sqrt(p**2 + me**2) / T) + 1)
    e = quad(lambda p: p**2 * np.sqrt(p**2 + me**2) / (np.exp(np.sqrt(p**2 + me**2) / T) + 1), 0, 60 * T + 10)
     [0] * 4 / (2 * np.pi**2)
    P = quad(lambda p: p**4 / np.sqrt(p**2 + me**2) / (np.exp(np.sqrt(p**2 + me**2) / T) + 1), 0, 60 * T + 10)
     [0] * 4 / (6 * np.pi**2)
    return e + np.pi**2 / 15 * T**4, P + np.pi**2 / 45 * T**4
def sdens(T): e, P = eP(T); return (e + P) / T
s_hi_coef = 2 * np.pi**2 / 45 * 5.5
def Tnu(T): return (sdens(T) / s_hi_coef)**(1 / 3)
def Rnu(T): enu = 6 * 7 / 8 * np.pi**2 / 30 * Tnu(T)**4; e, P = eP(T); return enu / (enu + e)
def cs2(T): h = 1e-4 * T; (e1, P1), (e2, P2) = eP(T - h), eP(T + h); return (P2 - P1) / (e2 - e1)
Ts = np.geomspace(0.03, 20, 160); pe = np.array([eP(T)[1] / eP(T)[0] for T in Ts]); i = np.argmin(pe)
print('KC0 validation: min p/e = %.4f at T = %.3f MeV (Steve: 0.274 at ~0.16)' % (pe[i], Ts[i]))
print('   R_nu(T=20) = %.4f (Steve 0.488) ; R_nu(T=0.03) = %.4f (Steve 0.405 for Neff=3)' % (Rnu(20.0), Rnu
    (0.03)))
c2 = np.array([cs2(T) for T in Ts]); j = np.argmin(c2); print('   min c_s^2 = dp/de = %.4f (c_s = %.4f) at T =
     %.3f MeV ; 1/3 = 0.3333' % (c2[j], c2[j]**0.5, Ts[j]))
```

```
ok0 = abs(pe[i] - 0.274) < 0.003 and abs(Rnu(20.0) - 0.488) < 0.002 and abs(Rnu(0.03) - 0.405) < 0.002
print('KC0 PASS' if ok0 else 'KC0 FAIL')
# horizon temperature from mass: M_H = 1.5e5 (T/MeV)^-2 (g*/10.75)^-1/2 ; g* defined by rho = (pi^2/30) g* T^4
      (plasma + nu)
def gstar(T):
    e, P = eP(T); enu = 6 * 7 / 8 * np.pi**2 / 30 * Tnu(T)**4; return (e + enu) / (np.pi**2 / 30 * T**4)
def TH_of(M):
    from scipy.optimize import brentq
    return brentq(lambda T: 1.5e5 * T**-2 * (gstar(T) / 10.75)**-0.5 - M, 0.02, 30)
Gtab = {0.25: 0.959, 0.5: 0.867, 1: 0.683, 2: 0.431, 4: 0.190, 8: 0.063}
def G(ed):
    x = np.log(list(Gtab.keys())); y = list(Gtab.values()); return float(np.interp(np.log(ed), x, y, left=1.0,
      right=0.063 * (8 / ed)**2))
print('\n  M_H[Msun]   T_H[MeV]  R_nu    c_s     f      0.15 R G   with f     with 1/f')
rows = []
for M in [3e5, 1e6, 3e6, 7e6, 1.6e7, 3e7, 7e7]:
    T = TH_of(M); R = Rnu(T); c = cs2(T)**0.5; f = (1 - c) / (1 - 3**-0.5); g_ = G(T / 1.0)
    base = 0.15 * R * g_ * 100; rows.append((M, T, base, base * f, base / f))
    print('  %8.1e  %8.3f  %.3f  %.4f  %.3f  %6.2f%%   %6.2f%%   %6.2f%%' % (M, T, R, c, f, base, base * f,
     base / f))
heavy = [r for r in rows if r[0] >= 3e6]
print('KC1: max shift_pred(with f) for M >= 3e6 = %.2f%% (kill if > 6.5%%) -> %s' % (max(r[3] for r in heavy),
      'FAIL' if max(r[3] for r in heavy) > 6.5 else 'PASS'))
print('KC2: min shift_pred(with 1/f) over all = %.2f%% (kill if < 5.3%%) -> %s' % (min(r[4] for r in rows), '
     FAIL' if min(r[4] for r in rows) < 5.3 else 'PASS'))
print('baseline (no f): range %.2f%% - %.2f%% ; envelope 5.3 - 6.2' % (min(r[2] for r in rows), max(r[2] for r
      in rows)))
```

```
KC0 validation: min p/e = 0.2736 at T = 0.167 MeV (Steve: 0.274 at ~0.16)
   R_nu(T=20) = 0.4884 (Steve 0.488) ; R_nu(T=0.03) = 0.4052 (Steve 0.405 for Neff=3)
   min c_s^2 = dp/de = 0.2697 (c_s = 0.5193) at T = 0.131 MeV ; 1/3 = 0.3333
KC0 PASS

  M_H[Msun]   T_H[MeV]  R_nu    c_s     f      0.15 R G   with f     with 1/f
   3.0e+05     0.713  0.483  0.5709  1.015    5.60%     5.68%     5.52%
   1.0e+06     0.398  0.474  0.5591  1.043    6.38%     6.65%     6.11%
   3.0e+06     0.240  0.457  0.5391  1.091    6.85%     7.47%     6.28%
   7.0e+06     0.166  0.439  0.5235  1.128    6.58%     7.42%     5.83%
   1.6e+07     0.118  0.421  0.5205  1.135    6.31%     7.16%     5.56%
   3.0e+07     0.090  0.411  0.5337  1.103    6.17%     6.81%     5.59%
   7.0e+07     0.061  0.406  0.5642  1.031    6.09%     6.28%     5.90%
KC1: max shift_pred(with f) for M >= 3e6 = 7.47% (kill if > 6.5%) -> FAIL
KC2: min shift_pred(with 1/f) over all = 5.52% (kill if < 5.3%) -> PASS
baseline (no f): range 5.60% - 6.85% ; envelope 5.3 - 6.2
```

B.10 Linear first maximum under gradual decoupling

```
# SCRIPT: LIN-FS-01  linear free-streaming + relaxation collisions, radiation era, Newtonian gauge (Ma-
     Bertschinger)
import numpy as np
from scipy.integrate import solve_ivp
LMAX=36
def build(k,Rnu,eps_dec,r_m=1.0):
    n=4+(LMAX-1)  # dg,tg,dn,tn,F2..FLMAX
    iF=lambda l:4+(l-2)
    def A(t):
        M=np.zeros((n,n))
        kt2=(k*t)**2
        # phi, thetaT, deltaT, sigma as linear functionals
        dT=np.zeros(n); dT[0]=1-Rnu; dT[2]=Rnu
        thT=np.zeros(n); thT[1]=1-Rnu; thT[3]=Rnu
        sig=np.zeros(n); sig[iF(2)]=0.5
        phi=-1.5/kt2*(dT+4*thT/(k*k*t))
        psi=phi-6*Rnu*sig/kt2
        phid=-psi/t+2*thT/(t*t*k*k)
```

```
        aK=0.0 if eps_dec is None else (1.0/t)*((eps_dec*r_m/t)**3)
        M[0]=4*phid; M[0,1]+=-4/3.
        M[1]=k*k*psi; M[1,0]+=k*k/4
        if Rnu>0:
            c=aK*Rnu/(1-Rnu); M[1,3]+=c; M[1,1]+=-c
        M[2]=4*phid; M[2,3]+=-4/3.
        M[3]=k*k*psi; M[3,2]+=k*k/4; M[3]-=k*k*sig; M[3,3]+=-aK; M[3,1]+=aK
        # F2
        M[iF(2),3]+=8/15.; M[iF(2),iF(3)]+=-0.6*k; M[iF(2),iF(2)]+=-aK
        for l in range(3,LMAX+1):
            r=iF(l)
            M[r,iF(l-1)]+=k*l/(2*l+1.)
            if l<LMAX: M[r,iF(l+1)]+=-k*(l+1)/(2*l+1.)
            else:
                # closure F_{L+1}=(2L+1)/(k t) F_L - F_{L-1}
                M[r,iF(l)]+=-k*(l+1)/(2*l+1.)*((2*l+1)/(k*t)); M[r,iF(l-1)]+=k*(l+1)/(2*l+1.)
            M[r,r]+=-aK
        return M,dT,thT,phi
    return n,A
def run(k,Rnu,eps_dec,t0,tmax,r_m=1.0,npts=400):
    n,A=build(k,Rnu,eps_dec,r_m)
    zeta=1.0; psi0=2/3.*zeta
    th0=0.5*k*k*t0*psi0
    d0=-2*psi0-(2/3.)*(k*t0)**2*psi0
    y0=np.zeros(n); y0[0]=d0; y0[1]=th0; y0[2]=d0; y0[3]=th0
    f=lambda t,y:A(t)[0]@y
    J=lambda t,y:A(t)[0]
    ts=np.linspace(t0,tmax,npts)
    s=solve_ivp(f,(t0,tmax),y0,method='Radau',jac=J,t_eval=ts,rtol=1e-8,atol=1e-11)
    Y=s.y; out={}
    dT=(1-Rnu)*Y[0]+Rnu*Y[2]; thT=(1-Rnu)*Y[1]+Rnu*Y[3]
    out['t']=s.t
    out['Delta']=dT+4*thT/(k*k*s.t)           # zero-momentum slices
    out['dpl']=dT+4*Y[1]/(k*k*s.t)            # plasma-orthogonal slices
    return out
def firstmax(t,y):
    i=np.where((y[1:-1]>y[:-2])&(y[1:-1]>=y[2:]))[0]
    if len(i)==0: return None
    j=i[0]+1
    # parabolic refinement
    a,b,c=y[j-1],y[j],y[j+1]
    d=0.5*(a-c)/(a-2*b+c); return t[j]+d*(t[1]-t[0]), b-0.25*(a-c)*d
if __name__=="__main__":
    # validation: single mode, nu decoupled from start (instant) vs perfect fluid
    for k in [1.0]:
        for Rnu in [0.409,0.488]:
            pf=run(k,0.0,None,0.01/k,12/k)
            fs=run(k,Rnu,None,0.01/k,12/k)
            for key in ['Delta','dpl']:
                m1=firstmax(pf['t'],pf[key]); m2=firstmax(fs['t'],fs[key])
                print(k,Rnu,key,m1,m2,'ratio',m2[1]/m1[1])
```

```
# SCRIPT: LIN-FS-03  central first-max factor vs gradual decoupling kappa=H(eps_dec/eps)^3 (linear theory,
    Gaussian profile)
import numpy as np, time, sys
from multiprocessing import Pool
from lin import run, firstmax
xk,wk=np.polynomial.legendre.leggauss(28)
kmax=8.0
ks=0.5*kmax*(xk+1); ws=0.5*kmax*wk
T=np.linspace(0.1,4.0,586)
def job(a):
    Rnu,ed,k=a
    r=run(k,Rnu,ed,0.1,4.0,npts=len(T))
    return (-r['Delta'],-r['dpl'])
def centre(Rnu,ed,pool):
    out=pool.map(job,[(Rnu,ed,k) for k in ks])
    D=sum(w*k*k*np.exp(-k*k/4.)*o[0] for o,k,w in zip(out,ks,ws))/(2*np.pi**2)
```

```
    P=sum(w*k*k*np.exp(-k*k/4.)*o[1] for o,k,w in zip(out,ks,ws))/(2*np.pi**2)
    return D,P
if __name__=="__main__":
    t0=time.time()
    with Pool(4) as pool:
        pf=centre(0.0,None,pool); mp=[firstmax(T,x)[1] for x in pf]
        res=[]
        for ed in [None,0.25,0.5,1,2,4,8]:
            fs=centre(0.409,ed,pool)
            mf=[firstmax(T,x) for x in fs]
            sh=[100*(mp[i]/mf[i][1]-1) for i in range(2)]
            print("edec",ed,"Delta shift %.3f  dpl shift %.3f  tmax %.3f %.3f  t=%.0f"%(sh[0],sh[1],mf[0][0],
    mf[1][0],time.time()-t0),flush=True)
```

```
edec None Delta shift 5.962  dpl shift 6.663  tmax 1.711 1.744  t=119
edec 0.25 Delta shift 5.124  dpl shift 5.878  tmax 1.711 1.743  t=182
edec 0.5 Delta shift 3.498  dpl shift 4.233  tmax 1.714 1.740  t=233
edec 1 Delta shift 1.109  dpl shift 1.423  tmax 1.730 1.746  t=280
edec 2 Delta shift -0.285  dpl shift -0.474  tmax 1.748 1.757  t=327
edec 4 Delta shift -0.145  dpl shift -0.197  tmax 1.737 1.739  t=369
edec 8 Delta shift -0.022  dpl shift -0.023  tmax 1.731 1.731  t=411
```

B.11 Rate-ratio and wall readings

```
# SCRIPT: CONN-ENERGY-STEVE-04  corrected scaling k/(aH) = k r_m * eps
import numpy as np
from scipy.optimize import brentq
G=np.array([5.90,5.33,4.20,2.65,1.17,0.39])/6.15; ed=np.array([.25,.5,1,2,4,8])
print('EP crossing (k r_m=2): kappa/H=(edec/eps)^3 equals (2/sqrt3)*2*eps')
print([round(brentq(lambda e:(x/e)**3-(4/np.sqrt(3))*e,1e-3,100),3) for x in ed], ' vs 0.811*edec^0.75 =',np.
    round(0.811*ed**0.75,3))
forms={'wall sqrt(1-3kap^2/4k^2)':lambda kap,k:np.sqrt(np.clip(1-0.75*kap**2/k**2,0,1)),
       'A k/(k+kappa)':lambda kap,k:k/(k+kap),'B exp(-kappa/k)':lambda kap,k:np.exp(-kap/k),'C 1-exp(-k/kappa)
    ':lambda kap,k:1-np.exp(-k/kap)}
for lo in [1.0,0.1]:
  eps=np.linspace(lo,4,6001)
  for kr in [1.0,2.0]:
    for name,f in forms.items():
      for wname,wt in [('uniform',np.ones_like(eps)),('log',1/eps)]:
        pred=np.array([np.sum(f((e/eps)**3,kr*eps)*wt)/np.sum(wt) for e in ed]); err=np.max(np.abs(pred-G))
        print('win %.1f-4 k r_m=%.0f %-26s %-8s %s maxerr %.3f %s'%(lo,kr,name,wname,np.round(pred,3),err,'
    PASS' if err<=0.10 else ''))
```

```
EP crossing (k r_m=2): kappa/H=(edec/eps)^3 equals (2/sqrt3)*2*eps
[0.287, 0.482, 0.811, 1.364, 2.294, 3.859]  vs 0.811*edec^0.75 = [0.287 0.482 0.811 1.364 2.294 3.858]
win 1.0-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   uniform  [1.    1.    0.98  0.746 0.35  0.   ] maxerr 0.315
win 1.0-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   log      [1.    0.999 0.962 0.596 0.223 0.   ] maxerr 0.279
win 1.0-4 k r_m=1 A k/(k+kappa)              uniform  [0.998 0.987 0.92  0.717 0.386 0.105] maxerr 0.286
win 1.0-4 k r_m=1 A k/(k+kappa)              log      [0.997 0.979 0.876 0.609 0.287 0.073] maxerr 0.193
win 1.0-4 k r_m=1 B exp(-kappa/k)            uniform  [0.998 0.987 0.91  0.66  0.283 0.015] maxerr 0.229
win 1.0-4 k r_m=1 B exp(-kappa/k)            log      [0.997 0.978 0.857 0.526 0.188 0.009] maxerr 0.174
win 1.0-4 k r_m=1 C 1-exp(-k/kappa)          uniform  [1.    1.    0.979 0.817 0.479 0.117] maxerr 0.386
win 1.0-4 k r_m=1 C 1-exp(-k/kappa)          log      [1.    1.    0.96  0.707 0.352 0.08 ] maxerr 0.277
win 1.0-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   uniform  [1.    1.    0.995 0.839 0.504 0.016] maxerr 0.408
win 1.0-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   log      [1.    1.    0.991 0.72  0.346 0.009] maxerr 0.308
win 1.0-4 k r_m=2 A k/(k+kappa)              uniform  [0.999 0.993 0.954 0.802 0.503 0.177] maxerr 0.371
win 1.0-4 k r_m=2 A k/(k+kappa)              log      [0.999 0.989 0.927 0.712 0.391 0.124] maxerr 0.282
win 1.0-4 k r_m=2 B exp(-kappa/k)            uniform  [0.999 0.993 0.951 0.762 0.416 0.066] maxerr 0.331
win 1.0-4 k r_m=2 B exp(-kappa/k)            log      [0.999 0.989 0.92  0.648 0.293 0.04 ] maxerr 0.237
win 1.0-4 k r_m=2 C 1-exp(-k/kappa)          uniform  [1.    1.    0.996 0.89  0.613 0.207] maxerr 0.460
win 1.0-4 k r_m=2 C 1-exp(-k/kappa)          log      [1.    1.    0.991 0.812 0.473 0.143] maxerr 0.381
win 0.1-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   uniform  [0.931 0.866 0.757 0.574 0.269 0.   ] maxerr 0.143
win 0.1-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   log      [0.646 0.505 0.364 0.224 0.084 0.   ] maxerr 0.361
win 0.1-4 k r_m=1 A k/(k+kappa)              uniform  [0.925 0.856 0.742 0.557 0.298 0.081] maxerr 0.126
```

```
win 0.1-4 k r_m=1 A k/(k+kappa)              log      [0.657 0.516 0.376 0.237 0.109 0.027] maxerr 0.350
win 0.1-4 k r_m=1 B exp(-kappa/k)            uniform  [0.915 0.839 0.713 0.508 0.218 0.012] maxerr 0.077 PASS
win 0.1-4 k r_m=1 B exp(-kappa/k)            log      [0.618 0.477 0.337 0.198 0.071 0.003] maxerr 0.389
win 0.1-4 k r_m=1 C 1-exp(-k/kappa)          uniform  [0.943 0.887 0.793 0.635 0.369 0.09 ] maxerr 0.204
win 0.1-4 k r_m=1 C 1-exp(-k/kappa)          log      [0.696 0.555 0.415 0.274 0.133 0.03 ] maxerr 0.311
win 0.1-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   uniform  [0.946 0.891 0.8   0.646 0.387 0.013] maxerr 0.215
win 0.1-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   log      [0.693 0.552 0.411 0.271 0.13  0.003] maxerr 0.314
win 0.1-4 k r_m=2 A k/(k+kappa)              uniform  [0.941 0.883 0.787 0.628 0.389 0.136] maxerr 0.199
win 0.1-4 k r_m=2 A k/(k+kappa)              log      [0.703 0.563 0.423 0.283 0.149 0.047] maxerr 0.304
win 0.1-4 k r_m=2 B exp(-kappa/k)            uniform  [0.932 0.869 0.762 0.587 0.32  0.051] maxerr 0.156
win 0.1-4 k r_m=2 B exp(-kappa/k)            log      [0.665 0.524 0.383 0.244 0.11  0.015] maxerr 0.343
win 0.1-4 k r_m=2 C 1-exp(-k/kappa)          uniform  [0.956 0.909 0.83  0.697 0.473 0.159] maxerr 0.283
win 0.1-4 k r_m=2 C 1-exp(-k/kappa)          log      [0.742 0.602 0.461 0.321 0.18  0.054] maxerr 0.264
```

```
# SCRIPT: PBH-NU-LONE-PASS  the one rate-ratio form that passed (1 of 32) tested on Test B, which it never saw
# Kill condition (written first): dies if |G_pred - 0.317| > 0.08 at n=6, eps_dec=2.
import numpy as np
eps=np.linspace(0.1,4,6001); kr=1.0
f=lambda kap,k:np.exp(-kap/k)
for n,ed,meas in [(3,2.0,0.431),(6,2.0,0.317)]:
    pred=np.mean(f((ed/eps)**n,kr*eps)); print('n=%d eps_dec=%.0f  pred %.3f  measured %.3f  diff %.3f  %s'%(n
     ,ed,pred,meas,pred-meas,'dies' if abs(pred-meas)>0.08 else 'survives'))
```

```
n=3 eps_dec=2  pred 0.508  measured 0.431  diff 0.077  survives
n=6 eps_dec=2  pred 0.513  measured 0.317  diff 0.196  dies
```

B.12 Diffusive escape

```
# SCRIPT: K6-DIFFUSIVE-LEADER-G-CURVE
# STATUS DISCLOSURE: this model was written AFTER the Table II G column had been read (the log-log slope of G
     /(1-G) ~ -1.6 suggested the
# exponent); the derivation below is independent of the data except for the choice of decision time eps* and
     of the interpolation rule.
# Derivation (all from Steve's Eq.7,10 and the paper's gap-clock reading, Sec.6/8): units r_m = 1, comoving,
     conformal time tau = eps.
#   kappa = H (e_dec/e)^3  ->  scattering rate a*kappa = sigma(tau) = e_dec^3 / tau^4          [Steve Eq.10 +
     gradual-decoupling law]
#   neutrinos (speed 1) diffuse with D = 1/(3 sigma) while sigma*L >> 1, so the NEUTRINO channel's decay rate
     (its principal Metzler
#   eigenvalue, K1-H2 above) is gamma ~ D (pi/L)^2 and its displacement by tau is  l_diff^2 = c * Int D dtau =
      c * tau^5/(15 e_dec^3).
#   Free-streaming (instantaneous decoupling) displaces by l_ball = tau.  Series (resistance) interpolation 1/
     l = 1/l_diff + 1/l_ball and
#   G := l/l_ball = 1/(1 + l_ball/l_diff) = 1/(1 + sqrt(15/c) (e_dec/e*)^(3/2)).        [c = 1: l^2 = Int D
     dtau ; c = 2: 1-D rms ; c = 6: 3-D rms]
# PRE-STATED PASS (this script): max|G_pred - G_pub| <= 0.10 with eps* = 4.0 (Steve's PF apparent-horizon time
     ) and c = 1. Because c is a convention
#   the other c are reported and the one-parameter best fit is reported so that the size of the freedom is
     visible.
# KILL for the exponent claim: a free-exponent fit must give n = 3/2 within its own 2-sigma bootstrap/fit
     uncertainty, else the diffusion scaling is dead.
# OPPOSITE: ballistic-limited regime reading G = 1 - l_diff/l_ball clipped, and G = exp(-l_ball/l_diff).
import numpy as np
from scipy.optimize import least_squares
ed=np.array([.25,.5,1,2,4,8]); G=np.array([.959,.867,.683,.431,.190,.063])
def V1(e,es=4.0,c=1.0): return 1/(1+np.sqrt(15/c)*(e/es)**1.5)
print('published   ',G)
for es in (3.5,4.0,4.5):
    for c in (1.0,2.0,6.0):
        g=V1(ed,es,c); print('V1 e*=%.1f c=%.0f'%(es,c),np.round(g,3),'max|err| %.3f'%np.max(np.abs(g-G)))
# one-parameter fits
r=least_squares(lambda p:1/(1+p[0]*(ed/4.0)**1.5)-G,[3.0]); A=r.x[0]; print('1-param (exponent 3/2, e*=4):
     A_fit=%.3f  (c-equivalent %.2f)  max|err| %.3f'%(A,15/A**2,np.max(np.abs(r.fun))))
r=least_squares(lambda p:1/(1+np.sqrt(15)*(ed/p[0])**1.5)-G,[4.0]); print('1-param (c=1): e*_fit=%.3f max|err|
      %.3f'%(r.x[0],np.max(np.abs(r.fun))))
```

```
# free exponent with covariance
def res(p): return 1/(1+p[0]*(ed/4.0)**p[1])-G
r=least_squares(res,[3.0,1.5]); J=r.jac; s2=np.sum(r.fun**2)/(len(ed)-2); cov=np.linalg.inv(J.T@J)*s2
print('free exponent fit: A=%.3f+-%.3f  n=%.3f+-%.3f  max|err| %.3f'%(r.x[0],np.sqrt(cov[0,0]),r.x[1],np.sqrt(
    cov[1,1]),np.max(np.abs(r.fun))))
print('KILL (exponent): n=1.5 is %.1f sd from fitted n'%(abs(r.x[1]-1.5)/np.sqrt(cov[1,1])))
# opposites
l=np.sqrt(1/15)*(ed/4.0)**-1.5*0+ (4.0**2.5)/np.sqrt(15*ed**3)   # l_diff with c=1, eps*=4
lb=4.0
print('OPP1 G=1-l_diff/l_ball clipped ',np.round(np.clip(1-l/lb,0,1),3),'max|err| %.3f'%np.max(np.abs(np.clip
    (1-l/lb,0,1)-G)))
print('OPP2 G=exp(-l_ball/l_diff)     ',np.round(np.exp(-lb/l),3),'max|err| %.3f'%np.max(np.abs(np.exp(-lb/l)-
    G)))
print('OPP3 quadrature G=(1+15 x^3)^-1/2',np.round((1+15*(ed/4.0)**3)**-.5,3),'max|err| %.3f'%np.max(np.abs
    ((1+15*(ed/4.0)**3)**-.5-G)))
# predictions (new, untested)
for es,tag in ((4.0,'Gaussian'),(8.0,'alpha=0.3 (decided at eps~7-9, Steve Sec.III B)')):
    print('PREDICT %-48s'%tag,{e:round(float(V1(e,es,1.0)),3) for e in (0.1,1,2,4,8,16)})
# Rnu check: odds linear in R? G is defined as ratio so the model predicts the R_nu-independence of G
print('R_nu enters only through the overall (rate) scale -> G(e_dec) curve independent of R_nu (UNTESTED;
    Steve ran gradual decoupling only at R=0.409)')
```

```
published    [0.959 0.867 0.683 0.431 0.19  0.063]
V1 e*=3.5 c=1 [0.931 0.827 0.628 0.374 0.174 0.07 ] max|err| 0.057
V1 e*=3.5 c=2 [0.95  0.871 0.705 0.458 0.23  0.096] max|err| 0.040
V1 e*=3.5 c=6 [0.971 0.921 0.805 0.594 0.341 0.155] max|err| 0.163
V1 e*=4.0 c=1 [0.943 0.854 0.674 0.422 0.205 0.084] max|err| 0.021
V1 e*=4.0 c=2 [0.959 0.892 0.745 0.508 0.267 0.114] max|err| 0.077
V1 e*=4.0 c=6 [0.976 0.935 0.835 0.641 0.387 0.183] max|err| 0.210
V1 e*=4.5 c=1 [0.952 0.875 0.711 0.466 0.236 0.098] max|err| 0.046
V1 e*=4.5 c=2 [0.965 0.908 0.777 0.552 0.303 0.133] max|err| 0.121
V1 e*=4.5 c=6 [0.98  0.945 0.858 0.681 0.43  0.211] max|err| 0.250
1-param (exponent 3/2, e*=4): A_fit=3.811  (c-equivalent 1.03)  max|err| 0.022
1-param (c=1): e*_fit=4.043 max|err| 0.022
free exponent fit: A=4.204+-0.151  n=1.610+-0.032  max|err| 0.010
KILL (exponent): n=1.5 is 3.4 sd from fitted n
OPP1 G=1-l_diff/l_ball clipped  [0.    0.    0.    0.27  0.742 0.909] max|err| 0.959
OPP2 G=exp(-l_ball/l_diff)      [0.941 0.843 0.616 0.254 0.021 0.   ] max|err| 0.177
OPP3 quadrature G=(1+15 x^3)^-1/2 [0.998 0.986 0.9   0.59  0.25  0.091] max|err| 0.217
PREDICT Gaussian                                        {0.1: 0.985, 1: 0.674, 2: 0.422, 4: 0.205, 8: 0.084,
    16: 0.031}
PREDICT alpha=0.3 (decided at eps~7-9, Steve Sec.III B)  {0.1: 0.995, 1: 0.854, 2: 0.674, 4: 0.422, 8: 0.205,
    16: 0.084}
R_nu enters only through the overall (rate) scale -> G(e_dec) curve independent of R_nu (UNTESTED; Steve ran
    gradual decoupling only at R=0.409)
```

B.13 Tests C and D and the controls: predictions, search script and run logs

```
# SCRIPT: PREREGISTERED_NEXT  (written 2026-10-31T17:09:37Z, before any run below)
Follow-up to "When the Neutrinos Leave" v2.  Frozen predictor: kept-sector memory model, two-epoch weight w
    =0.890 (fitted on n=3 only),
values taken unchanged from paper/code/main/memory_curve.json (file written 2026-10-30 13:47 CDT, never
    refitted).

TEST C (fresh history, out of sample): Gaussian profile (alpha=1), rate kappa/H=(eps_dec/eps)^6, eps_dec=1.
  Memory model predicts G = 0.6625.  History-blind rule (Steve's n=3 measured G at the same eps_dec, Table II)
     = 0.683.
  Kill: memory model dies if |G_measured - 0.6625| > 0.08.  History-blind dies if |G - 0.683| > 0.08.
  (The two are only 0.02 apart here, so this test checks the model's survival, not discrimination.)
TEST D (fresh history, discriminating): Gaussian, n=6, eps_dec=3.
  Memory model predicts G = 0.1176.  History-blind (Steve n=3, eps_dec interpolated at 3: between 0.431 (2)
    and 0.190 (4)) ~ 0.30.
  Kill: memory model dies if |G - 0.1176| > 0.08.  History-blind dies if |G - 0.30| > 0.08.

CONTROLS on Test B (n=6, eps_dec=2, measured G=0.317+-0.004 at N=400, K=32, kapcap=5e5):
```

```
  B-res: N=600, K=48.  B-cap: kapcap=5e6.  Pass if each shifts delta_c by less than 0.0005 (i.e. G by < 0.017)
     .
  If a control fails, Test B's G is reported as resolution/cap dependent and the 0.071 pass is suspended until
      resolved.
```

```
# SCRIPT: TRISECT3  three-valued two-point bracket search for delta_c (collapse / disperse / undecided-or-
    error)
# Usage: trisect3.py TAG qshape eps_dec nexp lo hi tol eps_max kapcap N K
# Outcome of one run: 'C' (collapse = status BH), 'D' (disperse = status dispersed), 'U' (anything else:
    undecided at eps_max, crash).
# An 'U' point NEVER moves the bracket. It is retried once with eps_max*1.5; if still 'U' it is recorded in '
    undecided' and reported.
import sys, json, time, os
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))

def classify(status):
    return 'C' if status == 'BH' else ('D' if status == 'dispersed' else 'U')

def one(a):
    from run_nuc_n3 import run
    d, q, ed, n, em, kc, N, K = a
    t1 = time.time()
    try:
        o = run(d, fnu=0.409, N=N, K=K, eps0=0.1, eps_dec=ed, eps_max=em, qshape=q, nexp=n, kapcap=kc)
        info = dict(d=d, status=o['status'], eps=float(o['eps']), eps_max=em, rawmax=float(o['rawmax']),
    kapcap=kc,
                    capsteps=int(o['capsteps']), steps=int(o['steps']), eps_cap_last=float(o['eps_cap_last']),
     sec=time.time()-t1)
    except Exception as ex:
        info = dict(d=d, status='error:' + type(ex).__name__ + ':' + str(ex)[:80], eps=float('nan'), eps_max=
    em, rawmax=float('nan'),
                    kapcap=kc, capsteps=-1, steps=-1, eps_cap_last=float('nan'), sec=time.time()-t1)
    info['out'] = classify(info['status'])
    return info

def fmt(tag, i):
    return (f"{tag} delta={i['d']:.5f}: {i['out']} [{i['status']}] eps={i['eps']:.2f} eps_max={i['eps_max']} "
            f"rawmax(kappa/H)={i['rawmax']:.3e} cap={i['kapcap']:.1e} capped_steps={i['capsteps']}/{i['steps']
    } "
            f"cap_released_at_eps~{i['eps_cap_last']:.3f} ({i['sec']:.0f}s)")

def search(tag, q, ed, n, lo, hi, tol, em, kc, N, K, mapper, log_path=None, retry_factor=1.5):
    log = []; undecided = []; nonmono = False; t0 = time.time()
    while hi - lo > tol:
        a = lo + (hi - lo) / 3; b = lo + 2 * (hi - lo) / 3
        res = mapper(one, [(a, q, ed, n, em, kc, N, K), (b, q, ed, n, em, kc, N, K)])
        for i in res: print(fmt(tag, i), flush=True); log.append(i)
        # retry each undecided/error point ONCE with raised eps_max (the other core is used for the second one
    )
        bad = [k for k, i in enumerate(res) if i['out'] == 'U']
        if bad:
            rr = mapper(one, [(res[k]['d'], q, ed, n, em * retry_factor, kc, N, K) for k in bad])
            for k, i in zip(bad, rr):
                i['retry'] = True; print('RETRY ' + fmt(tag, i), flush=True); log.append(i)
                if i['out'] == 'U':
                    undecided.append((i['d'], i['status'], i['eps']))
                res[k] = i
        ia, ib = res
        oa, ob = ia['out'], ib['out']
        if oa == 'U' and ob == 'U':
            print(f'{tag} STOP: both points undecided/error after retry; bracket [{lo:.5f},{hi:.5f}] unchanged
    ', flush=True); break
        if oa == 'D' and ob == 'C':     # ordinary case, delta_c between a and b
            lo, hi = a, b
        elif oa == 'C' and ob == 'D':
            print(f'{tag} STOP: non-monotonic outcomes (collapse at a, disperse at b); bracket unchanged',
    flush=True); nonmono = True; break
```

```
        elif oa == 'C':                 # a collapses (b is C or U): threshold below a
            hi = a
        elif ob == 'D':                 # b disperses (a is D or U): threshold above b
            lo = b
        elif oa == 'D':                 # a disperses, b undecided: threshold above a
            lo = a
        elif ob == 'C':                 # b collapses, a undecided: threshold below b
            hi = b
        if log_path:
            json.dump(dict(tag=tag, lo=lo, hi=hi, log=log, undecided=undecided), open(log_path, 'w'))
        print(f'{tag} bracket now [{lo:.5f},{hi:.5f}]', flush=True)
    res = dict(tag=tag, qshape=q, eps_dec=ed, nexp=n, kapcap=kc, N=N, K=K, lo=lo, hi=hi, dc=0.5 * (lo + hi), w
     =0.5 * (hi - lo),
               minutes=(time.time() - t0) / 60, undecided=undecided, nonmonotonic=nonmono, log=log)
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} bracket [{lo:.5f},{hi:.5f}] ({res["
     minutes"]:.1f} min) undecided_points={undecided}', flush=True)
    return res

if __name__ == '__main__':
    tag = sys.argv[1]; q = float(sys.argv[2]); ed = float(sys.argv[3]); n = float(sys.argv[4])
    lo, hi, tol = float(sys.argv[5]), float(sys.argv[6]), float(sys.argv[7]); em = float(sys.argv[8])
    kc = float(sys.argv[9]); N = int(sys.argv[10]); K = int(sys.argv[11])
    with Pool(2) as p:
        res = search(tag, q, ed, n, lo, hi, tol, em, kc, N, K, lambda f, args: p.map(f, args), log_path=f'
     state_{tag}.json')
    json.dump(res, open(f'res_{tag}.json', 'w'))
```

```
Bcap delta=0.50707: D [dispersed] eps=6.25 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250337/5968022 cap_released_at_eps~0.153 (4450s)
Bcap delta=0.50773: C [BH] eps=6.33 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250338/5968766 cap_released_at_eps~0.153 (4465s)
Bcap bracket now [0.50707,0.50773]
```

```
Bcap2 delta=0.50729: D [dispersed] eps=6.54 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250337/5970579 cap_released_at_eps~0.153 (4206s)
Bcap2 delta=0.50751: C [BH] eps=7.29 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250338/5977391 cap_released_at_eps~0.153 (4297s)
Bcap2 bracket now [0.50729,0.50751]
RESULT Bcap2: delta_c = 0.50740 +- 0.00011 bracket [0.50729,0.50751] (71.6 min) undecided_points=[]
```

```
Bres delta=0.50723: D [dispersed] eps=6.37 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809052/1092016 cap_released_at_eps~0.224 (1680s)
Bres delta=0.50757: C [BH] eps=7.32 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809053/1111131 cap_released_at_eps~0.224 (1708s)
Bres bracket now [0.50723,0.50757]
RESULT Bres: delta_c = 0.50740 +- 0.00017 bracket [0.50723,0.50757] (28.5 min) undecided_points=[]
```

```
Bres2 delta=0.50734: D [dispersed] eps=6.53 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809052/1095261 cap_released_at_eps~0.224 (1711s)
Bres2 delta=0.50746: D [dispersed] eps=6.81 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809053/1100808 cap_released_at_eps~0.224 (1712s)
Bres2 bracket now [0.50746,0.50757]
RESULT Bres2: delta_c = 0.50751 +- 0.00006 bracket [0.50746,0.50757] (28.5 min) undecided_points=[]
```

```
C delta=0.50793: D [dispersed] eps=4.77 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/321640 cap_released_at_eps~0.112 (231s)
C delta=0.51812: C [BH] eps=5.91 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps=115550/331789
      cap_released_at_eps~0.112 (240s)
C bracket now [0.50793,0.51812]
C delta=0.51133: D [dispersed] eps=5.00 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/323700 cap_released_at_eps~0.112 (240s)
C delta=0.51472: D [dispersed] eps=5.42 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/327372 cap_released_at_eps~0.112 (246s)
C bracket now [0.51472,0.51812]
```

```
C delta=0.51586: D [dispersed] eps=5.68 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
    =115550/329712 cap_released_at_eps~0.112 (235s)
C delta=0.51699: D [dispersed] eps=6.23 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
    =115550/334620 cap_released_at_eps~0.112 (239s)
C bracket now [0.51699,0.51812]
C delta=0.51737: D [dispersed] eps=6.81 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
    =115550/339823 cap_released_at_eps~0.112 (243s)
C delta=0.51775: C [BH] eps=6.32 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps=115550/335493
     cap_released_at_eps~0.112 (242s)
C bracket now [0.51737,0.51775]
RESULT C: delta_c = 0.51756 +- 0.00019 bracket [0.51737,0.51775] (16.1 min) undecided_points=[]
```

```
D delta=0.50793: C [BH] eps=4.60 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215105/1415099 cap_released_at_eps~0.337 (1047s)
D delta=0.51812: C [BH] eps=3.86 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215124/1408481 cap_released_at_eps~0.337 (1039s)
D bracket now [0.49773,0.50793]
D delta=0.50113: D [dispersed] eps=5.74 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215092/1425242 cap_released_at_eps~0.337 (1049s)
D delta=0.50453: C [BH] eps=5.23 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215098/1420654 cap_released_at_eps~0.337 (1054s)
D bracket now [0.50113,0.50453]
D delta=0.50226: D [dispersed] eps=6.42 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215094/1431257 cap_released_at_eps~0.337 (1062s)
D delta=0.50339: C [BH] eps=5.70 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215096/1424836 cap_released_at_eps~0.337 (1069s)
D bracket now [0.50226,0.50339]
D delta=0.50264: C [BH] eps=6.65 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215095/1433339 cap_released_at_eps~0.337 (1064s)
D delta=0.50302: C [BH] eps=5.99 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
    =1215095/1427441 cap_released_at_eps~0.337 (1068s)
D bracket now [0.50226,0.50264]
RESULT D: delta_c = 0.50245 +- 0.00019 bracket [0.50226,0.50264] (70.6 min) undecided_points=[]
```

B.14 The matched-pair witness

```
# SCRIPT: PREREGISTERED_WITNESS (preregistration text, not code)
Written: 2026-10-31T17:12:31Z  (UTC, before any production run; no witness numbers computed yet)

MODEL (identical to paper/code/main/memory_curve.py)
 dv/dtau = -(i k M + sigma(tau)(I-P0)) v, orthonormal Legendre moments v_l, l=0..LM,
 M_{l,l-1}=M_{l-1,l}=l/sqrt(4l^2-1), sigma = eps^n / tau^(n+1), n in {3,6}.
 k-weights: 36-node Gauss-Legendre on [0.25,9], weight k^2 exp(-k^2/2), normalised.
 eps_dec fixed at 0.5 (inside the paper's scanned range) so all plants sit before tau=1.7.

PAIR
 base b at tau_s: v0=1, v1=0.2, v2=0.1, rest 0. Pair: b +/- delta e_L, delta=0.05,
 hidden plant L=3 (main) and L=4; opposite check L=1, L=2 (retained plant, hidden matched).
 Both members scaled by the SAME constant to unit norm (keeps the match; needed for the bound).
 Positivity: f(mu)=sum v_l sqrt((2l+1)/2) P_l(mu) >= 0 on a 2001-point mu grid, both members.
 tau_s / eps in {0.1, 0.25, 0.5, 1, 2}  ->  tau_s in {0.05, 0.125, 0.25, 0.5, 1.0}.
 Readout epochs tau = 1.7, 4.0, late 6.0.

READOUTS (difference = member+ minus member-, exact by linearity = 2 delta U(t,s) e_L)
 R_ret  = sqrt(sum_k w ||Dv_{0,1,2}||^2) / sqrt(sum_k w ||b_{0,1,2}||^2)   (PRIMARY)
 R_A    = |A_w+ - A_w-| / mean(A_w+,A_w-),  A_w = sum_k w |v0|^2       (k-weighted monopole)
 R_F    = same with |v1|^2                                              (k-weighted flux)
 WITNESS ON at an epoch  <=>  R_ret > 1e-3 at that epoch.

DERIVED BEFORE RUNNING (block elimination, a=v0, h=(v1,...)):
 In the hidden block the collision term is sigma*I (scalar), so EXACTLY
   U_Q(t,r) = exp(-(Sig(t)-Sig(r))) exp(-i k (t-r) M_QQ),  Sig(t) = -eps^n/(n t^n).
 a' = S(t) - k^2 int_s^t K(t,r) a(r) dr,  K = M01^2 [U_Q(t,r)]_{11},
 S(t) = -i k M01 [U_Q(t,s) h0]_1.  The pair has Da(s)=0, Dh0 = 2 delta e_L, so the
 difference obeys the same Volterra equation driven ONLY by S[Dh0]: no source, no witness.
```

```
PREDICTION (stated before running)
 Two channels carry a hidden difference into the retained moments:
  (i) remnant: the hidden plant survives decoupling with factor exp(-(eps/tau_s)^n / n)
      (int_s^inf sigma converges because collisions die off) and then free-streams: O(delta*remnant);
  (ii) leakage during coupling through L damped rungs: ~ delta*(k/sigma(tau_s))^L.
 Expected witness strength ~ delta*max(remnant, leakage), so with delta=0.05:
   tau_s/eps=0.1 : OFF for n=3 and n=6 (remnant e^-333 / e^-1e6, leakage <1e-11)
   tau_s/eps=0.25: OFF both (n=3 remnant 7e-10, leakage ~2e-7)
   tau_s/eps=0.5 : n=3 ON (remnant e^-2.67=0.07), n=6 OFF or marginal (remnant 2e-5)
   tau_s/eps=1,2 : ON both, O(delta)
  n=6 suppresses more than n=3 at the same tau_s/eps for tau_s<eps, less (closer to 1) for tau_s>=eps.
  Opposite check: deep in coupling L=1 > L=2 > L=3 > L=4 (one factor k/sigma per rung);
  after decoupling all four O(delta) and comparable.

KILL / SUCCESS CONDITIONS
 K1 (witness dead): R_ret <= 1e-3 at both tau=1.7 and 4.0 for every plant with tau_s >= eps
     -> the retained (density/flux/stress) closure suffices; witness killed in this model.
 K2 (not converged): for any ON case, LM 60->120 or rtol 1e-9->1e-11 (atol 1e-12->1e-14)
     changes R_ret, R_A or R_F by >= 1% -> that entry is not reported as a pass.
 K3 (invalid pair): positivity fails for either member -> pair discarded.
 K4 (prediction killed): any tau_s/eps<=0.25 entry ON, or any tau_s/eps>=1 entry OFF.
 K5 (elimination claim killed): Da computed from the Volterra equation driven by the
     independently computed source S (closed-form U_Q, no full-system solve) differs from the
     full-system Da by > 1% of max|Da|, or fails to converge with grid refinement.
 K6 (speed bound violated): max over all trajectories/times of (A')^2/(4 (k^2/3) A(1-A)) > 1+1e-6.
 SUCCESS: K1-K6 all not triggered.
```

```
# SCRIPT: PBH-NU-WITNESS-RUN  matched retained-state witness in the kept-sector model (same object as
    memory_curve.py)
import os, sys, json, time
os.environ['OMP_NUM_THREADS']='1'
import numpy as np, scipy.sparse as sp
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre, eval_legendre
kn,kw=roots_legendre(36); kmin,kmax=0.25,9.0; ks=0.5*(kmax-kmin)*(kn+1)+kmin; wk=0.5*(kmax-kmin)*kw
wE=wk*ks**2*np.exp(-ks**2/2); wE/=wE.sum()
EPS=0.5; DELTA=0.05; TEP=[1.7,4.0,6.0]
def Mmat(LM):
    d=np.array([l/np.sqrt(4.0*l*l-1) for l in range(1,LM+1)])
    return sp.diags([d,d],[-1,1],format='csr')
def evolve(k,n,ts,LM,rtol,atol,cols):
    N=LM+1; M=Mmat(LM); Q=sp.diags(np.r_[0.0,np.ones(LM)],format='csr'); I=sp.identity(len(cols),format='csr')
    def A(t): return (-1j*k*M - (EPS**n/t**(n+1))*Q)
    def rhs(t,y): return (A(t)@y.reshape(len(cols),N).T).T.ravel()
    def jac(t,y): return sp.kron(I,A(t),format='csc')
    Y0=np.zeros((len(cols),N),complex)
    for j,c in enumerate(cols): Y0[j,:len(c)]=c
    tg=np.unique(np.r_[np.geomspace(ts,6.0,500),TEP])
    s=solve_ivp(rhs,(ts,6.0),Y0.ravel(),method='BDF',t_eval=tg,jac=jac,rtol=rtol,atol=atol)
    assert s.success, s.message
    return s.t, s.y.reshape(len(cols),N,-1)   # [col, l, t]
def positivity(v):
    mu=np.linspace(-1,1,2001)
    return float(min(sum(v[l]*np.sqrt((2*l+1)/2)*eval_legendre(l,mu) for l in range(len(v)))))
def run(n,ratio,L,LM=60,rtol=1e-9,atol=1e-12):
    ts=ratio*EPS; b=np.array([1.0,0.2,0.1,0,0]); e=np.zeros(5); e[L]=1
    c=1/np.sqrt(b@b+DELTA**2)
    pos=(positivity(c*(b+DELTA*e)),positivity(c*(b-DELTA*e)))
    rows={tt:dict(dret=0.0,bret=0.0,Ap=0.0,Am=0.0,Fp=0.0,Fm=0.0) for tt in TEP}
    worst=0.0; worstN=0.0
    for k,w in zip(ks,wE):
        t,Y=evolve(k,n,ts,LM,rtol,atol,[c*b,c*DELTA*e])
        U_b,U_d=Y[0],Y[1]
        for sgn in (1,-1):
            v=U_b+sgn*U_d; A=np.abs(v[0])**2; Ap=2*np.real(np.conj(v[0])*(-1j*k*(1/np.sqrt(3))*v[1]))
            Nrm=np.sum(np.abs(v)**2,axis=0)
            den=4*(k*k/3)*A*(1-A); ok=den>1e-300
```

```
            worst=max(worst,float(np.max(Ap[ok]**2/den[ok])))
            denN=4*(k*k/3)*A*(Nrm-A); ok=denN>1e-300
            worstN=max(worstN,float(np.max(Ap[ok]**2/denN[ok])))
        for tt in TEP:
            i=np.argmin(abs(t-tt)); r=rows[tt]
            r['dret']+=w*np.sum(np.abs(2*U_d[:3,i])**2); r['bret']+=w*np.sum(np.abs(U_b[:3,i])**2)
            vp=U_b[:,i]+U_d[:,i]; vm=U_b[:,i]-U_d[:,i]
            r['Ap']+=w*abs(vp[0])**2; r['Am']+=w*abs(vm[0])**2; r['Fp']+=w*abs(vp[1])**2; r['Fm']+=w*abs(vm
    [1])**2
    res={}
    for tt in TEP:
        r=rows[tt]
        res[str(tt)]=dict(R_ret=float(np.sqrt(r['dret']/r['bret'])),
            R_A=float(abs(r['Ap']-r['Am'])/(0.5*(r['Ap']+r['Am']))),
            R_F=float(abs(r['Fp']-r['Fm'])/(0.5*(r['Fp']+r['Fm']))))
    return dict(n=n,ratio=ratio,tau_s=ts,L=L,LM=LM,rtol=rtol,pos_min=pos,bound_worst=worst,bound_worst_N=
    worstN,res=res)
if __name__=='__main__':
    mode=sys.argv[1]; out=[]
    if mode=='prod':
        jobs=[(n,r,L) for n in (3,6) for r in (0.1,0.25,0.5,1,2) for L in (3,4,1,2)]
        kw_={}
    else:
        jobs=[(n,r,L) for n in (3,6) for r in (0.5,1,2) for L in (3,4)]
        kw_=dict(LM=120) if mode=='convLM' else dict(rtol=1e-11,atol=1e-14)
    for j in jobs:
        t0=time.time(); d=run(*j,**kw_); out.append(d)
        print(j,'%.1fs'%(time.time()-t0),' '.join('%s:%.3e/%.3e/%.3e'%(tt,v['R_ret'],v['R_A'],v['R_F']) for tt
    ,v in d['res'].items()),
              'pos %.3f %.3f bound %.6f %.6f'%(*d['pos_min'],d['bound_worst'],d['bound_worst_N']),flush=True)
    json.dump(out,open('witness_%s.json'%mode,'w'),indent=1)
```

```
# SCRIPT: PBH-NU-WITNESS-KERNEL  exact block elimination: hidden source S = -ik B U_Q(t,s) Dh0 drives the
    whole retained difference
import os; os.environ['OMP_NUM_THREADS']='1'
import numpy as np, json, scipy.sparse as sp
from scipy.integrate import solve_ivp
EPS=0.5; DELTA=0.05; LM=60; N=LM+1
M=np.zeros((N,N))
for l in range(1,N): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
MQQ=M[1:,1:]; lam,V=np.linalg.eigh(MQQ); M01=M[0,1]
Sig=lambda t,n: -EPS**n/(n*t**n)            # Sig' = sigma
def full_Da(k,n,ts,L,tg,c):
    Q=np.diag(np.r_[0.0,np.ones(LM)])
    A=lambda t: -1j*k*M-(EPS**n/t**(n+1))*Q
    y0=np.zeros(N,complex); y0[L]=2*DELTA*c
    s=solve_ivp(lambda t,y:A(t)@y,(ts,tg[-1]),y0,method='BDF',jac=lambda t,y:A(t),t_eval=tg,rtol=1e-12,atol=1e
    -15)
    return s.y[0]
def memory_Da(k,n,ts,L,tg,c):
    # closed-form hidden propagator, no full-system solve
    h=tg[1]-tg[0]; Nt=len(tg); Sg=Sig(tg,n)
    Dh0=np.zeros(LM,complex); Dh0[L-1]=2*DELTA*c
    coef=V.T@Dh0
    S=-1j*k*M01*np.exp(-(Sg-Sg[0]))*((V[0,:]*coef)[None,:]*np.exp(-1j*k*np.outer(tg-ts,lam))).sum(1)
    g=M01**2*((V[0,:]**2)[None,:]*np.exp(-1j*k*np.outer(np.arange(Nt)*h,lam))).sum(1)   # g(t-r)
    a=np.zeros(Nt,complex); F=np.zeros(Nt,complex); F[0]=S[0]
    for i in range(Nt-1):
        j=np.arange(i+2); Kr=np.exp(-(Sg[i+1]-Sg[j]))*g[i+1-j]
        w=np.full(i+2,h); w[0]=w[-1]=h/2
        Jknown=np.sum((w*Kr*a[j])[:-1])
        # a_{i+1} = a_i + h/2 (F_i + S_{i+1} - k^2 (Jknown + w_last K_ii a_{i+1}))
        a[i+1]=(a[i]+h/2*(F[i]+S[i+1]-k*k*Jknown))/(1+h/2*k*k*w[-1]*Kr[-1])
        F[i+1]=S[i+1]-k*k*(Jknown+w[-1]*Kr[-1]*a[i+1])
    Sint=np.concatenate([[0],np.cumsum(h/2*(S[1:]+S[:-1]))])
    return a,Sint
out=[]
b=np.array([1.0,0.2,0.1]); c=1/np.sqrt(b@b+DELTA**2)
```

```
for (n,ratio,L,k) in [(3,1,3,2.0),(3,0.5,3,1.5),(6,1,4,3.0),(6,2,3,1.0)]:
    ts=ratio*EPS; row=dict(n=n,ratio=ratio,L=L,k=k)
    for h in (4e-3,2e-3,1e-3):
        tg=ts+np.arange(int(round((4.0-ts)/h))+1)*h
        ref=full_Da(k,n,ts,L,tg,c); a,Sint=memory_Da(k,n,ts,L,tg,c)
        err=np.max(abs(a-ref))/np.max(abs(ref))
        row['h=%g'%h]=dict(rel_err=float(err),max_Da=float(np.max(abs(ref))),
            src_only_rel=float(np.max(abs(Sint-ref))/np.max(abs(ref))))
        print(n,ratio,L,k,h,'max|Da|=%.3e  memory-vs-full rel err=%.2e  (source integral alone, no kernel: %.2
    e)'%(np.max(abs(ref)),err,row['h=%g'%h]['src_only_rel']),flush=True)
    out.append(row)
json.dump(out,open('witness_kernel.json','w'),indent=1)
```

```
# SCRIPT: PBH-NU-WITNESS-CONV  compare production vs LM=120 and tight-tolerance runs (K2)
import json
P={(d['n'],d['ratio'],d['L']):d for d in json.load(open('witness_prod.json'))}
worst=0
for mode in ('convLM','convTol'):
    for d in json.load(open('witness_%s.json'%mode)):
        p=P[(d['n'],d['ratio'],d['L'])]; ch=[]
        for tt in ('1.7','4.0','6.0'):
            for q in ('R_ret','R_A','R_F'):
                ch.append(abs(d['res'][tt][q]-p['res'][tt][q])/abs(p['res'][tt][q]))
        worst=max(worst,max(ch)); print(mode,d['n'],d['ratio'],d['L'],'max rel change %.2e'%max(ch),'bound %.6
    f'%d['bound_worst'])
print('WORST',worst)
```


```
[
 {
  "n": 3,
  "ratio": 1,
  "L": 3,
  "k": 2.0,
  "h=0.004": {
   "rel_err": 4.742985300596294e-06,
   "max_Da": 0.04492003160306953,
   "src_only_rel": 1.9909068121545908
  },
  "h=0.002": {
   "rel_err": 1.1856044767257576e-06,
   "max_Da": 0.04492007612793319,
   "src_only_rel": 1.990905398421171
  },
  "h=0.001": {
   "rel_err": 2.9625562593651493e-07,
   "max_Da": 0.044920090604062296,
   "src_only_rel": 1.9909048967372047
  }
 },
 {
  "n": 3,
  "ratio": 0.5,
  "L": 3,
  "k": 1.5,
  "h=0.004": {
   "rel_err": 1.9702037965960922e-06,
   "max_Da": 0.00434791177399913,
   "src_only_rel": 1.2648986553695796
  },
  "h=0.002": {
   "rel_err": 4.932862271426013e-07,
   "max_Da": 0.00434791177399913,
   "src_only_rel": 1.2630671055573819
  },
  "h=0.001": {
   "rel_err": 1.2404572232928826e-07,
   "max_Da": 0.00434791177399913,
```

```
   "src_only_rel": 1.2630672812226218
  }
 },
 {
  "n": 6,
  "ratio": 1,
  "L": 4,
  "k": 3.0,
  "h=0.004": {
   "rel_err": 1.1862372761170309e-05,
   "max_Da": 0.049899459055607595,
   "src_only_rel": 2.3445679331226095
  },
  "h=0.002": {
   "rel_err": 2.965509149910457e-06,
   "max_Da": 0.049899739333988655,
   "src_only_rel": 2.3445545251342654
  },
  "h=0.001": {
   "rel_err": 7.412824292084345e-07,
   "max_Da": 0.049899739333988655,
   "src_only_rel": 2.3445558282676617
  }
 },
 {
  "n": 6,
  "ratio": 2,
  "L": 3,
  "k": 1.0,
  "h=0.004": {
   "rel_err": 7.922658929918443e-07,
   "max_Da": 0.03911120365950552,
   "src_only_rel": 0.1881606736764026
  },
  "h=0.002": {
   "rel_err": 1.9809644902127958e-07,
   "max_Da": 0.03911120365950552,
   "src_only_rel": 0.18816047472879305
  },
  "h=0.001": {
   "rel_err": 4.955392667016778e-08,
   "max_Da": 0.03911120365950552,
   "src_only_rel": 0.18816042499176522
  }
 }
]
```


```
(3, 0.1, 3) 18.4s 1.7:1.445e-14/0.000e+00/0.000e+00 4.0:7.318e-15/0.000e+00/0.000e+00 6.0:6.950e-15/0.000e
    +00/0.000e+00 pos 0.513 0.510 bound 0.788639 1.000001
(3, 0.1, 4) 20.3s 1.7:9.639e-19/0.000e+00/0.000e+00 4.0:4.764e-19/0.000e+00/0.000e+00 6.0:4.547e-19/0.000e
    +00/0.000e+00 pos 0.510 0.501 bound 0.788639 1.000001
(3, 0.1, 1) 18.2s 1.7:4.619e-06/4.227e-11/8.959e-11 4.0:3.230e-06/3.279e-11/3.619e-11 6.0:2.986e-06/3.647e
    -11/3.323e-11 pos 0.516 0.578 bound 0.788639 1.000001
(3, 0.1, 2) 17.8s 1.7:2.415e-10/1.898e-10/4.023e-10 4.0:1.349e-10/1.472e-10/1.625e-10 6.0:1.264e-10/1.638e
    -10/1.492e-10 pos 0.533 0.528 bound 0.788639 1.000001
(3, 0.25, 3) 16.1s 1.7:1.326e-09/2.388e-13/8.056e-13 4.0:6.311e-10/1.919e-13/2.070e-13 6.0:5.913e-10/2.028e
    -13/1.873e-13 pos 0.513 0.510 bound 0.788648 1.000000
(3, 0.25, 4) 15.3s 1.7:2.953e-11/8.962e-12/3.087e-11 4.0:6.182e-11/9.579e-11/6.599e-11 6.0:8.659e-11/1.151e
    -10/6.062e-11 pos 0.510 0.501 bound 0.788648 1.000000
(3, 0.25, 1) 16.6s 1.7:1.924e-04/7.336e-08/1.555e-07 4.0:1.345e-04/5.690e-08/6.280e-08 6.0:1.244e-04/6.329e
    -08/5.767e-08 pos 0.516 0.578 bound 0.788648 1.000000
(3, 0.25, 2) 16.3s 1.7:4.516e-07/3.551e-07/7.523e-07 4.0:2.522e-07/2.754e-07/3.039e-07 6.0:2.364e-07/3.062e
    -07/2.790e-07 pos 0.533 0.528 bound 0.788648 1.000000
(3, 0.5, 3) 15.7s 1.7:4.924e-03/1.185e-04/9.109e-05 4.0:8.557e-03/5.292e-05/2.266e-04 6.0:1.120e-02/3.412e
    -04/3.867e-04 pos 0.513 0.510 bound 0.794143 0.991388
(3, 0.5, 4) 16.6s 1.7:3.480e-03/6.867e-04/4.006e-03 4.0:7.905e-03/1.143e-02/7.323e-03 6.0:1.101e-02/1.203e
    -02/5.594e-03 pos 0.510 0.501 bound 0.794396 0.991416
```

```
(3, 0.5, 1) 15.1s 1.7:9.487e-03/5.249e-04/2.917e-04 4.0:1.096e-02/7.344e-04/4.928e-04 6.0:1.163e-02/8.143e
    -04/5.617e-04 pos 0.516 0.578 bound 0.794207 0.994604
(3, 0.5, 2) 16.0s 1.7:5.897e-03/7.879e-03/7.402e-03 4.0:8.996e-03/2.506e-03/8.059e-03 6.0:1.094e-02/1.278e
    -02/1.289e-02 pos 0.533 0.528 bound 0.796846 0.993249
(3, 1, 3) 12.1s 1.7:4.654e-02/7.218e-03/1.065e-03 4.0:8.123e-02/7.397e-03/1.277e-02 6.0:1.052e-01/1.536e
    -02/2.445e-02 pos 0.513 0.510 bound 0.828287 0.891815
(3, 1, 4) 11.4s 1.7:2.895e-02/3.765e-03/2.858e-02 4.0:7.420e-02/1.019e-01/5.072e-02 6.0:1.028e-01/7.643e
    -02/2.026e-02 pos 0.510 0.501 bound 0.828860 0.892318
(3, 1, 1) 10.9s 1.7:7.130e-02/2.942e-02/1.165e-02 4.0:8.961e-02/5.652e-02/3.277e-02 6.0:9.713e-02/5.837e
    -02/3.891e-02 pos 0.516 0.578 bound 0.836234 0.932473
(3, 1, 2) 9.7s 1.7:5.852e-02/5.858e-02/9.228e-02 4.0:8.650e-02/7.613e-02/1.029e-01 6.0:1.038e-01/1.652e
    -01/1.477e-01 pos 0.533 0.528 bound 0.854802 0.906232
(3, 2, 3) 8.5s 1.7:4.719e-02/1.900e-03/6.876e-03 4.0:9.529e-02/2.314e-02/1.613e-02 6.0:1.291e-01/1.467e
    -02/3.407e-02 pos 0.513 0.510 bound 0.834221 0.844598
(3, 2, 4) 7.9s 1.7:1.832e-02/1.597e-03/1.464e-02 4.0:8.500e-02/1.176e-01/5.982e-02 6.0:1.244e-01/9.771e
    -02/2.459e-02 pos 0.510 0.501 bound 0.834124 0.844526
(3, 2, 1) 8.3s 1.7:9.340e-02/1.884e-02/4.561e-02 4.0:1.096e-01/8.991e-02/4.796e-02 6.0:1.210e-01/9.091e
    -02/5.989e-02 pos 0.516 0.578 bound 0.848836 0.902645
(3, 2, 2) 7.7s 1.7:8.247e-02/3.742e-02/1.390e-01 4.0:1.039e-01/8.610e-02/1.267e-01 6.0:1.292e-01/2.062e
    -01/1.870e-01 pos 0.533 0.528 bound 0.868966 0.860951
(6, 0.1, 3) 17.4s 1.7:1.473e-23/0.000e+00/0.000e+00 4.0:7.797e-24/0.000e+00/0.000e+00 6.0:7.425e-24/0.000e
    +00/0.000e+00 pos 0.513 0.510 bound 0.795402 1.479631
(6, 0.1, 4) 18.4s 1.7:9.800e-31/0.000e+00/0.000e+00 4.0:5.108e-31/0.000e+00/0.000e+00 6.0:4.880e-31/0.000e
    +00/0.000e+00 pos 0.510 0.501 bound 0.795402 1.479636
(6, 0.1, 1) 17.5s 1.7:4.672e-09/0.000e+00/0.000e+00 4.0:3.300e-09/2.080e-16/0.000e+00 6.0:3.075e-09/0.000e
    +00/0.000e+00 pos 0.516 0.578 bound 0.795402 1.479637
(6, 0.1, 2) 18.3s 1.7:2.458e-16/0.000e+00/2.907e-16 4.0:1.414e-16/0.000e+00/1.377e-16 6.0:1.333e-16/1.393e
    -16/0.000e+00 pos 0.533 0.528 bound 0.795402 1.479631
(6, 0.25, 3) 17.0s 1.7:3.384e-15/0.000e+00/0.000e+00 4.0:1.791e-15/0.000e+00/0.000e+00 6.0:1.706e-15/0.000e
    +00/0.000e+00 pos 0.513 0.510 bound 0.795405 1.000002
(6, 0.25, 4) 16.4s 1.7:1.383e-19/0.000e+00/0.000e+00 4.0:7.210e-20/0.000e+00/0.000e+00 6.0:6.890e-20/0.000e
    +00/0.000e+00 pos 0.510 0.501 bound 0.795405 1.000002
(6, 0.25, 1) 17.2s 1.7:2.857e-06/1.570e-11/3.354e-11 4.0:2.018e-06/1.325e-11/1.429e-11 6.0:1.880e-06/1.457e
    -11/1.326e-11 pos 0.516 0.578 bound 0.795405 1.000002
(6, 0.25, 2) 17.6s 1.7:9.203e-11/7.033e-11/1.502e-10 4.0:5.293e-11/5.935e-11/6.400e-11 6.0:4.993e-11/6.526e
    -11/5.941e-11 pos 0.533 0.528 bound 0.795405 1.000002
(6, 0.5, 3) 15.0s 1.7:1.589e-06/6.042e-10/2.030e-09 4.0:2.816e-06/3.803e-09/2.994e-09 6.0:3.699e-06/4.842e
    -09/3.566e-09 pos 0.513 0.510 bound 0.795417 0.999989
(6, 0.5, 4) 15.4s 1.7:1.126e-06/2.047e-07/1.299e-06 4.0:2.609e-06/3.835e-06/2.372e-06 6.0:3.643e-06/3.727e
    -06/1.678e-06 pos 0.510 0.501 bound 0.795417 0.999989
(6, 0.5, 1) 15.2s 1.7:4.181e-04/3.412e-07/7.185e-07 4.0:2.961e-04/2.866e-07/3.083e-07 6.0:2.759e-04/3.142e
    -07/2.859e-07 pos 0.516 0.578 bound 0.795417 0.999992
(6, 0.5, 2) 14.2s 1.7:2.880e-06/4.291e-06/1.372e-06 4.0:2.592e-06/3.594e-07/1.237e-06 6.0:3.051e-06/2.823e
    -06/2.960e-06 pos 0.533 0.528 bound 0.795418 0.999992
(6, 1, 3) 11.7s 1.7:5.345e-02/9.702e-03/8.884e-04 4.0:9.377e-02/1.032e-02/1.629e-02 6.0:1.207e-01/1.853e
    -02/3.100e-02 pos 0.513 0.510 bound 0.835375 0.867593
(6, 1, 4) 11.3s 1.7:3.326e-02/3.456e-03/3.198e-02 4.0:8.570e-02/1.157e-01/5.204e-02 6.0:1.179e-01/7.385e
    -02/1.193e-02 pos 0.510 0.501 bound 0.835639 0.867877
(6, 1, 1) 11.0s 1.7:8.003e-02/3.822e-02/1.457e-02 4.0:1.010e-01/7.239e-02/4.155e-02 6.0:1.094e-01/7.339e
    -02/4.928e-02 pos 0.516 0.578 bound 0.846711 0.917310
(6, 1, 2) 11.3s 1.7:6.703e-02/6.418e-02/1.060e-01 4.0:9.957e-02/1.071e-01/1.271e-01 6.0:1.189e-01/2.012e
    -01/1.769e-01 pos 0.533 0.528 bound 0.866573 0.883317
(6, 2, 3) 8.6s 1.7:4.848e-02/1.999e-03/7.186e-03 4.0:9.826e-02/2.450e-02/1.702e-02 6.0:1.329e-01/1.523e
    -02/3.585e-02 pos 0.513 0.510 bound 0.836364 0.836927
(6, 2, 4) 8.6s 1.7:1.882e-02/1.634e-03/1.490e-02 4.0:8.768e-02/1.221e-01/6.027e-02 6.0:1.282e-01/9.738e
    -02/2.238e-02 pos 0.510 0.501 bound 0.836144 0.836710
(6, 2, 1) 8.5s 1.7:9.572e-02/1.969e-02/4.752e-02 4.0:1.123e-01/9.466e-02/5.035e-02 6.0:1.240e-01/9.520e
    -02/6.286e-02 pos 0.516 0.578 bound 0.852239 0.897805
(6, 2, 2) 8.2s 1.7:8.471e-02/3.832e-02/1.416e-01 4.0:1.070e-01/9.479e-02/1.327e-01 6.0:1.329e-01/2.159e
    -01/1.947e-01 pos 0.533 0.528 bound 0.872473 0.853435
```

```
(3, 0.5, 3) 13.1s 1.7:4.924e-03/1.185e-04/9.109e-05 4.0:8.557e-03/5.292e-05/2.266e-04 6.0:1.120e-02/3.412e
    -04/3.867e-04 pos 0.513 0.510 bound 0.794143 0.991388
(3, 0.5, 4) 14.8s 1.7:3.480e-03/6.867e-04/4.006e-03 4.0:7.905e-03/1.143e-02/7.323e-03 6.0:1.101e-02/1.203e
    -02/5.594e-03 pos 0.510 0.501 bound 0.794396 0.991416
(3, 1, 3) 12.2s 1.7:4.654e-02/7.218e-03/1.065e-03 4.0:8.123e-02/7.397e-03/1.277e-02 6.0:1.052e-01/1.536e
    -02/2.445e-02 pos 0.513 0.510 bound 0.828287 0.891815
```

```
(3, 1, 4) 12.1s 1.7:2.895e-02/3.765e-03/2.858e-02 4.0:7.420e-02/1.019e-01/5.072e-02 6.0:1.028e-01/7.643e
    -02/2.026e-02 pos 0.510 0.501 bound 0.828860 0.892318
(3, 2, 3) 9.8s 1.7:4.719e-02/1.900e-03/6.876e-03 4.0:9.529e-02/2.314e-02/1.613e-02 6.0:1.291e-01/1.467e
    -02/3.407e-02 pos 0.513 0.510 bound 0.834221 0.844598
(3, 2, 4) 9.4s 1.7:1.832e-02/1.597e-03/1.464e-02 4.0:8.500e-02/1.176e-01/5.982e-02 6.0:1.244e-01/9.771e
    -02/2.459e-02 pos 0.510 0.501 bound 0.834124 0.844526
(6, 0.5, 3) 16.4s 1.7:1.589e-06/6.042e-10/2.030e-09 4.0:2.816e-06/3.803e-09/2.994e-09 6.0:3.699e-06/4.842e
    -09/3.566e-09 pos 0.513 0.510 bound 0.795417 0.999989
(6, 0.5, 4) 17.0s 1.7:1.126e-06/2.047e-07/1.299e-06 4.0:2.609e-06/3.835e-06/2.372e-06 6.0:3.643e-06/3.727e
    -06/1.678e-06 pos 0.510 0.501 bound 0.795417 0.999989
(6, 1, 3) 11.9s 1.7:5.345e-02/9.702e-03/8.884e-04 4.0:9.377e-02/1.032e-02/1.629e-02 6.0:1.207e-01/1.853e
    -02/3.100e-02 pos 0.513 0.510 bound 0.835375 0.867593
(6, 1, 4) 13.1s 1.7:3.326e-02/3.456e-03/3.198e-02 4.0:8.570e-02/1.157e-01/5.204e-02 6.0:1.179e-01/7.385e
    -02/1.193e-02 pos 0.510 0.501 bound 0.835639 0.867877
(6, 2, 3) 9.3s 1.7:4.848e-02/1.999e-03/7.186e-03 4.0:9.826e-02/2.450e-02/1.702e-02 6.0:1.329e-01/1.523e
    -02/3.585e-02 pos 0.513 0.510 bound 0.836364 0.836927
(6, 2, 4) 9.1s 1.7:1.882e-02/1.634e-03/1.490e-02 4.0:8.768e-02/1.221e-01/6.027e-02 6.0:1.282e-01/9.738e
    -02/2.238e-02 pos 0.510 0.501 bound 0.836144 0.836710
```

```
(3, 0.5, 3) 30.5s 1.7:4.924e-03/1.185e-04/9.109e-05 4.0:8.557e-03/5.292e-05/2.266e-04 6.0:1.120e-02/3.412e
    -04/3.867e-04 pos 0.513 0.510 bound 0.794143 0.991388
(3, 0.5, 4) 31.6s 1.7:3.480e-03/6.867e-04/4.006e-03 4.0:7.905e-03/1.143e-02/7.323e-03 6.0:1.101e-02/1.203e
    -02/5.594e-03 pos 0.510 0.501 bound 0.794396 0.991416
(3, 1, 3) 19.7s 1.7:4.654e-02/7.218e-03/1.065e-03 4.0:8.123e-02/7.397e-03/1.277e-02 6.0:1.052e-01/1.536e
    -02/2.445e-02 pos 0.513 0.510 bound 0.828287 0.891815
(3, 1, 4) 19.6s 1.7:2.895e-02/3.765e-03/2.858e-02 4.0:7.420e-02/1.019e-01/5.072e-02 6.0:1.028e-01/7.643e
    -02/2.026e-02 pos 0.510 0.501 bound 0.828860 0.892318
(3, 2, 3) 17.9s 1.7:4.719e-02/1.900e-03/6.876e-03 4.0:9.529e-02/2.314e-02/1.613e-02 6.0:1.291e-01/1.467e
    -02/3.407e-02 pos 0.513 0.510 bound 0.834221 0.844598
(3, 2, 4) 18.8s 1.7:1.832e-02/1.597e-03/1.464e-02 4.0:8.500e-02/1.176e-01/5.982e-02 6.0:1.244e-01/9.771e
    -02/2.459e-02 pos 0.510 0.501 bound 0.834124 0.844526
(6, 0.5, 3) 29.7s 1.7:1.589e-06/6.042e-10/2.030e-09 4.0:2.816e-06/3.803e-09/2.994e-09 6.0:3.699e-06/4.842e
    -09/3.566e-09 pos 0.513 0.510 bound 0.795417 0.999989
(6, 0.5, 4) 28.4s 1.7:1.126e-06/2.047e-07/1.299e-06 4.0:2.609e-06/3.834e-06/2.372e-06 6.0:3.643e-06/3.727e
    -06/1.678e-06 pos 0.510 0.501 bound 0.795417 0.999989
(6, 1, 3) 19.3s 1.7:5.345e-02/9.702e-03/8.884e-04 4.0:9.377e-02/1.032e-02/1.629e-02 6.0:1.207e-01/1.853e
    -02/3.100e-02 pos 0.513 0.510 bound 0.835375 0.867593
(6, 1, 4) 21.4s 1.7:3.326e-02/3.456e-03/3.198e-02 4.0:8.570e-02/1.157e-01/5.204e-02 6.0:1.179e-01/7.385e
    -02/1.193e-02 pos 0.510 0.501 bound 0.835639 0.867877
(6, 2, 3) 17.5s 1.7:4.848e-02/1.999e-03/7.186e-03 4.0:9.826e-02/2.450e-02/1.702e-02 6.0:1.329e-01/1.523e
    -02/3.585e-02 pos 0.513 0.510 bound 0.836364 0.836927
(6, 2, 4) 18.5s 1.7:1.882e-02/1.634e-03/1.490e-02 4.0:8.768e-02/1.221e-01/6.027e-02 6.0:1.282e-01/9.738e
    -02/2.238e-02 pos 0.510 0.501 bound 0.836144 0.836710
```

B.15 The flux–shear link and the neutrino fraction: predictions, changed solvers, drivers, run logs

Every prediction and kill line of Section 7 was written before the run it judges; the full pre-registration file, with each result appended under its test and including the intermediate readings that the final runs superseded, is `code/v4_runs/PREREGISTERED_V4.txt` in the archive and on GitHub. The final runs are the ones listed here. The changed solvers: McCormick's `ms_nuc.py` (MIT licence, copyright S. McCormick; the licence file ships with the archive) with the collision term split, and the shear-only variant; each differs from McCormick's file in its header line, the signature of `rhs`, the lines that compute the discrete projection and κ_2, and the collision term. The runners label a breakdown after $2M/R > 0.95$ separately from a detected horizon and log the minimum intensity; the driver moves a bracket only on a detected horizon or a dispersal.

```
# SCRIPT: NU-V4-MS-NUC-SPLIT  McCormick ms_nuc.py with ONE change: flux relaxes at kap, l>=2 at kap2 = rr*kap
    (dipole split off by discrete projection)
"""
Misner-Sharp perfect fluid (p = e/3) + massless free-streaming neutrinos (energy-integrated intensity
I(t, r, mu) in the fluid frame, finite volume in (r, mu), MUSCL/MC slopes, upwind).  Fluid-comoving gauge.
```

```
Transport (non-conservative form for I, using dV/dt = A theta V with V = int R^2 B dr):
  dI/dt = -A th I - [F_r(i+1)-F_r(i)]/V - [F_mu(k+1/2)-F_mu(k-1/2)]/(dmu V) - A (U/R + mu a + mu^2 S) I
  F_r = A R^2 mu I (edges),  F_mu = A (1-mu^2) [G1 - a V - mu S V] I,  G1 = (R_{i+1}^2 - R_i^2)/2
"""
import numpy as np
from numba import njit

PI = np.pi

@njit(cache=True)
def rho_b(t):
    return 3.0/(32.0*PI*t*t)

@njit(cache=True)
def mc(dm, dp):
    if dm*dp <= 0.0:
        return 0.0
    s = 1.0 if dm > 0 else -1.0
    return s*min(2*abs(dm), 2*abs(dp), 0.5*abs(dm+dp))

@njit(cache=True)
def state(t, R, U, Gam, e, I, mue, fnu, kap):
    """Derived quantities. I has shape (N, K)."""
    N, K = I.shape
    dmu = mue[1]-mue[0]
    eps = np.zeros(N); q = np.zeros(N); Pr = np.zeros(N)
    for i in range(N):
        s0 = 0.0; s1 = 0.0; s2 = 0.0
        for k in range(K):
            m0 = mue[k]; m1 = mue[k+1]
            s0 += I[i, k]
            s1 += I[i, k]*0.5*(m0+m1)
            s2 += I[i, k]*(m1**3-m0**3)/(3*dmu)
        eps[i] = 2*PI*s0*dmu; q[i] = 2*PI*s1*dmu; Pr[i] = 2*PI*s2*dmu
    M = np.zeros(N+1)
    for i in range(N):
        dV = (4.0*PI/3.0)*(R[i+1]**3 - R[i]**3)
        Ubar = 0.5*(U[i]+U[i+1]); Gbar = 0.5*(Gam[i]+Gam[i+1])
        M[i+1] = M[i] + dV*(e[i] + eps[i] + q[i]*Ubar/Gbar)
    eb = (1.0-fnu)*rho_b(t)
    Ac = np.empty(N); Ae = np.empty(N+1); p = np.empty(N)
    for i in range(N):
        p[i] = e[i]/3.0
    # fluid acceleration a = (-Gam dp/dR + kap q)/(e+p) at edges (0 at the centre; outer edge: drag only)
    ae = np.zeros(N+1)
    for i in range(1, N):
        Rc0 = 0.5*(R[i-1]+R[i]); Rc1 = 0.5*(R[i]+R[i+1])
        dpdR = (p[i]-p[i-1])/(Rc1-Rc0)
        ae[i] = (-Gam[i]*dpdR + kap*0.5*(q[i-1]+q[i]))/(0.5*(e[i-1]+p[i-1]+e[i]+p[i]))
    ae[N] = kap*q[N-1]/(e[N-1]+p[N-1])
    # lapse: A = (eb/e)^(1/4) at the outer cell, then d ln A = a * (proper distance) inward
    Ac[N-1] = (eb/e[N-1])**0.25
    for i in range(N-2, -1, -1):
        Rc0 = 0.5*(R[i]+R[i+1]); Rc1 = 0.5*(R[i+1]+R[i+2])
        Ac[i] = Ac[i+1]*np.exp(-ae[i+1]*(Rc1-Rc0)/Gam[i+1])
    Ae[0] = Ac[0]; Ae[N] = Ac[N-1]
    for i in range(1, N):
        Ae[i] = np.sqrt(Ac[i-1]*Ac[i])
    return eps, q, Pr, M, p, Ac, Ae, ae

@njit(cache=True)
def rhs(t, R, U, Gam, e, I, mue, fnu, Ib_out, kap, rr, dR, dUt, dG, de, dI):
    N, K = I.shape
    dmu = mue[1]-mue[0]
    eps, q, Pr, M, p, Ac, Ae, ae = state(t, R, U, Gam, e, I, mue, fnu, kap)
    # ---- fluid + metric ----
    dR[0] = 0.0; dUt[0] = 0.0; dG[0] = 0.0
    for i in range(1, N+1):
```

```
        if i < N:
            pe = 0.5*(p[i-1]+p[i]); Pre = 0.5*(Pr[i-1]+Pr[i]); qe = 0.5*(q[i-1]+q[i])
        else:
            pe = p[N-1]; Pre = Pr[N-1]; qe = q[N-1]
        dR[i] = Ae[i]*U[i]
        dUt[i] = Ae[i]*(-M[i]/R[i]**2 - 4.0*PI*R[i]*(pe+Pre) + Gam[i]*ae[i])
        dG[i] = Ae[i]*(4.0*PI*R[i]*qe + ae[i]*U[i])      # D_t Gamma = 4 pi R q + a U (evolved: avoids U^2 - 2
     M/R cancellation)
    th = np.empty(N); Gc = np.empty(N); V = np.empty(N); S = np.empty(N); ac = np.empty(N); UR = np.empty(N)
    for i in range(N):
        dR3 = R[i+1]**3 - R[i]**3
        Rc = 0.5*(R[i]+R[i+1]); Gc[i] = 0.5*(Gam[i]+Gam[i+1])
        th[i] = 3.0*(R[i+1]**2*U[i+1] - R[i]**2*U[i])/dR3 - 4.0*PI*Rc*q[i]/Gc[i]
        de[i] = -Ac[i]*(e[i]+p[i])*th[i]
        V[i] = dR3/(3.0*Gc[i])
        UR[i] = (U[i+1]+U[i])/(R[i+1]+R[i])
        S[i] = (U[i+1]-U[i])/(R[i+1]-R[i]) - 4.0*PI*Rc*q[i]/Gc[i] - UR[i]
        ac[i] = 0.5*(ae[i]+ae[i+1])
    # ---- neutrinos ----
    # radial fluxes at edges
    Fr = np.zeros((N+1, K))
    for k in range(K):
        mu = 0.5*(mue[k]+mue[k+1])
        for i in range(1, N+1):
            if mu > 0:
                j = i-1
                if j >= 1 and j <= N-2:
                    sl = mc(I[j, k]-I[j-1, k], I[j+1, k]-I[j, k])
                else:
                    sl = 0.0
                Iface = I[j, k] + 0.5*sl
            else:
                if i == N:
                    Iface = Ib_out
                else:
                    j = i
                    if j >= 1 and j <= N-2:
                        sl = mc(I[j, k]-I[j-1, k], I[j+1, k]-I[j, k])
                    else:
                        sl = 0.0
                    Iface = I[j, k] - 0.5*sl
            Fr[i, k] = Ae[i]*R[i]**2*mu*Iface
    smu2 = 0.0
    for k in range(K):
        mk = 0.5*(mue[k]+mue[k+1]); smu2 += mk*mk*dmu
    kap2 = rr*kap
    for i in range(N):
        G1 = 0.5*(R[i+1]**2 - R[i]**2)
        # angular fluxes at interior bin edges
        Fm = np.zeros(K+1)
        for k in range(1, K):
            m = mue[k]
            c = Ac[i]*(1.0-m*m)*(G1 - ac[i]*V[i] - m*S[i]*V[i])
            if c > 0:
                sl = mc(I[i, k-1]-I[i, k-2], I[i, k]-I[i, k-1]) if k-1 >= 1 else 0.0
                Iface = I[i, k-1] + 0.5*sl
            else:
                sl = mc(I[i, k]-I[i, k-1], I[i, k+1]-I[i, k]) if k+1 <= K-1 else 0.0
                Iface = I[i, k] - 0.5*sl
            Fm[k] = c*Iface
        for k in range(K):
            mu = 0.5*(mue[k]+mue[k+1]); mu2 = (mue[k+1]**3-mue[k]**3)/(3*dmu)
            dI[i, k] = (-Ac[i]*th[i]*I[i, k] - (Fr[i+1, k]-Fr[i, k])/V[i] - (Fm[k+1]-Fm[k])/(dmu*V[i])
                        - Ac[i]*(UR[i] + mu*ac[i] + mu2*S[i])*I[i, k]
                        + Ac[i]*(kap2*(eps[i]/(4*PI) - I[i, k]) + (kap2-kap)*q[i]*mu/(2*PI*smu2)))
    return M, Ac, ac, S, V

@njit(cache=True)
```

```
def timestep(R, U, e, I, mue, Gam, Ac, ac, S, cfl_r, cfl_mu):
    N, K = I.shape
    dmu = mue[1]-mue[0]
    dt = 1e30
    for i in range(N):
        dRc = R[i+1]-R[i]; Gc = 0.5*(Gam[i]+Gam[i+1])
        dt = min(dt, cfl_r*dRc/(abs(Gc)*Ac[i]))
        dR3 = R[i+1]**3 - R[i]**3
        V = dR3/(3.0*Gc); G1 = 0.5*(R[i+1]**2 - R[i]**2)
        cmax = Ac[i]*(abs(G1/V) + abs(ac[i]) + abs(S[i]))
        dt = min(dt, cfl_mu*dmu/cmax)
    return dt
```

```
# SCRIPT: NU-V4-MS-NUC-SHEAR  McCormick ms_nuc.py with ONE change: only the l = 2 (shear) part relaxes at kap2
      = rr*kap; flux and l >= 3 at kap
"""
Misner-Sharp perfect fluid (p = e/3) + massless free-streaming neutrinos (energy-integrated intensity
I(t, r, mu) in the fluid frame, finite volume in (r, mu), MUSCL/MC slopes, upwind).  Fluid-comoving gauge.
Transport (non-conservative form for I, using dV/dt = A theta V with V = int R^2 B dr):
  dI/dt = -A th I - [F_r(i+1)-F_r(i)]/V - [F_mu(k+1/2)-F_mu(k-1/2)]/(dmu V) - A (U/R + mu a + mu^2 S) I
  F_r = A R^2 mu I (edges),  F_mu = A (1-mu^2) [G1 - a V - mu S V] I,  G1 = (R_{i+1}^2 - R_i^2)/2
"""
import numpy as np
from numba import njit

PI = np.pi

@njit(cache=True)
def rho_b(t):
    return 3.0/(32.0*PI*t*t)

@njit(cache=True)
def mc(dm, dp):
    if dm*dp <= 0.0:
        return 0.0
    s = 1.0 if dm > 0 else -1.0
    return s*min(2*abs(dm), 2*abs(dp), 0.5*abs(dm+dp))

@njit(cache=True)
def state(t, R, U, Gam, e, I, mue, fnu, kap):
    """Derived quantities. I has shape (N, K)."""
    N, K = I.shape
    dmu = mue[1]-mue[0]
    eps = np.zeros(N); q = np.zeros(N); Pr = np.zeros(N)
    for i in range(N):
        s0 = 0.0; s1 = 0.0; s2 = 0.0
        for k in range(K):
            m0 = mue[k]; m1 = mue[k+1]
            s0 += I[i, k]
            s1 += I[i, k]*0.5*(m0+m1)
            s2 += I[i, k]*(m1**3-m0**3)/(3*dmu)
        eps[i] = 2*PI*s0*dmu; q[i] = 2*PI*s1*dmu; Pr[i] = 2*PI*s2*dmu
    M = np.zeros(N+1)
    for i in range(N):
        dV = (4.0*PI/3.0)*(R[i+1]**3 - R[i]**3)
        Ubar = 0.5*(U[i]+U[i+1]); Gbar = 0.5*(Gam[i]+Gam[i+1])
        M[i+1] = M[i] + dV*(e[i] + eps[i] + q[i]*Ubar/Gbar)
    eb = (1.0-fnu)*rho_b(t)
    Ac = np.empty(N); Ae = np.empty(N+1); p = np.empty(N)
    for i in range(N):
        p[i] = e[i]/3.0
    # fluid acceleration a = (-Gam dp/dR + kap q)/(e+p) at edges (0 at the centre; outer edge: drag only)
    ae = np.zeros(N+1)
    for i in range(1, N):
        Rc0 = 0.5*(R[i-1]+R[i]); Rc1 = 0.5*(R[i]+R[i+1])
        dpdR = (p[i]-p[i-1])/(Rc1-Rc0)
        ae[i] = (-Gam[i]*dpdR + kap*0.5*(q[i-1]+q[i]))/(0.5*(e[i-1]+p[i-1]+e[i]+p[i]))
    ae[N] = kap*q[N-1]/(e[N-1]+p[N-1])
```

```
    # lapse: A = (eb/e)^(1/4) at the outer cell, then d ln A = a * (proper distance) inward
    Ac[N-1] = (eb/e[N-1])**0.25
    for i in range(N-2, -1, -1):
        Rc0 = 0.5*(R[i]+R[i+1]); Rc1 = 0.5*(R[i+1]+R[i+2])
        Ac[i] = Ac[i+1]*np.exp(-ae[i+1]*(Rc1-Rc0)/Gam[i+1])
    Ae[0] = Ac[0]; Ae[N] = Ac[N-1]
    for i in range(1, N):
        Ae[i] = np.sqrt(Ac[i-1]*Ac[i])
    return eps, q, Pr, M, p, Ac, Ae, ae

@njit(cache=True)
def rhs(t, R, U, Gam, e, I, mue, fnu, Ib_out, kap, rr, dR, dUt, dG, de, dI):
    N, K = I.shape
    dmu = mue[1]-mue[0]
    eps, q, Pr, M, p, Ac, Ae, ae = state(t, R, U, Gam, e, I, mue, fnu, kap)
    # ---- fluid + metric ----
    dR[0] = 0.0; dUt[0] = 0.0; dG[0] = 0.0
    for i in range(1, N+1):
        if i < N:
            pe = 0.5*(p[i-1]+p[i]); Pre = 0.5*(Pr[i-1]+Pr[i]); qe = 0.5*(q[i-1]+q[i])
        else:
            pe = p[N-1]; Pre = Pr[N-1]; qe = q[N-1]
        dR[i] = Ae[i]*U[i]
        dUt[i] = Ae[i]*(-M[i]/R[i]**2 - 4.0*PI*R[i]*(pe+Pre) + Gam[i]*ae[i])
        dG[i] = Ae[i]*(4.0*PI*R[i]*qe + ae[i]*U[i])      # D_t Gamma = 4 pi R q + a U (evolved: avoids U^2 - 2
     M/R cancellation)
    th = np.empty(N); Gc = np.empty(N); V = np.empty(N); S = np.empty(N); ac = np.empty(N); UR = np.empty(N)
    for i in range(N):
        dR3 = R[i+1]**3 - R[i]**3
        Rc = 0.5*(R[i]+R[i+1]); Gc[i] = 0.5*(Gam[i]+Gam[i+1])
        th[i] = 3.0*(R[i+1]**2*U[i+1] - R[i]**2*U[i])/dR3 - 4.0*PI*Rc*q[i]/Gc[i]
        de[i] = -Ac[i]*(e[i]+p[i])*th[i]
        V[i] = dR3/(3.0*Gc[i])
        UR[i] = (U[i+1]+U[i])/(R[i+1]+R[i])
        S[i] = (U[i+1]-U[i])/(R[i+1]-R[i]) - 4.0*PI*Rc*q[i]/Gc[i] - UR[i]
        ac[i] = 0.5*(ae[i]+ae[i+1])
    # ---- neutrinos ----
    # radial fluxes at edges
    Fr = np.zeros((N+1, K))
    for k in range(K):
        mu = 0.5*(mue[k]+mue[k+1])
        for i in range(1, N+1):
            if mu > 0:
                j = i-1
                if j >= 1 and j <= N-2:
                    sl = mc(I[j, k]-I[j-1, k], I[j+1, k]-I[j, k])
                else:
                    sl = 0.0
                Iface = I[j, k] + 0.5*sl
            else:
                if i == N:
                    Iface = Ib_out
                else:
                    j = i
                    if j >= 1 and j <= N-2:
                        sl = mc(I[j, k]-I[j-1, k], I[j+1, k]-I[j, k])
                    else:
                        sl = 0.0
                    Iface = I[j, k] - 0.5*sl
            Fr[i, k] = Ae[i]*R[i]**2*mu*Iface
    smu2 = 0.0
    for k in range(K):
        mk = 0.5*(mue[k]+mue[k+1]); smu2 += mk*mk*dmu
    c2 = smu2/2.0                       # P2_k = mu_k^2 - c2 is orthogonal to 1 and to mu_k on the midpoint
     bins
    sp2 = 0.0
    for k in range(K):
        mk = 0.5*(mue[k]+mue[k+1]); sp2 += (mk*mk - c2)**2*dmu
```

```
    kap2 = rr*kap
    for i in range(N):
        G1 = 0.5*(R[i+1]**2 - R[i]**2)
        # angular fluxes at interior bin edges
        Fm = np.zeros(K+1)
        for k in range(1, K):
            m = mue[k]
            c = Ac[i]*(1.0-m*m)*(G1 - ac[i]*V[i] - m*S[i]*V[i])
            if c > 0:
                sl = mc(I[i, k-1]-I[i, k-2], I[i, k]-I[i, k-1]) if k-1 >= 1 else 0.0
                Iface = I[i, k-1] + 0.5*sl
            else:
                sl = mc(I[i, k]-I[i, k-1], I[i, k+1]-I[i, k]) if k+1 <= K-1 else 0.0
                Iface = I[i, k] - 0.5*sl
            Fm[k] = c*Iface
        a2 = 0.0
        for k in range(K):
            mk = 0.5*(mue[k]+mue[k+1]); a2 += I[i, k]*(mk*mk - c2)*dmu
        a2 = a2/sp2
        for k in range(K):
            mu = 0.5*(mue[k]+mue[k+1]); mu2 = (mue[k+1]**3-mue[k]**3)/(3*dmu)
            dI[i, k] = (-Ac[i]*th[i]*I[i, k] - (Fr[i+1, k]-Fr[i, k])/V[i] - (Fm[k+1]-Fm[k])/(dmu*V[i])
                        - Ac[i]*(UR[i] + mu*ac[i] + mu2*S[i])*I[i, k]
                        + Ac[i]*(kap*(eps[i]/(4*PI) - I[i, k]) + (kap-kap2)*a2*(mu*mu - c2)))
    return M, Ac, ac, S, V

@njit(cache=True)
def timestep(R, U, e, I, mue, Gam, Ac, ac, S, cfl_r, cfl_mu):
    N, K = I.shape
    dmu = mue[1]-mue[0]
    dt = 1e30
    for i in range(N):
        dRc = R[i+1]-R[i]; Gc = 0.5*(Gam[i]+Gam[i+1])
        dt = min(dt, cfl_r*dRc/(abs(Gc)*Ac[i]))
        dR3 = R[i+1]**3 - R[i]**3
        V = dR3/(3.0*Gc); G1 = 0.5*(R[i+1]**2 - R[i]**2)
        cmax = Ac[i]*(abs(G1/V) + abs(ac[i]) + abs(S[i]))
        dt = min(dt, cfl_mu*dmu/cmax)
    return dt
```

```
# SCRIPT: NU-V4-RUN-SPLIT-V2  run_nuc_n.py importing ms_nuc_split and passing rr; nothing else changed
import numpy as np, time
from ms_nuc_split import rhs, state, timestep, rho_b
from ms_fluid import initial_data, make_grid

def run(delta_m, fnu=0.409, N=400, K=32, rmax=9.0, beta=3.0, eps0=0.1, cfl_r=0.4, cfl_mu=0.4, cH=0.005,
        eps_max=30.0, qshape=1.0, snap_eps=(), hist_every=20, stop_rule=True, disperse_rule=True, kap=0.0,
     eps_dec=None, nexp=3, kapcap=5.0e5, rr=1.0):
    r = make_grid(N, rmax, beta)
    t0, R, U, rho = initial_data(r, delta_m, eps0, 0.0, qshape)
    e = (1-fnu)*rho
    mue = np.linspace(-1, 1, K+1)
    I = np.repeat((fnu*rho/(4*np.pi))[:, None], K, axis=1)
    Mi = np.concatenate([[0.0], np.cumsum((4*np.pi/3)*np.diff(R**3)*rho)])
    Gam = np.ones(N+1); Gam[1:] = np.sqrt(1.0 + U[1:]**2 - 2*Mi[1:]/R[1:])
    t = t0
    dR = np.zeros(N+1); dU = np.zeros(N+1); dG = np.zeros(N+1); de = np.zeros(N); dI = np.zeros((N, K))
    Imin = float(I.min()); mtwo = 0.0
    peakC = -1.0; hist = []; snaps = {}; step = 0; status = 'undecided'
    snap_eps = list(snap_eps)
    while True:
        Ib = fnu/(1-fnu)*e[-1]/(4*np.pi)      # incoming background tied to the outer cell's fluid density
        if eps_dec is not None:
            kap = (1/(2*t))*min((eps_dec/(2*np.sqrt(t*t0)))**nexp, kapcap)
        try:
            M, Ac, ac, S, V = rhs(t, R, U, Gam, e, I, mue, fnu, Ib, kap, rr, dR, dU, dG, de, dI)
        except (ZeroDivisionError, FloatingPointError):
            status = 'crash'; break
```

```
        epsn = 2*np.sqrt(t*t0)
        twoMR = 2*M[1:]/R[1:]
        C = 2*(M[1:] - (4*np.pi/3)*rho_b(t)*R[1:]**3)/R[1:]
        iC = np.argmax(C); peakC = max(peakC, C[iC])
        if step % hist_every == 0:
            dmu = mue[1]-mue[0]
            inner = r[1:] < 3.0
            hcv = np.max(np.abs(Gam[1:]**2 - 1 - U[1:]**2 + twoMR)[inner])     # Hamiltonian-constraint check
            hist.append((t, epsn, C[iC], r[1+iC], e[0]/rho_b(t), 2*np.pi*I[0].sum()*dmu/rho_b(t),
                         e[-1]/rho_b(t), 2*np.pi*I[-1].sum()*dmu/rho_b(t), hcv))
        if snap_eps and epsn >= snap_eps[0]:
            snaps[snap_eps.pop(0)] = dict(t=t, eps=epsn, R=R.copy(), U=U.copy(), e=e.copy(), I=I.copy(), M=M.
    copy())
        if stop_rule:
            if np.any((twoMR >= 1.0) & (U[1:] < 0)):
                status = 'BH'; break
            if disperse_rule and epsn > 3.0 and C[iC] < 0.5*peakC:
                status = 'dispersed'; break
        if epsn > eps_max: break
        try:
            dt = min(timestep(R, U, e, I, mue, Gam, Ac, ac, S, cfl_r, cfl_mu), cH*2*t, (0.5/(kap*max(1.0, rr)*
    np.max(Ac)) if kap > 0 else 1e30))
            # SSP RK3 (the outer-boundary intensity is evaluated at each stage time)
            R1 = R + dt*dR; U1 = U + dt*dU; G1 = Gam + dt*dG; e1 = e + dt*de; I1 = I + dt*dI
            rhs(t+dt, R1, U1, G1, e1, I1, mue, fnu, fnu/(1-fnu)*e1[-1]/(4*np.pi), kap, rr, dR, dU, dG, de, dI)
            R2 = 0.75*R + 0.25*(R1+dt*dR); U2 = 0.75*U + 0.25*(U1+dt*dU); G2 = 0.75*Gam + 0.25*(G1+dt*dG)
            e2 = 0.75*e + 0.25*(e1+dt*de); I2 = 0.75*I + 0.25*(I1+dt*dI)
            rhs(t+0.5*dt, R2, U2, G2, e2, I2, mue, fnu, fnu/(1-fnu)*e2[-1]/(4*np.pi), kap, rr, dR, dU, dG, de,
     dI)
            R = R/3 + 2/3*(R2+dt*dR); U = U/3 + 2/3*(U2+dt*dU); Gam = Gam/3 + 2/3*(G2+dt*dG)
            e = e/3 + 2/3*(e2+dt*de); I = I/3 + 2/3*(I2+dt*dI)
        except (ZeroDivisionError, FloatingPointError):
            coll = U[1:] < 0
            status = 'BH_breakdown' if (coll.any() and np.max(twoMR[coll]) > 0.95) else 'crash'   # v2:
    inferred, not a horizon   # breakdown just before collapse
            break
        t += dt; step += 1
        Imin = min(Imin, float(I.min())); mtwo = max(mtwo, float(np.max(twoMR)))
        if np.any(e <= 0) or not np.all(np.isfinite(U)) or not np.all(np.isfinite(I)):
            status = 'crash'; break
    return dict(Imin=Imin, max2MR=mtwo, status=status, t=t, t0=t0, eps=2*np.sqrt(t*t0), steps=step, peakC=
     peakC, hist=np.array(hist),
                snaps=snaps, r=r, mue=mue, R=R, U=U, e=e, I=I, Gam=Gam)

if __name__ == '__main__':
    t1 = time.time()
    out = run(0.0, fnu=0.409, N=200, K=16, eps_max=5.0, stop_rule=False)
    t = out['t']; Ib = 0.409*rho_b(t)/(4*np.pi); eb = 0.591*rho_b(t)
    print(f"FRW test to eps={out['eps']:.2f} in {out['steps']} steps ({time.time()-t1:.1f}s)")
    print("  max|I/Ib-1| =", np.abs(out['I']/Ib-1).max(), " max|e/eb-1| =", np.abs(out['e']/eb-1).max())
    a = np.sqrt(t/out['t0']); print("  max|R/(a r)-1| =", np.abs(out['R'][1:]/(a*out['r'][1:])-1).max())
```

```
# SCRIPT: NU-V4-TRISECT-V2  trisection in which only a detected horizon ('BH') or 'dispersed' moves the
     bracket.
# 'BH_breakdown' (numerical breakdown after 2M/R > 0.95) and 'crash' get one retry at CFL 0.3, then are
     reported and skipped.
# Every run reports min I over the run (positivity monitor) and max 2M/R.
import sys, json, time, os, importlib
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
def one(a):
    d,kw=a; mod=importlib.import_module(RUNNER); o=mod.run(d,**kw); s=o['status']; note=''
    if s not in ('BH','dispersed'):
        kw2=dict(kw,cfl_r=0.3,cfl_mu=0.3); o2=mod.run(d,**kw2); note=' (retry at CFL 0.3 after %s)'%s; o=o2; s
     =o['status']
    return d,s,float(o['eps']),float(o['Imin']),float(o['max2MR']),note
if __name__=='__main__':
```

```
    tag=sys.argv[1]; RUNNER=sys.argv[2]; lo,hi,tol=float(sys.argv[3]),float(sys.argv[4]),float(sys.argv[5]);
     kw=json.loads(sys.argv[6])
    log=[]; t0=time.time()
    with Pool(2) as p:
        while hi-lo>tol:
            a=lo+(hi-lo)/3; b=lo+2*(hi-lo)/3
            res=p.map(one,[(a,kw),(b,kw)])
            for d,s,e,im,m2,note in res: print(f'{tag} delta={d:.8f}: {s} eps={e:.2f} minI={im:.3e} max2M/R={
     m2:.4f}{note}',flush=True); log.append((d,s,e,im,m2,note))
            ok=[(d,s) for d,s,*_ in res if s in ('BH','dispersed')]
            if not ok: print('both points undecided; stop',flush=True); break
            for d,s in ok:
                if s=='BH': hi=min(hi,d)
                else: lo=max(lo,d)
            if lo>=hi: print('inconsistent outcomes; stop',flush=True); break
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.8f} +- {0.5*(hi-lo):.8f}  bracket [{lo:.8f}, {hi:.8f}] ({(
     time.time()-t0)/60:.1f} min)',flush=True)
    json.dump(dict(tag=tag,dc=0.5*(lo+hi),w=hi-lo,lo=lo,hi=hi,log=log),open(f'res_{tag}.json','w'))
```

```
# SCRIPT: NU-V4-TRISECT-CLEAN  trisect in which a crash never moves the bracket: one retry at CFL 0.3, else
     reported and skipped
import sys, json, time, os
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_split import run
def one(a):
    d,q,ed,n,em=a
    o=run(d,fnu=FNU,N=400,K=32,eps0=0.1,eps_dec=ed,eps_max=em,qshape=q,nexp=n,rr=RR); s=o['status']; note=''
    if s not in ('BH','dispersed'):
        o=run(d,fnu=FNU,N=400,K=32,eps0=0.1,eps_dec=ed,eps_max=em,qshape=q,nexp=n,rr=RR,cfl_r=0.3,cfl_mu=0.3);
      note=' (retry at CFL 0.3 after %s)'%s; s=o['status']
    return d,s,float(o['eps']),note
if __name__=='__main__':
    tag=sys.argv[1]; q=float(sys.argv[2]); ed=float(sys.argv[3]); n=float(sys.argv[4]); lo,hi,tol=float(sys.
     argv[5]),float(sys.argv[6]),float(sys.argv[7]); em=float(sys.argv[8]); FNU=float(sys.argv[9]); RR=float(
     sys.argv[10])
    log=[]; t0=time.time()
    with Pool(2) as p:
        while hi-lo>tol:
            a=lo+(hi-lo)/3; b=lo+2*(hi-lo)/3
            res=p.map(one,[(a,q,ed,n,em),(b,q,ed,n,em)])
            for d,s,e,note in res: print(f'{tag} delta={d:.8f}: {s} eps={e:.2f}{note}',flush=True); log.append
     ((d,s,e,note))
            ok=[(d,s) for d,s,e,_ in res if s in ('BH','dispersed')]
            if not ok: print('both points undecided; stop',flush=True); break
            for d,s in ok:
                if s=='BH': hi=min(hi,d)
                else: lo=max(lo,d)
            if lo>=hi: print('inconsistent outcomes; stop',flush=True); break
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.8f} +- {0.5*(hi-lo):.8f}  bracket [{lo:.8f}, {hi:.8f}] ({(
     time.time()-t0)/60:.1f} min)',flush=True)
    json.dump(dict(tag=tag,dc=0.5*(lo+hi),w=hi-lo,lo=lo,hi=hi,log=log),open(f'res_{tag}.json','w'))
```

```
# SCRIPT: NU-V4-TRISECT-SPLIT  two trial amplitudes per round in parallel (thirds of the bracket)
import sys, json, time, os
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_split import run
def one(a):
    d,q,ed,n,em=a; o=run(d,fnu=FNU,N=400,K=32,eps0=0.1,eps_dec=ed,eps_max=em,qshape=q,nexp=n,rr=RR); return d,
     o['status'],float(o['eps'])
if __name__=='__main__':
    tag=sys.argv[1]; q=float(sys.argv[2]); ed=float(sys.argv[3]); n=float(sys.argv[4]); lo,hi,tol=float(sys.
     argv[5]),float(sys.argv[6]),float(sys.argv[7]); em=float(sys.argv[8]); FNU=float(sys.argv[9]); RR=float(
     sys.argv[10])
    log=[]; t0=time.time()
    with Pool(2) as p:
```

```
        while hi-lo>tol:
            a=lo+(hi-lo)/3; b=lo+2*(hi-lo)/3
            res=p.map(one,[(a,q,ed,n,em),(b,q,ed,n,em)])
            for d,s,e in res: print(f'{tag} delta={d:.5f}: {s} eps={e:.2f}',flush=True); log.append((d,s,e))
            (da,sa,_),(db,sb,_)=res
            if 'BH' not in (sa,sb) and 'dispersed' not in (sa,sb): break
            if sa=='BH': hi=da
            elif sb=='BH': lo,hi=da,db
            else: lo=db
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} ({(time.time()-t0)/60:.1f} min)',
     flush=True)
    json.dump(dict(tag=tag,dc=0.5*(lo+hi),w=hi-lo,log=log),open(f'res_{tag}.json','w'))
```

```
# SCRIPT: NU-V4-TRISECT-FNU  two trial amplitudes per round in parallel (thirds of the bracket)
import sys, json, time, os
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_nuc_n import run
def one(a):
    d,q,ed,n,em=a; o=run(d,fnu=FNU,N=400,K=32,eps0=0.1,eps_dec=ed,eps_max=em,qshape=q,nexp=n); return d,o['
     status'],float(o['eps'])
if __name__=='__main__':
    tag=sys.argv[1]; q=float(sys.argv[2]); ed=float(sys.argv[3]); n=float(sys.argv[4]); lo,hi,tol=float(sys.
     argv[5]),float(sys.argv[6]),float(sys.argv[7]); em=float(sys.argv[8]); FNU=float(sys.argv[9])
    log=[]; t0=time.time()
    with Pool(2) as p:
        while hi-lo>tol:
            a=lo+(hi-lo)/3; b=lo+2*(hi-lo)/3
            res=p.map(one,[(a,q,ed,n,em),(b,q,ed,n,em)])
            for d,s,e in res: print(f'{tag} delta={d:.5f}: {s} eps={e:.2f}',flush=True); log.append((d,s,e))
            (da,sa,_),(db,sb,_)=res
            if 'BH' not in (sa,sb) and 'dispersed' not in (sa,sb): break
            if sa=='BH': hi=da
            elif sb=='BH': lo,hi=da,db
            else: lo=db
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} ({(time.time()-t0)/60:.1f} min)',
     flush=True)
    json.dump(dict(tag=tag,dc=0.5*(lo+hi),w=hi-lo,log=log),open(f'res_{tag}.json','w'))
```

```
# SCRIPT: NU-V4-MEMORY-SPLIT  frozen memory form (w = 0.890, unchanged) with flux rate kappa1 and shear+ rate
     kappa2 = r kappa1
# Same object as PBH-NU-MEMORY-CURVE; the only change is the collision diagonal (0, 1, r, r, ...) in place of
     (0, 1, 1, 1, ...).
import numpy as np, json, sys
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
kn,kw=roots_legendre(36); kmin,kmax=0.25,9.0; ks=0.5*(kmax-kmin)*(kn+1)+kmin; wk=0.5*(kmax-kmin)*kw
wE=wk*ks**2*np.exp(-ks**2/2); wE/=wE.sum()
LM=60; M=np.zeros((LM+1,LM+1))
for l in range(1,LM+1): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
tau0=0.02; eg=np.linspace(tau0,6.0,1200); n=LM+1
def surv(k,ed,p,r):
    d=np.full(n,r); d[0]=0.0; d[1]=1.0
    def rhs(t,v):
        K=0.0 if ed is None else ed**p/t**(p+1)
        x,y=v[:n],v[n:]
        return np.concatenate([-K*d*x+k*(M@y),-K*d*y-k*(M@x)])
    v0=np.zeros(2*n); v0[0]=1
    s=solve_ivp(rhs,(tau0,eg[-1]),v0,method='BDF',t_eval=eg,rtol=1e-9,atol=1e-12)
    return s.y[0]**2+s.y[n]**2
f=lambda ed,p,r: sum(w_*surv(kk,ed,p,r) for kk,w_ in zip(ks,wE))
i17,i40=np.searchsorted(eg,1.7),np.searchsorted(eg,4.0); w=0.890; Ai=f(None,3,1.0)
out={}
for r in [1.0,0.5,2.0]:
    g=(1-f(2.0,3,r))/(1-Ai); G=float(w*g[i17]+(1-w)*g[i40]); out[str(r)]=G
    print(f'n=3 eps_dec=2 r=kappa2/kappa1={r}: Gamma12={(r-1)/(r+1):+.4f}  memory-form G={G:.4f}',flush=True)
json.dump(out,open('memory_split.json','w'))
```

```
REPRO delta=0.5109 n=3 ed=2: BH 6.61 75s  (paper: BH, inside 0.51074-0.51109)
```

```
REGRESSION rr=1 delta=0.5109: BH 6.61 79s (unmodified: BH 6.61)
```

```
REGRESSION shear-only rr=1 delta=0.5109: BH 6.61 90s (unmodified: BH 6.61)
```

```
E_fnu020_ed2_n3 delta=0.50280: dispersed eps=5.72
E_fnu020_ed2_n3 delta=0.50790: BH eps=4.81
E_fnu020_ed2_n3 delta=0.50450: BH eps=5.90
E_fnu020_ed2_n3 delta=0.50620: BH eps=5.15
E_fnu020_ed2_n3 delta=0.50337: dispersed eps=6.00
E_fnu020_ed2_n3 delta=0.50393: dispersed eps=6.84
E_fnu020_ed2_n3 delta=0.50412: BH eps=6.44
E_fnu020_ed2_n3 delta=0.50431: BH eps=6.10
E_fnu020_ed2_n3 delta=0.50400: BH eps=7.23
E_fnu020_ed2_n3 delta=0.50406: BH eps=6.66
RESULT E_fnu020_ed2_n3: delta_c = 0.50396 +- 0.00003 (6.8 min)
```

```
F_r025 delta=0.50843: dispersed eps=4.87
F_r025 delta=0.51457: dispersed eps=5.60
F_r025 delta=0.51661: BH eps=6.73
F_r025 delta=0.51866: BH eps=5.25
F_r025 delta=0.51525: dispersed eps=5.81
F_r025 delta=0.51593: dispersed eps=6.16
F_r025 delta=0.51616: dispersed eps=6.37
F_r025 delta=0.51638: dispersed eps=6.76
F_r025 delta=0.51646: dispersed eps=7.06
F_r025 delta=0.51654: BH eps=7.21
RESULT F_r025: delta_c = 0.51650 +- 0.00004 (6.1 min)
```

```
F_r4 delta=0.50843: BH eps=5.01
F_r4 delta=0.51457: BH eps=4.26
F_r4 delta=0.50434: dispersed eps=5.85
F_r4 delta=0.50639: BH eps=5.69
F_r4 delta=0.50503: dispersed eps=6.24
F_r4 delta=0.50571: BH eps=6.35
F_r4 delta=0.50525: dispersed eps=6.51
F_r4 delta=0.50548: BH eps=7.36
F_r4 delta=0.50533: dispersed eps=6.65
F_r4 delta=0.50540: dispersed eps=6.88
RESULT F_r4: delta_c = 0.50544 +- 0.00004 (7.4 min)
```

```
F2_r1o16 delta=0.51610: dispersed eps=5.07
F2_r1o16 delta=0.52220: BH eps=5.90
F2_r1o16 delta=0.51813: dispersed eps=5.32
F2_r1o16 delta=0.52017: dispersed eps=5.78
F2_r1o16 delta=0.52084: dispersed eps=6.11
F2_r1o16 delta=0.52152: BH eps=7.25
F2_r1o16 delta=0.52107: dispersed eps=6.30
F2_r1o16 delta=0.52130: dispersed eps=6.60
F2_r1o16 delta=0.52137: dispersed eps=6.77
F2_r1o16 delta=0.52145: dispersed eps=7.07
RESULT F2_r1o16: delta_c = 0.52148 +- 0.00004 (6.6 min)
```

```
F2_r16 delta=0.50180: BH eps=6.59
F2_r16 delta=0.50590: BH eps=4.76
F2_r16 delta=0.49907: dispersed eps=5.46
F2_r16 delta=0.50043: dispersed eps=5.80
F2_r16 delta=0.50089: dispersed eps=6.00
F2_r16 delta=0.50134: dispersed eps=6.34
F2_r16 delta=0.50150: dispersed eps=6.56
F2_r16 delta=0.50165: dispersed eps=7.09
F2_r16 delta=0.50170: BH eps=7.25
```

```
F2_r16 delta=0.50175: BH eps=6.80
RESULT F2_r16: delta_c = 0.50167 +- 0.00003 (14.1 min)
```

```
H1_r1 delta=0.51050: dispersed eps=6.45
H1_r1 delta=0.51150: BH eps=5.83
H1_r1 delta=0.51083: BH eps=6.89
H1_r1 delta=0.51117: BH eps=6.13
H1_r1 delta=0.51061: dispersed eps=6.65
H1_r1 delta=0.51072: dispersed eps=7.03
H1_r1 delta=0.51076: BH eps=7.36
H1_r1 delta=0.51080: BH eps=7.19
RESULT H1_r1: delta_c = 0.51074 +- 0.00002 (5.0 min)
```

```
I_r025_dt4 delta=0.51600: dispersed eps=6.21
I_r025_dt4 delta=0.51700: BH eps=6.05
I_r025_dt4 delta=0.51633: dispersed eps=6.63
I_r025_dt4 delta=0.51667: BH eps=6.56
I_r025_dt4 delta=0.51644: dispersed eps=6.97
I_r025_dt4 delta=0.51656: BH eps=7.02
I_r025_dt4 delta=0.51648: dispersed eps=7.22
I_r025_dt4 delta=0.51652: BH eps=7.40
RESULT I_r025_dt4: delta_c = 0.51650 +- 0.00002 (6.5 min)
```

```
J_r2 delta=0.50667: dispersed eps=5.78
J_r2 delta=0.50833: BH eps=6.18
J_r2 delta=0.50722: dispersed eps=6.02
J_r2 delta=0.50778: dispersed eps=6.54
J_r2 delta=0.50796: dispersed eps=7.19
J_r2 delta=0.50815: BH eps=6.52
J_r2 delta=0.50802: BH eps=7.07
J_r2 delta=0.50809: BH eps=6.71
J_r2 delta=0.50798: BH eps=7.36
J_r2 delta=0.50800: BH eps=7.34
RESULT J_r2: delta_c = 0.50797 +- 0.00001 (6.3 min)
```

```
J_r05 delta=0.51367: BH eps=7.38
J_r05 delta=0.51533: BH eps=5.39
J_r05 delta=0.51256: dispersed eps=5.86
J_r05 delta=0.51311: dispersed eps=6.17
J_r05 delta=0.51330: dispersed eps=6.34
J_r05 delta=0.51348: dispersed eps=6.61
J_r05 delta=0.51354: dispersed eps=6.75
J_r05 delta=0.51360: dispersed eps=6.98
J_r05 delta=0.51363: dispersed eps=7.09
J_r05 delta=0.51365: dispersed eps=7.28
RESULT J_r05: delta_c = 0.51366 +- 0.00001 (6.5 min)
```

```
K_r125_clean delta=0.50984047: dispersed eps=7.35
K_r125_clean delta=0.50984184: BH eps=7.35
K_r125_clean delta=0.50984092: BH eps=7.35
K_r125_clean delta=0.50984138: BH eps=7.35
RESULT K_r125_clean: delta_c = 0.50984070 +- 0.00000023  bracket [0.50984047, 0.50984092] (3.4 min)
```

```
K_r08_clean delta=0.51170069: BH eps=7.37
K_r08_clean delta=0.51170754: BH eps=7.38
K_r08_clean delta=0.51169611: BH eps=7.36
K_r08_clean delta=0.51169840: BH eps=7.37
K_r08_clean delta=0.51169459: dispersed eps=7.36 (retry at CFL 0.3 after crash)
K_r08_clean delta=0.51169535: dispersed eps=7.36
RESULT K_r08_clean: delta_c = 0.51169573 +- 0.00000038  bracket [0.51169535, 0.51169611] (7.0 min)
```

```
L_r4 delta=0.50833333: BH eps=4.98
L_r4 delta=0.51366667: BH eps=4.31
```

```
L_r4 delta=0.50477778: dispersed eps=6.22
L_r4 delta=0.50655556: BH eps=5.49
L_r4 delta=0.50537037: BH eps=6.60
L_r4 delta=0.50596296: BH eps=5.81
L_r4 delta=0.50497531: dispersed eps=6.44
L_r4 delta=0.50517284: dispersed eps=6.88
L_r4 delta=0.50523868: BH eps=7.34
L_r4 delta=0.50530453: BH eps=6.89
L_r4 delta=0.50519479: dispersed eps=6.98
L_r4 delta=0.50521674: dispersed eps=7.13
L_r4 delta=0.50522405: dispersed eps=7.19
L_r4 delta=0.50523137: dispersed eps=7.28
RESULT L_r4: delta_c = 0.50523503 +- 0.00000366  bracket [0.50523137, 0.50523868] (12.2 min)
```

```
L_r025 delta=0.50833333: dispersed eps=4.83
L_r025 delta=0.51366667: dispersed eps=5.31
L_r025 delta=0.51544444: dispersed eps=5.67
L_r025 delta=0.51722222: BH eps=6.71
L_r025 delta=0.51603704: dispersed eps=5.88
L_r025 delta=0.51662963: dispersed eps=6.24
L_r025 delta=0.51682716: dispersed eps=6.46
L_r025 delta=0.51702469: dispersed eps=6.89
L_r025 delta=0.51709053: dispersed eps=7.31
L_r025 delta=0.51715638: BH eps=7.03
L_r025 delta=0.51711248: BH eps=7.40
L_r025 delta=0.51713443: BH eps=7.24
L_r025 delta=0.51709785: BH eps=7.39
L_r025 delta=0.51710517: BH eps=7.40
RESULT L_r025: delta_c = 0.51709419 +- 0.00000366  bracket [0.51709053, 0.51709785] (9.8 min)
```

```
M_r4_K48 delta=0.50516667: dispersed eps=6.36
M_r4_K48 delta=0.50583333: BH eps=6.19
M_r4_K48 delta=0.50538889: dispersed eps=6.75
M_r4_K48 delta=0.50561111: BH eps=6.66
M_r4_K48 delta=0.50546296: dispersed eps=7.07
M_r4_K48 delta=0.50553704: BH eps=7.07
M_r4_K48 delta=0.50548765: dispersed eps=7.32
M_r4_K48 delta=0.50551235: crash eps=7.36 (retry at CFL 0.3 after crash)
M_r4_K48 delta=0.50550412: BH eps=7.36
M_r4_K48 delta=0.50552058: BH eps=7.27
RESULT M_r4_K48: delta_c = 0.50549588 +- 0.00000823  bracket [0.50548765, 0.50550412] (21.0 min)
```

```
M_r025_K48 delta=0.51616667: dispersed eps=6.31
M_r025_K48 delta=0.51683333: BH eps=6.35
M_r025_K48 delta=0.51638889: dispersed eps=6.62
M_r025_K48 delta=0.51661111: BH eps=7.11
M_r025_K48 delta=0.51646296: dispersed eps=6.80
M_r025_K48 delta=0.51653704: dispersed eps=7.13
M_r025_K48 delta=0.51656173: BH eps=7.39
M_r025_K48 delta=0.51658642: BH eps=7.40
M_r025_K48 delta=0.51654527: dispersed eps=7.20
M_r025_K48 delta=0.51655350: dispersed eps=7.28
RESULT M_r025_K48: delta_c = 0.51655761 +- 0.00000412  bracket [0.51655350, 0.51656173] (14.1 min)
```

```
A1_shear_r12 delta=0.50883333: dispersed eps=5.85 minI=1.930e-08 max2M/R=8100.0000
A1_shear_r12 delta=0.50966667: dispersed eps=6.41 minI=1.379e-08 max2M/R=8100.0000
A1_shear_r12 delta=0.50994444: dispersed eps=7.22 minI=3.378e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.51022222: BH eps=6.33 minI=1.444e-08 max2M/R=8100.0000
A1_shear_r12 delta=0.51003704: BH eps=6.85 minI=8.923e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.51012963: BH eps=6.52 minI=1.295e-08 max2M/R=8100.0000
A1_shear_r12 delta=0.50997531: BH eps=7.37 minI=5.965e-10 max2M/R=8100.0000
A1_shear_r12 delta=0.51000617: BH eps=7.05 minI=3.935e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.50995473: BH eps=7.35 minI=1.270e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.50996502: BH eps=7.37 minI=8.737e-10 max2M/R=8100.0000
A1_shear_r12 delta=0.50994787: dispersed eps=7.26 minI=2.493e-09 max2M/R=8100.0000
```

```
A1_shear_r12 delta=0.50995130: dispersed eps=7.31 minI=1.913e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.50995245: dispersed eps=7.32 minI=1.570e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.50995359: dispersed eps=7.34 minI=1.379e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.50995397: dispersed eps=7.35 minI=1.281e-09 max2M/R=8100.0000
A1_shear_r12 delta=0.50995435: BH eps=7.35 minI=1.273e-09 max2M/R=8100.0000
RESULT A1_shear_r12: delta_c = 0.50995416 +- 0.00000019  bracket [0.50995397, 0.50995435] (7.6 min)
```

```
A1_shear_r083 delta=0.51183333: BH eps=6.35 minI=1.407e-08 max2M/R=8100.0000
A1_shear_r083 delta=0.51266667: BH eps=5.62 minI=2.173e-08 max2M/R=8100.0000
A1_shear_r083 delta=0.51127778: dispersed eps=6.40 minI=1.369e-08 max2M/R=8100.0000
A1_shear_r083 delta=0.51155556: dispersed eps=7.15 minI=5.669e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51164815: BH eps=6.91 minI=7.048e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51174074: BH eps=6.55 minI=1.258e-08 max2M/R=8100.0000
A1_shear_r083 delta=0.51158642: BH eps=7.37 minI=9.559e-10 max2M/R=8100.0000
A1_shear_r083 delta=0.51161728: BH eps=7.16 minI=2.294e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51156584: dispersed eps=7.24 minI=3.090e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51157613: BH eps=7.36 minI=1.237e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51156927: dispersed eps=7.28 minI=2.168e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51157270: dispersed eps=7.33 minI=1.575e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51157385: dispersed eps=7.34 minI=1.389e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51157499: BH eps=7.36 minI=1.219e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51157423: dispersed eps=7.35 minI=1.383e-09 max2M/R=8100.0000
A1_shear_r083 delta=0.51157461: dispersed eps=7.36 minI=1.225e-09 max2M/R=8100.0000
RESULT A1_shear_r083: delta_c = 0.51157480 +- 0.00000019  bracket [0.51157461, 0.51157499] (7.4 min)
```

```
A1_all_r12 delta=0.50916667: dispersed eps=5.98 minI=1.786e-08 max2M/R=8100.0000
A1_all_r12 delta=0.50983333: dispersed eps=6.59 minI=1.250e-08 max2M/R=8100.0000
A1_all_r12 delta=0.51005556: BH eps=7.09 minI=3.266e-09 max2M/R=8100.0000
A1_all_r12 delta=0.51027778: BH eps=6.32 minI=1.450e-08 max2M/R=8100.0000
A1_all_r12 delta=0.50990741: dispersed eps=6.76 minI=1.135e-08 max2M/R=8100.0000
A1_all_r12 delta=0.50998148: dispersed eps=7.09 minI=7.702e-09 max2M/R=8100.0000
A1_all_r12 delta=0.51000617: dispersed eps=7.33 minI=1.519e-09 max2M/R=8100.0000
A1_all_r12 delta=0.51003086: BH eps=7.38 minI=5.240e-10 max2M/R=8100.0000
A1_all_r12 delta=0.51001440: dispersed eps=7.36 minI=1.084e-09 max2M/R=8100.0000
A1_all_r12 delta=0.51002263: BH eps=7.36 minI=8.092e-10 max2M/R=8100.0000
A1_all_r12 delta=0.51001715: BH eps=7.37 minI=9.355e-10 max2M/R=8100.0000
A1_all_r12 delta=0.51001989: crash eps=7.36 minI=8.780e-10 max2M/R=8100.0000 (retry at CFL 0.3 after crash)
A1_all_r12 delta=0.51001532: BH eps=7.36 minI=1.033e-09 max2M/R=8100.0000
A1_all_r12 delta=0.51001623: BH eps=7.36 minI=9.955e-10 max2M/R=8100.0000 (retry at CFL 0.3 after crash)
RESULT A1_all_r12: delta_c = 0.51001486 +- 0.00000046  bracket [0.51001440, 0.51001532] (9.6 min)
```

```
A1_all_r083 delta=0.51146667: dispersed eps=6.97 minI=1.011e-08 max2M/R=8100.0000
A1_all_r083 delta=0.51213333: BH eps=5.93 minI=1.799e-08 max2M/R=8100.0000
A1_all_r083 delta=0.51168889: BH eps=6.53 minI=1.272e-08 max2M/R=8100.0000
A1_all_r083 delta=0.51191111: BH eps=6.15 minI=1.578e-08 max2M/R=8100.0000
A1_all_r083 delta=0.51154074: crash eps=7.37 minI=5.274e-10 max2M/R=8100.0000 (retry at CFL 0.3 after crash)
A1_all_r083 delta=0.51161481: BH eps=6.77 minI=1.088e-08 max2M/R=8100.0000
A1_all_r083 delta=0.51151605: BH eps=7.36 minI=1.246e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51156543: BH eps=7.07 minI=3.547e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51148313: dispersed eps=7.06 minI=8.900e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51149959: dispersed eps=7.18 minI=4.545e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51150508: dispersed eps=7.23 minI=3.165e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51151056: dispersed eps=7.30 minI=1.958e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51151239: dispersed eps=7.32 minI=1.628e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51151422: dispersed eps=7.35 minI=1.303e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51151483: BH eps=7.36 minI=1.223e-09 max2M/R=8100.0000
A1_all_r083 delta=0.51151544: BH eps=7.36 minI=1.234e-09 max2M/R=8100.0000
RESULT A1_all_r083: delta_c = 0.51151453 +- 0.00000030  bracket [0.51151422, 0.51151483] (8.4 min)
```

The endpoint audit: every decisive bracket edge of every threshold above, rerun with the labelled runners.

```
E fnu=0.2        lower edge delta=0.50393333: dispersed    eps=6.84 minI=5.281e-09 max2M/R=8100.0000
E fnu=0.2        upper edge delta=0.50399630: BH           eps=7.23 minI=3.051e-10 max2M/R=8100.0000
F r=1/4          lower edge delta=0.51645967: dispersed    eps=7.06 minI=9.434e-09 max2M/R=8100.0000
F r=1/4          upper edge delta=0.51653539: BH           eps=7.21 minI=1.827e-09 max2M/R=8100.0000
```

```
F r=4            lower edge delta=0.50540453: dispersed    eps=6.88 minI=1.130e-08 max2M/R=8100.0000
F r=4            upper edge delta=0.50548025: BH           eps=7.36 minI=5.897e-10 max2M/R=8100.0000
F2 r=1/16        lower edge delta=0.52144691: dispersed    eps=7.07 minI=9.623e-09 max2M/R=8100.0000
F2 r=1/16        upper edge delta=0.52152222: BH           eps=7.25 minI=1.633e-09 max2M/R=8100.0000
F2 r=16          lower edge delta=0.50164815: dispersed    eps=7.09 minI=9.617e-09 max2M/R=8100.0000
F2 r=16          upper edge delta=0.50169877: BH           eps=7.25 minI=-2.378e-10 max2M/R=8100.0000
H1 r=1           lower edge delta=0.51072222: dispersed    eps=7.03 minI=9.787e-09 max2M/R=8100.0000
H1 r=1           upper edge delta=0.51075926: BH           eps=7.36 minI=1.254e-09 max2M/R=8100.0000
I r=1/4 dt/4     lower edge delta=0.51648148: dispersed    eps=7.22 minI=4.008e-09 max2M/R=8100.0000
I r=1/4 dt/4     upper edge delta=0.51651852: BH           eps=7.40 minI=6.240e-10 max2M/R=8100.0000
J r=2            lower edge delta=0.50796296: dispersed    eps=7.19 minI=4.276e-09 max2M/R=8100.0000
J r=2            upper edge delta=0.50798354: BH           eps=7.36 minI=1.007e-09 max2M/R=8100.0000
J r=1/2          lower edge delta=0.51364609: dispersed    eps=7.28 minI=2.467e-09 max2M/R=8100.0000
J r=1/2          upper edge delta=0.51366667: BH           eps=7.38 minI=9.064e-10 max2M/R=8100.0000
K r=1.25         lower edge delta=0.50984047: dispersed    eps=7.35 minI=1.323e-09 max2M/R=8100.0000
K r=1.25         upper edge delta=0.50984092: BH           eps=7.35 minI=1.250e-09 max2M/R=8100.0000
K r=0.8          lower edge delta=0.51169535: dispersed    eps=7.36 minI=9.809e-10 max2M/R=8100.0000
K r=0.8          upper edge delta=0.51169611: BH           eps=7.36 minI=9.598e-10 max2M/R=8100.0000
L r=4 shear      lower edge delta=0.50523137: dispersed    eps=7.28 minI=2.033e-09 max2M/R=8100.0000
L r=4 shear      upper edge delta=0.50523868: BH           eps=7.34 minI=7.621e-10 max2M/R=8100.0000
L r=1/4 shear    lower edge delta=0.51709053: dispersed    eps=7.31 minI=2.017e-09 max2M/R=8100.0000
L r=1/4 shear    upper edge delta=0.51709785: BH           eps=7.39 minI=1.066e-09 max2M/R=8100.0000
M r=4 K48        lower edge delta=0.50548765: dispersed    eps=7.32 minI=1.547e-09 max2M/R=8100.0000
M r=4 K48        upper edge delta=0.50550412: BH           eps=7.36 minI=4.809e-10 max2M/R=8100.0000
M r=1/4 K48      lower edge delta=0.51655350: dispersed    eps=7.28 minI=2.540e-09 max2M/R=8100.0000
M r=1/4 K48      upper edge delta=0.51656173: BH           eps=7.39 minI=1.198e-09 max2M/R=8100.0000
```

B.16 The check of the 3 November review

```
# SCRIPT: NU-V4-RUN-STEP  run_nuc_n.py with the review's step history (kappa/H = 1e4 until eps_b, then 0 for
     good)
import numpy as np, time
from ms_nuc import rhs, state, timestep, rho_b
from ms_fluid import initial_data, make_grid

def run(delta_m, fnu=0.409, N=400, K=32, rmax=9.0, beta=3.0, eps0=0.1, cfl_r=0.4, cfl_mu=0.4, cH=0.005,
        eps_max=30.0, qshape=1.0, snap_eps=(), hist_every=20, stop_rule=True, disperse_rule=True, kap=0.0,
     eps_dec=None, nexp=3, kapcap=5.0e5):
    r = make_grid(N, rmax, beta)
    t0, R, U, rho = initial_data(r, delta_m, eps0, 0.0, qshape)
    e = (1-fnu)*rho
    mue = np.linspace(-1, 1, K+1)
    I = np.repeat((fnu*rho/(4*np.pi))[:, None], K, axis=1)
    Mi = np.concatenate([[0.0], np.cumsum((4*np.pi/3)*np.diff(R**3)*rho)])
    Gam = np.ones(N+1); Gam[1:] = np.sqrt(1.0 + U[1:]**2 - 2*Mi[1:]/R[1:])
    t = t0
    dR = np.zeros(N+1); dU = np.zeros(N+1); dG = np.zeros(N+1); de = np.zeros(N); dI = np.zeros((N, K))
    peakC = -1.0; hist = []; snaps = {}; step = 0; status = 'undecided'
    snap_eps = list(snap_eps)
    while True:
        Ib = fnu/(1-fnu)*e[-1]/(4*np.pi)      # incoming background tied to the outer cell's fluid density
        if eps_dec is not None:
            kap = (1/(2*t))*(1.0e4 if 2*np.sqrt(t*t0) < eps_dec else 0.0)   # step: kappa/H = 1e4 until eps_b
     (= eps_dec), then 0
        try:
            M, Ac, ac, S, V = rhs(t, R, U, Gam, e, I, mue, fnu, Ib, kap, dR, dU, dG, de, dI)
        except (ZeroDivisionError, FloatingPointError):
            status = 'crash'; break
        epsn = 2*np.sqrt(t*t0)
        twoMR = 2*M[1:]/R[1:]
        C = 2*(M[1:] - (4*np.pi/3)*rho_b(t)*R[1:]**3)/R[1:]
        iC = np.argmax(C); peakC = max(peakC, C[iC])
        if step % hist_every == 0:
            dmu = mue[1]-mue[0]
            inner = r[1:] < 3.0
            hcv = np.max(np.abs(Gam[1:]**2 - 1 - U[1:]**2 + twoMR)[inner])     # Hamiltonian-constraint check
```

```
            hist.append((t, epsn, C[iC], r[1+iC], e[0]/rho_b(t), 2*np.pi*I[0].sum()*dmu/rho_b(t),
                         e[-1]/rho_b(t), 2*np.pi*I[-1].sum()*dmu/rho_b(t), hcv))
        if snap_eps and epsn >= snap_eps[0]:
            snaps[snap_eps.pop(0)] = dict(t=t, eps=epsn, R=R.copy(), U=U.copy(), e=e.copy(), I=I.copy(), M=M.
    copy())
        if stop_rule:
            if np.any((twoMR >= 1.0) & (U[1:] < 0)):
                status = 'BH'; break
            if disperse_rule and epsn > 3.0 and C[iC] < 0.5*peakC:
                status = 'dispersed'; break
        if epsn > eps_max: break
        try:
            dt = min(timestep(R, U, e, I, mue, Gam, Ac, ac, S, cfl_r, cfl_mu), cH*2*t, (0.5/(kap*np.max(Ac))
    if kap > 0 else 1e30))
            # SSP RK3 (the outer-boundary intensity is evaluated at each stage time)
            R1 = R + dt*dR; U1 = U + dt*dU; G1 = Gam + dt*dG; e1 = e + dt*de; I1 = I + dt*dI
            rhs(t+dt, R1, U1, G1, e1, I1, mue, fnu, fnu/(1-fnu)*e1[-1]/(4*np.pi), kap, dR, dU, dG, de, dI)
            R2 = 0.75*R + 0.25*(R1+dt*dR); U2 = 0.75*U + 0.25*(U1+dt*dU); G2 = 0.75*Gam + 0.25*(G1+dt*dG)
            e2 = 0.75*e + 0.25*(e1+dt*de); I2 = 0.75*I + 0.25*(I1+dt*dI)
            rhs(t+0.5*dt, R2, U2, G2, e2, I2, mue, fnu, fnu/(1-fnu)*e2[-1]/(4*np.pi), kap, dR, dU, dG, de, dI)
            R = R/3 + 2/3*(R2+dt*dR); U = U/3 + 2/3*(U2+dt*dU); Gam = Gam/3 + 2/3*(G2+dt*dG)
            e = e/3 + 2/3*(e2+dt*de); I = I/3 + 2/3*(I2+dt*dI)
        except (ZeroDivisionError, FloatingPointError):
            coll = U[1:] < 0
            status = 'BH' if (coll.any() and np.max(twoMR[coll]) > 0.95) else 'crash'   # breakdown just
    before collapse
            break
        t += dt; step += 1
        if np.any(e <= 0) or not np.all(np.isfinite(U)) or not np.all(np.isfinite(I)):
            status = 'crash'; break
    return dict(status=status, t=t, t0=t0, eps=2*np.sqrt(t*t0), steps=step, peakC=peakC, hist=np.array(hist),
                snaps=snaps, r=r, mue=mue, R=R, U=U, e=e, I=I, Gam=Gam)

if __name__ == '__main__':
    t1 = time.time()
    out = run(0.0, fnu=0.409, N=200, K=16, eps_max=5.0, stop_rule=False)
    t = out['t']; Ib = 0.409*rho_b(t)/(4*np.pi); eb = 0.591*rho_b(t)
    print(f"FRW test to eps={out['eps']:.2f} in {out['steps']} steps ({time.time()-t1:.1f}s)")
    print("  max|I/Ib-1| =", np.abs(out['I']/Ib-1).max(), " max|e/eb-1| =", np.abs(out['e']/eb-1).max())
    a = np.sqrt(t/out['t0']); print("  max|R/(a r)-1| =", np.abs(out['R'][1:]/(a*out['r'][1:])-1).max())
```

```
# SCRIPT: NU-V4-QUEUE-G  independent check of the Open Math OS review's measurements: two runs either side of
     each quoted threshold
: "${SIMDIR:?set SIMDIR to PBH_neutrino_simulation/simulation (McCormick, doi:10.5281/zenodo.23808376)}"
cd "$(dirname "$0")"
python3 - > log_G.txt 2>&1 <<'PY'
import sys,os;sys.path.insert(0,os.environ['SIMDIR']);sys.path.insert(0,'.')
from multiprocessing import Pool
def step(d):
    from run_step import run; o=run(d,fnu=0.409,N=400,K=32,eps0=0.1,eps_dec=1.0,eps_max=15.0,nexp=3); return
     ('step eps_b=1',d,o['status'],float(o['eps']))
def k5(d):
    from run_nuc_n import run; o=run(d,fnu=0.409,N=400,K=32,eps0=0.1,eps_dec=1.20,eps_max=15.0,nexp=3); return
      ('K5 n=3 eps_dec=1.20',d,o['status'],float(o['eps']))
jobs=[(step,0.5132),(step,0.5138),(k5,0.5165),(k5,0.5171)]
def go(a): return a[0](a[1])
with Pool(2) as p:
    for t,d,s,e in p.map(go,jobs): print(f'{t} delta={d}: {s} eps={e:.2f}',flush=True)
print('review: step eps_b=1 delta_c=0.513512 (expect 0.5132 dispersed, 0.5138 BH); K5 n=3 eps_dec=1.20 delta_c
     =0.516797 (expect 0.5165 dispersed, 0.5171 BH)')
PY
```

```
step eps_b=1 delta=0.5132: dispersed eps=6.37
step eps_b=1 delta=0.5138: BH eps=6.30
K5 n=3 eps_dec=1.20 delta=0.5165: dispersed eps=6.42
K5 n=3 eps_dec=1.20 delta=0.5171: BH eps=6.29
```

```
review: step eps_b=1 delta_c=0.513512 (expect 0.5132 dispersed, 0.5138 BH); K5 n=3 eps_dec=1.20 delta_c
    =0.516797 (expect 0.5165 dispersed, 0.5171 BH)
```

B.17 The coordinate of the law: the intrinsic-ruler check

```
# SCRIPT: PREREGISTERED_RULER  (written 2026-11-05 before any number below was computed)
The intrinsic-ruler check: are the measured threshold offsets a law in the kept sector's own geometry
(IQGT, Paper I Sec. 4: "the parameter manifold does not generate the geometry") rather than in the parameter
    chart x = (1/2) ln r?

DATA (fixed, no new collapse runs): the 15 thresholds of v4.1 + A1, read as tau_G = ln((1-G)/G), G = (dc -
    0.49773)/(0.52832 - 0.49773),
offsets taken relative to the measured anchor r = 1 (dc = 0.51074074).
  all-higher family (every l >= 2 at kappa2 = r kappa1): r = 1/16, 1/4, 1/2, 1/1.2, 0.8, 1, 1.2, 1.25, 2, 4,
    16  (clean K points).
  shear-only family (l = 2 alone at kappa2):             r = 1/4, 1/1.2, 1.2, 4.
Bracket corners carried through every statistic.

THE OBJECT.  The moment chain of the kept sector, L(r) = -i k M - kappa D_r, M the Legendre streaming matrix
(M[l,l-1] = l/sqrt(4l^2-1)), D_r = diag(0,1,r,r,...) (all-higher) or diag(0,1,r,1,1,...) (shear-only), LM+1 =
    61 moments
(the chain of memory_curve.py).  L is non-normal.  Projector: the Riesz projector Pi_r onto the least-damped
    eigenvalue
(largest real part), the mode the density survives in; built from right/left eigenvectors, Pi = |R><L|/<L|R>,
    Pi^2 = Pi.
Fork (both reported, decided now): rank-1 (least damped) is PRIMARY; rank-2 (two least damped) is the
    comparison.
Scale: kappa/k at readout.  The memory form reads at eps* = 1.7 where kappa = eps_dec^3/tau^4 = 8/1.7^4 =
    0.958 (n = 3,
eps_dec = 2); k runs over the profile nodes with weights k^2 exp(-k^2/2) (memory_curve.py, 36 Gauss nodes on
    [0.25, 9]).
PRIMARY: kappa = 0.958, profile-weighted.  Sensitivity (reported, not used for verdicts): kappa = 0.3, 3.0;
    single k = 2; LM = 20.

THE RULER.  s(r) = signed intrinsic distance of Pi_r from Pi_1 along the family, three readings computed side
    by side:
  (a) arc length of the Hermitian pullback metric, g_rr = (1/2)||d Pi/dr||_F^2, s = int sqrt(g) dr   [IQGT Sec
    . 4, Hermitian form]
  (b) chord: d_HS(r) = ||Pi_r - Pi_1||_F / sqrt 2, signed by the side of r = 1                          [
    control on (a)]
  (c) his biorthogonal trace form q(V) = (1/2) Re Tr V^2 with V = d Pi/dr                               [cone
     / B57: may be <= 0]
  (d) principal angle theta(r) between the right eigenvectors R_r and R_1 (and the left pair)            [Gr
    (1,n) geodesic]
Profile average: s is averaged over k with the same weights as the memory form (that is how the threshold saw
    the chain).
Each reading is a separate ruler; the verdicts below are stated for (a) and repeated for (b),(d).  (c) is
    reported with
its sign; if q < 0 anywhere on the family the Hermitian arc length is not a length there and that is stated.

WHAT IS ALREADY KNOWN IN THE CHART (the thing to beat):
  all-higher odd rates in x: 1.1128 (|x| = 0.112), 1.1006 (0.091, A1), 1.1093 (0.347), 1.1177 (0.693), 1.1384
    (1.386):
    spread 0.038 (3.4 % of the mean), a monotone rise beyond |x| = 0.35 ("the bend").
  shear-only odd rate in x: 1.2038 (0.693), 1.1895 (0.091).  Family ratio shear/all-higher: 1.077 (0.693),
    1.081 (0.091).
  Linear fit of all 11 all-higher offsets against x: residual RMS to be computed and reported as the chart
    baseline.

READINGS AND KILL LINES (fixed now):
  R1 "the chart was the whole discrepancy": in ruler (a) the all-higher and shear-only families fall on ONE
    line.
     Test: slope_shear / slope_all in s lies in [0.97, 1.03] (it is 1.08 in x).  Dies otherwise.
  R2 "the chart explains the bend": in ruler (a) the all-higher odd rates (same pairs as above, now per unit
    of s)
```

```
      have spread below 1.2 % of their mean (it is 3.4 % in x), i.e. less than 0.35 of the chart spread.  Dies
      otherwise.
  R3 "the ruler is not it": R1 and R2 both die.  Then the discrepancy is not the chart (failed this way; other
      projectors / scales remain open and are listed, not run).
  Controls that must come out as stated or the check is void:
  C1 positive control (the pipeline can see a chart effect): synthetic offsets tau_syn = 1.0 * s(r) (one law
     in s, both
      families), read back in x, must show a family split and a bend of the measured kind (ratio != 1, spread
     > 1.2 %);
      read back in s must give ratio 1 and spread 0 to machine precision.
  C2 negative control (the pipeline can fail): the wrong ruler, s built from the OTHER family's projector
      (shear-only data read on the all-higher projector and vice versa), must NOT pass R1 better than the
     right ruler.
  C3 chart control: the ruler Gamma12 = (r-1)/(r+1) (also a function of parameters only) must not pass R1/R2
     either;
      if it does, the rulers are not discriminating and the verdict is "undecided".
  C4 bracket corners: every statistic is recomputed at both corners of every bracket; a verdict counts only if
      it
      holds at every corner.
  C5 eigen-solve control: Pi^2 = Pi to 1e-10, Tr Pi = 1, left/right biorthonormality to 1e-10 at every r, k.
Opposite built in: if the slope ratio in s moves AWAY from 1 (beyond 1.08) the chart was hiding a larger
     family split.
Nothing above is edited after the run; results are appended below.

AMENDMENT (before any run; a definition, not a data decision).  L = -ikM - kappa D has S L S = conj(L) with S
     = diag((-1)^l),
so eigenvalues come in conjugate pairs with EQUAL real part whenever they are complex (the two-moment pair -
     kappa/2 +- sqrt(kappa^2/4 - k^2/3)
is the model case).  "The least-damped eigenvalue" is therefore a conjugate PAIR wherever it is complex and
     rank-1 is not well
defined there.  PRIMARY projector := the Riesz projector onto that pair (rank 2 when complex, rank 1 when real
     ; it is real and
tracks continuously).  The rank-1 projector onto the +Im member is the comparison.  Minimum gap (real part) to
      the next mode
is logged at every r, k; a crossing voids the tracking at that k and is reported.

RESULT 1 (primary run, kappa = 0.958, LM = 60, profile-weighted; out_primary.txt).
  Chart baseline: all-higher slope 1.1328, rms 0.0142, odd-rate spread 3.39 % (corners 2.90-3.87); shear-only
     slope 1.2036; ratio 1.0624 (1.058-1.067).
  C3 (Gamma12 chart): spread 53 %, ratio 0.877: not discriminating in favour of Gamma12; fine.
  Rulers (a),(b),(d) on the least-damped Riesz projector: R1 DIES (ratios 1.27, 0.67, 0.73), R2 DIES (spreads
     195 %, 255 %, 201 %).
  C1 passes (the pipeline can see a chart effect), C2 passes (wrong ruler no better), eigen checks 1e-13.
  INSTRUMENT FINDING (diag_modes.py), which voids this ruler rather than the reading: at kappa = 0.958 the
     least-damped
  eigenvalue is isolated only for k < 1.4 (gap 0.9 -> 0.06); for k >= 1.4, which carries ~85 % of the profile
     weight, it is a
  REAL eigenvalue at about -kappa with real-part gap 1e-4 or less, Petermann factor 374 at the transition, and
      3 % weight on
  l = 0.  That is the paper's own Sec. 2 statement (the hydrodynamic pole meets the branch cut at kappa ~
     0.646 k): there is NO
  isolated kept eigenprojector at readout for most of the profile, so "the kept-sector Riesz projector" is not
      a well-posed
  object there and its distance is not a ruler.  R3 is therefore NOT concluded from this run: the ruler was
     ill-posed
  ("failed this way" at the instrument, not at the reading).  The biorthogonal form q was indefinite along the
      path
  (-1e2 .. +4e3), as B57/the cone paper allow; the Hermitian arc length is not a length there, as stated above
     .

RULER (e), written now, before running it.  The object that IS well-posed through decoupling is the state
     itself: the
kept-sector trajectory v_r(tau) of memory_curve.py (61 moments, start in |0>, evolve under -(ikM + kappa(tau)
     D_r), the
same history as the collapse runs: kappa = 8/tau^4, n = 3, eps_dec = 2, tau0 = 0.02).  Ruler (e): the
     principal angle
theta_r = arccos |<v_1(tau*)|v_r(tau*)>| / (norms) between the state at ratio r and at r = 1, read at tau* =
     1.7 (PRIMARY,
```

```
the memory form's first-maximum readout) and tau* = 4.0 (sensitivity), profile-averaged with the memory-form
     weights,
signed by the side of r = 1.  This is Paper I Sec. 10's own instrument (raw data -> intrinsic angles): gauge-
     free, scale-
free, needs no isolated eigenvalue.  Control alongside: the memory form's own number A_r(tau*) = |v_0|^2 (the
     kept
population), which is what the frozen memory form reads and which FAILED Test F (predicted +-0.004).
Same readings R1, R2, R3 and controls C1-C4 as above, verbatim, applied to ruler (e).  Expected under his
     framework: R1
and/or R2 survive.  Expected if the threshold reads something the single-mode state does not carry (nonlinear
     collapse
response): both die and the reading is R3 for this ruler too.

RESULT 2 (ruler (e), out_state.txt, ruler_state.json).
  tau* = 1.7 (primary): all-higher slope 21.20 rad^-1, rms 0.130 (chart rms 0.014); odd rates 16.7 -> 23.4,
     spread 35.9 % (corners 35.3-36.4);
    shear-only slope 20.07; RATIO 0.9465 (corners 0.942-0.950).  R1 DIES (outside [0.97, 1.03]); R2 DIES (36 %
      vs 3.4 % in the chart).
  tau* = 4.0: ratio 1.668, spread 22 %: worse.  Memory-form A0 log-odds as ruler: ratio 0.885 / 1.84, spread
     40 % / 97 %: worse.
  C1 passes (theta read in x shows ratio 1.15, spread 28 %); C2: the swapped ruler is nearly as good (0.921)
     because the two
    families' state angles are almost equal (theta_all(4) = 0.0475, theta_shear(4) = 0.0467: 2 % apart, while
     the measured
    offsets are 8 % apart) -- the linear kept-sector state at readout barely distinguishes the two collision
     models.
  Corner sampling (2000 draws over every bracket incl. the anchor): every verdict holds at every corner.
VERDICT: R3 for both rulers tried -- the intrinsic distance of the kept sector, whether as the Riesz projector
      of the
  least-damped mode (ill-posed at readout; kappa ~ k puts the pole in the branch cut) or as the trajectory's
     principal
  angle (well-posed), does NOT explain the 1.06-1.08 family split or the 3.4 % bend.  The chart x = (1/2) ln r
      = (1/2) ln(kappa2/kappa1)
  remains by far the most linear description of the data (rms 0.014 against 0.13).  Failed this way: other
     projectors
  (the full angular moment set of the collapse run, the nonlinear state at the horizon) and other readout
     epochs remain open and are not run.
  What this establishes, in the author's coordinate: the threshold's offset tracks the rapidity of the damping
      OPERATOR's
  flux-shear link (B189: ln(kappa2/kappa1) is a property of D, not of a state), and the linear kept-sector
     state at readout
  does not carry the difference between damping the shear alone and damping every l >= 2 (2 % in angle vs 8 %
     in threshold):
  that difference is written after readout, in the nonlinear collapse, which agrees with the external review's
      "nonlinear in
  the ledger" reading and with K5.  No rate constant is claimed.

CORRECTION (2026-11-05, after the verdict; numbers only, verdict unchanged).  The share "~85 % of the profile
     weight" at
  k >= 1.4 in RESULT 1 was misread from the diag_modes.py table.  Summing its weight column with the memory-
     form weights
  (36 Gauss nodes, k^2 exp(-k^2/2)): k < 1.4 carries 0.362, k >= 1.4 carries 0.638.  The isolated-mode region
     is therefore
  about a third of the profile, not 15 %.  The instrument finding stands (no isolated kept eigenvalue on most
     of the profile,
  Petermann 374 at the transition) and the VERDICT rests on ruler (e), which is well-posed on the whole
     profile.
```

```
# SCRIPT: NU-V4-RULER-CHECK  intrinsic-ruler check of the measured threshold offsets (see PREREGISTERED_RULER.
     txt)
import numpy as np, json, os, sys, math
from scipy.special import roots_legendre
here=os.path.dirname(os.path.abspath(__file__)); up=os.path.join(here,'..')
rng=np.random.default_rng(1)
FL,INS=0.49773,0.52832
def tau_of(dc): G=(dc-FL)/(INS-FL); return math.log((1-G)/G)
def load(tag):
```

```
    d=json.load(open(os.path.join(up,'res_%s.json'%tag))); lo=d.get('lo',d['dc']-d['w']/2); hi=d.get('hi',d['
     dc']+d['w']/2); return d['dc'],lo,hi
ALL={1/16:'F2_r1o16',0.25:'F_r025',0.5:'J_r05',1/1.2:'A1_all_r083',0.8:'K_r08_clean',1.0:'H1_r1',1.2:'
     A1_all_r12',1.25:'K_r125_clean',2.0:'J_r2',4.0:'F_r4',16.0:'F2_r16'}
SHR={0.25:'L_r025',1/1.2:'A1_shear_r083',1.2:'A1_shear_r12',4.0:'L_r4'}
anc=load('H1_r1')
# ---------- chain ----------
LM=int(os.environ.get('LM','60')); n=LM+1
M=np.zeros((n,n))
for l in range(1,n): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
S=np.diag([(-1.0)**l for l in range(n)])
kn,kw=roots_legendre(36); ks=0.5*(9.0-0.25)*(kn+1)+0.25; wk=0.5*(9.0-0.25)*kw; wE=wk*ks**2*np.exp(-ks**2/2);
     wE/=wE.sum()
if os.environ.get('SINGLEK'): ks=np.array([float(os.environ['SINGLEK'])]); wE=np.array([1.0])
KAP=float(os.environ.get('KAP','0.958'))
def Dr(r,fam):
    d=np.full(n,r if fam=='all' else 1.0); d[0]=0.0; d[1]=1.0
    if fam=='shear': d[2]=r
    return d
def proj(k,r,fam,rank):
    L=-1j*k*M-KAP*np.diag(Dr(r,fam))
    ev,R=np.linalg.eig(L); Lf=np.linalg.inv(R)       # rows of Lf are left eigenvectors (biorthonormal to R)
    o=np.argsort(-ev.real); ev,R,Lf=ev[o],R[:,o],Lf[o,:]
    if rank=='pair':
        idx=[0] if abs(ev[0].imag)<1e-9 else ([0,1] if abs(ev[0].real-ev[1].real)<1e-7 else [0])
    elif rank=='one': idx=[0] if abs(ev[0].imag)<1e-9 else [0 if ev[0].imag>0 else 1]
    P=sum(np.outer(R[:,i],Lf[i,:]) for i in idx)
    m=len(idx); gap=ev[m].real-ev[0].real if m<n else 0.0
    return P,gap,ev[0]
def rulers(fam,rank,us):
    """s_a (arc length, Hermitian), s_b (chord), q (biorth form along the path), theta (principal angle), each
      profile-averaged; indexed by u=ln r."""
    sa=np.zeros(len(us)); sb=np.zeros(len(us)); qmin=np.zeros(len(us)); th=np.zeros(len(us)); worst_gap=1e9;
     chk=0.0
    i1=int(np.argmin(abs(us)))
    for k,w in zip(ks,wE):
        Ps=[];
        for u in us:
            P,gap,ev=proj(k,math.exp(u),fam,rank); Ps.append(P); worst_gap=min(worst_gap,gap)
            chk=max(chk,np.abs(P@P-P).max(),abs(np.trace(P).real-round(np.trace(P).real)))
        Ps=np.array(Ps); du=us[1]-us[0]
        dP=np.gradient(Ps,du,axis=0)
        g=0.5*np.sum(np.abs(dP)**2,axis=(1,2))            # (1/2)||dP/du||_F^2
        q=0.5*np.real(np.einsum('uij,uji->u',dP,dP))       # (1/2) Re Tr V^2
        arc=np.concatenate([[0],np.cumsum(np.sqrt(g[1:])*du)]); arc-=arc[i1]
        sa+=w*arc
        chord=np.array([np.linalg.norm(P-Ps[i1])/math.sqrt(2) for P in Ps]); sb+=w*np.sign(us)*chord
        qmin+=w*q
        # principal angle via the Hermitian part: use range of P (right eigvecs) vs range at r=1
        for j,P in enumerate(Ps):
            A=np.linalg.svd(P,full_matrices=False)[0][:,:int(round(np.trace(P).real))]
            B=np.linalg.svd(Ps[i1],full_matrices=False)[0][:,:int(round(np.trace(Ps[i1]).real))]
            sv=np.linalg.svd(A.conj().T@B,compute_uv=False); th[j]+=w*np.sign(us[j])*math.sqrt(np.sum(np.
     arccos(np.clip(sv,0,1))**2))
    return dict(a=sa,b=sb,q=qmin,theta=th,gap=worst_gap,chk=chk)
us=np.linspace(math.log(1/16),math.log(16),641)
def interp(arr,r): return float(np.interp(math.log(r),us,arr))
# ---------- data ----------
def family_points(tab):
    rs=sorted(tab); return rs,[load(tab[r]) for r in rs]
def tau_off(pt,a): return tau_of(pt)-tau_of(a)
def stats(rs,pts,xfun,corner=None):
    """slope of tau against xfun(r) (least squares with intercept) and odd rates per unit of xfun; corner:
     None=mid, else dict r->dc"""
    xs=np.array([xfun(r) for r in rs]); ts=np.array([tau_off(corner[r] if corner else p[0], corner.get(1.0,anc
     [0]) if corner else anc[0]) for r,p in zip(rs,pts)])
    A=np.vstack([np.ones(len(xs)),xs]).T; (a,b),res,_,_=np.linalg.lstsq(A,ts,rcond=None); rms=float(np.sqrt(np
     .mean((ts-a-b*xs)**2)))
```

```
    odd={}
    for r in rs:
        if r>1 and (1/r) in rs or any(abs(1/r-q)<1e-9 for q in rs):
            q=[q for q in rs if abs(1/r-q)<1e-9]
            if r>1 and q:
                i,j=rs.index(r),rs.index(q[0]); odd[r]=(ts[i]-ts[j])/(xs[i]-xs[j])
    return b,rms,odd
def spread(odd): v=np.array(list(odd.values())); return float((v.max()-v.min())/abs(v.mean()))
def report(name,xall,xshr):
    ra,pa=family_points(ALL); rsh,psh=family_points(SHR)
    ba,rmsa,odda=stats(ra,pa,xall); bs,rmss,odds=stats(rsh,psh,xshr)
    # corner sampling
    ratios=[];spreads=[]
    for _ in range(2000):
        ca={r:rng.uniform(p[1],p[2]) for r,p in zip(ra,pa)}; cs={r:rng.uniform(p[1],p[2]) for r,p in zip(rsh,
     psh)}; ca[1.0]=rng.uniform(anc[1],anc[2]); cs[1.0]=ca[1.0]
        b1,_,o1=stats(ra,pa,xall,ca); b2,_,o2=stats(rsh,psh,xshr,cs); ratios.append(b2/b1); spreads.append(
     spread(o1))
    print(f'[{name}] all-higher: slope {ba:.4f} rms {rmsa:.4f}  odd rates '+' '.join(f'r={r:g}:{v:.4f}' for r,
     v in sorted(odda.items()))+f'  spread {100*spread(odda):.2f}% (corners {100*min(spreads):.2f}-{100*max(
     spreads):.2f}%)')
    print(f'[{name}] shear-only: slope {bs:.4f} rms {rmss:.4f}  odd rates '+' '.join(f'r={r:g}:{v:.4f}' for r,
     v in sorted(odds.items())))
    print(f'[{name}] slope ratio shear/all = {bs/ba:.4f}  (corners {min(ratios):.4f}-{max(ratios):.4f})')
    return bs/ba,spread(odda),(min(ratios),max(ratios)),(min(spreads),max(spreads))
print('kappa',KAP,'LM',LM,'profile',len(ks),'k nodes')
X=lambda r: 0.5*math.log(r)
base=report('CHART x=1/2 ln r',X,X)
GAM=lambda r:(r-1)/(r+1); report('C3 chart control Gamma12',GAM,GAM)
RA=rulers('all','pair',us); RS=rulers('shear','pair',us)
print(f'eigen controls: worst |P^2-P|,trace dev {RA["chk"]:.2e}/{RS["chk"]:.2e}; min real-part gap to next
     mode {RA["gap"]:.4f}/{RS["gap"]:.4f}')
print('biorthogonal form q along the path (profile avg): all-higher min %.3e max %.3e; shear-only min %.3e max
      %.3e'%(RA['q'].min(),RA['q'].max(),RS['q'].min(),RS['q'].max()))
for key,lab in [('a','(a) arc length'),('b','(b) chord'),('theta','(d) principal angle')]:
    print('--- ruler',lab,'  s(r) all-higher:',' '.join(f'{r:g}:{interp(RA[key],r):+.4f}' for r in
     [1/16,.25,.5,1/1.2,.8,1.2,1.25,2,4,16]),' shear-only:',' '.join(f'{r:g}:{interp(RS[key],r):+.4f}' for r
     in [.25,1/1.2,1.2,4]))
    res=report('RULER '+lab,lambda r,k=key:interp(RA[k],r),lambda r,k=key:interp(RS[k],r))
    print('   C2 wrong ruler:'); report('   swapped',lambda r,k=key:interp(RS[k],r),lambda r,k=key:interp(RA[k
     ],r))
    # C1 positive control
    rs=sorted(ALL); ts=np.array([interp(RA[key],r) for r in rs]); xs=np.array([X(r) for r in rs]); A=np.vstack
     ([np.ones(len(xs)),xs]).T; b1=np.linalg.lstsq(A,ts,rcond=None)[0][1]
    rsh=sorted(SHR); ts2=np.array([interp(RS[key],r) for r in rsh]); xs2=np.array([X(r) for r in rsh]); A2=np.
     vstack([np.ones(len(xs2)),xs2]).T; b2=np.linalg.lstsq(A2,ts2,rcond=None)[0][1]
    od={r:(interp(RA[key],r)-interp(RA[key],1/r))/(X(r)-X(1/r)) for r in [1.2,1.25,2,4,16]}
    print(f'   C1 positive control (tau = s read in x): family ratio {b2/b1:.4f}, odd-rate spread {100*spread(
     od):.2f}%  -> must differ from 1 and exceed 1.2%')
    key2=key
# rank-1 comparison on ruler (a)
RA1=rulers('all','one',us); RS1=rulers('shear','one',us)
report('RULER (a) rank-1 comparison',lambda r:interp(RA1['a'],r),lambda r:interp(RS1['a'],r))
```

```
# SCRIPT: NU-V4-RULER-DIAG  per-k diagnostics: least-damped pair, real-part gap (correct sign), Petermann size
     , principal angle at r = 1.2 and 1/1.2
import numpy as np, math
from scipy.special import roots_legendre
n=61; M=np.zeros((n,n))
for l in range(1,n): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
kn,kw=roots_legendre(36); ks=0.5*(9.0-0.25)*(kn+1)+0.25; wk=0.5*(9.0-0.25)*kw; wE=wk*ks**2*np.exp(-ks**2/2);
     wE/=wE.sum()
KAP=0.958
def modes(k,r):
    d=np.full(n,r); d[0]=0; d[1]=1
    L=-1j*k*M-KAP*np.diag(d); ev,R=np.linalg.eig(L); Lf=np.linalg.inv(R); o=np.argsort(-ev.real); ev,R,Lf=ev[o
     ],R[:,o],Lf[o,:]
    pair=[0,1] if abs(ev[0].imag)>1e-9 and abs(ev[0].real-ev[1].real)<1e-7 else [0]
```

```
    m=len(pair); P=sum(np.outer(R[:,i],Lf[i,:]) for i in pair)
    gap=ev[0].real-ev[m].real; pet=np.linalg.norm(P)**2/m
    U=np.linalg.svd(P)[0][:,:m]
    return ev[:3],gap,pet,U,abs(R[0,0])**2/np.sum(abs(R[:,0])**2)
print(' k     w      ev0 (r=1)              gap    Petermann  w0(l=0)  angle(1.2) angle(1/1.2)')
for k,w in zip(ks,wE):
    e1,g1,p1,U1,w0=modes(k,1.0); e2,g2,p2,U2,_=modes(k,1.2); e3,g3,p3,U3,_=modes(k,1/1.2)
    ang=lambda A,B: math.sqrt(np.sum(np.arccos(np.clip(np.linalg.svd(A.conj().T@B,compute_uv=False),0,1))**2))
    print(f'{k:5.2f} {w:.4f}  {e1[0]:+.4f}  {g1:7.4f} {p1:9.2f} {w0:7.3f}  {ang(U1,U2):8.4f} {ang(U1,U3):8.4f}
     ')
```

```
# SCRIPT: NU-V4-RULER-STATE  ruler (e): principal angle between kept-sector trajectories v_r(tau*) and v_1(tau
     *) (see PREREGISTERED_RULER.txt)
import numpy as np, json, os, math, sys
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
here=os.path.dirname(os.path.abspath(__file__)); up=os.path.join(here,'..')
FL,INS=0.49773,0.52832
def tau_of(dc): G=(dc-FL)/(INS-FL); return math.log((1-G)/G)
def load(tag):
    d=json.load(open(os.path.join(up,'res_%s.json'%tag))); return d['dc'],d.get('lo',d['dc']-d['w']/2),d.get('
     hi',d['dc']+d['w']/2)
ALL={1/16:'F2_r1o16',0.25:'F_r025',0.5:'J_r05',1/1.2:'A1_all_r083',0.8:'K_r08_clean',1.0:'H1_r1',1.2:'
     A1_all_r12',1.25:'K_r125_clean',2.0:'J_r2',4.0:'F_r4',16.0:'F2_r16'}
SHR={0.25:'L_r025',1/1.2:'A1_shear_r083',1.2:'A1_shear_r12',4.0:'L_r4'}
anc=load('H1_r1'); rng=np.random.default_rng(1)
kn,kw=roots_legendre(36); ks=0.5*(9.0-0.25)*(kn+1)+0.25; wk=0.5*(9.0-0.25)*kw; wE=wk*ks**2*np.exp(-ks**2/2);
     wE/=wE.sum()
LM=60; n=LM+1; M=np.zeros((n,n))
for l in range(1,n): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
tau0=0.02; TS=[1.7,4.0]; ed,p=2.0,3
def Dr(r,fam):
    d=np.full(n,r if fam=='all' else 1.0); d[0]=0.0; d[1]=1.0
    if fam=='shear': d[2]=r
    return d
def state(k,r,fam):
    d=Dr(r,fam)
    def rhs(t,v):
        K=ed**p/t**(p+1); x,y=v[:n],v[n:]
        return np.concatenate([-K*d*x+k*(M@y),-K*d*y-k*(M@x)])
    v0=np.zeros(2*n); v0[0]=1
    s=solve_ivp(rhs,(tau0,TS[-1]),v0,method='BDF',t_eval=TS,rtol=1e-9,atol=1e-12)
    return {t:(s.y[:n,i]+1j*s.y[n:,i]) for i,t in enumerate(TS)}
cache={}
def V(k,r,fam):
    key=(k,round(r,9),fam)
    if key not in cache: cache[key]=state(k,r,fam)
    return cache[key]
rs_all=sorted(ALL); rs_sh=sorted(SHR)
theta={t:{'all':{},'shear':{}} for t in TS}; A0={t:{'all':{},'shear':{}} for t in TS}
for fam,rs in [('all',rs_all),('shear',rs_sh)]:
    for r in rs:
        for t in TS:
            th=0.0; a=0.0
            for k,w in zip(ks,wE):
                v1=V(k,1.0,'all')[t]; vr=V(k,r,fam)[t]
                c=abs(np.vdot(v1,vr))/(np.linalg.norm(v1)*np.linalg.norm(vr)); th+=w*math.acos(min(1.0,c)); a
     +=w*abs(vr[0])**2
            theta[t][fam][r]=math.copysign(th,math.log(r)) if r!=1 else 0.0; A0[t][fam][r]=a
        print(f'{fam:5s} r={r:8.4f}  theta(1.7)={theta[1.7][fam][r]:+.5f} theta(4.0)={theta[4.0][fam][r]:+.5f}
       A0(1.7)={A0[1.7][fam][r]:.5f} A0(4.0)={A0[4.0][fam][r]:.5f}',flush=True)
json.dump({'theta':{str(t):{f:{str(r):v for r,v in d.items()} for f,d in theta[t].items()} for t in TS},'A0':{
     str(t):{f:{str(r):v for r,v in d.items()} for f,d in A0[t].items()} for t in TS}},open(os.path.join(here
     ,'ruler_state.json'),'w'),indent=1)
def stats(rs,tab,xf,corner=None):
    xs=np.array([xf(r) for r in rs]); ts=np.array([tau_of(corner[r] if corner else load(tab[r])[0])-tau_of(
     corner[1.0] if corner else anc[0]) for r in rs])
```

```
    A=np.vstack([np.ones(len(xs)),xs]).T; (a,b)=np.linalg.lstsq(A,ts,rcond=None)[0]; rms=float(np.sqrt(np.mean
     ((ts-a-b*xs)**2)))
    odd={r:(ts[rs.index(r)]-ts[[rs.index(q) for q in rs if abs(q-1/r)<1e-9][0]])/(xs[rs.index(r)]-xs[[rs.index
     (q) for q in rs if abs(q-1/r)<1e-9][0]]) for r in rs if r>1 and any(abs(q-1/r)<1e-9 for q in rs)}
    return b,rms,odd
spread=lambda o:(lambda v:(v.max()-v.min())/abs(v.mean()))(np.array(list(o.values())))
def report(name,xa,xs_):
    ba,ra,oa=stats(rs_all,ALL,xa); bs,rs_,os_=stats(rs_sh,SHR,xs_); ratios=[];sp=[]
    for _ in range(2000):
        ca={r:rng.uniform(*load(ALL[r])[1:]) for r in rs_all}; cs={r:rng.uniform(*load(SHR[r])[1:]) for r in
     rs_sh}; ca[1.0]=rng.uniform(anc[1],anc[2]); cs[1.0]=ca[1.0]
        b1,_,o1=stats(rs_all,ALL,xa,ca); b2,_,_=stats(rs_sh,SHR,xs_,cs); ratios.append(b2/b1); sp.append(
     spread(o1))
    print(f'[{name}] all-higher slope {ba:.4f} rms {ra:.4f} odd '+' '.join(f'{r:g}:{v:.4f}' for r,v in sorted(
     oa.items()))+f' spread {100*spread(oa):.2f}% ({100*min(sp):.2f}-{100*max(sp):.2f})')
    print(f'[{name}] shear-only slope {bs:.4f} rms {rs_:.4f} odd '+' '.join(f'{r:g}:{v:.4f}' for r,v in sorted
     (os_.items()))+f'   RATIO {bs/ba:.4f} ({min(ratios):.4f}-{max(ratios):.4f})')
X=lambda r:0.5*math.log(r)
report('CHART',X,X)
for t in TS:
    ta=theta[t]['all']; tsh=theta[t]['shear']
    report(f'RULER (e) theta at tau*={t}',lambda r:ta[r],lambda r:tsh[r])
    report(f'   C2 swapped tau*={t}',lambda r:tsh[r] if r in tsh else ta[r],lambda r:ta[r])
    # C1: synthetic tau = theta read in x
    xa=np.array([X(r) for r in rs_all]); ya=np.array([ta[r] for r in rs_all]); b1=np.linalg.lstsq(np.vstack([
     np.ones(len(xa)),xa]).T,ya,rcond=None)[0][1]
    xs2=np.array([X(r) for r in rs_sh]); ys=np.array([tsh[r] for r in rs_sh]); b2=np.linalg.lstsq(np.vstack([
     np.ones(len(xs2)),xs2]).T,ys,rcond=None)[0][1]
    od={r:(ta[r]-ta[1/r])/(2*X(r)) for r in [1.2,1.25,2,4,16]}
    print(f'   C1 tau*={t}: theta read in x -> family ratio {b2/b1:.4f}, odd spread {100*spread(od):.2f}%')
    # memory-form control: offset of A0 (log-odds of kept population) as a ruler
    aa=A0[t]['all']; ash=A0[t]['shear']; lo=lambda a: math.log((1-a)/a)
    report(f'   control: memory-form A0 log-odds as ruler, tau*={t}',lambda r:lo(aa[r])-lo(aa[1.0]),lambda r:
     lo(ash[r])-lo(aa[1.0]))
```

```
[CHART] all-higher slope 1.1328 rms 0.0142 odd 1.2:1.1006 1.25:1.1128 2:1.1093 4:1.1177 16:1.1384 spread 3.39%
      (2.90-3.87)
[CHART] shear-only slope 1.2036 rms 0.0048 odd 1.2:1.1895 4:1.2038   RATIO 1.0624 (1.0580-1.0672)
[RULER (e) theta at tau*=1.7] all-higher slope 21.2034 rms 0.1299 odd 1.2:16.7185 1.25:16.9164 2:17.1901
     4:18.4351 16:23.3653 spread 35.88% (35.30-36.43)
[RULER (e) theta at tau*=1.7] shear-only slope 20.0700 rms 0.0602 odd 1.2:18.5248 4:20.2932   RATIO 0.9465
     (0.9424-0.9504)
[   C2 swapped tau*=1.7] all-higher slope 21.3339 rms 0.1269 odd 1.2:17.1394 1.25:16.9164 2:17.1901 4:18.8417
     16:23.3653 spread 34.50% (33.95-35.04)
[   C2 swapped tau*=1.7] shear-only slope 19.6502 rms 0.0579 odd 1.2:18.0699 4:19.8552   RATIO 0.9211
     (0.9173-0.9252)
   C1 tau*=1.7: theta read in x -> family ratio 1.1455, odd spread 28.00%
[   control: memory-form A0 log-odds as ruler, tau*=1.7] all-higher slope 50.8990 rms 0.0990 odd 1.2:38.3644
     1.25:38.8248 2:39.6135 4:42.9600 16:55.6947 spread 40.22% (39.67-40.78)
[   control: memory-form A0 log-odds as ruler, tau*=1.7] shear-only slope 45.0634 rms 0.0085 odd 1.2:40.6395
     4:45.1552   RATIO 0.8853 (0.8812-0.8891)
[RULER (e) theta at tau*=4.0] all-higher slope 3.0661 rms 0.3495 odd 1.2:4.7133 1.25:4.7598 2:4.6021 4:4.2381
     16:3.7766 spread 22.26% (21.83-22.68)
[RULER (e) theta at tau*=4.0] shear-only slope 5.1143 rms 0.1723 odd 1.2:6.0555 4:5.5612   RATIO 1.6680
     (1.6599-1.6747)
[   C2 swapped tau*=4.0] all-higher slope 3.0977 rms 0.3631 odd 1.2:5.6026 1.25:4.7598 2:4.6021 4:5.1635
     16:3.7766 spread 38.20% (37.76-38.66)
[   C2 swapped tau*=4.0] shear-only slope 4.1660 rms 0.1797 odd 1.2:5.0943 4:4.5646   RATIO 1.3449
     (1.3390-1.3513)
   C1 tau*=4.0: theta read in x -> family ratio 0.7429, odd spread 26.67%
[   control: memory-form A0 log-odds as ruler, tau*=4.0] all-higher slope -3.2912 rms 0.5019 odd 1.2:-17.4233
     1.25:-17.4823 2:-14.5603 4:-9.6170 16:-5.0894 spread 96.56% (96.26-96.91)
[   control: memory-form A0 log-odds as ruler, tau*=4.0] shear-only slope -6.0464 rms 0.3134 odd 1.2:-13.7295
     4:-8.2729   RATIO 1.8372 (1.8288-1.8456)
```

B.18 Figures

```
# SCRIPT: PBH-NU-FIGURES  draws the three figures from the run results and the memory curves
import json, numpy as np, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
plt.rcParams.update({'font.family':'DejaVu Serif','font.size':11})
mc=json.load(open('code/main/memory_curve.json'))
ed3=[0.25,0.5,0.75,1,1.5,2,3,4,6,8]; ed6=[0.5,1,1.5,2,3,4]
G3=[mc['3_%g'%e]['two_epoch'] for e in ed3]; G6=[mc['6_%g'%e]['two_epoch'] for e in ed6]
steve_e=[0.25,0.5,1,2,4,8]; steve_G=[0.959,0.867,0.683,0.431,0.190,0.063]
# Fig 1
fig,ax=plt.subplots(figsize=(7.2,4.6))
ax.plot(ed3,G3,'-',color='#1f5fa8',lw=2,label='kept-sector memory, rate $\\propto\\epsilon^{-3}$ (one weight
    fitted here)')
ax.plot(ed6,G6,'--',color='#c0392b',lw=2,label='kept-sector memory, rate $\\propto\\epsilon^{-6}$ (prediction)
    ')
ax.plot(steve_e,steve_G,'o',color='k',ms=7,label='McCormick, Gaussian, rate $\\propto\\epsilon^{-3}$')
ax.plot([4,8],[0.181,0.045],'s',color='#2e8b57',ms=8,label='this work, broad profile $\\alpha=0.3$, rate $\\
    propto\\epsilon^{-3}$')
ax.errorbar([2],[0.317],yerr=[0.004],fmt='D',color='#c0392b',ms=8,label='this work, Gaussian, rate $\\propto\\
    epsilon^{-6}$')
ax.plot([2],[0.431],'x',color='gray',ms=10,mew=2); ax.annotate('if only $\\epsilon_{dec}$ mattered',(2,0.431)
    ,(2.6,0.52),fontsize=9,color='gray',arrowprops=dict(arrowstyle='->',color='gray'))
ax.set_xscale('log'); ax.set_xticks([0.25,0.5,1,2,4,8]); ax.set_xticklabels(['0.25','0.5','1','2','4','8']);
    ax.minorticks_off(); ax.set_xlabel('decoupling time $\\epsilon_{dec}$'); ax.set_ylabel('surviving share
    of the free-streaming shift, $G$')
ax.set_ylim(0,1.02); ax.legend(fontsize=8.2,loc='lower left',frameon=False); ax.set_title('How much of the
    neutrino effect survives gradual decoupling',fontsize=11)
fig.tight_layout(); fig.savefig('figs/fig1_G.png',dpi=220)
# Fig 2: Legendre chain
fig,ax=plt.subplots(figsize=(7.2,2.6)); ax.axis('off')
xs=np.arange(6)*1.2
for i,x in enumerate(xs):
    ax.add_patch(plt.Circle((x,0),0.28,color=['#f4d03f','#e67e22','#c0392b','#8e44ad','#8e44ad','#8e44ad'][i],
    zorder=3))
    ax.text(x,0,'$\\ell=%d$'%i if i<5 else '$\\cdots$',ha='center',va='center',fontsize=10,zorder=4,color='
    white' if i>1 else 'k')
for i in range(5):
    lw=[6,4,0.8,0.8,0.8][i]; col=['#c0392b','#e67e22','#bbbbbb','#bbbbbb','#bbbbbb'][i]
    ax.plot([xs[i]+0.28,xs[i+1]-0.28],[0,0],color=col,lw=lw,zorder=2)
ax.text((xs[0]+xs[1])/2,0.45,'$k^2\\kappa_1^2/3$',ha='center',fontsize=10,color='#c0392b')
ax.text((xs[1]+xs[2])/2,0.45,'$\\frac{4k^2}{15}(\\kappa_2-\\kappa_1)^2$',ha='center',fontsize=10,color='#
    e67e22')
ax.text((xs[3]+xs[4])/2,0.45,'0',ha='center',fontsize=10,color='gray')
ax.text(xs[0],-0.6,'density\n(kept)',ha='center',fontsize=8.5); ax.text(xs[1],-0.6,'flux\nrate $\\kappa_1$',ha
    ='center',fontsize=8.5); ax.text(xs[2],-0.6,'shear\nrate $\\kappa_2$',ha='center',fontsize=8.5); ax.text(
    xs[4],-0.6,'higher moments\nrate $\\kappa_2$',ha='center',fontsize=8.5)
ax.set_xlim(-0.5,xs[-1]+0.5); ax.set_ylim(-1.0,0.85)
ax.set_title('Where the non-normality of the neutrino collision operator lives',fontsize=11)
fig.tight_layout(); fig.savefig('figs/fig2_chain.png',dpi=220)
# Fig 3: clock set early
fig,ax=plt.subplots(figsize=(7.2,3.6))
ax.bar([0.8,2.8],[0.190,0.063],0.35,color='#555555',label='Gaussian (McCormick): collapses at $\\epsilon\\
    approx4$')
ax.bar([1.2,3.2],[0.181,0.045],0.35,color='#2e8b57',label='broad profile $\\alpha=0.3$ (this work): collapses
    at $\\epsilon\\approx7$-$9$')
ax.plot([0.8,1.2,2.8,3.2],[0.38,0.38,0.17,0.17],'_',ms=0)
ax.hlines([0.42],1.02,1.38,colors='#c0392b',linestyles='dotted',lw=2); ax.hlines([0.205],3.02,3.38,colors='#
    c0392b',linestyles='dotted',lw=2)
ax.text(1.2,0.44,'if the clock were the collapse time',ha='center',fontsize=8.5,color='#c0392b'); ax.text
    (3.38,0.235,'if the clock were the collapse time',ha='right',fontsize=8.5,color='#c0392b')
ax.set_xticks([1,3]); ax.set_xticklabels(['$\\epsilon_{dec}=4$','$\\epsilon_{dec}=8$']); ax.set_ylabel('$G$');
     ax.set_ylim(0,0.6)
ax.legend(fontsize=8.5,frameon=False,loc='upper right',bbox_to_anchor=(1.0,0.98)); ax.set_title('The clock is
    set early: a late collapse does not change $G$',fontsize=11)
fig.tight_layout(); fig.savefig('figs/fig3_clock.png',dpi=220)
print('ok')
```

```
# SCRIPT: PBH-NU-FIGURES-V3
# Figures 4, 5, 6 for the PBH-NU paper. Uses only the numbers and data files listed below.
# Writes fig4_tests.png, fig5_killlines.png, fig6_door.png into figs/. Never touches fig1-fig3.
import json
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib.colors import Normalize
from matplotlib.lines import Line2D
from matplotlib.patches import Patch

BASE = '/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad'
PAPER = BASE + '/paper'
MEM = PAPER + '/code/main/memory_curve.json'
WIT = BASE + '/next/witness/witness_prod.json'
OUT = PAPER + '/figs/'
PROTECTED = {'fig1_G.png', 'fig2_chain.png', 'fig3_clock.png'}

plt.rcParams.update({
    'font.family': 'DejaVu Sans',
    'font.size': 9.5,
    'axes.labelsize': 10,
    'xtick.labelsize': 9,
    'ytick.labelsize': 9,
    'legend.fontsize': 8,
    'axes.spines.top': False,
    'axes.spines.right': False,
})

# Okabe-Ito based colour-blind-safe palette
BLUE = '#0072B2'
ORANGE = '#E69F00'
BLACK = '#000000'
GREY = '#7a7a7a'
BAND = '#d6e6f3'


def save(fig, name, **kw):
    assert name not in PROTECTED, 'refusing to touch an existing figure'
    path = OUT + name
    fig.savefig(path, dpi=300, facecolor='white', **kw)
    plt.close(fig)
    print('wrote', path)


# ---------- data ----------
mc = json.load(open(MEM))


def memory_curve(n):
    pts = []
    for key, val in mc.items():
        nn, ee = key.split('_')
        if int(nn) == n:
            pts.append((float(ee), val['two_epoch']))
    return sorted(pts)


c3 = memory_curve(3)
c6 = memory_curve(6)

# McCormick measured n=3 (history-blind rule)
mc_e = [0.25, 0.5, 1, 2, 4, 8]
mc_G = [0.959, 0.867, 0.683, 0.431, 0.190, 0.063]

# New measured n=6 runs (C, B, D)
```

```
new6_e = [1, 2, 3]
new6_G = [0.648, 0.317, 0.154]
new6_err = [0.006, 0.004, 0.006]
new6_lab = ['C', 'B', 'D']

# Broad profile alpha=0.3, n=3
broad_e = [4, 8]
broad_G = [0.181, 0.045]

# ---------- FIG 4 ----------
fig, ax = plt.subplots(figsize=(7.8, 4.6))

ax.plot(mc_e, mc_G, linestyle='--', color=GREY, lw=0.9, zorder=1,
        label='history-blind rule (uses the n=3 curve for any history)')
ax.plot(mc_e, mc_G, 'o', color=BLACK, ms=5.5, zorder=3,
        label='McCormick, n=3 (measured)')

ax.plot([p[0] for p in c3], [p[1] for p in c3], '-', color=BLUE, lw=1.8, zorder=2,
        label='memory model, n=3 (fitted, one weight)')
ax.plot([p[0] for p in c6], [p[1] for p in c6], '-', color=ORANGE, lw=1.8, zorder=2,
        label='memory model, n=6 (frozen, predicted before the runs)')

ax.errorbar(new6_e, new6_G, yerr=new6_err, fmt='D', color=ORANGE, mec=ORANGE,
            ms=6, capsize=3, elinewidth=1.0, zorder=4,
            label="McCormick's code, n=6 (new runs)")
for x, y, lab in zip(new6_e, new6_G, new6_lab):
    ax.annotate(lab, (x, y), xytext=(7, -9), textcoords='offset points',
                fontsize=9, fontweight='bold', color=BLACK)

ax.plot(broad_e, broad_G, 's', mfc='none', mec=BLACK, mew=1.2, ms=6.5, zorder=4,
        label='broad profile, n=3 (new runs)')

ax.set_xscale('log')
ax.set_xlim(0.2, 10)
ax.set_ylim(0, 1)
ticks = [0.2, 0.5, 1, 2, 5, 10]
ax.set_xticks(ticks)
ax.set_xticklabels(['0.2', '0.5', '1', '2', '5', '10'])
ax.minorticks_off()
ax.set_xlabel(r'decoupling time  $\varepsilon_{\mathrm{dec}}$')
ax.set_ylabel(r'share of the neutrino effect kept, $G$')
ax.legend(loc='upper right', frameon=False, handlelength=2.4)
fig.tight_layout()
save(fig, 'fig4_tests.png')

# ---------- FIG 5 ----------
rows = [
    # label, measured, meas_err, memory, history_blind, annotation
    ('Test B  (n=6, $\\varepsilon_{\\mathrm{dec}}$=2)', 0.317, 0.004, 0.246, 0.431,
     'memory off 0.071 . history-blind off 0.114'),
    ('Test C  (n=6, $\\varepsilon_{\\mathrm{dec}}$=1)', 0.648, 0.006, 0.6625, 0.683,
     'memory off 0.014 . history-blind off 0.035'),
    ('Test D  (n=6, $\\varepsilon_{\\mathrm{dec}}$=3)', 0.154, 0.006, 0.1176, 0.30,
     'memory off 0.037 . history-blind off 0.146 (dies)'),
]
KILL = 0.08

fig, ax = plt.subplots(figsize=(10.5, 3.8))
fig.subplots_adjust(left=0.24, right=0.6, bottom=0.3, top=0.95)
half_h = 0.26
for k, (lab, meas, merr, mem, hb, note) in enumerate(rows):
    y = 2 - k
    ax.fill_betweenx([y - half_h, y + half_h], meas - KILL, meas + KILL,
                     color=BAND, lw=0, zorder=1)
    ax.errorbar([meas], [y], xerr=[merr], fmt='D', color=BLACK, ms=6.5,
                capsize=3, elinewidth=1.0, zorder=5)
    ax.plot([mem], [y], marker='|', color=ORANGE, ms=17, mew=3, ls='none', zorder=4)
    ax.plot([hb], [y], 'o', mfc='none', mec=GREY, mew=1.8, ms=9, ls='none', zorder=4)
```

```
    ax.text(1.03, y, note, transform=ax.get_yaxis_transform(),
            va='center', ha='left', fontsize=9, color=BLACK)

ax.set_xlim(0, 0.8)
ax.set_ylim(-0.6, 2.6)
ax.set_yticks([2, 1, 0])
ax.set_yticklabels([r[0] for r in rows], fontsize=9)
ax.set_xlabel('G')
ax.spines['left'].set_visible(False)
ax.tick_params(axis='y', length=0)

handles = [
    Patch(facecolor=BAND, edgecolor='none', label='kill band: +-0.08 around the measurement'),
    Line2D([], [], marker='D', color=BLACK, ls='none', ms=6, label='measured (bar = stated uncertainty)'),
    Line2D([], [], marker='|', color=ORANGE, ls='none', ms=12, mew=3, label='memory-model prediction'),
    Line2D([], [], marker='o', mfc='none', mec=GREY, mew=1.8, ls='none', ms=8,
           label='history-blind prediction'),
]
ax.legend(handles=handles, loc='upper center', bbox_to_anchor=(0.4, -0.2), ncol=2,
          frameon=False)
save(fig, 'fig5_killlines.png')

# ---------- FIG 6 ----------
wit = json.load(open(WIT))
RATIOS = [0.1, 0.25, 0.5, 1, 2]
LS = [1, 2, 3, 4]
FLOOR = -32.0


def height_grid(n):
    H = np.zeros((len(LS), len(RATIOS)))
    for rec in wit:
        if rec['n'] != n:
            continue
        i = int(np.argmin([abs(rec['ratio'] - r) for r in RATIOS]))
        j = LS.index(rec['L'])
        R = rec['res']['4.0']['R_ret']
        H[j, i] = max(np.log10(max(R, 1e-40)) - FLOOR, 0.0)
    return H


H = {3: height_grid(3), 6: height_grid(6)}
hmax = max(H[3].max(), H[6].max())
ZTOP = max(34.0, np.ceil(hmax) + 2)
norm = Normalize(vmin=0, vmax=ZTOP)
cmap = plt.cm.viridis
WALL_LOG = -3.0
WALL_H = WALL_LOG - FLOOR

fig = plt.figure(figsize=(11.5, 5.4))
axes = []
for k, n in enumerate([3, 6]):
    ax = fig.add_subplot(1, 2, k + 1, projection='3d')
    axes.append(ax)
    Hn = H[n]
    xs, ys, zs, dxs, dys, dzs, cols = [], [], [], [], [], [], []
    for j, L in enumerate(LS):
        for i in range(len(RATIOS)):
            h = Hn[j, i]
            xs.append(i - 0.3)
            ys.append(L - 0.3)
            zs.append(0.0)
            dxs.append(0.6)
            dys.append(0.6)
            dzs.append(h)
            cols.append(cmap(norm(h)))
    ax.bar3d(xs, ys, zs, dxs, dys, dzs, color=cols, shade=True, alpha=1.0,
             edgecolor='none')
```

```
    # translucent "witness on" plane at log10 R = -3
    gx, gy = np.meshgrid([-0.6, len(RATIOS) - 0.4], [0.5, 4.5])
    gz = np.full_like(gx, WALL_H, dtype=float)
    ax.plot_surface(gx, gy, gz, color='#555555', alpha=0.22, shade=False,
                    linewidth=0, zorder=0)

    ax.set_xticks(range(len(RATIOS)))
    ax.set_xticklabels(['0.1', '0.25', '0.5', '1', '2'], fontsize=8)
    ax.set_yticks(LS)
    ax.set_yticklabels([str(L) for L in LS], fontsize=8)
    ax.set_zlim(0, ZTOP)
    ax.set_zticks([2, 12, 22, 32])
    ax.set_zticklabels(['-30', '-20', '-10', '0'], fontsize=8)
    ax.set_xlabel('plant time / decoupling time', fontsize=9, labelpad=6)
    ax.set_ylabel('moment that holds the difference, $l$', fontsize=9, labelpad=6)
    ax.set_zlabel(r'$\log_{10} R_{\mathrm{ret}}$', fontsize=9, labelpad=4)
    ax.view_init(elev=28, azim=-32)
    ax.text2D(0.04, 0.9, 'n = %d' % n, transform=ax.transAxes, fontsize=10,
              fontweight='bold', color=BLACK)
    ax.set_box_aspect((1.3, 1.0, 0.95))

fig.subplots_adjust(left=0.03, right=0.97, bottom=0.04, top=0.97, wspace=0.28)
save(fig, 'fig6_door.png', bbox_inches='tight', pad_inches=0.15)
```

```
# SCRIPT: PBH-NU-FIG6-DOOR  witness strength terrain (log10 R_ret at tau=4.0) from witness_prod.json
import json, math, numpy as np, matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib import cm
d=json.load(open('/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/next/witness/
    witness_prod.json'))
ratios=[0.1,0.25,0.5,1,2]; Ls=[1,2,3,4]; FLOOR=-32
fig=plt.figure(figsize=(12,5.2))
for p,n in enumerate((3,6)):
    ax=fig.add_subplot(1,2,p+1,projection='3d')
    for r in d:
        if r['n']!=n: continue
        i=ratios.index(r['ratio']); j=Ls.index(r['L'])
        z=math.log10(r['res']['4.0']['R_ret'])
        col=cm.viridis((z-FLOOR)/(0-FLOOR))
        ax.bar3d(i-0.35,j-0.35,FLOOR,0.7,0.7,z-FLOOR,color=col,shade=True,edgecolor='k',linewidth=0.3)
    X,Y=np.meshgrid([-0.6,4.6],[-0.6,3.6]); ax.plot_surface(X,Y,np.full_like(X,-3.0),color='red',alpha=0.18)
    ax.set_xticks(range(5)); ax.set_xticklabels([str(x) for x in ratios])
    ax.set_yticks(range(4)); ax.set_yticklabels([str(l) for l in Ls])
    ax.set_zlim(FLOOR,0); ax.set_zticks([-30,-20,-10,-3,0])
    ax.set_xlabel('plant time / decoupling time',labelpad=8); ax.set_ylabel('moment holding the difference, l
    ',labelpad=8); ax.set_zlabel('log$_{10}$ R$_{ret}$ at $\\tau$=4')
    ax.view_init(elev=22,azim=-128)
    ax.text2D(0.03,0.92,'n = %d'%n,transform=ax.transAxes,fontsize=13,weight='bold')
fig.text(0.5,0.02,'red plane: witness on (R$_{ret}$ = 10$^{-3}$)',ha='center',fontsize=10)
plt.tight_layout(); plt.savefig('figs/fig6_door.png',dpi=300,facecolor='white')
```

```
# SCRIPT: NU-V5-FIG7-LINK  the flux-shear link: G against Gamma12, and the offset of G against the link
    rapidity (eleven thresholds + four shear-only)
import json, math, numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt, os
here=os.path.dirname(os.path.abspath(__file__)); fl,ins=0.49773,0.52832
runs={1/16:'F2_r1o16',0.25:'F_r025',0.5:'J_r05',1/1.2:'A1_all_r083',0.8:'K_r08_clean',1.0:'H1_r1',1.2:'
    A1_all_r12',1.25:'K_r125_clean',2.0:'J_r2',4.0:'F_r4',16.0:'F2_r16'}
Ls={0.25:'L_r025',1/1.2:'A1_shear_r083',1.2:'A1_shear_r12',4.0:'L_r4'}
dc=lambda t: json.load(open(os.path.join(here,'res_%s.json'%t)))['dc']
r=np.array(sorted(runs)); G=np.array([(dc(runs[x])-fl)/(ins-fl) for x in r]); Gam=(r-1)/(r+1); x=0.5*np.log(r)
    ; tau=np.log((1-G)/G); t1=tau[list(r).index(1.0)]
rl=np.array(sorted(Ls)); Gl=np.array([(dc(Ls[x])-fl)/(ins-fl) for x in rl])
fig,ax=plt.subplots(1,2,figsize=(11,4.3))
ax[0].plot(Gam,G,'o',color='#1f5fa8',ms=7); ax[0].axhline(G[list(r).index(1.0)],color='0.6',lw=0.8,ls=':')
ax[0].axvspan((0.75-1)/1.75,(1.5-1)/2.5,color='#1f5fa8',alpha=0.08)
```

```
ax[0].set_xlabel(r'reflection coefficient of the flux--shear link, $\Gamma_{12}=(\kappa_2-\kappa_1)/(\kappa_2+
    \kappa_1)$'); ax[0].set_ylabel(r'surviving share $G$'); ax[0].set_title('The threshold reads the link')
xx=np.linspace(-1.45,1.45,200)
ax[1].axvspan(0.5*math.log(0.75),0.5*math.log(1.5),color='#1f5fa8',alpha=0.08,label='admissible window')
ax[1].plot(xx,1.11*xx,color='#d9822b',lw=1.6,label='rate 1.11 (inner pairs)'); ax[1].plot(xx,xx,color='0.5',lw
    =1,ls='--',label='rate 1')
ax[1].plot(x,tau-t1,'o',color='#1f5fa8',ms=7,label=r'all $\ell\geq2$ at $\kappa_2$')
ax[1].plot(0.5*np.log(rl),np.log((1-Gl)/Gl)-t1,'s',color='#c0392b',ms=7,label=r'shear ($\ell=2$) only')
ax[1].set_xlabel(r'link rapidity $\frac{1}{2}\ln(\kappa_2/\kappa_1)$'); ax[1].set_ylabel(r'$\ln\frac{1-G}{G}-\
    ln\frac{1-G_1}{G_1}$'); ax[1].set_title('Offset against rapidity'); ax[1].legend(fontsize=8,loc='upper
    left')
for a in ax: a.grid(alpha=0.25)
plt.tight_layout(); plt.savefig(os.path.join(here,'..','..','figs','fig7_link_v5.png'),dpi=160)
for a,b,c in zip(r,G,tau-t1): print(f'r={a:g} G={b:.4f} offset shift={c:+.4f}')
for a,b in zip(rl,Gl): print(f'shear-only r={a:g} G={b:.4f} offset shift={math.log((1-b)/b)-t1:+.4f}')
```

```
r=0.0625 G=0.7765 offset shift=-1.5466
r=0.25 G=0.6135 offset shift=-0.7631
r=0.5 G=0.5206 offset shift=-0.3836
r=0.8 G=0.4565 offset shift=-0.1267
r=0.833333 G=0.4506 offset shift=-0.1028
r=1 G=0.4253 offset shift=+0.0000
r=1.2 G=0.4016 offset shift=+0.0979
r=1.25 G=0.3959 offset shift=+0.1216
r=2 G=0.3349 offset shift=+0.3854
r=4 G=0.2521 offset shift=+0.7864
r=16 G=0.1289 offset shift=+1.6097
shear-only r=0.25 G=0.6330 offset shift=-0.8462
shear-only r=0.833333 G=0.4526 offset shift=-0.1107
shear-only r=1.2 G=0.3996 offset shift=+0.1061
shear-only r=4 G=0.2453 offset shift=+0.8227
```

B.19 The stage-correct runner and the reconvergence of Section 7

`A1_stage_fix.diff`: the one change to the v5.1 runner.

```
1c1
< # SCRIPT: NU-V4-RUN-SPLIT-V2  run_nuc_n.py importing ms_nuc_split and passing rr; nothing else changed
---
> # SCRIPT: NU-V6-RUN-SPLIT-STAGE  run_split_v2.py with ONE change: kappa(t) is evaluated at each SSPRK3 stage
      time (t, t+dt, t+dt/2); the stiff dt cap still uses the stage-start kappa
19a20,21
>     def kapf(tt):                           # v6: the collision rate at any time tt (same formula as v5.1)
>         return (1/(2*tt))*min((eps_dec/(2*np.sqrt(tt*t0)))**nexp, kapcap) if eps_dec is not None else kap
23c25
<             kap = (1/(2*t))*min((eps_dec/(2*np.sqrt(t*t0)))**nexp, kapcap)
---
>             kap = kapf(t)
50c52
<             rhs(t+dt, R1, U1, G1, e1, I1, mue, fnu, fnu/(1-fnu)*e1[-1]/(4*np.pi), kap, rr, dR, dU, dG, de,
    dI)
---
>             rhs(t+dt, R1, U1, G1, e1, I1, mue, fnu, fnu/(1-fnu)*e1[-1]/(4*np.pi), kapf(t+dt), rr, dR, dU, dG
    , de, dI)
53c55
<             rhs(t+0.5*dt, R2, U2, G2, e2, I2, mue, fnu, fnu/(1-fnu)*e2[-1]/(4*np.pi), kap, rr, dR, dU, dG,
    de, dI)
---
>             rhs(t+0.5*dt, R2, U2, G2, e2, I2, mue, fnu, fnu/(1-fnu)*e2[-1]/(4*np.pi), kapf(t+0.5*dt), rr, dR
    , dU, dG, de, dI)
```

`A1_PREREGISTERED.txt`: predictions, kill lines and appended results.

```
PREREGISTERED.txt -- v6 task A1: the 1.11 law on stage-correct collision histories
Written 2026-11-11T14:05Z, before any threshold search of this file was run.
```

```
Results are appended below each test after it runs; nothing above an appended result is edited.

SETUP (fixed before the runs)
  Solver: ms_nuc_split.py and ms_nuc_shear.py, byte copies of v5.1 (v4_runs), with McCormick's ms_fluid.py at
    commit 41a0f83.
  Runner: run_split_stage.py / run_shear_stage.py = v5.1's run_split_v2.py / run_shear_v2.py with ONE change:
    kappa(t) = (1/2t) min[(eps_dec/eps(t))^3, 5e5] is evaluated at each SSPRK3 stage time (t, t+dt, t+dt/2)
     instead of
    once per step at t.  The stiff time-step cap 0.5/(kappa max(1,r) max A) still uses the stage-start kappa(t
    ).
  Physics fixed as v5.1: eps_dec = 2, n = 3, Gaussian profile (q = 1), R_nu = 0.409, eps0 = 0.1, eps_max = 15,
     N = 400, Nmu = 32,
    Courant 0.4/0.4, cH = 0.005.
  Endpoint rule (v5.1): a run moves a bracket only if it ends in a detected horizon (2M/R >= 1 with U < 0) ->
    'BH', or in
    dispersal.  'crash' or 'BH_breakdown' (breakdown after 2M/R > 0.95 without a detected horizon) is retried
     once at
    Courant 0.3; if it fails again it is reported and never moves the bracket.
  Search: ONE simulation process at a time (house rule), so the two-points-per-round trisection is replaced by
     sequential
    bisection with the same endpoint rule.  Each search first runs both ends of its starting bracket: the
     lower end must
    disperse and the upper end must form a horizon; if an end gives the other outcome the bracket is extended
     outward by
    3e-4 (that run's outcome is kept as a bracket edge) and the new end is run.  Bisection stops at bracket
     width <= 1e-6.
    An undecided midpoint is replaced by the points at 0.4 and then 0.6 of the bracket; if all three fail the
     search stops
    and is reported.
  Starting brackets: v5.1 midpoints +- 3e-4.
    r = 1      : 0.51074074 (v5.1 H1 bracket [0.51072222, 0.51075926])
    r = 1.2    : 0.51001486      r = 1/1.2 : 0.51151453      (all l >= 2 family)
    r = 1.25   : 0.50984070      r = 0.8   : 0.51169573      (all l >= 2 family)
    r = 1.2    : 0.50995416      r = 1/1.2 : 0.51157480      (shear-only family)
  References (v5.1, McCormick's archive): delta_fluid = 0.49773, delta_inst = 0.52832.
    G = (dc - df)/(di - df); tau_G = ln[(1-G)/G]; odd rate = [tau_G(r) - tau_G(1/r)]/ln r.
    Ranges are hard four-corner ranges over the two brackets (v5.1 App. A.11).  Every rate is also reported
     with the fluid
    reference shifted by +7.8e-5 and -7.8e-5 (archive uncertainty).
  Check before the runs (oddrate_v51_check.txt): oddrate.py reproduces v5.1's 1.1006 [1.1000,1.1011], 1.1128
     [1.1124,1.1131]
    and shear 1.1895 [1.1892,1.1898] from v5.1's stored brackets.

REGRESSION (run before this file was written, at 14:00Z; it decides nothing and is recorded here for
     completeness)
  r = 1, delta = 0.5109, v5.1 runner against the stage-correct runner, same process.  Result in log_regress.
     txt:
  v5.1 runner: BH at eps = 6.6147 (v5.1 reports 6.61); stage-correct: BH at eps = 6.6176.  Move +2.9e-3 in eps
     at stop.
  Max relative field differences: 1e-6 at eps = 1-2, 1.2e-4 (e, I) at eps = 4, 3.4e-3 (e) at eps = 6, 4.7e-3
     at stop.

TEST S1 (the inner pairs, all l >= 2 family, and the r = 1 anchor)
  Thresholds: r = 1, 1.2, 1/1.2, 1.25, 0.8, each to bracket width <= 1e-6.
  Kill lines for the statement "inside the admissible window the odd rate is 1.11 +- 0.01":
   (a) the statement is withdrawn if either inner odd rate (central value, v5.1 references) moves by more than
      0.01 from
       v5.1: 1.1006 at r = 1.2 pair, 1.1128 at r = 1.25 pair;
   (b) the statement is withdrawn if the two inner pairs' odd rates (central values) still differ by more than
      0.01 after
       the stage fix.
  Opposite reading, stated before the runs: if the stage fix pulls the two pairs together (their difference
     falls well
   below v5.1's 0.0122, i.e. below 0.005), the old spread was the first-order collision-history error, not a
     per-point
   systematic of the threshold itself; if it leaves the difference near 0.012, the spread is not from the
     history order.
  Prediction (expectation, not a kill line): each threshold moves by less than 2e-5 from v5.1 (the history
```

```
    error is
   O(dt) in kappa, and dt/t = 0.01); the r = 1 threshold stays inside v5.1's bracket [0.51072222, 0.51075926].
  The anchor tau_G(1) cancels from the odd rate; r = 1 enters only G(1) and the even parts.

TEST S2 (shear-only pair, r = 1.2 and 1/1.2)
  Statement under test: "shear-only damping gives 1.19-1.20" at the inner pair (v5.1: 1.1895 [1.1892, 1.1898])
    .
  Kill: withdrawn at the inner pair if the stage-correct shear-only odd rate moves by more than 0.01 from
    1.1895.
  Second line: the gap shear-only minus all-l>=2 at the same pair (v5.1: 0.089) must stay above 0.05; else the
     statement
   "damping the shear alone moves the threshold more than damping the shear and every higher moment together"
    is withdrawn
   at the inner pair.

TEST S3 (simultaneous refinement, r = 1.25 and 0.8 only)
  N = 600, Nmu = 48, cH (the H time-step limit) and the Courant numbers halved together: cH = 0.0025, Courant
    0.2/0.2;
   the retry rule halves too (Courant 0.15).  The stiff collision cap is left as it is (task specification:
    only the H and
   Courant limits are halved).  Stage-correct runner.  Starting brackets: the S1 stage-correct thresholds +- 3
    e-4.
  Prediction / kill: refinement shifts the r = 1.25 pair's odd rate by less than 0.005 from its S1 stage-
    correct value.
   Fail: the shift is >= 0.005; then that shift is carried as a resolution error on every inner rate, and the
    +-0.01 is
   re-judged with it added.

TEST S1 RESULT (appended 2026-11-11 ~15:20Z; all five searches to width <= 1e-6, logs log_S1_*.txt, res_S1_*.
    json)
  r = 1     : [0.51077824, 0.51077883]  dc = 0.51077853  (v5.1 0.51074074; shift +3.8e-5, OUTSIDE v5.1's
    bracket)  G(1) = 0.42656
  r = 1.2   : [0.51000783, 0.51000841]  dc = 0.51000812  (shift -6.7e-6)
  r = 1/1.2 : [0.51151547, 0.51151641]  dc = 0.51151594  (shift +1.4e-6; one point 0.51152391 crashed twice,
    reported, skipped)
  r = 1.25  : [0.50984129, 0.50984187]  dc = 0.50984158  (shift +0.9e-6)
  r = 0.8   : [0.51169198, 0.51169268]  dc = 0.51169233  (shift -3.4e-6; one point crashed twice, reported,
    skipped)
  Odd rates: r = 1.2 pair 1.1066, range [1.1061, 1.1072] (v5.1 1.1006: change +0.0060)
             r = 1.25 pair 1.1102, range [1.1098, 1.1106] (v5.1 1.1128: change -0.0026)
  (a) both changes < 0.01: the +-0.01 statement survives line (a).
  (b) the pairs now differ by 0.0036 (v5.1: 0.0122), < 0.01: survives line (b).
  Opposite reading: the difference fell below 0.005, so on the pre-stated reading the old 0.012 spread was
    mostly the
   first-order collision-history error.  The expectation "each threshold moves < 2e-5" held for the four inner
     thresholds
   and FAILED at r = 1 (+3.8e-5); not a kill line (the anchor cancels from the odd rate) but it is flagged and
     checked below.

CONTROL C1 (written 2026-11-11 ~15:22Z, before it runs; queued after S3): is the r = 1 shift the stage fix?
  Rerun the r = 1 threshold with v5.1's own runner (orig/run_split_v2.py, kappa once per step), same bisection
     driver, from
  [0.51072222, 0.51078883], to width <= 1e-6.  Reading: if the v5.1-runner threshold lies inside v5.1's
    bracket
  [0.51072222, 0.51075926], the +3.8e-5 is (nearly all) the stage fix at r = 1; if it lands within 5e-6 of
    0.51077853,
  the v5.1 r = 1 bracket (tolerance 1e-4 era) was off and the stage fix is small at r = 1 too.  Affects G(1)
    and the even
  parts only; the odd rates do not use r = 1.

TEST S2 RESULT (appended ~15:41Z; logs log_S2_*.txt)
  shear only r = 1.2   : [0.50995416, 0.50995510]  dc = 0.50995463 (shift +0.5e-6; one point crashed twice,
    reported, skipped)
  shear only r = 1/1.2 : [0.51157949, 0.51158007]  dc = 0.51157978 (shift +5.0e-6)
  Odd rate 1.1928, range [1.1922, 1.1933] (v5.1 1.1895: change +0.0033 < 0.01): survives.
  Shear-only minus all-l>=2 at the same pair: 0.0862 (v5.1 0.0889) > 0.05: survives.

TEST S3 RESULT (appended ~20:35Z; logs log_S3_*.txt, point_S3_r08.txt)
```

```
  refined r = 1.25 : [0.50991658, 0.50991716]  dc = 0.50991687  (N=400 stage value 0.50984158: shift +7.53e-5)
  refined r = 0.8  : [0.51175608, 0.51175678]  dc = 0.51175643  (N=400 stage value 0.51169233: shift +6.41e-5)
    One point (0.51175620) never ended: Courant 0.2 attempt not decisive, Courant 0.15 retry cut at 33 min by
     the queue
    window, rerun from scratch stopped by hand at 70 min.  Reported undecided; per the pre-stated rule the 0.4
      point
    (0.51175608) was run instead and dispersed (run_point.py, one attempt, Courant 0.2).
  Odd rate refined 1.1020, range [1.1016, 1.1024]; N = 400 stage value 1.1102.  Shift -0.0082 (corner range of
     the shift
   [-0.0090, -0.0074]).  |shift| >= 0.005: the refinement prediction FAILS.
  Per the pre-stated fail branch, the shift is carried as a resolution error on every inner rate and the
     +-0.01 re-judged:
   central inner values across stage fix and refinement: 1.1066 (r=1.2, N=400), 1.1102 (r=1.25, N=400), 1.1020
      (r=1.25,
   refined).  Errors: pair spread 0.0036, fluid reference +-0.0038 (common), resolution 0.0082.  In quadrature
      0.0097,
   linear 0.0156.  The band 1.11 +- 0.01 still contains every measured value, but the +-0.01 is no longer an
     error bar
   with margin: it is the size of the error budget itself, now dominated by resolution, and one refinement
     level cannot
   say whether the rate has converged (untested).  The two thresholds both rose by 6.4-7.5e-5 under refinement
     ; the
   rate moved because they rose by different amounts (1.1e-5).

CONTROL C1 RESULT (appended ~20:45Z; log_C1_r1_v51runner.txt)
  v5.1 runner (kappa once per step), r = 1, same bisection driver: [0.51075657, 0.51075709], dc = 0.51075683.
  It lies inside v5.1's bracket [0.51072222, 0.51075926].  So of the +3.78e-5 move at r = 1, +2.17e-5 is the
     stage fix
  (0.51075683 -> 0.51077853) and +1.61e-5 is the offset of v5.1's tolerance-1e-4 midpoint from the threshold.
  The stage fix still moves r = 1 more (2.2e-5) than any inner threshold (0.9-6.7e-6).  No effect on any odd
     rate.
```

A1_search.py: bisection driver.

```
# SCRIPT: NU-V6-SEARCH  sequential bisection (one simulation at a time) with v5.1's endpoint rule: only 'BH' (
    detected
# horizon) or 'dispersed' moves a bracket; 'crash' / 'BH_breakdown' get one retry at the lower Courant number,
     then are
# reported and skipped.  Both ends of the starting bracket are run first; an end with the other outcome
    extends the bracket.
# Resumable: progress is saved after every run in prog_<tag>.json.
import sys, os, json, time, importlib
HERE = os.path.dirname(os.path.abspath(__file__)); sys.path.insert(0, HERE)

def main():
    tag, runner = sys.argv[1], sys.argv[2]
    center, half, tol = float(sys.argv[3]), float(sys.argv[4]), float(sys.argv[5])
    kw = json.loads(sys.argv[6]); cretry = float(sys.argv[7])
    mod = importlib.import_module(runner)
    pf = os.path.join(HERE, f'prog_{tag}.json'); lf = os.path.join(HERE, f'log_{tag}.txt'); master = os.path.
     join(HERE, 'RUNLOG.txt')
    if os.path.exists(pf):
        P = json.load(open(pf))
    else:
        P = dict(tag=tag, runner=runner, kw=kw, lo=None, hi=None, lo_try=center-half, hi_try=center+half, log
    =[], t_used=0.0)
    def save(): json.dump(P, open(pf, 'w'), indent=1)
    def say(s):
        line = time.strftime('%H:%M:%S ') + s
        print(line, flush=True); open(lf, 'a').write(line + '\n'); open(master, 'a').write(f'[{tag}] ' + line
    + '\n')
    def one(d):
        t1 = time.time(); o = mod.run(d, **kw); s = o['status']; note = ''
        if s not in ('BH', 'dispersed'):
            o = mod.run(d, **dict(kw, cfl_r=cretry, cfl_mu=cretry)); note = f' (retry at Courant {cretry}
     after {s})'; s = o['status']
        dtm = time.time() - t1; P['t_used'] += dtm
        rec = (d, s, float(o['eps']), float(o['Imin']), float(o['max2MR']), note, round(dtm, 1))
```

```
        P['log'].append(rec); save()
        say(f'{tag} delta={d:.8f}: {s} eps={o["eps"]:.4f} minI={o["Imin"]:.3e} steps={o["steps"]}{note} ({dtm
    :.0f}s)')
        return s
    # 1) the lower end must disperse
    while P['lo'] is None:
        d = P['lo_try']; s = one(d)
        if s == 'dispersed': P['lo'] = d
        elif s == 'BH': P['hi'] = d if P['hi'] is None else min(P['hi'], d); P['lo_try'] = d - 2*half
        else: say('lower end undecided after retry; stop'); save(); return
        save()
    # 2) the upper end must form a horizon
    while P['hi'] is None:
        d = P['hi_try']; s = one(d)
        if s == 'BH': P['hi'] = d
        elif s == 'dispersed': P['lo'] = max(P['lo'], d); P['hi_try'] = d + 2*half
        else: say('upper end undecided after retry; stop'); save(); return
        save()
    # 3) bisect
    while P['hi'] - P['lo'] > tol:
        lo, hi = P['lo'], P['hi']; moved = False
        for f in (0.5, 0.4, 0.6):
            d = lo + f*(hi - lo); s = one(d)
            if s == 'BH': P['hi'] = d; moved = True; break
            if s == 'dispersed': P['lo'] = d; moved = True; break
        save()
        if not moved: say('three points undecided; stop'); return
        if P['lo'] >= P['hi']: say('inconsistent outcomes; stop'); return
    lo, hi = P['lo'], P['hi']
    say(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.8f} +- {0.5*(hi-lo):.8f}  bracket [{lo:.8f}, {hi:.8f}]  ({P["
    t_used"]/60:.1f} min of runs)')
    json.dump(dict(tag=tag, runner=runner, kw=kw, dc=0.5*(lo+hi), w=hi-lo, lo=lo, hi=hi, log=P['log']), open(
    os.path.join(HERE, f'res_{tag}.json'), 'w'), indent=1)

if __name__ == '__main__':
    main()
```

A1_oddrate.py: odd rates and four-corner ranges.

```
# SCRIPT: NU-V6-ODDRATE  odd rate [tau(r)-tau(1/r)]/ln r with hard four-corner bracket ranges (v5.1 App. A.9,
    A.11)
import math, json, sys
DF, DI = 0.49773, 0.52832
def tau(d, df=DF, di=DI): return math.log((di-d)/(d-df))
def G(d, df=DF, di=DI): return (d-df)/(di-df)
def rate(dr, dir_, r, df=DF, di=DI): return (tau(dr,df,di)-tau(dir_,df,di))/math.log(r)
def corners(br, bir, r, df=DF, di=DI):
    # br, bir = (lo, hi) brackets for r and 1/r; tau decreases in delta
    vals = [rate(a, b, r, df, di) for a in br for b in bir]
    return min(vals), max(vals)
def even(dr, dir_, d1, df=DF, di=DI): return 0.5*(tau(dr,df,di)+tau(dir_,df,di))-tau(d1,df,di)
def summary(name, br, bir, r, d1=None):
    mr = 0.5*sum(br); mi = 0.5*sum(bir); out = dict(pair=name, r=r, dc_r=mr, dc_inv=mi, G_r=G(mr), G_inv=G(mi)
    )
    out['rate'] = rate(mr, mi, r); out['range'] = corners(br, bir, r)
    for s, lab in ((+7.8e-5, 'df+7.8e-5'), (-7.8e-5, 'df-7.8e-5')):
        out['rate_'+lab] = rate(mr, mi, r, DF+s); out['range_'+lab] = corners(br, bir, r, DF+s)
    if d1 is not None: out['even'] = even(mr, mi, d1)
    return out
if __name__ == '__main__':
    # reproduce v5.1's inner rates from its stored brackets
    for nm, br, bir, r in (('v5.1 all r=1.2', (0.51001440, 0.51001532), (0.51151422, 0.51151483), 1.2),
                           ('v5.1 all r=1.25', (0.50984047, 0.50984092), (0.51169535, 0.51169611), 1.25),
                           ('v5.1 shear r=1.2', (0.50995397, 0.50995435), (0.51157461, 0.51157499), 1.2)):
        s = summary(nm, br, bir, r, d1=0.510741)
        print(f"{nm}: rate {s['rate']:.4f} range [{s['range'][0]:.4f}, {s['range'][1]:.4f}] even {s['even
    ']:+.4f}  "
              f"df+: {s['rate_df+7.8e-5']:.4f}  df-: {s['rate_df-7.8e-5']:.4f}")
```

A1_RUNLOG.txt: every run.

```
[S1_r1] 08:06:02 S1_r1 delta=0.51044074: dispersed eps=6.3764 minI=1.398e-08 steps=59501 (63s)
[S1_r1] 08:07:08 S1_r1 delta=0.51104074: BH eps=6.3065 minI=1.452e-08 steps=58891 (67s)
[S1_r1] 08:08:32 S1_r1 delta=0.51074074: dispersed eps=7.1578 minI=5.273e-09 steps=66524 (83s)
[S1_r1] 08:09:22 S1_r1 delta=0.51089074: BH eps=6.6472 minI=1.201e-08 steps=61943 (50s)
[S1_r1] 08:10:37 S1_r1 delta=0.51081574: BH eps=7.0114 minI=4.796e-09 steps=65222 (75s)
[S1_r1] 08:12:11 S1_r1 delta=0.51077824: dispersed eps=7.3726 minI=6.468e-10 steps=69789 (93s)
[S1_r1] 08:13:26 S1_r1 delta=0.51079699: BH eps=7.1951 minI=1.736e-09 steps=66895 (75s)
[S1_r1] 08:14:52 S1_r1 delta=0.51078762: BH eps=7.3441 minI=5.725e-10 steps=68243 (86s)
[S1_r1] 08:16:14 S1_r1 delta=0.51078293: BH eps=7.3822 minI=4.855e-10 steps=69823 (82s)
[S1_r1] 08:19:18 S1_r1 delta=0.51078058: BH eps=7.3771 minI=5.615e-10 steps=93126 (retry at Courant 0.3 after
    crash) (184s)
[S1_r1] 08:20:27 S1_r1 delta=0.51077941: BH eps=7.3745 minI=6.030e-10 steps=69763 (69s)
[S1_r1] 08:22:00 S1_r1 delta=0.51077883: BH eps=7.3735 minI=6.246e-10 steps=69817 (93s)
[S1_r1] 08:22:00 RESULT S1_r1: delta_c = 0.51077853 +- 0.00000029  bracket [0.51077824, 0.51077883]  (17.0 min
     of runs)
[S1_all_r12] 08:23:01 S1_all_r12 delta=0.50971486: dispersed eps=6.4033 minI=1.387e-08 steps=60708 (60s)
[S1_all_r12] 08:23:50 S1_all_r12 delta=0.51031486: BH eps=6.2651 minI=1.500e-08 steps=59487 (49s)
[S1_all_r12] 08:25:00 S1_all_r12 delta=0.51001486: BH eps=7.3603 minI=1.109e-09 steps=70540 (70s)
[S1_all_r12] 08:26:17 S1_all_r12 delta=0.50986486: dispersed eps=6.6523 minI=1.206e-08 steps=62936 (77s)
[S1_all_r12] 08:27:44 S1_all_r12 delta=0.50993986: dispersed eps=6.8708 minI=1.072e-08 steps=64897 (87s)
[S1_all_r12] 08:29:14 S1_all_r12 delta=0.50997736: dispersed eps=7.0550 minI=8.927e-09 steps=66558 (90s)
[S1_all_r12] 08:30:44 S1_all_r12 delta=0.50999611: dispersed eps=7.2012 minI=3.950e-09 steps=67886 (90s)
[S1_all_r12] 08:32:17 S1_all_r12 delta=0.51000548: dispersed eps=7.3099 minI=1.866e-09 steps=68865 (93s)
[S1_all_r12] 08:33:50 S1_all_r12 delta=0.51001017: BH eps=7.3545 minI=1.244e-09 steps=70582 (93s)
[S1_all_r12] 08:35:22 S1_all_r12 delta=0.51000783: dispersed eps=7.3470 minI=1.321e-09 steps=69226 (92s)
[S1_all_r12] 08:36:28 S1_all_r12 delta=0.51000900: BH eps=7.3531 minI=1.239e-09 steps=70525 (66s)
[S1_all_r12] 08:37:26 S1_all_r12 delta=0.51000841: BH eps=7.3523 minI=1.241e-09 steps=70502 (58s)
[S1_all_r12] 08:37:26 RESULT S1_all_r12: delta_c = 0.51000812 +- 0.00000029  bracket [0.51000783, 0.51000841]
     (15.4 min of runs)
[S1_all_r083] 08:38:15 S1_all_r083 delta=0.51121453: dispersed eps=6.3985 minI=1.371e-08 steps=59685 (48s)
[S1_all_r083] 08:39:02 S1_all_r083 delta=0.51181453: BH eps=6.2840 minI=1.460e-08 steps=58676 (47s)
[S1_all_r083] 08:39:55 S1_all_r083 delta=0.51151453: dispersed eps=7.3409 minI=1.429e-09 steps=68152 (53s)
[S1_all_r083] 08:40:43 S1_all_r083 delta=0.51166453: BH eps=6.6004 minI=1.225e-08 steps=61509 (48s)
[S1_all_r083] 08:41:34 S1_all_r083 delta=0.51158953: BH eps=6.9061 minI=7.198e-09 steps=64257 (51s)
[S1_all_r083] 08:42:27 S1_all_r083 delta=0.51155203: BH eps=7.2338 minI=1.335e-09 steps=67236 (53s)
[S1_all_r083] 08:43:23 S1_all_r083 delta=0.51153328: BH eps=7.3722 minI=7.535e-10 steps=69854 (56s)
[S1_all_r083] 08:46:14 S1_all_r083 delta=0.51152391: crash eps=7.3712 minI=1.006e-09 steps=120537 (retry at
    Courant 0.3 after crash) (171s)
[S1_all_r083] 08:47:09 S1_all_r083 delta=0.51152203: BH eps=7.3678 minI=1.106e-09 steps=70416 (56s)
[S1_all_r083] 08:48:04 S1_all_r083 delta=0.51151828: BH eps=7.3604 minI=1.264e-09 steps=69615 (55s)
[S1_all_r083] 08:48:59 S1_all_r083 delta=0.51151641: BH eps=7.3582 minI=1.234e-09 steps=69661 (55s)
[S1_all_r083] 08:49:56 S1_all_r083 delta=0.51151547: dispersed eps=7.3572 minI=1.218e-09 steps=68511 (58s)
[S1_all_r083] 08:49:56 RESULT S1_all_r083: delta_c = 0.51151594 +- 0.00000047  bracket [0.51151547,
    0.51151641]  (12.5 min of runs)
[S1_all_r125] 08:50:53 S1_all_r125 delta=0.50954070: dispersed eps=6.3933 minI=1.398e-08 steps=60862 (56s)
[S1_all_r125] 08:51:52 S1_all_r125 delta=0.51014070: BH eps=6.2762 minI=1.494e-08 steps=59830 (59s)
[S1_all_r125] 08:52:47 S1_all_r125 delta=0.50984070: dispersed eps=7.3349 minI=1.529e-09 steps=69324 (55s)
[S1_all_r125] 08:53:36 S1_all_r125 delta=0.50999070: BH eps=6.5924 minI=1.248e-08 steps=62661 (49s)
[S1_all_r125] 08:54:28 S1_all_r125 delta=0.50991570: BH eps=6.8977 minI=7.530e-09 steps=65405 (52s)
[S1_all_r125] 08:55:22 S1_all_r125 delta=0.50987820: BH eps=7.2246 minI=1.375e-09 steps=68377 (54s)
[S1_all_r125] 08:56:20 S1_all_r125 delta=0.50985945: BH eps=7.3663 minI=7.446e-10 steps=70932 (58s)
[S1_all_r125] 08:57:22 S1_all_r125 delta=0.50985008: BH eps=7.3631 minI=1.016e-09 steps=77173 (62s)
[S1_all_r125] 08:58:19 S1_all_r125 delta=0.50984539: BH eps=7.3557 minI=1.271e-09 steps=70875 (57s)
[S1_all_r125] 08:59:14 S1_all_r125 delta=0.50984304: BH eps=7.3531 minI=1.244e-09 steps=70775 (56s)
[S1_all_r125] 09:00:10 S1_all_r125 delta=0.50984187: BH eps=7.3516 minI=1.250e-09 steps=70735 (55s)
[S1_all_r125] 09:01:04 S1_all_r125 delta=0.50984129: dispersed eps=7.3451 minI=1.349e-09 steps=69446 (55s)
[S1_all_r125] 09:01:04 RESULT S1_all_r125: delta_c = 0.50984158 +- 0.00000029  bracket [0.50984129,
    0.50984187]  (11.1 min of runs)
[S1_all_r08] 09:01:51 S1_all_r08 delta=0.51139573: dispersed eps=6.4126 minI=1.358e-08 steps=59808 (46s)
[S1_all_r08] 09:02:35 S1_all_r08 delta=0.51199573: BH eps=6.2681 minI=1.471e-08 steps=58531 (45s)
[S1_all_r08] 09:04:48 S1_all_r08 delta=0.51169573: crash eps=7.3622 minI=9.915e-10 steps=95691 (retry at
    Courant 0.3 after crash) (132s)
[S1_all_r08] 09:05:38 S1_all_r08 delta=0.51163573: dispersed eps=6.9552 minI=1.016e-08 steps=64673 (50s)
[S1_all_r08] 09:06:26 S1_all_r08 delta=0.51181573: BH eps=6.6618 minI=1.183e-08 steps=62057 (49s)
[S1_all_r08] 09:07:19 S1_all_r08 delta=0.51172573: BH eps=7.1911 minI=1.880e-09 steps=66841 (53s)
[S1_all_r08] 09:08:12 S1_all_r08 delta=0.51168073: dispersed eps=7.2826 minI=2.162e-09 steps=67642 (53s)
```

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[S1_all_r08] 09:09:07 S1_all_r08 delta=0.51170323: BH eps=7.3723 minI=7.725e-10 steps=69851 (55s)
[S1_all_r08] 09:10:03 S1_all_r08 delta=0.51169198: dispersed eps=7.3677 minI=1.128e-09 steps=72384 (56s)
[S1_all_r08] 09:10:57 S1_all_r08 delta=0.51169760: BH eps=7.3642 minI=9.452e-10 steps=69597 (54s)
[S1_all_r08] 09:11:54 S1_all_r08 delta=0.51169479: BH eps=7.3728 minI=9.793e-10 steps=72171 (57s)
[S1_all_r08] 09:12:49 S1_all_r08 delta=0.51169339: BH eps=7.3701 minI=1.052e-09 steps=70384 (56s)
[S1_all_r08] 09:13:47 S1_all_r08 delta=0.51169268: BH eps=7.3690 minI=1.090e-09 steps=69666 (57s)
[S1_all_r08] 09:13:47 RESULT S1_all_r08: delta_c = 0.51169233 +- 0.00000035  bracket [0.51169198, 0.51169268]
     (12.7 min of runs)
[S2_shear_r12] 09:14:39 S2_shear_r12 delta=0.50965416: dispersed eps=6.3939 minI=1.392e-08 steps=60625 (52s)
[S2_shear_r12] 09:15:27 S2_shear_r12 delta=0.51025416: BH eps=6.2758 minI=1.487e-08 steps=59584 (48s)
[S2_shear_r12] 09:16:21 S2_shear_r12 delta=0.50995416: dispersed eps=7.3383 minI=1.512e-09 steps=69118 (54s)
[S2_shear_r12] 09:17:12 S2_shear_r12 delta=0.51010416: BH eps=6.5916 minI=1.244e-08 steps=62411 (51s)
[S2_shear_r12] 09:18:04 S2_shear_r12 delta=0.51002916: BH eps=6.8962 minI=7.601e-09 steps=65149 (52s)
[S2_shear_r12] 09:18:57 S2_shear_r12 delta=0.50999166: BH eps=7.2215 minI=1.400e-09 steps=68106 (54s)
[S2_shear_r12] 09:19:54 S2_shear_r12 delta=0.50997291: BH eps=7.3657 minI=7.242e-10 steps=70677 (57s)
[S2_shear_r12] 09:22:14 S2_shear_r12 delta=0.50996354: crash eps=7.3622 minI=9.995e-10 steps=97995 (retry at
    Courant 0.3 after crash) (141s)
[S2_shear_r12] 09:23:12 S2_shear_r12 delta=0.50996166: BH eps=7.3586 minI=1.106e-09 steps=70606 (58s)
[S2_shear_r12] 09:24:11 S2_shear_r12 delta=0.50995791: BH eps=7.3544 minI=1.269e-09 steps=70582 (59s)
[S2_shear_r12] 09:25:10 S2_shear_r12 delta=0.50995604: BH eps=7.3521 minI=1.266e-09 steps=70499 (59s)
[S2_shear_r12] 09:26:10 S2_shear_r12 delta=0.50995510: BH eps=7.3508 minI=1.275e-09 steps=70452 (60s)
[S2_shear_r12] 09:26:10 RESULT S2_shear_r12: delta_c = 0.50995463 +- 0.00000047  bracket [0.50995416,
    0.50995510]  (12.4 min of runs)
[S2_shear_r083] 09:26:59 S2_shear_r083 delta=0.51127480: dispersed eps=6.3987 minI=1.373e-08 steps=59686 (49s)
[S2_shear_r083] 09:27:46 S2_shear_r083 delta=0.51187480: BH eps=6.2849 minI=1.462e-08 steps=58683 (47s)
[S2_shear_r083] 09:28:39 S2_shear_r083 delta=0.51157480: dispersed eps=7.3435 minI=1.390e-09 steps=68173 (54s)
[S2_shear_r083] 09:29:29 S2_shear_r083 delta=0.51172480: BH eps=6.6011 minI=1.226e-08 steps=61515 (50s)
[S2_shear_r083] 09:30:18 S2_shear_r083 delta=0.51164980: BH eps=6.9067 minI=7.131e-09 steps=64262 (49s)
[S2_shear_r083] 09:31:12 S2_shear_r083 delta=0.51161230: BH eps=7.2338 minI=1.356e-09 steps=67235 (53s)
[S2_shear_r083] 09:32:09 S2_shear_r083 delta=0.51159355: BH eps=7.3734 minI=7.545e-10 steps=69833 (57s)
[S2_shear_r083] 09:33:06 S2_shear_r083 delta=0.51158417: BH eps=7.3730 minI=1.003e-09 steps=72302 (57s)
[S2_shear_r083] 09:34:04 S2_shear_r083 delta=0.51157949: dispersed eps=7.3641 minI=1.250e-09 steps=71867 (58s)
[S2_shear_r083] 09:34:59 S2_shear_r083 delta=0.51158183: BH eps=7.3689 minI=1.126e-09 steps=69656 (55s)
[S2_shear_r083] 09:36:00 S2_shear_r083 delta=0.51158066: BH eps=7.3667 minI=1.192e-09 steps=74874 (61s)
[S2_shear_r083] 09:37:01 S2_shear_r083 delta=0.51158007: BH eps=7.3654 minI=1.225e-09 steps=75067 (61s)
[S2_shear_r083] 09:37:01 RESULT S2_shear_r083: delta_c = 0.51157978 +- 0.00000029  bracket [0.51157949,
    0.51158007]  (10.8 min of runs)
[S3_fine_r125] 09:44:34 S3_fine_r125 delta=0.50954158: dispersed eps=6.3238 minI=1.302e-08 steps=251302 (433s)
[S3_fine_r125] 09:51:56 S3_fine_r125 delta=0.51014158: BH eps=6.3877 minI=1.252e-08 steps=253893 (442s)
[S3_fine_r125] 10:00:02 S3_fine_r125 delta=0.50984158: dispersed eps=6.8938 minI=9.478e-09 steps=274210 (486s)
[S3_fine_r125] 10:07:55 S3_fine_r125 delta=0.50999158: BH eps=6.8336 minI=8.469e-09 steps=271826 (473s)
[S3_fine_r125] 10:16:26 S3_fine_r125 delta=0.50991658: dispersed eps=7.3615 minI=7.648e-10 steps=300480 (511s)
[S3_fine_r125] 10:24:19 S3_fine_r125 delta=0.50995408: BH eps=7.0871 minI=3.013e-09 steps=282065 (472s)
[S3_fine_r125] 10:32:32 S3_fine_r125 delta=0.50993533: BH eps=7.3334 minI=4.901e-10 steps=292023 (494s)
[S3_fine_r125] 10:40:50 S3_fine_r125 delta=0.50992595: BH eps=7.3673 minI=5.145e-10 steps=297801 (498s)
[S3_fine_r125] 10:49:20 S3_fine_r125 delta=0.50992127: BH eps=7.3601 minI=6.299e-10 steps=299550 (509s)
[S3_fine_r125] 10:57:40 S3_fine_r125 delta=0.50991892: BH eps=7.3664 minI=6.888e-10 steps=297945 (501s)
[S3_fine_r125] 11:06:00 S3_fine_r125 delta=0.50991775: BH eps=7.3635 minI=7.313e-10 steps=297430 (500s)
[S3_fine_r125] 11:14:26 S3_fine_r125 delta=0.50991716: BH eps=7.3628 minI=7.434e-10 steps=297096 (506s)
[S3_fine_r125] 11:14:26 RESULT S3_fine_r125: delta_c = 0.50991687 +- 0.00000029  bracket [0.50991658,
    0.50991716]  (97.1 min of runs)
[S3_fine_r08] 11:21:52 S3_fine_r08 delta=0.51139233: dispersed eps=6.3282 minI=1.250e-08 steps=250428 (425s)
[S3_fine_r08] 11:28:56 S3_fine_r08 delta=0.51199233: BH eps=6.3986 minI=1.199e-08 steps=253282 (424s)
[S3_fine_r08] 11:36:41 S3_fine_r08 delta=0.51169233: dispersed eps=6.8969 minI=9.201e-09 steps=273283 (465s)
[S3_fine_r08] 11:44:04 S3_fine_r08 delta=0.51184233: BH eps=6.8483 minI=7.920e-09 steps=271369 (443s)
[S3_fine_r08] 11:52:12 S3_fine_r08 delta=0.51176733: BH eps=7.3688 minI=8.229e-10 steps=296923 (488s)
[S3_fine_r08] 11:59:55 S3_fine_r08 delta=0.51172983: dispersed eps=7.0930 minI=7.610e-09 steps=281199 (463s)
[S3_fine_r08] 12:07:51 S3_fine_r08 delta=0.51174858: dispersed eps=7.2554 minI=2.622e-09 steps=287785 (476s)
[S3_fine_r08] 12:16:15 S3_fine_r08 delta=0.51175796: BH eps=7.3536 minI=1.182e-09 steps=295477 (503s)
[S3_fine_r08] 12:24:29 S3_fine_r08 delta=0.51175327: dispersed eps=7.3136 minI=1.695e-09 steps=290102 (495s)
[S3_fine_r08] 12:32:57 S3_fine_r08 delta=0.51175561: dispersed eps=7.3492 minI=1.263e-09 steps=291526 (508s)
[S3_fine_r08] 12:41:39 S3_fine_r08 delta=0.51175678: BH eps=7.3597 minI=1.168e-09 steps=296395 (522s)
13:15:02 [note] queue background window expired during the retry of S3_fine_r08 delta=0.51175620 (first
    attempt at Courant 0.2 did not end in BH/dispersed); relaunched, the point is rerun from scratch
14:25:06 [note] S3_fine_r08 delta=0.51175620 stopped by hand after 70 min on the relaunch (and 33 min before
    the first window expired) without ending; reported as UNDECIDED (did not finish), never moves the bracket
    . Next: the 0.4 point of the bracket, one attempt at Courant 0.2 with a 25-min wall limit, retry at 0.15
    with 40 min.
```

```
14:33:37 [S3_fine_r08] S3_fine_r08 delta=0.51175608: dispersed eps=7.3579 minI=1.182e-09 steps=292476 (0.4
    point, Courant 0.2, 487s)
[S3_fine_r08] 14:33:38 RESULT S3_fine_r08: delta_c = 0.51175643 +- 0.00000035  bracket [0.51175608,
    0.51175678]  (95.0 min of runs)
QUEUE DONE (S1, S2, S3 all landed; written by hand after the S3_fine_r08 closure)
[C1_r1_v51runner] 14:35:15 C1_r1_v51runner delta=0.51072222: dispersed eps=7.0339 minI=9.787e-09 steps=65400
    (50s)
[C1_r1_v51runner] 14:36:12 C1_r1_v51runner delta=0.51078883: BH eps=7.3019 minI=8.007e-10 steps=67874 (57s)
[C1_r1_v51runner] 14:37:04 C1_r1_v51runner delta=0.51075552: dispersed eps=7.3312 minI=1.477e-09 steps=68083
    (52s)
[C1_r1_v51runner] 14:37:57 C1_r1_v51runner delta=0.51077218: BH eps=7.3660 minI=8.286e-10 steps=69690 (53s)
[C1_r1_v51runner] 14:38:54 C1_r1_v51runner delta=0.51076385: BH eps=7.3647 minI=1.091e-09 steps=71895 (57s)
[C1_r1_v51runner] 14:39:47 C1_r1_v51runner delta=0.51075969: BH eps=7.3575 minI=1.263e-09 steps=69668 (53s)
[C1_r1_v51runner] 14:40:42 C1_r1_v51runner delta=0.51075761: BH eps=7.3554 minI=1.225e-09 steps=69621 (55s)
[C1_r1_v51runner] 14:41:36 C1_r1_v51runner delta=0.51075657: dispersed eps=7.3489 minI=1.328e-09 steps=68285
    (54s)
[C1_r1_v51runner] 14:42:33 C1_r1_v51runner delta=0.51075709: BH eps=7.3549 minI=1.220e-09 steps=69599 (57s)
[C1_r1_v51runner] 14:42:33 RESULT C1_r1_v51runner: delta_c = 0.51075683 +- 0.00000026  bracket [0.51075657,
    0.51075709]  (8.1 min of runs)
QUEUE2 DONE
```

B.20 Self-scattering as the flux–shear link: algebra and frozen prediction (Section 10)

A2_PREREGISTERED.txt: method, predictions and comparison.

```
PREREGISTERED --- Task A2 (self-scattering and the flux-shear link law)
Written 2026-11-11T14:04Z, before the script that computes r_s(t) along McCormick's background was written or
    run.

DISCLOSURE (what was already seen when this was written)
- The two thresholds to be predicted are in the task text and in McCormick's App. C(d): with nu-nu 0.51941 +-
    0.00016
  (res_mg_phys6_10.0.json), without 0.52639 +- 0.00011 (res_mg_nonunu6_10.0.json), fluid 0.49742 (
    res_eos_pf_10.0.json).
- The paper quotes nu-nu faster than nu-e+- by 1.5 (nu_e) and 7 (nu_mu,tau). While reading, a rough mental
    estimate with
  those two quoted ratios and flavour weights 1/3, 2/3 was made. It is NOT the prediction below; the
    prediction uses the
  rates exactly as run_mg.py evaluates them along the background, with the weighting fixed here and not
    changed afterwards.
- McCormick's single-rate thresholds at T_H = 1 MeV (res_eosc_<Tdec>_10.0.json) were listed: Tdec 0.75:
    0.51586,
  1.0: 0.52076, 1.5: 0.52583, 2.0: 0.52837, 3.0: 0.53065. These are used by the opposite (timing) reading
    below.

REFERENCES FOR G AT T_H = 1 MeV
- delta_fluid = 0.49742 (Table VII; res_eos_pf_10.0.json, full EOS).
- delta_inst: McCormick's archive has NO instantaneous free-streaming threshold at T_H = 1 MeV with the full
    EOS
  (res_eos_fs_*.json exist only for T0 <= 5 MeV, T_H <= 0.506 MeV; earlier_start/ has T_H = 0.06, 0.12 only).
  The nearest instantaneous references are pure radiation, eps0 = 0.1 (his Table II):
  R_nu = 0.409 -> 0.52832, R_nu = 0.488 -> 0.53379.  0.52832 is ruled out as a reference at this mass because
    his own
  single-rate run with Tdec = 3 MeV at T_H = 1 MeV already sits higher (0.53065), which would put G > 1.
  PRIMARY: delta_inst = 0.53379 (R_nu = 0.488, the neutrino fraction while e+- are relativistic, which is the
    epoch
  T >~ 1 MeV in which these neutrinos decouple). SENSITIVITY: delta_inst interpolated linearly in R_nu (his Eq
    . 18
  says the shift is linear in R_nu) at the background R_nu at the e+- flux-link last-scattering epoch,
    computed by the
  script from his thermo table.
- G = (delta_c - delta_fluid)/(delta_inst - delta_fluid), tau_G = ln[(1-G)/G].

READING 1 (the frozen link law, primary)
```

```
McCormick's C3 with the first-order target is the v5.1 all-l>=2 family exactly: flux at k1 = k_e,s, every l>=2
     at
k2 = k_e,s + k_n,s, so r_s = (k_e,s + k_n,s)/k_e,s >= 1, with k_e,s IDENTICAL in his two runs (nunu only
    zeroes k_n).
That is the v5.1 intervention (flux rate held, shear+higher rate changed), with r now depending on group,
    flavour and time.
- r_s(t): from run_mg.rates(Tb) exactly as the code evaluates it on the background (FD rates, chi_e at the
    background
  temperature, both rates capped at 100 H, 6 groups 0.4..20, two flavours), in the transport phase (after the
    switch
  from the perfect fluid at total nu_mu,tau rate at x=1 = 100 H).
- Weighting (fixed now): the link rapidity u_s(t) = (1/2) ln r_s(t) is averaged
  (a) over time with the VISIBILITY of the flux link of that group, g_s = (k_e,s/H) exp(-tau_e,s) per d ln a,
      tau_e,s = integral of k_e,s dt to the future. Why: in v5.1 the law was measured with the flux rate's
    decoupling
      history held fixed and r constant; the flux link is what lets the neutrinos go, so the rapidity it reads
     is the
      one in force while the flux link releases --- the last-scattering density of that link.
  (b) over groups and flavours with the background energy weights (flavour 1/3, 2/3 times the Fermi-Dirac
    energy
      fraction of each group on the 0.4..20 grid, as run_mg.spectral_weights at tau = 1). Why: the threshold
    reads the
      neutrino anisotropic stress, which is energy-weighted.
  The rapidity is averaged (not r), because the v5.1 law is linear in the rapidity.
- Prediction: tau_G(with) = tau_G(without) + rate * u_eff, rate = 1.11 (all-l>=2 family; primary) and
    1.19-1.20
  (shear-only bracket), with tau_G(without) from McCormick's measured 0.52639 and the references above.
    Converted back
  to delta_c(with) and to a predicted shift in G.
- Sensitivity weightings, reported but NOT deciding: visibility of the total rate k_e+k_n; uniform in ln a
    while
  0.1 < k_e,s/H < 10; v5.1's memory-form readout epochs eps* = 1.7 and 4.0 mixed 0.89/0.11.

KILL LINE (Reading 1, primary weighting, primary reference, rate 1.11):
  KILLED if the predicted shift in G has the wrong sign relative to the measured shift, or if
  |predicted shift in G - measured shift in G| > 0.5 * |measured shift in G|.
  Measured shift in G = G(0.51941) - G(0.52639) with the same references.

READING 2 (the opposite: timing, not link rapidity)
Self-scattering keeps the neutrinos isotropic longer: it delays decoupling, as if it were part of the flux
    rate.
Then the threshold follows a single-rate decoupling temperature set by the TOTAL rate, and the link rapidity
    plays no role.
- Tdec_eff is where the energy-weighted mean rate over groups and flavours (same weights as above) equals H on
     the
  background, computed from the same rate functions: from k_e alone (no nu-nu) and from k_e + k_n (with nu-nu)
    .
- Each Tdec_eff is mapped to a threshold by McCormick's single-rate family at T_H = 1 MeV (listed above),
    linear in
  ln Tdec. Predicts delta_c(with), delta_c(without) and their shift in G.
- Same kill line applied to Reading 2's predicted shift in G (and its two absolute thresholds reported).
Both readings predict the same direction (nu-nu lowers delta_c); the size separates them, and if both pass,
    the
numbers do not separate them and the decisive simulation is stated instead.

Results are appended below this line; nothing above is edited afterwards.
-------------------------------------------------------------------------------------------------------------


[2026-11-11T14:06Z] PREDICTIONS (step3_predict.py, step3_out.txt; the with-nu-nu threshold is not used by it)
Setup notes found while computing: on McCormick's background r_s is almost constant in time and energy
(nu_e 2.43-2.45, nu_mu,tau 7.69-7.77 for 0.3 < T_b < 5 MeV; caps make r = 2 only for the highest groups above
    ~5 MeV,
where they carry no visibility weight). So every weighting gives the same rapidity to 0.2%.
READING 1, primary (visibility of k_e, energy-weighted, d_inst = 0.53379):
  u_eff = 0.8313 (r_eff = 5.27); flavour means u = 0.4468 (nu_e), 1.0235 (nu_mu,tau).
  G(without) = 0.7965 (from 0.52639).
  rate 1.11:      G(with) = 0.6087, predicted shift in G = -0.1878, predicted delta_c(with) = 0.51956
```

```
  rate 1.19-1.20: G(with) = 0.5928-0.5908, shift -0.2037 to -0.2057, delta_c(with) = 0.51898-0.51891
  Sensitivity weightings and the R_nu-interpolated d_inst (0.53377) move delta_c(with) by <= 0.00002.
READING 2 (timing): Tdec_eff = 1.552 MeV without nu-nu, 0.979 MeV with.
  Single-rate family at T_H = 1 MeV: delta_c(without) = 0.52613, delta_c(with) = 0.52040; shift -0.00573;
  shift in G = -0.1576 (d_inst 0.53379). Anchored on the measured 0.52639: delta_c(with) = 0.52066.

[2026-11-11T14:06Z] COMPARISON (step4_compare.py, step4_out.txt)
Measured (d_inst 0.53379): G(without) = 0.7966, G(with) = 0.6045, shift in G = -0.1920 +- 0.0052; kill
    tolerance 0.0960.
READING 1, rate 1.11 (deciding): predicted shift -0.1878, miss +0.0043 (2.2% of the shift); delta_c(with)
    0.51956 vs
  0.51941 +- 0.00016 (miss +0.00015). SURVIVES. Rate 1.19 / 1.20: misses 6.1% / 7.1%, delta_c 0.51898 /
    0.51891. SURVIVE.
READING 2 (timing): predicted shift -0.1576, miss +0.0344 (17.9%). SURVIVES. Absolute: without 0.52613 (miss
    -0.00026),
  with 0.52040 (miss +0.00099); anchored 0.52066 (miss +0.00125).
Not pre-registered, reported as sensitivity: Reading 1's absolute prediction moves with the d_inst proxy
  (0.5315 -> miss +0.00162; 0.5370 -> miss -0.00139); the kill line holds over that whole scan. Using v5.1's
    eleven
  measured offsets directly instead of the linear law: delta_c(with) = 0.51930 (miss -0.00011).
```

A2_step2_algebra.py: the operator on the bins, from McCormick's compiled right-hand side.

```
# SCRIPT: A2_STEP2_SELFSCATTER_ALGEBRA
"""Exact algebra of McCormick's multigroup collision term (App. C, Eq. C3; ms_mg.py, nnmode=0) for one frozen
    group.

Checks, on his own 32-bin angle grid (mue = linspace(-1,1,33), midpoints, weight dmu):
 1. the collision operator C[I] = k_e (eps/4pi - I) + k_n (eps/4pi + c mu - I), with c = q/(2pi sum mu^2 dmu),
    built line-for-line from ms_mg.py, equals -diag(0, k_e, k_e+k_n, ..., k_e+k_n) in the discrete orthonormal
    Legendre basis (Gram-Schmidt of 1, mu, mu^2, ... under <f,g> = sum f g dmu);
 2. the k_n part of that operator, taken from McCormick's compiled rhs() itself (two calls differing only in
     kbn,
    with kbe = 0, so nothing else in the right-hand side moves), equals the formula of 1;
 3. (1/8)||[L,L*]||_F^2 = k^2 (c0^2 k_e^2 + c1^2 k_n^2) exactly for L = -i k M - D, M the discrete mu operator
    ,
    with c0^2, c1^2 the discrete Jacobi weights, and these tend to 1/3 and 4/15 as N_mu grows.
"""
import sys, os
import numpy as np
SIM = '/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/steve/src/simulation'

K = 32
mue = np.linspace(-1, 1, K+1); dmu = mue[1]-mue[0]; mu = 0.5*(mue[:-1]+mue[1:])
sum_mu2 = np.sum(mu*mu*dmu)

def coll(I, ke, kn):
    """ms_mg.py collision lines (nnmode = 0), lapse A = 1."""
    eps = 2*np.pi*np.sum(I)*dmu; q = 2*np.pi*np.sum(I*mu)*dmu
    cs = q/(2*np.pi*sum_mu2)
    tgt = eps/(4*np.pi) + cs*mu
    return ke*(eps/(4*np.pi) - I) + kn*(tgt - I)

def legendre_basis(K, mu, dmu):
    """Discrete orthonormal polynomials on the bins: columns of Q, <Q_l, Q_m> = delta (weight dmu)."""
    V = np.vander(mu, K, increasing=True)
    W = np.sqrt(dmu)
    Q, Rr = np.linalg.qr(W*V)          # orthonormal in the plain dot product of sqrt(dmu)*f
    sgn = np.sign(np.diag(Rr)); Q = Q*sgn
    return Q                            # columns are sqrt(dmu) * P_l(mu_k)

Q = legendre_basis(K, mu, dmu)
ke, kn = 0.7, 3.1
Cmat = np.column_stack([coll(np.eye(K)[:, j], ke, kn) for j in range(K)])      # acts on I (bin values)
# in the orthonormal basis: f = Q^T sqrt(dmu) I  ->  C_hat = Q^T diag(sqrt dmu) C diag(1/sqrt dmu) Q
Wh = np.sqrt(dmu)
Chat = Q.T @ (Wh*Cmat/Wh) @ Q
target = -np.diag(np.r_[0.0, ke, np.full(K-2, ke+kn)])
```

```
err1 = np.max(np.abs(Chat - target))
print(f'1. max |C_hat - (-diag(0,k_e,k_e+k_n,...))| on 32 bins = {err1:.2e}  (orthonormality error {np.max(np.
    abs(Q.T@Q-np.eye(K))):.1e})')
# Note: high-degree monomials make QR ill-conditioned; the claim needs only span{1}, span{1,mu} to be exact.
# Basis-free check: C = -k_e (1 - P0) - k_n (1 - P01) with P0, P01 the bin-orthogonal projectors.
one = np.ones(K)
P0 = np.outer(one, one*dmu)/np.sum(one*one*dmu)
P1 = np.outer(mu, mu*dmu)/np.sum(mu*mu*dmu)
Cproj = -ke*(np.eye(K)-P0) - kn*(np.eye(K)-P0-P1)
print(f'   basis-free: max |C - [-k_e(1-P0) - k_n(1-P0-P1)]| = {np.max(np.abs(Cmat-Cproj)):.2e};'
      f' <1,mu> = {np.sum(mu*dmu):.1e}')
ev = np.sort(np.linalg.eigvals(Cmat).real)
print(f'   eigenvalues: one at {ev[-1]:.3e}, one at {ev[-2]:.15f} (-k_e = {-ke}), {np.sum(np.abs(ev[:-2]+ke+kn
    ) < 1e-12)} at -(k_e+k_n)')

# 2. the k_n part from McCormick's compiled rhs()
sys.path.insert(0, SIM)
try:
    from ms_mg import rhs
    N = 4; G = 1
    R = np.linspace(1.0, 2.0, N+1); U = np.full(N+1, 0.3); Gam = np.ones(N+1); e = np.full(N, 1.0)
    rng = np.random.default_rng(1)
    I = 1.0 + 0.3*rng.standard_normal((1, N, K))
    le = np.log(np.array([1e-3, 1e3])); lp = le - np.log(3.0); lg = np.zeros(2); lT = 0.25*le
    lchi = np.zeros(2)
    out = []
    for kbn in (0.0, 1.0):
        dR = np.zeros(N+1); dU = np.zeros(N+1); dG = np.zeros(N+1); de = np.zeros(N); dI = np.zeros(I.shape)
        res = rhs(0.0, R, U, Gam, e, I, mue, le, lp, lg, lT, np.zeros(1), np.full(1, kbn), 1.0, 0.0, 4.0, G,
     1.0, 0.0,
                  np.zeros(1), dR, dU, dG, de, dI, lchi, 0.0, 0)
        out.append((dI.copy(), res[1].copy(), res[6].copy()))
    # enub = 0 makes fn = 0 in state(); use enub > 0 instead so kap_n = kbn * eps/enub
    out = []
    for kbn in (0.0, 1.0):
        dR = np.zeros(N+1); dU = np.zeros(N+1); dG = np.zeros(N+1); de = np.zeros(N); dI = np.zeros(I.shape)
        res = rhs(0.0, R, U, Gam, e, I, mue, le, lp, lg, lT, np.zeros(1), np.full(1, kbn), 1.0, 1.0, 4.0, G,
     1.0, 0.0,
                  np.zeros(1), dR, dU, dG, de, dI, lchi, 0.0, 0)
        out.append((dI.copy(), res[1].copy(), res[6].copy()))
    diff = out[1][0] - out[0][0]
    Ac = out[1][1]; kapn = out[1][2]
    err2 = 0.0
    for i in range(N):
        pred = Ac[i]*coll(I[0, i], 0.0, kapn[0, i])
        err2 = max(err2, np.max(np.abs(diff[0, i] - pred)))
    print(f'2. McCormick rhs(): k_n part minus formula, max abs = {err2:.2e} (scale of I ~ 1; kap_n = {kapn[0]
     })')
except Exception as ex:
    print('2. could not call ms_mg.rhs:', repr(ex))

# 3. connection energy on the discrete chain
def energy(K, ke, kn, k=1.3):
    mue = np.linspace(-1, 1, K+1); dmu = mue[1]-mue[0]; mu = 0.5*(mue[:-1]+mue[1:])
    # discrete Jacobi (Lanczos) on weights dmu: stable orthonormal basis
    w = np.full(K, dmu)
    q_prev = np.zeros(K); q = np.ones(K)/np.sqrt(np.sum(w)); a = []; b = []
    Qs = [q]
    for l in range(K-1):
        v = mu*q; al = np.sum(w*q*v); v = v - al*q - (b[-1] if b else 0.0)*q_prev
        for qq in Qs: v -= np.sum(w*qq*v)*qq          # full reorthogonalisation
        be = np.sqrt(np.sum(w*v*v)); a.append(al); b.append(be)
        q_prev, q = q, v/be; Qs.append(q)
    M = np.diag(b, 1) + np.diag(b, -1)                  # alpha = 0 by symmetry
    Dd = np.diag(np.r_[0.0, ke, np.full(K-2, ke+kn)])
    L = -1j*k*M - Dd
    Cm = L@L.conj().T - L.conj().T@L
    lhs = 0.125*np.sum(np.abs(Cm)**2)
```

```
    rhs_ = k*k*(b[0]**2*ke**2 + b[1]**2*kn**2)
    return lhs, rhs_, b[0]**2, b[1]**2, k*k*(ke**2/3 + 4*kn**2/15)
for KK in (32, 64, 128, 1024):
    lhs, r_, c0, c1, cont = energy(KK, ke, kn)
    print(f'3. N_mu={KK:5d}: (1/8)||[L,L*]||^2 = {lhs:.15f}, k^2(c0^2 ke^2 + c1^2 kn^2) = {r_:.15f}, rel diff
     {abs(lhs-r_)/r_:.1e};'
          f' c0^2 = {c0:.10f} (1/3 - dmu^2/12 = {1/3-(2/KK)**2/12:.10f}), c1^2 = {c1:.10f} (4/15 = {4/15:.10f}
     );'
          f' continuum form k^2(ke^2/3+4kn^2/15) = {cont:.10f}, rel {abs(lhs-cont)/cont:.1e}')
```

`A2_step2_out.txt`: output.

```
1. max |C_hat - (-diag(0,k_e,k_e+k_n,...))| on 32 bins = 3.55e-15  (orthonormality error 8.9e-16)
   basis-free: max |C - [-k_e(1-P0) - k_n(1-P0-P1)]| = 1.11e-16; <1,mu> = 0.0e+00
   eigenvalues: one at -1.647e-16, one at -0.700000000000000 (-k_e = -0.7), 30 at -(k_e+k_n)
2. McCormick rhs(): k_n part minus formula, max abs = 4.88e-15 (scale of I ~ 1; kap_n = [12.4512372
    12.12843302 12.39337775 12.44816255])
3. N_mu=   32: (1/8)||[L,L*]||^2 = 4.589752832031250, k^2(c0^2 ke^2 + c1^2 kn^2) = 4.589752832031252, rel diff
     3.9e-16; c0^2 = 0.3330078125 (1/3 - dmu^2/12 = 0.3330078125), c1^2 = 0.2656250000 (4/15 = 0.2666666667);
     continuum form k^2(ke^2/3+4kn^2/15) = 4.6069400000, rel 3.7e-03
3. N_mu=   64: (1/8)||[L,L*]||^2 = 4.602643208007812, k^2(c0^2 ke^2 + c1^2 kn^2) = 4.602643208007815, rel diff
     5.8e-16; c0^2 = 0.3332519531 (1/3 - dmu^2/12 = 0.3332519531), c1^2 = 0.2664062500 (4/15 = 0.2666666667);
     continuum form k^2(ke^2/3+4kn^2/15) = 4.6069400000, rel 9.3e-04
3. N_mu=  128: (1/8)||[L,L*]||^2 = 4.605865802001954, k^2(c0^2 ke^2 + c1^2 kn^2) = 4.605865802001953, rel diff
     1.9e-16; c0^2 = 0.3333129883 (1/3 - dmu^2/12 = 0.3333129883), c1^2 = 0.2666015625 (4/15 = 0.2666666667);
     continuum form k^2(ke^2/3+4kn^2/15) = 4.6069400000, rel 2.3e-04
3. N_mu= 1024: (1/8)||[L,L*]||^2 = 4.606923215656280, k^2(c0^2 ke^2 + c1^2 kn^2) = 4.606923215656279, rel diff
     1.9e-16; c0^2 = 0.3333330154 (1/3 - dmu^2/12 = 0.3333330154), c1^2 = 0.2666656494 (4/15 = 0.2666666667);
     continuum form k^2(ke^2/3+4kn^2/15) = 4.6069400000, rel 3.6e-06
```

`A2_step3_predict.py`: link rapidity along his background and the frozen predictions.

```
# SCRIPT: A2_STEP3_LINK_PREDICTION
"""Prediction of McCormick's T_H = 1 MeV multigroup threshold WITH nu-nu scattering, from the one WITHOUT it,
by the frozen v5.1 flux-shear link law (Reading 1) and by the opposite timing reading (Reading 2).
Rates exactly as run_mg.rates(Tb) evaluates them on the background (G=6, 0.4..20, FD, chi_e at T_b, cap 100 H)
    .
Weighting fixed in PREREGISTERED.txt before this script was written. Does NOT use the with-nu-nu threshold."""
import sys, json
import numpy as np
SIM = '/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/steve/src/simulation'
sys.path.insert(0, SIM)
import run_mg as M

G, XLO, XHI, CAP, SWITCH = 6, 0.4, 20.0, 100.0, 100.0
edges, xg, dl = M.groups(G, XLO, XHI)
NF = 2
wE = M.spectral_weights(edges, 1.0)[:, 0]
W = np.concatenate([M.FR[0]*wE, M.FR[1]*wE])               # energy weight of each (flavour, group), sums to 1
ce_e = M.coef('e_FD', xg)('nu_e'); ce_m = M.coef('mu_FD', xg)('nu_e'); cn = M.coef('e_FD', xg)('nu_nu')
c1 = M.coef('mu_FD', np.array([1.0])); c_sw = (c1('nu_e')[0], c1('nu_nu')[0])

def rates_uncapped(Tb, nunu=True):
    H, Tnu, chi = M.background(Tb)
    re = np.concatenate([ce_e, ce_m])*M.GF2*np.tile(xg, NF)*Tnu*Tb**4*chi/H
    rn = np.tile(cn, NF)*M.GF2*np.tile(xg, NF)*Tnu**5/H*(1.0 if nunu else 0.0)
    return re, rn
def rates(Tb, nunu=True):                                   # == run_mg.rates (mode 'phys', scale 1)
    re, rn = rates_uncapped(Tb, nunu)
    return np.minimum(re, CAP), np.minimum(rn, CAP)
def tight_at(Tb, nunu=True):
    H, Tnu, chi = M.background(Tb)
    tot = (c_sw[0]*Tb**4*chi + c_sw[1]*Tnu**4*(1.0 if nunu else 0.0))*M.GF2*Tnu/H
    return tot > SWITCH

# background clock: ln a = -ln T_nu ; eps = (T_nu/H)/(T_nu/H)|_{T_H}
TH = 1.00296                                                # T_b at eps = 1 in res_mg_*_10.0.json
Tb = np.exp(np.linspace(np.log(10.0), np.log(0.02), 40001))
```

```
H, Tnu, chi = M.background(Tb)
lna = -np.log(Tnu)
HH, TnuH, _ = M.background(TH)
eps = (Tnu/H)/(TnuH/HH)
RE = np.zeros((len(Tb), 2*G)); RN = np.zeros((len(Tb), 2*G)); RE0 = np.zeros_like(RE); RN0 = np.zeros_like(RE)
for j, T in enumerate(Tb):
    RE[j], RN[j] = rates(T); RE0[j], RN0[j] = rates_uncapped(T)
tight_nn = np.array([tight_at(T, True) for T in Tb]); tight_no = np.array([tight_at(T, False) for T in Tb])
jsw_nn = np.argmax(~tight_nn); jsw_no = np.argmax(~tight_no)
print(f'switch to transport: with nu-nu at T_b = {Tb[jsw_nn]:.3f} MeV (eps {eps[jsw_nn]:.3f}); without at {Tb[
    jsw_no]:.3f} MeV (eps {eps[jsw_no]:.3f})')
print(f'eps at T_b = 1.6, 1.0, 0.5, 0.2 MeV: ' + ', '.join(f'{np.interp(-np.log(T), -np.log(Tb), eps):.3f}'
    for T in (1.6, 1.0, 0.5, 0.2)))

r = (RE + RN)/RE
u = 0.5*np.log(r)
print('\nr_s on the background (rates as the code evaluates them, capped):')
print('  T_b   eps   ' + ' '.join(f'e{g}:{xg[g]:.2f}' for g in range(G)) + ' | ' + ' '.join(f'm{g}' for g in
    range(G)))
for T in (8.0, 5.0, 3.0, 2.0, 1.6, 1.3, 1.0, 0.8, 0.6, 0.4, 0.3, 0.2, 0.15):
    j = np.argmin(np.abs(Tb - T))
    print(f'  {Tb[j]:5.2f} {eps[j]:5.2f}  ' + ' '.join(f'{v:7.3f}' for v in r[j]))
print('\nk_e/H and k_n/H at x_s (uncapped) at T_b = 2, 1, 0.5 MeV:')
for T in (2.0, 1.0, 0.5):
    j = np.argmin(np.abs(Tb - T))
    print(f'  T={Tb[j]:.2f}: k_e/H ' + ' '.join(f'{v:8.3f}' for v in RE0[j]) + '\n          k_n/H ' + ' '.join
    (f'{v:8.3f}' for v in RN0[j]))

def visibility(rate, j0):
    """g = (k/H) exp(-tau) per d ln a, tau = integral of k/H d ln a to the future; zero before the switch j0
    ."""
    d = np.diff(lna)
    mid = 0.5*(rate[1:] + rate[:-1])
    tau = np.zeros(len(rate)); tau[:-1] = np.cumsum((mid*d)[::-1])[::-1]
    g = rate*np.exp(-tau); g[:j0] = 0.0
    return g, tau
def wavg(f, g):
    num = np.trapezoid(f*g, lna); den = np.trapezoid(g, lna)
    return num/den, den

U = {}
# (a) primary: visibility of the flux link k_e,s
ueff_s = np.zeros(2*G); norm_s = np.zeros(2*G); Teps = np.zeros(2*G)
for s in range(2*G):
    g, tau = visibility(RE[:, s], jsw_nn)
    ueff_s[s], norm_s[s] = wavg(u[:, s], g)
    Teps[s] = np.exp(-wavg(-np.log(Tb), g)[0])
U['primary: visibility of k_e'] = float(np.sum(W*ueff_s))
print('\nper (flavour, group): energy weight, <u>_vis(k_e), r_eff = exp(2<u>), visibility norm, last-
    scattering T_b')
for s in range(2*G):
    print(f'  {"nu_e " if s < G else "nu_mt"} x={xg[s % G]:5.2f}  w={W[s]:.4f}  <u>={ueff_s[s]:.4f}  r_eff={np
    .exp(2*ueff_s[s]):.3f}  norm={norm_s[s]:.4f}  T_ls={Teps[s]:.3f}')
for f, nm in ((0, 'nu_e'), (1, 'nu_mt')):
    sl = slice(f*G, (f+1)*G)
    print(f'  flavour {nm}: <u> = {np.sum(W[sl]*ueff_s[sl])/np.sum(W[sl]):.4f}')
# sensitivity weightings
ueff_t = np.zeros(2*G)
for s in range(2*G):
    g, _ = visibility(RE[:, s] + RN[:, s], jsw_nn); ueff_t[s] = wavg(u[:, s], g)[0]
U['sens: visibility of k_e+k_n'] = float(np.sum(W*ueff_t))
ueff_w = np.zeros(2*G)
for s in range(2*G):
    g = ((RE0[:, s] > 0.1) & (RE0[:, s] < 10.0)).astype(float); g[:jsw_nn] = 0.0; ueff_w[s] = wavg(u[:, s], g)
    [0]
U['sens: uniform in ln a, 0.1<k_e/H<10'] = float(np.sum(W*ueff_w))
uu17 = np.array([np.interp(1.7, eps, u[:, s]) for s in range(2*G)]); uu40 = np.array([np.interp(4.0, eps, u[:,
    s]) for s in range(2*G)])
```

```
U['sens: v5.1 readout eps*=1.7/4.0 (0.89/0.11)'] = float(np.sum(W*(0.89*uu17 + 0.11*uu40)))

# R_nu at the flux-link last-scattering epoch (for the sensitivity reference)
rpl = np.interp(Tb, M._T, M._rpl); rnu = np.interp(Tb, M._T, M._rnu); Rnu = rnu/(rnu + rpl)
T_ls = float(np.exp(np.sum(W*np.log(Teps))))
Rnu_ls = float(np.interp(-np.log(T_ls), -np.log(Tb), Rnu))
print(f'\nenergy-weighted flux-link last-scattering T_b = {T_ls:.3f} MeV, R_nu there = {Rnu_ls:.4f}; R_nu(T_H)
      = {np.interp(-np.log(TH), -np.log(Tb), Rnu):.4f}')

d_fluid = 0.49742; d_without = 0.52639
d_inst_P = 0.53379
d_inst_S = 0.52832 + (0.53379 - 0.52832)*(Rnu_ls - 0.409)/(0.488 - 0.409)
def Gof(d, di): return (d - d_fluid)/(di - d_fluid)
def tau(Gv): return np.log((1 - Gv)/Gv)
def inv(t, di):
    Gv = 1/(1 + np.exp(t)); return d_fluid + Gv*(di - d_fluid), Gv
out = dict(U=U, d_inst_primary=d_inst_P, d_inst_sens=d_inst_S, Rnu_ls=Rnu_ls, T_ls=T_ls, reading1={}, reading2
    ={})
print('\nREADING 1 (frozen link law) predictions for delta_c WITH nu-nu, from delta_c without = 0.52639:')
for refname, di in (('primary d_inst=0.53379', d_inst_P), (f'sens d_inst={d_inst_S:.5f}', d_inst_S)):
    G0 = Gof(d_without, di); t0 = tau(G0)
    for wname, ue in U.items():
        for rate in (1.11, 1.19, 1.20):
            dpred, Gp = inv(t0 + rate*ue, di)
            key = f'{refname} | {wname} | rate {rate}'
            out['reading1'][key] = dict(u_eff=ue, r_eff=float(np.exp(2*ue)), G_without=G0, G_with_pred=Gp,
     dG_pred=Gp - G0, delta_with_pred=dpred)
            print(f'  {key}: u_eff={ue:.4f} (r_eff {np.exp(2*ue):.3f})  G_without={G0:.4f} -> G_with={Gp:.4f}
      dG={Gp-G0:+.4f}  delta_c(with)={dpred:.5f}')

# READING 2: timing. Energy-weighted mean rate = H.
def Tdec_eff(nunu):
    m = np.array([np.sum(W*(RE0[j] + (RN0[j] if nunu else 0.0))) for j in range(len(Tb))])
    j = np.argmax(m < 1.0)
    # log interpolation between j-1 and j
    x0, x1 = np.log(m[j-1]), np.log(m[j]); t0_, t1_ = np.log(Tb[j-1]), np.log(Tb[j])
    return float(np.exp(t0_ + (0 - x0)*(t1_ - t0_)/(x1 - x0)))
Td_with = Tdec_eff(True); Td_without = Tdec_eff(False)
fam_T = np.array([0.75, 1.0, 1.5, 2.0, 3.0]); fam_d = np.array([0.51586, 0.52076, 0.52583, 0.52837, 0.53065])
def fam(T): return float(np.interp(np.log(T), np.log(fam_T), fam_d))
dw, dwo = fam(Td_with), fam(Td_without)
print(f'\nREADING 2 (timing): Tdec_eff with nu-nu = {Td_with:.3f} MeV, without = {Td_without:.3f} MeV')
print(f'  single-rate family at T_H = 1 MeV -> delta_c(with) = {dw:.5f}, delta_c(without) = {dwo:.5f}, shift {
     dw-dwo:+.5f}')
for refname, di in (('primary', d_inst_P), ('sens', d_inst_S)):
    print(f'  [{refname} d_inst={di:.5f}] G(with) = {Gof(dw, di):.4f}, G(without) = {Gof(dwo, di):.4f}, dG = {
     Gof(dw, di)-Gof(dwo, di):+.4f}')
    out['reading2'][refname] = dict(Tdec_with=Td_with, Tdec_without=Td_without, d_with=dw, d_without=dwo,
                                    dG_pred=Gof(dw, di) - Gof(dwo, di), G_with=Gof(dw, di), G_without=Gof(dwo,
      di))
# Reading 2, variant: anchored on the measured no-nu-nu threshold (shift only, from the family's slope)
print(f'  anchored on measured 0.52639: delta_c(with) = {0.52639 + dw - dwo:.5f}')
json.dump(out, open('step3_predictions.json', 'w'), indent=1)
```

A2_step4_compare.py: comparison.

```
# SCRIPT: A2_STEP4_COMPARE
"""Compare the step-3 predictions with McCormick's measured T_H = 1 MeV multigroup thresholds; kill lines from
PREREGISTERED.txt. Plus sensitivity: bracket errors, d_inst scan, and v5.1's measured eleven-point table used
     directly
in place of the linear law."""
import json
import numpy as np
P = json.load(open('step3_predictions.json'))
d_fluid = 0.49742
d_with, e_with = 0.51940625, 0.00031249999999993783/2        # res_mg_phys6_10.0.json (midpoint, half-width)
d_wo, e_wo = 0.526390625, 0.00021874999999993427/2           # res_mg_nonunu6_10.0.json
def Gof(d, di): return (d - d_fluid)/(di - d_fluid)
```

```
def tau(G): return np.log((1 - G)/G)
di = 0.53379
dG_meas = Gof(d_with, di) - Gof(d_wo, di)
dG_err = np.hypot(e_with, e_wo)/(di - d_fluid)
dtau_meas = tau(Gof(d_with, di)) - tau(Gof(d_wo, di))
print(f'measured: G(without) = {Gof(d_wo, di):.4f}, G(with) = {Gof(d_with, di):.4f}, shift in G = {dG_meas:+.4
    f} +- {dG_err:.4f} (brackets);'
     f' shift in tau_G = {dtau_meas:+.4f}; implied rapidity at rate 1.11: {dtau_meas/1.11:.4f}')
print(f'kill tolerance = 0.5*|shift| = {0.5*abs(dG_meas):.4f}\n')
k1 = 'primary d_inst=0.53379 | primary: visibility of k_e | rate 1.11'
for name, pr in (('READING 1, rate 1.11 (decides)', P['reading1'][k1]),
                 ('READING 1, rate 1.19', P['reading1'][k1.replace('1.11', '1.19')]),
                 ('READING 1, rate 1.20', P['reading1'][k1.replace('1.11', '1.2')])):
    miss = pr['dG_pred'] - dG_meas
    verdict = 'KILLED' if (np.sign(pr['dG_pred']) != np.sign(dG_meas) or abs(miss) > 0.5*abs(dG_meas)) else '
    SURVIVES'
    print(f'{name}: predicted shift {pr["dG_pred"]:+.4f}, miss {miss:+.4f} ({100*abs(miss)/abs(dG_meas):.1f}%
    of shift); '
          f'delta_c(with) predicted {pr["delta_with_pred"]:.5f} vs {d_with:.5f} +- {e_with:.5f}: miss {pr["
    delta_with_pred"]-d_with:+.5f} -> {verdict}')
r2 = P['reading2']['primary']
miss2 = r2['dG_pred'] - dG_meas
v2 = 'KILLED' if (np.sign(r2['dG_pred']) != np.sign(dG_meas) or abs(miss2) > 0.5*abs(dG_meas)) else 'SURVIVES'
print(f'READING 2 (timing): predicted shift {r2["dG_pred"]:+.4f}, miss {miss2:+.4f} ({100*abs(miss2)/abs(
    dG_meas):.1f}% of shift) -> {v2}')
print(f'   absolute: without {r2["d_without"]:.5f} vs {d_wo:.5f} (miss {r2["d_without"]-d_wo:+.5f}); '
     f'with {r2["d_with"]:.5f} vs {d_with:.5f} (miss {r2["d_with"]-d_with:+.5f}); anchored with {d_wo + r2["
    d_with"] - r2["d_without"]:.5f} (miss {d_wo + r2["d_with"] - r2["d_without"] - d_with:+.5f})')
# equivalent single-rate Tdec of the measured thresholds (McCormick's family at T_H = 1 MeV)
fam_T = np.array([0.75, 1.0, 1.5, 2.0, 3.0]); fam_d = np.array([0.51586, 0.52076, 0.52583, 0.52837, 0.53065])
def Teq(d): return float(np.exp(np.interp(d, fam_d, np.log(fam_T))))
print(f'   measured thresholds as single-rate Tdec: without {Teq(d_wo):.3f} MeV, with {Teq(d_with):.3f} MeV '
     f'(Reading 2 rate-based: {r2["Tdec_without"]:.3f}, {r2["Tdec_with"]:.3f})')

# d_inst scan
print('\nd_inst scan (Reading 1, rate 1.11, u_eff = %.4f):' % P['U']['primary: visibility of k_e'])
u = P['U']['primary: visibility of k_e']
for dI in (0.5315, 0.5325, 0.53379, 0.5345, 0.5355, 0.5370):
    G0 = Gof(d_wo, dI); t = tau(G0) + 1.11*u; Gp = 1/(1 + np.exp(t)); dp = d_fluid + Gp*(dI - d_fluid)
    dGm = Gof(d_with, dI) - G0
    print(f'  d_inst={dI:.5f}: delta_c(with) pred {dp:.5f} (miss {dp-d_with:+.5f}); shift pred {Gp-G0:+.4f} vs
     meas {dGm:+.4f}')

# v5.1's eleven measured offsets used directly (per flavour, then energy-averaged in tau_G), instead of the
    linear law
uu = np.array([-1.386, -0.693, -0.347, -0.112, -0.091, 0.0, 0.091, 0.112, 0.347, 0.693, 1.386])
tt = np.array([-1.547, -0.763, -0.384, -0.127, -0.103, 0.0, 0.098, 0.122, 0.385, 0.786, 1.610])
o = np.argsort(uu); uu, tt = uu[o], tt[o]
ue, um = 0.4468, 1.0235
dt = (1/3)*np.interp(ue, uu, tt) + (2/3)*np.interp(um, uu, tt)
G0 = Gof(d_wo, di); Gp = 1/(1 + np.exp(tau(G0) + dt)); dp = d_fluid + Gp*(di - d_fluid)
print(f'\nv5.1 table used directly: d tau_G = {dt:.4f} (linear law: {1.11*u:.4f}); delta_c(with) = {dp:.5f} (
    miss {dp-d_with:+.5f}); shift {Gp-G0:+.4f}')
# v5.1 shear-only family has no table beyond two pairs; rate 1.19-1.20 above is its stand-in.
```

B.21 The deciding run (Section 10)

`A2b_ms_mg_r1.diff`: the change to McCormick's solver.

```
1c1
< # SCRIPT: ORIG_MS_MG (unmodified copy of McCormick's file, eb7c4de; only this line added)
---
> # SCRIPT: MS_MG_R1 (copy of McCormick ms_mg.py, eb7c4de; adds nnmode = 2: nu-nu relaxes to plasma-frame
    isotropy, link ratio r = 1)
28c28
< def state(t, R, U, Gam, e, I, mue, le, lp, lg, lT, kbe, kbn, Tb, enub, pe, lchi, chib):
```

```
---
> def state(t, R, U, Gam, e, I, mue, le, lp, lg, lT, kbe, kbn, Tb, enub, pe, lchi, chib, nnmode):
62a63,64
>             if nnmode == 2:
>                 kq[i] += kap_n[s, i]*q_s[s, i]          # r1 variant: nu-nu also hands its momentum to the
    plasma
85c87
<                                                             Tb, enub, pe, lchi, chib)
---
>                                                             Tb, enub, pe, lchi, chib, nnmode)
172c174,179
<                 tgt = eps_s[s, i]*gn[i, k] if nnmode == 1 else eps_s[s, i]/(4*PI) + cs*mu
---
>                 if nnmode == 1:
>                     tgt = eps_s[s, i]*gn[i, k]
>                 elif nnmode == 2:
>                     tgt = eps_s[s, i]/(4*PI)                # r1 variant: isotropy in the plasma frame
>                 else:
>                     tgt = eps_s[s, i]/(4*PI) + cs*mu
```

A2b_gate_operator.py: operator, energy and momentum gates.

```
# SCRIPT: A2B_GATE_OPERATOR
"""Gate for the r1 variant (nnmode = 2), single frozen group, through the compiled rhs()/state() of the copies
    .
Why this checks what it claims: the collision term is the only part of dI that depends on (kbe, kbn) except
    through the
plasma momentum kq -> ae -> lapse A and the angular term mu*ac. With a uniform, enormous plasma density (e ~ 1
    e60) the
pressure gradient is zero and ae = kq/(e+p) ~ 1e-60, so A = 1 exactly in double precision and mu*ac*I is ~1e
    -60 of the
collision term. Then dI(kbe,kbn) - dI(0,0) IS the collision term, from the compiled code itself. Columns are
    taken on
perturbations orthogonal to the constant (so eps and hence kap_n are unchanged), and the constant column is C
    (1)."""
import sys, numpy as np
which = sys.argv[1]                     # 'r1' or 'orig'
sys.path.insert(0, f'/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/nu6/
    A2b_decisive/{which}')
import ms_mg
nn = 2 if which == 'r1' else 0
K = 32; N = 4; PI = np.pi
mue = np.linspace(-1, 1, K+1); dmu = mue[1]-mue[0]; mu = 0.5*(mue[:-1]+mue[1:])
le = np.linspace(100.0, 180.0, 50); lp = le + np.log(1/3); lg = le/4; lT = le/4 - 30.0
lchi = np.zeros(50)
R = np.linspace(1.0, 2.0, N+1); U = np.zeros(N+1); Gam = np.ones(N+1)
e = np.full(N, np.exp(140.0)); Tb = float(np.exp(140.0/4 - 30.0)); pe = 4.0
rng = np.random.default_rng(1)
def dI_of(I, kbe, kbn, enub):
    dR = np.zeros(N+1); dU = np.zeros(N+1); dG = np.zeros(N+1); de = np.zeros(N); dI = np.zeros(I.shape)
    ms_mg.rhs(0.0, R, U, Gam, e, I, mue, le, lp, lg, lT, np.array([kbe]), np.array([kbn]), Tb, enub, pe, 1,
     0.1, 0.0,
              np.zeros(1), dR, dU, dG, de, dI, lchi, 0.0, nn)
    return dI
I0 = (1.0 + 0.3*rng.standard_normal((1, N, K)))
eps0 = 2*PI*I0[0].sum(axis=1)*dmu
ke, kbn = 0.7, 1.9
enub = float(eps0[1])                    # kap_n = kbn * eps/enub; cell 1 then has kap_n = kbn exactly
kn = kbn*eps0/enub
def coll(I): return dI_of(I, ke, kbn, enub) - dI_of(I, 0.0, 0.0, enub)
# discrete orthonormal Legendre basis on the bin centres, inner product sum_k f g dmu
V = np.vander(mu, K, increasing=True); Q, _ = np.linalg.qr(V*np.sqrt(dmu)); Q = Q/np.sqrt(dmu)
for j in range(K):
    if Q[:, j] @ V[:, j] < 0: Q[:, j] *= -1
i = 1
Cmat = np.zeros((K, K))
const = np.ones((1, N, K)); Cmat[:, 0] = coll(const)[0, i]
for j in range(1, K):
```

```
    Ij = I0.copy(); Ij[0, i] = I0[0, i] + 1.0*Q[:, j]
    Cmat[:, j] = coll(Ij)[0, i] - coll(I0)[0, i]
# operator in the Legendre basis: L = Q^T W C Q with W = dmu (Q orthonormal under that product)
L = (Q.T*dmu) @ Cmat
kk = ke + kn[i]
target = -kk*np.diag([0.0] + [1.0]*(K-1)) if nn == 2 else -np.diag([0.0, ke] + [ke+kn[i]]*(K-2))
print(f'{which}: nnmode {nn}, k_e = {ke}, k_n(cell) = {kn[i]:.15f}')
print('  max |L - target| =', np.abs(L - target).max(), ' (|target| ~', np.abs(target).max(), ')')
print('  diag(L)[:5] =', np.round(np.diag(L)[:5], 15))
print('  eigenvalues (sorted, first 4):', np.round(np.sort(np.linalg.eigvals(L).real)[::-1][:4], 14), '... min
    ', np.linalg.eigvals(L).real.min())
# nonlinear check on random anisotropic I in every cell: C(I) vs closed form
C = coll(I0)[0]
if nn == 2:
    closed = -(ke + kn)[:, None]*(I0[0] - (eps0/(4*PI))[:, None])
else:
    q = 2*PI*(I0[0]*mu).sum(axis=1)*dmu; cs = q/(2*PI*(mu*mu*dmu).sum())
    closed = ke*((eps0/(4*PI))[:, None] - I0[0]) + kn[:, None]*((eps0/(4*PI))[:, None] + cs[:, None]*mu[None,
    :] - I0[0])
print('  C(I) vs closed form, all cells: max rel err =', np.abs(C - closed).max()/np.abs(closed).max())
# energy and momentum moments of the collision term (per cell)
dE = 2*PI*C.sum(axis=1)*dmu; dQ = 2*PI*(C*mu).sum(axis=1)*dmu
q0 = 2*PI*(I0[0]*mu).sum(axis=1)*dmu
print('  energy moment of collisions / (k eps):', np.abs(dE).max()/np.abs((ke+kn)*eps0).max())
# momentum handed to the plasma: from state(); moderate plasma density so ae = kq/(e+p) is resolved
e_mod = np.full(N, np.exp(110.0))
Tb_mod = float(np.exp(110.0/4 - 30.0))       # same local/background temperature ratio (= 1) as the collision
    run
args = (0.0, R, U, Gam, e_mod, I0, mue, le, lp, lg, lT, np.array([ke]), np.array([kbn]), Tb_mod, enub, pe,
    lchi, 0.0)
st = ms_mg.state(*args, nn) if which == 'r1' else ms_mg.state(*args)
ae = st[8]; p_mod = np.exp(np.interp(110.0, le, lp))
kq_code = ae[N]*(e_mod[-1] + p_mod)          # ae[N] = kq[N-1]/(e+p)
loss = -dQ[N-1]                              # neutrino momentum removed by collisions in the last cell
print(f'  momentum: neutrino collisional loss = {loss:.15e}, handed to plasma (code kq) = {kq_code:.15e}, '
      f'rel diff = {abs(loss-kq_code)/abs(loss):.2e}')
print(f'  (nu-e part alone k_e q = {ke*q0[N-1]:.15e})')
```

A2b_PREREGISTERED.txt: kill lines and results.

```
PREREGISTERED --- Task A2b (the two decisive simulations proposed by A2)
Written 2026-11-11T14:09Z, before any simulation in this folder was run and before the variant code was
    written.

DISCLOSURE: A2's numbers are known (its REPORT.md and PREREGISTERED.txt were read): McCormick's measured
    thresholds at
T_H = 1 MeV (T0 = 10, G = 6, x 0.4..20, N = 300, Nmu = 32): with nu-nu (nnmode 0) 0.51941 +- 0.00016, without
    0.52639
+- 0.00011; delta_fluid = 0.49742; A2's delta_inst stand-in 0.53379.

RUN 1 --- variant 'r1' of McCormick's multigroup solver (copies in r1/; originals in orig/, unchanged).
The nnmode-0 nu-nu term A kap_n (eps_s/4pi + c_s mu - I) is replaced (as nnmode = 2) by A kap_n (eps_s/4pi - I
    ), and
kap_n,s q_s is added to the momentum handed to the plasma (kq). Then for each group the collision operator is
-(kap_e + kap_n)(1 - P0): flux and every l >= 2 relax at the same rate (link ratio r = 1), while the l >= 2 (
    isotropy)
rate kap_e + kap_n is exactly the one of the archived nu-nu run, so the total isotropy history is unchanged.
Why this decides: the timing reading says the threshold follows only the isotropy (total-rate) history, which
    is the
same as in McCormick's nu-nu run, so it predicts the same threshold, 0.5194. The link law says the threshold
    reads the
flux-shear link rapidity (1/2) ln r; at r = 1 with flux rate kap_e + kap_n it predicts about 0.5204 +- 0.0003
    (A2).
Bisection: T0 = 10 MeV, mode = phys, G = 6, x_lo = 0.4, x_hi = 20, N = 300, K = 32, nnmode = 2, start bracket
0.517-0.523 (widened only if an endpoint check fails), tolerance width 2e-4 (+-1e-4), McCormick's bisect_mg
    logic:
only BH moves hi, only dispersed moves lo, anything else stops the bisection.
KILL LINES (A2's, copied):
```

```
  result above 0.5194 by more than 0.0005 (delta_c > 0.5199)       -> favours the link law (A2 predicts 0.5204
      +- 0.0003)
  result within +-0.0003 of 0.5194 (0.5191 <= delta_c <= 0.5197)  -> favours timing
  anything else (0.5197 < delta_c <= 0.5199, or delta_c < 0.5191) -> neither; reported as such.
  The bisection half-width is reported with the result; if the bracket straddles a kill line, that is said.
Gates before the bisection (setup, not deciding): single frozen group, discrete operator equals
-(k_e+k_n) diag(0,1,1,...) in the discrete Legendre basis; energy conserved and momentum exchange balancing (
     neutrino
momentum loss = momentum handed to plasma), to machine precision. Regression: with kappa_n = 0 (nunu = 0) the
     variant
(nnmode 2) reproduces the unmodified code bit for bit up to eps = 0.6.

RUN 2 --- instantaneous free-streaming threshold at T_H = 1 MeV with the full EOS (McCormick's run_eos /
     bisect_eos,
kind 'fs', T0 = 10, his defaults N = 400, K = 32), to +-2e-4 (tolerance width 4e-4). No kill line; it replaces
      A2's
delta_inst stand-in 0.53379, and A2's step3/step4 scripts (copied) are rerun with it. Expectation (untested):
     near
0.533, since R_nu = 0.488 at T > 1 MeV.

Results are appended below this line; nothing above is edited afterwards.
-------------------------------------------------------------------------------------------------------------------



[2026-11-11T14:50Z] GATES (setup; gate_operator.py -> gate_operator_r1.txt / gate_operator_orig.txt; regress.
     py, regress_compare.py)
  r1 (nnmode 2), one frozen group, Nmu = 32, k_e = 0.7, k_n = 1.9, operator from the compiled rhs():
    max |L - (-(k_e+k_n) diag(0,1,...,1))| = 5.3e-15 (entries 2.6); eigenvalues 0 and 31 x -2.6.
    C(I) vs -(k_e+k_n)(I - eps/4pi) on random anisotropic I: rel err 1.2e-15. Energy moment of collisions 1.5e
     -16 (rel).
    momentum: neutrino collisional loss -1.219742786974912e-01 = momentum handed to plasma -1.219742786974893e
     -01 (rel 1.5e-14).
  orig (nnmode 0), same test: -diag(0, k_e, k_e+k_n, ...) to 7.8e-15; plasma receives only k_e q (rel 3.3e-14)
     . (A2's result reproduced.)
  regression, T0 = 10, delta = 0.52, G = 6, N = 300, K = 32, nunu = 0, to eps = 0.6 (3340 steps): r1 nnmode 2
     vs unmodified
    copy: I, e, eps, steps, peakC all bit-identical (max |diff| = 0.0). Gates PASS; bisection launched next.

[2026-11-11T15:20Z] RUN 1 RESULT (r1/res_mg_r1_6_10.0.json, log_bisect_r1.txt; 5 runs, 28.9 min, first run 353
      s)
  log: 0.52000 BH (eps 6.10) | 0.51850 dispersed (5.53) | 0.51925 dispersed (5.85) | 0.519625 dispersed (6.23)
      | 0.5198125 BH (7.20)
  delta_c(r1) = 0.51972 +- 0.00009 (bracket 0.519625 - 0.5198125), TH = 1.0029 MeV.
  Against the kill lines: midpoint 0.51972 lies in the 'neither' band (0.5197 < d <= 0.5199). The bracket's
     lower end
  (0.519625) is inside the timing band's top edge, its upper end (0.519813) is below the link-law line
     (0.5199); the
  whole bracket lies below the link-law line. VERDICT: NEITHER (as pre-registered), link law NOT favoured; the
      result
  sits at the top edge of the timing band.
  Shift from McCormick's nnmode-0 bracket (0.51925 dispersed, 0.5195625 BH): r1 threshold is higher by
     +0.00006 to
  +0.00056 (strictly positive: his 0.5195625 is BH, here 0.519625 disperses); midpoint shift +0.00031. Link
     law
  predicted about +0.0010 (0.5204); timing predicted 0.

[2026-11-11T15:30Z] RUN 2 RESULT (orig/res_eos_fs_10.0.json, log_bisect_fs.txt; 5 runs, 55-76 s each)
  log: 0.533 dispersed | 0.535 BH | 0.534 BH | 0.5335 BH | 0.53325 BH.  delta_inst(T_H = 1.0030 MeV, full EOS)
      = 0.53313 +- 0.00012.
  A2's stand-in 0.53379 was high by 0.00066. A2 step3/step4 rerun (copies, d_inst = 0.533125;
     step3_out_newdinst.txt,
  step4_out_newdinst.txt): measured G 0.8114 -> 0.6158, shift -0.1956 +- 0.0053.
  Reading 1 rate 1.11: shift -0.1804 (miss 7.8%), delta_c(with) 0.51995 (miss +0.00054) SURVIVES; rate 1.19:
     0.51939
  (miss -0.00002) SURVIVES; rate 1.20: 0.51932 SURVIVES. Reading 2 (timing): shift -0.1606 (miss 17.9%)
     SURVIVES.
  Not pre-registered context (step5_r1_readout.py): G(r1) = 0.6245 vs G(nnmode0) = 0.6158: r1 undoes 4.5% of
     the
```

```
  full nu-nu shift in G; it recovers 31% (25% anchored) of the move the link law predicted for r1 (0.5204),
     and 4.7%
  of the move given by the tau route (removing u_eff = 0.8313 at rate 1.11 predicts 0.5260).
```

A2b_log_bisect_r1.txt: bisection log, r = 1.

```
Wed Nov 11 14:50:50 UTC 2026
   r1_6 T0=10.0 delta=0.52000: BH eps=6.10 (353s)
   r1_6 T0=10.0 delta=0.51850: dispersed eps=5.53 (300s)
   r1_6 T0=10.0 delta=0.51925: dispersed eps=5.85 (320s)
   r1_6 T0=10.0 delta=0.51963: dispersed eps=6.23 (343s)
   r1_6 T0=10.0 delta=0.51981: BH eps=7.20 (418s)
RESULT r1_6 T0=10.0: TH=1.0029 delta_c=0.51972 +- 0.00009 (28.9 min)
Wed Nov 11 15:19:45 UTC 2026
```

A2b_log_bisect_fs.txt: instantaneous free-streaming threshold at 1 MeV.

```
Wed Nov 11 15:19:45 UTC 2026
   fs T0=10.0 delta=0.53300: dispersed eps=6.29 TH=1.0029809368991307 (70s)
   fs T0=10.0 delta=0.53500: BH eps=5.22 TH=1.0029829979632972 (55s)
   fs T0=10.0 delta=0.53400: BH eps=5.64 TH=1.0029819676031162 (58s)
   fs T0=10.0 delta=0.53350: BH eps=6.11 TH=1.0029814522936875 (65s)
   fs T0=10.0 delta=0.53325: BH eps=6.94 TH=1.002981194607072 (76s)
RESULT fs T0=10.0: TH=1.0030 MeV delta_c=0.53313 +- 0.00012
Wed Nov 11 15:25:10 UTC 2026
```

B.22 The recorder in McCormick's multigroup solver (Section 10)

A3_PREREGISTERED.txt: definitions, readings and results.

```
A3 -- two-cut quantities read out of McCormick's multigroup model (nnmode=1). Written 2026-11-11T14:06:02Z,
     before any of the four runs.
(Only the eps=0.6 bit-for-bit regression has been run so far: identical I and e, 3342 steps; log_regress.txt.)

RUNS. nnmode=1, mode=phys, G=6, x 0.4-20, N=300, K=32, all other options at McCormick's defaults; recorder
     every 20 steps.
  T_H=1 MeV  (T0=10): archived bisection edges in res_mg_phys6_10.0.json: 0.51925 dispersed, 0.5195625 BH (
     nnmode=0).
  T_H=0.51 MeV (T0=5): archived edges in res_mg_phys6_5.0.json: 0.52475 dispersed, 0.525125 BH (nnmode=0).
  These edges were bisected with nnmode=0. Whether they still split under nnmode=1 is untested; Table VIII
     only shows
  the +-0.001 pair kept. Rule set now: if an edge pair does not split, that is reported as a setup result, and
      the pair is
  replaced by the archived nnmode=1 pair of Table VIII (validation/res_test_boosted_*: 0.518406/0.520406 at T0
     =10,
  0.523938/0.525938 at T0=5), which is known to split.

DEFINITIONS (recorder; cells: centre cell i=0, and the cell whose Lagrangian centre is nearest r=r_m=1, i=95).
  Discrete Legendre moments: p_l, l=0..4, Gram-Schmidt of 1,mu,..,mu^4 on McCormick's 32 midpoint bins with
    <f,g> = (1/2) sum_k f_k g_k dmu (so p_0=1 and I_0 = eps_s/4pi exactly); I_l = <p_l, I>.
  Kept share (v5.1 Sec. 6, A=|v_0|^2 with ||v||=1, read on the local angular chain instead of a Fourier mode):
    A = I_0^2 / ||I||^2, ||I||^2 = <I,I> over all 32 bins (all discrete moments). Also A_4 = I_0^2/sum_{l<=4}
     I_l^2.
    Computed per (flavour, group), per flavour (summed intensity), and for the whole gas (sum of all 12
     intensities).
  Link rapidity per group (v5.1 Sec. 7; brief item 2: k1=k_e, k2=k_e+k_n): rho_s = (1/2) ln[(k_e,s+k_n,s)/k_e,
     s],
    r_s = 1 + k_n,s/k_e,s, with the LOCAL rates kap_e, kap_n returned by McCormick's state().
    Caveat fixed now: with nnmode=1 the nu-nu target is a common boosted frame, so k_n leaves only the total
     momentum
    alone, not each group's dipole; D=diag(0,k_e,k_e+k_n,...) is exact for nnmode=0 and for the gas as a whole
     .
  Energy-frame velocity of the whole neutrino gas: v_L = 2q/[e+P_r+sqrt((e+P_r)^2-4q^2)] from the summed e, q,
      P_r.
  Speed-limit ratio: R_SL = (dA/dtau)^2 / (4 E_B A(1-A)) for the whole-gas kept share, tau = proper time of
     the cell
```

```
    (dtau = lapse*dt, finite difference between records), E_B = k^2/3 with k = sqrt(3)/R(r_m) (the rms k of v5
     .1's
    profile weight k^2 e^{-k^2/2} is sqrt(3)/r_m), so E_B = 1/R(r_m)^2. The bound's hypotheses (one Fourier
     mode,
    flat space, ||v|| non-increasing) do not hold in the GR model, so R_SL > 1 would not be a violation; it is
      a reading.

PREDICTION (i) -- where the collapsing and dispersing runs first separate (same T_H, values compared at equal
     eps by
  interpolating the BH run onto the dispersing run's eps records, at the r_m cell):
  separation of a quantity X at eps = first record where |X_BH - X_disp| > 0.01 * max_{eps'<=eps} |X_disp|.
  Predicted: for X = v_L and X = 1-A (whole gas) the first 1% separation lies in 3.0 <= eps <= 5.5 at both T_H
     , and the
  two quantities separate within 0.5 of each other in eps. Kept share A itself (not 1-A) differs by more than
     1%
  only after eps=4, if at all, before the earlier run stops.
  Why: the edges differ by 6e-4 relative in amplitude; before the critical phase every field differs at that
     level;
  a factor ~16 of growth is needed, which only the near-critical phase after the first compaction maximum (eps
     ~1.7)
  supplies; the archived edge runs stop at eps 6.17/6.35 (T0=10) and 6.03/6.46 (T0=5).
  KILL: first 1% separation of v_L or of 1-A before eps=3.0 or after 5.5 (or never before the earlier stop) at
      either
  T_H; or v_L and 1-A separating more than 0.5 apart in eps.

PREDICTION (ii) -- link rapidity of the dominant energy group (largest eps_s among the 12 slots at the r_m
     cell) over
  the readout epoch 1.7 <= eps <= 4.
  Predicted: the dominant group is a nu_mu,tau group; its r = 1+k_n/k_e is about 6-9 (rapidity 0.9-1.1),
  OUTSIDE v5.1's admissible window 3/4 <= r <= 3/2 (rapidity -0.144..+0.203), on the high side, at both T_H
     and in all
  four runs. The nu_e groups sit at r about 2.2-2.8, also outside.
  Why: McCormick App. C: nu-nu faster than nu-e+- by 1.5 (nu_e) and 7 (nu_mu,tau) at E~T; the local factor
  (T_nu/T)^4/chi_e grows as e+- annihilate below ~0.5 MeV, which pushes r up, not down.
  KILL: the dominant group's r inside [0.75, 1.5] at any record in 1.7<=eps<=4, in any run.

======================================================================
RESULTS (appended 2026-11-11T14:50:53Z; nothing above edited)
Runs (log_runs.txt): T10_lo 0.51925 dispersed eps 6.295; T10_hi 0.5195625 BH eps 6.139; T5_lo 0.52475
     dispersed eps 6.052;
  T5_hi 0.525125 BH eps 6.339. Both archived nnmode=0 edge pairs still split under nnmode=1; fallback pair not
      needed.
(i) first >1% separation at r_m (log_analyse.txt):
  T_H=1:    v_L 4.228; 1-A 2.907; A itself 4.488; (peak compaction Cmax 3.777)
  T_H=0.51: as written, v_L and 1-A trip at the FIRST record, eps=0.100, where both runs have X = 0 to
     roundoff (1e-19,
            1e-16) and the running-max scale is zero: a definition failure of my metric, reported as such.
     With that one
            record excluded: v_L 4.095; 1-A 2.926; A itself 4.157; (Cmax 3.645).
  VERDICT: KILLED. 1-A separates before eps=3.0 at both T_H (2.907, 2.926), and v_L and 1-A separate 1.32 and
     1.17 apart
  in eps (> 0.5). v_L itself is inside the predicted window at both; A itself separates after eps=4 as
     predicted.
(ii) dominant group at r_m over 1.7<=eps<=4: slots 9-10 (nu_mu,tau, groups 3-4) in all four runs.
  r_dom: T_H=1 [7.881, 8.575] (rapidity 1.032-1.074); T_H=0.51 [8.243, 12.144] (rapidity 1.055-1.248).
  nu_e groups: T_H=1 [2.466, 2.625]; T_H=0.51 [2.543, 3.390].
  VERDICT: SURVIVES (never inside [0.75,1.5]; outside on the high side in all four runs). The numerical ranges
      written
  (6-9 and 2.2-2.8) were too narrow at T_H=0.51 (up to 12.1 and 3.39): a miss of the range, not of the kill
     line.
```

`A3_analyse.py`: separation epochs and link rapidity.

```
# SCRIPT: analyse_A3
"""Derived two-cut quantities from raw_<label>.json (recorder output), the preregistered separation readings,
     the compact
JSON and the plots. usage: python3 analyse.py <label> [<label> ...]"""
import sys, os, json, numpy as np
```

```
HERE = os.path.dirname(os.path.abspath(__file__))
CELLS = ('centre', 'rm')

def derive(raw):
    G, NF = raw['G'], raw['NF']; rec = raw['rec']
    out = dict(eps=np.array([x['eps'] for x in rec]), t=np.array([x['t'] for x in rec]), Tb=np.array([x['Tb']
     for x in rec]),
               Rm=np.array([x['Rm'] for x in rec]), Cmax=np.array([x['Cmax'] for x in rec]))
    for c, name in enumerate(CELLS):
        g = lambda k: np.array([x['cells'][c][k] for x in rec])
        mom = g('mom'); n2 = g('norm2'); es = g('es'); qs = g('qs'); ps = g('ps'); ke = g('ke'); kn = g('kn')
        momT = g('momT'); nT = g('normT'); momF = g('momF'); nF = g('normF'); lapse = g('A')
        e, q, P = es.sum(1), qs.sum(1), ps.sum(1)
        d = {}
        d['vL'] = 2*q/(e + P + np.sqrt(np.maximum((e+P)**2 - 4*q*q, 0.0)))
        d['q_e'] = q/e; d['P_e'] = P/e
        d['A'] = momT[:, 0]**2/nT                                  # whole-gas kept share, full discrete norm
        d['one_minus_A'] = 1.0 - d['A']
        d['A4'] = momT[:, 0]**2/np.sum(momT**2, axis=1)
        d['ratio_l'] = np.abs(momT[:, 1:])/momT[:, :1]            # |I_l|/I_0, l=1..4, whole gas
        d['AF'] = momF[:, :, 0]**2/nF                              # per flavour
        d['As'] = mom[:, :, 0]**2/n2                               # per (flavour, group)
        d['qs_es'] = qs/es; d['ps_es'] = ps/es
        d['r_s'] = 1.0 + kn/ke; d['rho_s'] = 0.5*np.log(d['r_s'])
        d['ke_H'] = ke/np.array([x['Hb'] for x in rec])[:, None]; d['kn_H'] = kn/np.array([x['Hb'] for x in
     rec])[:, None]
        d['dom'] = np.argmax(es, axis=1)
        d['rho_dom'] = d['rho_s'][np.arange(len(rec)), d['dom']]; d['r_dom'] = d['r_s'][np.arange(len(rec)), d
     ['dom']]
        d['es_share'] = es/e[:, None]
        # speed-limit ratio for the whole-gas kept share, proper time of the cell, E_B = 1/R(r_m)^2
        tau = np.concatenate([[0.0], np.cumsum(0.5*(lapse[1:]+lapse[:-1])*np.diff(out['t']))])
        A = d['A']; dA = np.gradient(A, tau)
        EB = 1.0/out['Rm']**2
        d['R_SL'] = np.where(1-A > 1e-10, dA**2/(4*EB*A*np.maximum(1-A, 1e-300)), np.nan)   # undefined where
     1-A is at roundoff
        d['lapse'] = lapse
        out[name] = d
    return out

def first_sep(a, b, key, cell='rm', thr=0.01, rel_to_self=False, skip=0):
    """first eps (on a's records, within the common range) where |X_b - X_a| > thr * running max |X_a|"""
    ea, eb = a['eps'], b['eps']; m = ea <= min(ea.max(), eb.max())
    get = lambda z: z[cell][key] if key in z[cell] else z[key]
    Xa = get(a)[m]; Xb = np.interp(ea[m], eb, get(b))
    scale = np.abs(Xa) if rel_to_self else np.maximum.accumulate(np.abs(Xa))
    idx = np.nonzero(np.abs(Xb - Xa) > thr*scale)[0]
    idx = idx[idx >= skip]
    return (float(ea[m][idx[0]]) if len(idx) else None), float(ea[m].max())

if __name__ == '__main__':
    labels = sys.argv[1:]
    raws = {L: json.load(open(os.path.join(HERE, f'raw_{L}.json'))) for L in labels}
    D = {L: derive(raws[L]) for L in labels}
    for L in labels:
        r = raws[L]; d = D[L]; m = (d['eps'] >= 1.7) & (d['eps'] <= 4.0)
        print(f"{L}: delta={r['delta']} {r['status']} eps_end={r['eps_end']:.3f} TH={r['TH']:.4f} records={len
     (r['rec'])} wall={r['seconds']/60:.1f} min")
        for cell in CELLS:
            c = d[cell]
            doms = sorted(set(int(x) for x in c['dom'][m]))
            print(f"   {cell}: readout 1.7-4: dominant slots {doms} (s=f*G+g); r_dom range [{c['r_dom'][m].min
     ():.3f}, {c['r_dom'][m].max():.3f}] "
                  f"rho_dom [{c['rho_dom'][m].min():.3f}, {c['rho_dom'][m].max():.3f}]; nu_e r range [{c['r_s
     '][m][:, :6].min():.3f},{c['r_s'][m][:, :6].max():.3f}] "
                  f"nu_mutau r range [{c['r_s'][m][:, 6:].min():.3f},{c['r_s'][m][:, 6:].max():.3f}]")
            print(f"      max|v_L| {np.abs(c['vL']).max():.4f} at eps {d['eps'][np.argmax(np.abs(c['vL']))]:.3
     f}; 1-A max {c['one_minus_A'].max():.3e}; "
```

```
                f"min A {c['A'].min():.6f}; max R_SL (eps 0.2-end) {np.nanmax(c['R_SL'][d['eps'] > 0.2]):.3e
    }; "
                f"max R_SL in 1.7-4 {np.nanmax(c['R_SL'][m]):.3e}; max |q_s|/e_s {np.abs(c['qs_es']).max()
    :.3f}")
    pairs = [(a, b) for a, b in (('T10_lo', 'T10_hi'), ('T5_lo', 'T5_hi'), ('T10b_lo', 'T10b_hi'), ('T5b_lo',
     'T5b_hi')) if a in D and b in D]
    for a, b in pairs:
        for cell in CELLS:
            s = {k: first_sep(D[a], D[b], k, cell) for k in ('vL', 'one_minus_A', 'A', 'Cmax')}
            print(f"   separation {a} vs {b} [{cell}]: " + '; '.join(f"{k} first >1% at eps {v[0]} (common
     range to {v[1]:.3f})" for k, v in s.items()))
            s2 = {k: first_sep(D[a], D[b], k, cell, skip=1) for k in ('vL', 'one_minus_A', 'A', 'Cmax')}
            print(f"      same, first record (X = 0 to roundoff, metric undefined) excluded: " + '; '.join(f"{
    k} {v[0]}" for k, v in s2.items()))
```

B.23 The restart test (Section 9)

A4_PREREGISTERED.txt: every prediction, kill line and result, in time order.

```
# SCRIPT: A4-PREREGISTERED (preregistration text, not code)
Written: 2026-11-11T14:04Z (UTC). Before any threshold run and before the linear-level check.
Only the restart-surface verification (check_restart_out.txt: moments, positivity) had been run.

SETUP
 Solver: copies of S. McCormick's ms_nuc.py / ms_fluid.py (commit eb7c4de; ms_nuc physics identical to the
 41a0f83 era used by v5.1), single-rate collision model (the r=1 case), driver restart.py = McCormick's
     run_nuc.py loop with
 kappa(t) = (1/2t)(eps_dec/eps)^3 re-evaluated at each SSPRK3 stage time, Gaussian profile, R_nu = 0.409,
 eps_dec = 2, n = 3, N = 400, Nmu = 32, eps0 = 0.1. The step that would cross t_s is shortened to land on t_s
     exactly.
 Restart surface eps_s in {2.0, 0.5}. At eps_s, in every cell i:  I_ik -> I_ik + a (eps_i/4pi) p_l(mu_k),
 p_l the discrete orthogonal polynomial on the 32 bin midpoints (weight dmu), scaled to the leading
     coefficient of P_l.
 This is I(1 + a P_l) applied to the isotropic part of each cell; applying it to the full anisotropic I would
     move l=2
 (coupling of l=1 with P_3), so the carrier is the isotropic part. It changes only the discrete l component.
 Pair: a = +0.5 and a = -0.5, for l = 3 (main) and l = 4.
 Positivity bounds (from check_restart_out.txt): isotropic 32-bin grid |a| < 1/max|p_3| = 1.21498 for l=3;
 l=4: a < 2.35086, -a < 1.38855. On the actual delta=0.5109 states: eps_s=2: l=3 +0.8435/-0.8325, l=4
     +2.026/-0.951;
 eps_s=0.5: l=3 1.212/1.212, l=4 2.350/1.386. a = 0.5 is inside every bound.
 Verified at the restart: discrete l=0,1,2 components, e_nu, q, P_r of the two members agree to <= 9e-16
     relative per cell;
 fluid and metric arrays bit-identical.

THRESHOLD SEARCH
 Trisection, one process, strict endpoint rule: only a detected apparent horizon ('BH': 2M/R >= 1 with U < 0)
     lowers hi,
 only McCormick's dispersal rule ('dispersed') raises lo; 'BH_breakdown', 'crash', 'undecided' never move a
     bracket.
 Endpoints verified by runs for each member. Start bracket [0.5095, 0.5125] (width 3e-3), five rounds -> full
     width
 w = 3e-3/243 = 1.23e-5 (target <= 2e-5). eps_max = 15. Thresholds quoted as midpoint, full width w.

PREDICTIONS (from Jeromie's matched-pair witness: survival exp[-(eps_dec/eps_s)^n / n]; v5.1 table at tau_s/
     eps_dec)
 P1 eps_s = 2.0 (tau_s/eps_dec = 1, survival e^{-1/3} = 0.72; witness R_ret = 0.081 for l=3, 0.074 for l=4):
    the hidden difference returns into density/flux/stress and MOVES the threshold:
    |delta_c(+a) - delta_c(-a)| > 3 w  (with w = 1.23e-5: > 3.7e-5; if w reached is larger, 3 x that w).
    Kill line P1: |difference| <= 3 w at eps_s = 2  -> prediction P1 killed for that l.
 P2 eps_s = 0.5 (tau_s/eps_dec = 0.25, survival e^{-64/3} = 5e-10; witness R_ret = 6e-10 (l=3), 6e-11 (l=4)):
    the difference is erased in the coupled era; |delta_c(+a) - delta_c(-a)| <= 2 w.
    Kill line P2: |difference| > 2 w at eps_s = 0.5 -> prediction P2 killed.
 Between 2w and 3w at eps_s = 2: inconclusive, reported as such (bracket tightened if time allows).
 OPPOSITE READING (registered as the alternative): the nonlinear collapse washes out l >= 3 regardless of
     timing:
```

```
   both pairs agree within 2 w. P1's kill line is this reading's success line.
 SIGN: Jeromie's corpus does not give the sign of a threshold shift from hidden l=3 structure. The P3 slab
     result
    (audit: d/dt (P_r+ - P_r-) = -(4/35) e a') gives, for a centre-peaked e and +a, an early rise of core
     radial pressure
    in the + member; whether radial pressure here resists collapse (neutrino pressure gradient) or aids it (
     pressure
    gravitates, -4 pi R P_r in dU/dt) is not settled by it. NO SIGN IS PREDICTED.
 Magnitude (lean, not a kill line): Dc_3/c_0 = 0.14 at the restart, witness return ~8% of that, the full free-
     streaming
    shift ~0.03 in delta -> |difference| of order 1e-4 at eps_s = 2.

NUMERICAL GUARD
 Both members use identical code, grid and step rule; the pair at a = 0 vs no restart is not used for the
     verdict.
 Any member whose run logs min I < 0 is reported with that value; a crash/breakdown at a trial point is
     reported.

================== RESULTS (appended below; nothing above edited) ==================

--- 2026-11-11T14:09Z  RESULT of the uniform-a linear check (linear_check.py, run after the text above, before
      any threshold run)
 Uniform a = 0.5 in every cell, delta = 0.5109 (file linear_check_uniform_out.txt):
  eps_s=2, l=3: dCmax(+ minus -) = -2.3e-4 (eps 2.5), -2.6e-3 (eps 3), -6.5e-2 (eps 4); centre nu density
     differs by 91% of base at eps 2.5.
  eps_s=2, l=4: dCmax = +1.2e-4, +1.3e-3, +1.3e-2.
  eps_s=0.5, l=3: dCmax = -4.2e-5, -7.3e-5, -2.5e-4;  l=4: -2.1e-5, -2.9e-5, -8.3e-5;  max_cell |dP_r|/e_nu ~
     2e-3.
 The eps_s=0.5 entries are ~1e6 x above the survival law (5e-10). Diagnosis (computed, not a fit): a uniform a
      is irregular
 at r=0 (a regular l-th angular moment vanishes like r^l), and at the centre cells (dR ~ 0.03 at eps 0.5, 1/
     kappa ~ 0.04)
 streaming across one cell is as fast as collisions, so (k/sigma)^l ~ 1 at grid scale. The uniform-a pair
     therefore
 measures a centre artefact, not the hidden-moment door. It is NOT used for any threshold verdict.

--- 2026-11-11T14:09Z  DESIGN ADDITIONS (coordinator, adopted before any threshold run; none of the runs below
      had been made)
 D1 Third restart epoch eps_s = 4.0 (tau_s/eps_dec = 2, survival e^{-1/24} = 0.96, witness R_ret 0.095 (l=3),
     0.085 (l=4)).
    Why 4: near-threshold runs form the horizon at eps ~ 6-7.4 and disperse later than that (v5.1 log H1); at
     eps = 4 the
    compaction is still rising (Cmax 0.520 at delta 0.5109), so the outcome is not decided; 5 would leave <
     1.5 in eps.
    Prediction P3 (from the witness: planted after decoupling it returns): |delta_c(+)-delta_c(-)| > 3 w. Kill
     : <= 3 w.
    Lean (not a kill line): smaller than at eps_s = 2, because less time remains for the return before
     collapse.
 D2 Perturbation (replaces uniform a for every verdict):  dI_ik = f b(r_i) m_i (eps_i/4pi) p_l(mu_k),
    b = exp(1 - 1/(1-x^2)), x = (r - 1.15)/0.85: C-infinity bump, 1 at r = 1.15 r_m, identically 0 for r <=
     0.3 and r >= 2
    (Lagrangian r, r_m = 1; regular at the origin, far from the outer edge r = 9);
    m_i = the cell's positivity margin on the 32 bins, min over sign and bins of I_ik/(Ibar_i |p_l(mu_k)|);
    f = +0.5 and -0.5 (exact mirrors; every bin keeps at least half its margin, so I > 0 at the restart by
     construction).
 D3 At the restart, check +f vs -f: discrete l=0,1,2 components, e_nu, q, P_r, the Misner-Sharp mass M, the
     constraint
    residual Gamma^2 - (1 + U^2 - 2M/R), the lapse A (cells and edges) and the fluid acceleration: identical
     to roundoff.
 D4 SCREEN before trisection. Deltas 0.5100 and 0.5115 (v5.1 r=1 runs: 0.5105 dispersed, 0.5115 BH; checked
     with a=0).
    For each (eps_s, l) the pair (+f, -f) plus f=0 is run to the decision, recording Cmax(eps), centre P_r/
     e_nu(eps) and outcome.
    Noise floor at each delta: max over the common record of |Cmax(f=0 restart) - Cmax(continuous run, no
     restart)|
    (the restart's shortened step is the smallest genuine numerical perturbation of the same solver), floored
     at 1e-12.
    SELECTION RULE: trisect the pair iff at either screening delta (a) the two members' outcomes differ, or
```

```
    (b) max over eps <= 6 of |Cmax(+f) - Cmax(-f)| > 10 x noise floor.
    Not selected -> null result, reported with (i) the rigorous bound: if both members disperse at 0.5100 and
     both form a
    horizon at 0.5115, |shift| < 1.5e-3; (ii) the linearised estimate |dCmax| / (dCmax/d delta) at eps = 5
     from the two
    screening deltas, labelled an estimate.
    Order: (2, l3), (0.5, l3), (4, l3), then l = 4 at each epoch, as time allows.
 D5 A threshold difference counts as REAL only if the two brackets, each widened on both sides by a refinement
      drift
    allowance d_K = max over members |delta_c(Nmu=48) - delta_c(Nmu=32)| measured on ONE pair (the first
     selected pair,
    both members rerun at Nmu=48), are disjoint. Verdicts: P1/P3 pass iff REAL and |diff| > 3 w; P2 passes iff
    |diff| <= 2 w (or, if not selected, the linearised estimate <= 2 w). Opposite reading passes iff no pair
     is REAL.

--- 2026-11-11T14:5xZ  ADDITION (coordinator, from A5 kinetic Gramian), written before any run of this
     direction
 D6 Further pair at eps_s = 2: angular direction d = A5's L-infinity-optimal hidden direction (
     restart_direction_A5.json,
    ed2_n3_s2_T4 / Linf_optimal; odd; l-weights 3:0.851 5:0.117 7:0.010; l<=2 components on my p_l: <= 5e-17;
     max|d| = 1;
    orientation equal to +p_3: d(-0.969) = -1 like p_3). Applied exactly as D2 with p_l replaced by d (bump b,
      per-cell
    positivity margin m_i of d, f = +/-0.5), code label l = -1. So the comparison with p_3 is at equal
     positivity margin.
    Prediction P4: larger separation than p_3 at equal positivity margin: screen max|dCmax(+ - -)| (and, if
     both are
    trisected, |delta_c(+)-delta_c(-)|) for d exceeds that for p_3 at eps_s = 2. A5's linear chain says 2.47x;
      kill
    line: ratio d / p_3 <= 1. Lean (not a kill line): ratio between 1.5 and 3.
    Its screen runs after the (2, l3) pair; it takes p_4's place if time is short. Same selection rule D4,
     same verdicts D5.
 (timestamp of D6 above is 2026-11-11T14:38Z)

--- 2026-11-11T15:30Z  RESULT screen (2.0, l=3), bump, f=+/-0.5 (screen_out.txt)
 delta 0.5100: f=0 dispersed@6.03 (same as no-restart run); +f dispersed@5.47; -f BH@5.55. max|dCmax| 0.61,
     noise floor 3.1e-9.
 delta 0.5115: f=0 BH@5.83; +f DISPERSED@5.87; -f BH@5.08. max|dCmax| 0.54, noise floor 2.2e-8.
 dCmax(+ - -) at eps 2.5/3/4/5: -2.6e-4/-2.5e-3/-4.9e-2/-0.28 (0.5100). Outcomes flip at both deltas ->
     SELECTED (TRISECT).
 Already rigorous from these four outcomes: delta_c(+) > 0.5115 and delta_c(-) < 0.5100, so delta_c(+) -
     delta_c(-) > 1.5e-3
 (the -f member is more collapse-prone; +f, oriented like +p_3, raises the threshold). Linearised estimate
     +4.3e-3.

--- 2026-11-11T15:35Z  RESULT screen (2.0, d = A5 L-inf direction), bump, f=+/-0.5
 delta 0.5100: +f dispersed@5.32, -f BH@5.22; delta 0.5115: +f DISPERSED@5.59, -f BH@4.89 -> SELECTED. Same
     sign as p_3.
 dCmax(+ - -) at eps 2.5/3/4: d: -4.43e-4/-3.84e-3/-8.19e-2 (0.5100), -4.87e-4/-3.97e-3/-9.16e-2 (0.5115)
                              p_3: -2.64e-4/-2.54e-3/-4.91e-2 (0.5100), -2.91e-4/-2.62e-3/-5.50e-2 (0.5115)
 Ratio d/p_3 at fixed eps: 1.68, 1.51, 1.67 (0.5100); 1.67, 1.51, 1.67 (0.5115).  P4 (ratio > 1): PASSES at
     the screen level.
 Max over each record (windows end at different eps because outcomes differ, so these saturate): 0.693 vs
     0.608, 0.568 vs 0.545.
 Linearised threshold-shift estimates: d +6.6e-3, p_3 +4.3e-3 (ratio 1.54). A5's linear chain said 2.47x;
     measured ~1.5-1.7x.

--- 2026-11-11T15:38Z  RESULT trisection (2.0, l3, +f): delta_c = 0.5125833, bracket [0.5125741, 0.5125926], w
      = 1.85e-5 (log_es2_l3_plus.txt)
--- 2026-11-11T15:46Z  RESULT trisection (2.0, l3, -f): delta_c = 0.5088395, bracket [0.5088313, 0.5088477], w
      = 1.65e-5
 Pair (2.0, l3): delta_c(+) - delta_c(-) = +3.744e-3 = 202 x the larger bracket width (1.85e-5). Brackets
     disjoint by 3.7e-3.
 P1 (> 3 w) satisfied on the raw brackets; REAL/not-REAL call waits for the Nmu=48 drift allowance (D5).

--- 2026-11-11T15:57Z  RESULT screens, remaining pairs (all bump, f=+/-0.5; dCmax = + minus -; estimate =
     linearised threshold shift)
 (0.5, l3): no outcome flip; max|dCmax| 2.2e-3 / 1.3e-2 (0.5100/0.5115), noise 1.2e-8/1.1e-9; dCmax(eps 5)
```

```
    -8.5e-4/-1.4e-3; est +1.3e-5. SELECTED.
 (4.0, l3): no flip; max|dCmax| 4.5e-2 / 1.5e-1; dCmax(eps 5) -4.3e-3/-6.5e-3; est +6.2e-5. SELECTED.
 (2.0, l4): FLIP at both deltas, opposite sign to l3 (+f collapses more easily); est -2.2e-3. SELECTED.
 (0.5, l4): no flip; max|dCmax| 8.2e-4 / 4.7e-3; dCmax(eps 5) -3.2e-4/-4.9e-4; est +4.7e-6. SELECTED.
 (4.0, l4): no flip; max|dCmax| 2.1e-2 / 8.3e-2; dCmax(eps 5) +3.1e-4/+7.1e-4; est -5.9e-6. SELECTED.
 Note: at eps_s = 0.5 the separation is ~1e5 x the noise floor, and the estimate (1.3e-5 for l3) is ~3e-3 of
    the eps_s = 2
 shift, against the survival law's 5e-10. Whether this is a physical channel (e.g. the dipole's drag handing
    momentum to the
 plasma during coupling, absent from the witness chain) or an angular-discretisation leak is untested at this
    point; an
 Nmu = 48 rerun of the (0.5, l3) screen is added to separate them (added now, after this screen, before its
    trisection).

--- 2026-11-11T16:31Z  RESULT drift pair (2.0, l3) at Nmu = 48 (same recipe; p_3 and margins recomputed on 48
    bins)
 +f: 0.5125134 [0.5125082, 0.5125185] (Nmu=32: 0.5125833) -> drift 7.0e-5
 -f: 0.5090041 [0.5089959, 0.5090123] (Nmu=32: 0.5088395) -> drift 1.65e-4
 d_K = 1.65e-4 (D5). Difference at Nmu=48: +3.509e-3 (Nmu=32: +3.744e-3; changes by 6%).
 (2.0, l3) widened brackets: + [0.5124091, 0.5127576], - [0.5086663, 0.5090127]: DISJOINT (gap 3.40e-3) ->
    REAL.
 P1 for l=3: |diff| = 3.744e-3 > 3 w = 5.6e-5 and REAL -> P1 PASSES (l=3). Sign: +f (like +p_3) raises the
    threshold.
 Consequence of D5 for the small pairs: a difference can be REAL only if it exceeds about w + 2 d_K = 3.5e-4.

--- 2026-11-11T16:46Z  RESULT trisections (2.0, d = A5 L-inf direction)
 +f: 0.5134352 [0.5134259, 0.5134444] w 1.85e-5;  -f: 0.5079300 [0.5079259, 0.5079342] w 8.2e-6
 Difference +5.505e-3 (297 w); widened by d_K = 1.65e-4: disjoint -> REAL.
 Ratio to p_3 at equal positivity margin: 5.505e-3 / 3.744e-3 = 1.47. P4 PASSES (ratio > 1) at threshold level
    ;
 inside the lean 1.5-3 only at its edge (1.47 < 1.5); A5's linear chain gave 2.47.

--- 2026-11-11T16:57Z  RESULT f = 0 (no perturbation, restart at eps_s = 2): delta_c = 0.5107500 [0.5107407,
    0.5107593] w 1.85e-5
 (v5.1 r=1 with once-per-step kappa: 0.51074 +- 0.00002: the stage-time kappa moves it by less than one
    bracket here.)
 Shifts from f = 0: p_3 +f +1.833e-3, -f -1.911e-3 (nearly odd in f); d +f +2.685e-3, -f -3.820e-3 (less
    symmetric).

--- 2026-11-11T17:12Z  RESULT trisections (0.5, l3)
 +f: 0.5107685 [0.5107593, 0.5107778] w 1.85e-5;  -f: 0.5107510 [0.5107469, 0.5107551] w 8.2e-6
 (-f search: two trial points ended 'crash' at eps 7.37-7.38 just before the horizon; by the strict rule they
    did not move the bracket.)
 Difference +1.75e-5 = 0.95 w (w = 1.85e-5). P2 (|diff| <= 2 w = 3.7e-5): PASSES. Not REAL under D5 (widened
    brackets overlap).
 Plainly: the raw brackets are disjoint by 4e-6, the sign matches eps_s = 2, and the screen estimate was +1.3e
    -5. A residual
 of ~0.5% of the eps_s = 2 shift is consistent with every number here; the survival law alone predicts ~5e-10
    of it.
 At this bracket and drift resolution it cannot be told from zero.

--- 2026-11-11T17:16Z  RESULT Nmu = 48 screen of (0.5, l3), delta 0.5100 (screen48_out.txt)
 dCmax(+ - -) at eps 2.5/3/4/5/5.5: Nmu=32 -4.10e-5/-7.07e-5/-2.29e-4/-8.48e-4/-1.54e-3; Nmu=48 -3.60e-5/-6.26
    e-5/-2.03e-4/-7.48e-4/-1.32e-3.
 Ratio 0.86-0.89: the coupled-era separation does not vanish with angular refinement (an O(dmu^2) leak would
    fall to ~0.44).
 It is a channel of the GR solver, not an angular-grid artefact. Radial-resolution dependence untested.
    Candidate mechanism,
 untested: the fluid-frame angular terms (shear S, acceleration a) couple l=3 to l=2 at a rate ~ S rather than
     k, and l=2
 changes the energy through shear work, so the coupled-era suppression is ~ S/kappa per rung, not exp(-21) (
    the witness's
 flat chain has no such term).

--- 2026-11-11T17:28Z  RESULT trisections (4.0, l3)
 +f: 0.5108796 [0.5108704, 0.5108889] w 1.85e-5;  -f: 0.5106358 [0.5106296, 0.5106420] w 1.23e-5 (one 'crash'
    trial, not counted)
 Difference +2.438e-4 = 13 w, same sign as eps_s = 2 and 0.5, 6.5% of the eps_s = 2 shift (lean "smaller than
```

```
    at 2": holds).
 D5: widened brackets [0.5107054, 0.5110539] and [0.5104646, 0.5108070] OVERLAP -> not REAL. P3 as
    preregistered
 (REAL and > 3 w): NOT PASSED. It fails only on the drift allowance, which was measured on the (2, l3) pair
    where the
 difference itself moved by 2.4e-4 between Nmu = 32 and 48; a drift measured on this pair is untested.

--- 2026-11-11T17:44Z  RESULT trisections (2.0, l4)
 +f: 0.5096749 [0.5096708, 0.5096790] w 8.2e-6;  -f: 0.5117726 [0.5117675, 0.5117778] w 1.03e-5
 Difference (+ minus -) = -2.098e-3 = 204 w; widened by d_K: disjoint (gap 1.75e-3) -> REAL. P1 (l=4) PASSES.
 Sign opposite to l=3: +p_4 lowers the threshold. |shift| ratio p_4/p_3 = 0.56 at equal positivity margin (A5
    chain: 0.40).
 Shifts from f = 0: +f -1.075e-3, -f +1.023e-3.

--- 2026-11-11T18:12Z  RESULT trisections (0.5, l4) and (4.0, l4)
 (0.5, l4): +f 0.5107685 [0.5107593, 0.5107778]; -f 0.5107500 [0.5107407, 0.5107593] (first -f search was
    stopped by the
   2 h background limit mid-way, log kept as *_killed; rerun from scratch). Difference +1.85e-5 = 1.0 w -> P2
    (l=4) PASSES; not REAL.
 (4.0, l4): +f 0.5106759 [0.5106667, 0.5106852]; -f 0.5108241 [0.5108148, 0.5108333]. Difference -1.482e-4 = 8
     w, same sign
   as (2, l4). Widened brackets overlap -> not REAL. P3 (l=4): NOT PASSED (fails on the drift allowance only).

SUMMARY OF VERDICTS
 P1 eps_s=2 moves the threshold (> 3 w and REAL): l=3 PASS (+3.744e-3), l=4 PASS (-2.098e-3).
 P2 eps_s=0.5 agrees within 2 w: l=3 PASS (+1.75e-5 = 0.95 w), l=4 PASS (+1.85e-5 = 1.0 w).
 P3 eps_s=4 moves the threshold (> 3 w and REAL): l=3 NOT PASSED (+2.44e-4 = 13 w, not REAL), l=4 NOT PASSED
    (-1.48e-4 = 8 w, not REAL).
 P4 d moves more than p_3 at equal margin: PASS (ratio 1.47 in threshold, 1.5-1.7 in the screen).
 Opposite reading (collapse washes out l >= 3 regardless of timing): KILLED by the eps_s = 2 pairs.
 Sign: none was predicted. Measured: odd directions oriented like +p_3 raise the threshold; +p_4 lowers it.
```

A4_restart.py: driver: stage-time rate, exact landing on the restart surface, the hidden perturbation.

```
# SCRIPT: A4-RESTART  McCormick run_nuc.py loop (r=1 single-rate collision model, unchanged physics) split at
    a restart surface eps_s
"""
Driver for the matched-pair restart test.
Same loop as McCormick's run_nuc.py (ms_nuc.py rhs/state/timestep imported unchanged), with:
  (1) kappa(t) = (1/2t)(eps_dec/eps)^n re-evaluated at EACH SSPRK3 stage time (audit item 1), kapcap 5e5 (
    inactive here);
  (2) the step that would cross t_s is shortened to land exactly on t_s = eps_s^2/(4 t0);
  (3) at t_s the state (t, R, U, Gam, e, I, peakC) is saved; the continuation starts from a saved state, with
    an
     optional perturbation dI_ik = a * (eps_i/4pi) * p_l(mu_k) that changes only the discrete l component.
Status 'BH' only from a detected apparent horizon (2M/R >= 1 with U < 0); 'dispersed' from McCormick's rule
(eps > 3 and max C < peakC/2); a solver breakdown is 'BH_breakdown' or 'crash' and never moves a bracket.
"""
import numpy as np, os, time
from ms_nuc import rhs, state, timestep, rho_b
from ms_fluid import initial_data, make_grid

FNU = 0.409; N = 400; K = 32; EPS_DEC = 2.0; NEXP = 3; KAPCAP = 5.0e5
RMAX = 9.0; BETA = 3.0; EPS0 = 0.1; CFL_R = 0.4; CFL_MU = 0.4; CH = 0.005

def kap_of(t, t0):
    return (1/(2*t))*min((EPS_DEC/(2*np.sqrt(t*t0)))**NEXP, KAPCAP)

def discrete_legendre(mue, lmax=4):
    """Discrete orthogonal polynomials on the bin midpoints, weight dmu, leading coefficient of continuous P_l
     ."""
    mu = 0.5*(mue[1:]+mue[:-1]); dmu = mue[1]-mue[0]
    lead = {0: 1.0, 1: 1.0, 2: 1.5, 3: 2.5, 4: 35/8}
    P = []
    for l in range(lmax+1):
        v = mu**l
        for p in P:
```

```
            v = v - (np.sum(v*p)*dmu/(np.sum(p*p)*dmu))*p
        P.append(v*lead[l])    # mu^l minus lower part has leading coeff 1; scale to P_l's leading coefficient
    return mu, np.array(P)

def initial(delta):
    r = make_grid(N, RMAX, BETA)
    t0, R, U, rho = initial_data(r, delta, EPS0, 0.0, 1.0)
    e = (1-FNU)*rho
    mue = np.linspace(-1, 1, K+1)
    I = np.repeat((FNU*rho/(4*np.pi))[:, None], K, axis=1)
    Mi = np.concatenate([[0.0], np.cumsum((4*np.pi/3)*np.diff(R**3)*rho)])
    Gam = np.ones(N+1); Gam[1:] = np.sqrt(1.0 + U[1:]**2 - 2*Mi[1:]/R[1:])
    return dict(t=t0, t0=t0, r=r, mue=mue, R=R, U=U, Gam=Gam, e=e, I=I, peakC=-1.0, step=0, Imin=float(I.min()
    ))

def evolve(st, t_stop=None, eps_max=15.0, snap_eps=(), stop_rule=True):
    """Advance state dict st. If t_stop given, stop exactly at t_stop (no stop rules applied before it)."""
    t, t0 = st['t'], st['t0']; r, mue = st['r'], st['mue']
    R, U, Gam, e, I = st['R'].copy(), st['U'].copy(), st['Gam'].copy(), st['e'].copy(), st['I'].copy()
    peakC = st['peakC']; step = st['step']; Imin = st['Imin']
    dR = np.zeros(N+1); dU = np.zeros(N+1); dG = np.zeros(N+1); de = np.zeros(N); dI = np.zeros((N, K))
    snaps = {}; snap_eps = list(snap_eps); status = 'undecided'; mtwo = 0.0
    dmu_ = mue[1]-mue[0]; w2 = (mue[1:]**3-mue[:-1]**3)/(3*dmu_); rec = []
    while True:
        kap = kap_of(t, t0)
        Ib = FNU/(1-FNU)*e[-1]/(4*np.pi)
        try:
            M, Ac, ac, S, V = rhs(t, R, U, Gam, e, I, mue, FNU, Ib, kap, dR, dU, dG, de, dI)
        except (ZeroDivisionError, FloatingPointError):
            status = 'crash'; break
        epsn = 2*np.sqrt(t*t0)
        twoMR = 2*M[1:]/R[1:]
        C = 2*(M[1:] - (4*np.pi/3)*rho_b(t)*R[1:]**3)/R[1:]
        iC = np.argmax(C); peakC = max(peakC, C[iC])
        e0n = I[0].sum(); ePk = I[iC].sum()
        rec.append((epsn, C[iC], e[0]/rho_b(t), 2*np.pi*e0n*dmu_/rho_b(t), (I[0] @ w2)/e0n, (I[iC] @ w2)/ePk,
    r[1+iC]))
        while snap_eps and epsn >= snap_eps[0] - 1e-12:
            eps_c, q_c, Pr_c = state(t, R, U, Gam, e, I, mue, FNU, kap)[:3]
            snaps[snap_eps.pop(0)] = dict(eps=epsn, Cmax=float(C[iC]), rC=float(r[1+iC]), ec=float(e[0]/rho_b(
    t)),
                                          nuc=float(eps_c[0]/rho_b(t)), Prc=float(Pr_c[0]/rho_b(t)), q=q_c.
    copy(), Pr=Pr_c.copy(),
                                          C=C.copy(), e=e.copy(), nu=eps_c.copy())
        if t_stop is not None and t >= t_stop*(1-1e-15):
            status = 'surface'; break
        if stop_rule and t_stop is None:
            if np.any((twoMR >= 1.0) & (U[1:] < 0)):
                status = 'BH'; break
            if epsn > 3.0 and C[iC] < 0.5*peakC:
                status = 'dispersed'; break
        if epsn > eps_max: break
        try:
            dt = min(timestep(R, U, e, I, mue, Gam, Ac, ac, S, CFL_R, CFL_MU), CH*2*t, (0.5/(kap*np.max(Ac))
    if kap > 0 else 1e30))
            if t_stop is not None and t + dt > t_stop:
                dt = t_stop - t
            R1 = R + dt*dR; U1 = U + dt*dU; G1 = Gam + dt*dG; e1 = e + dt*de; I1 = I + dt*dI
            rhs(t+dt, R1, U1, G1, e1, I1, mue, FNU, FNU/(1-FNU)*e1[-1]/(4*np.pi), kap_of(t+dt, t0), dR, dU, dG
    , de, dI)
            R2 = 0.75*R + 0.25*(R1+dt*dR); U2 = 0.75*U + 0.25*(U1+dt*dU); G2 = 0.75*Gam + 0.25*(G1+dt*dG)
            e2 = 0.75*e + 0.25*(e1+dt*de); I2 = 0.75*I + 0.25*(I1+dt*dI)
            rhs(t+0.5*dt, R2, U2, G2, e2, I2, mue, FNU, FNU/(1-FNU)*e2[-1]/(4*np.pi), kap_of(t+0.5*dt, t0), dR
    , dU, dG, de, dI)
            R = R/3 + 2/3*(R2+dt*dR); U = U/3 + 2/3*(U2+dt*dU); Gam = Gam/3 + 2/3*(G2+dt*dG)
            e = e/3 + 2/3*(e2+dt*de); I = I/3 + 2/3*(I2+dt*dI)
        except (ZeroDivisionError, FloatingPointError):
            coll = U[1:] < 0
```

```
            status = 'BH_breakdown' if (coll.any() and np.max(twoMR[coll]) > 0.95) else 'crash'
            break
        t += dt; step += 1
        if t_stop is not None and abs(t - t_stop) < 1e-14*t_stop:
            t = t_stop
        Imin = min(Imin, float(I.min())); mtwo = max(mtwo, float(np.max(twoMR)))
        if np.any(e <= 0) or not np.all(np.isfinite(U)) or not np.all(np.isfinite(I)):
            status = 'crash'; break
    return dict(t=t, t0=t0, r=r, mue=mue, R=R, U=U, Gam=Gam, e=e, I=I, peakC=peakC, step=step, Imin=Imin,
                status=status, eps=2*np.sqrt(t*t0), snaps=snaps, max2MR=mtwo, rec=np.array(rec))

KEYS = ('t', 't0', 'r', 'mue', 'R', 'U', 'Gam', 'e', 'I', 'peakC', 'step', 'Imin')

def surface_state(delta, eps_s, cache_dir):
    global K
    """Run from eps0 to the restart surface eps_s (cached on disk)."""
    os.makedirs(cache_dir, exist_ok=True)
    fn = os.path.join(cache_dir, f'surf_d{delta:.10f}_es{eps_s:g}_K{K}.npz')
    if os.path.exists(fn):
        z = np.load(fn); return {k: (z[k] if z[k].ndim else z[k].item()) for k in KEYS}
    st = initial(delta)
    t_s = eps_s**2/(4*st['t0'])
    out = evolve(st, t_stop=t_s)
    assert out['status'] == 'surface', out['status']
    np.savez(fn, **{k: out[k] for k in KEYS})
    return {k: out[k] for k in KEYS}

def perturb(st, l, a):
    """dI_ik = a * (eps_i/4pi) * p_l(mu_k): changes only the discrete l component of every cell."""
    mu, P = discrete_legendre(st['mue'])
    dmu = st['mue'][1]-st['mue'][0]
    s = dict(st); I = st['I']
    Ibar = I.sum(axis=1)*dmu/2.0          # = eps/(4 pi) with eps = 2 pi sum I dmu
    s['I'] = I + a*Ibar[:, None]*P[l][None, :]
    return s

def positivity_bound(st, l):
    mu, P = discrete_legendre(st['mue']); p = P[l]
    dmu = st['mue'][1]-st['mue'][0]; I = st['I']; Ibar = I.sum(axis=1)*dmu/2.0
    ratio = I/Ibar[:, None]
    amax_plus = np.min(ratio[:, p < 0]/(-p[p < 0])[None, :])
    amax_minus = np.min(ratio[:, p > 0]/(p[p > 0])[None, :])
    return float(amax_plus), float(amax_minus), float(1/np.max(-p)), float(1/np.max(p))

def moments(st):
    """Discrete l=0..4 Legendre components per cell, plus McCormick's e, q, P_r."""
    mu, P = discrete_legendre(st['mue']); dmu = st['mue'][1]-st['mue'][0]
    comp = np.array([st['I'] @ (P[l]*dmu) for l in range(5)])
    t = st['t']; kap = kap_of(t, st['t0'])
    eps, q, Pr = state(t, st['R'], st['U'], st['Gam'], st['e'], st['I'], st['mue'], FNU, kap)[:3]
    return comp, eps, q, Pr

def bump(rc, r1=0.3, r2=2.0):
    """C-infinity bump, 1 at the middle of [r1, r2], identically 0 outside (regular at r=0, away from the
     outer edge)."""
    x = (rc - 0.5*(r1+r2))/(0.5*(r2-r1))
    b = np.zeros_like(rc); m = np.abs(x) < 1
    b[m] = np.exp(1.0 - 1.0/(1.0 - x[m]**2))
    return b

def angvec(mue, l):
    """Angular direction on the bins: l >= 0 -> discrete p_l; l == -1 -> A5 L-infinity-optimal hidden
     direction (eps_s=2, T=4)."""
    if l >= 0:
        return discrete_legendre(mue)[1][l]
    import json
    d = json.load(open(os.path.join(os.path.dirname(os.path.abspath(__file__)), 'restart_direction_A5.json')))
    v = np.array(d['directions']['ed2_n3_s2_T4']['candidates']['Linf_optimal']['d_on_32_bins'], float)
```

```
    assert len(v) == len(mue)-1
    return v

def cell_margin(st, l):
    """Per-cell positivity margin for +/- the l perturbation on the discrete bins: largest |a_i| keeping every
      bin >= 0 for both signs."""
    p = angvec(st['mue'], l)
    dmu = st['mue'][1]-st['mue'][0]; I = st['I']; Ibar = I.sum(axis=1)*dmu/2.0
    ratio = I/Ibar[:, None]
    mp = np.min(ratio[:, p < 0]/(-p[p < 0])[None, :], axis=1)
    mm = np.min(ratio[:, p > 0]/(p[p > 0])[None, :], axis=1)
    return np.minimum(mp, mm)

def perturb_bump(st, l, f):
    """dI_ik = f * b(r_i) * m_i * (eps_i/4pi) * p_l(mu_k); m_i the cell's positivity margin, b the compact
     bump in r.
    |f| < 1 keeps every bin positive; +f and -f are exact mirrors."""
    p = angvec(st['mue'], l)
    dmu = st['mue'][1]-st['mue'][0]; I = st['I']
    Ibar = I.sum(axis=1)*dmu/2.0
    rc = 0.5*(st['r'][1:]+st['r'][:-1])
    amp = f*bump(rc)*cell_margin(st, l)
    s = dict(st); s['I'] = I + (amp*Ibar)[:, None]*p[None, :]
    return s

def run_member(delta, eps_s, l, a, cache_dir, eps_max=15.0, snap_eps=(), mode='uniform'):
    st = surface_state(delta, eps_s, cache_dir)
    if a == 0:
        s2 = st
    elif mode == 'bump':
        s2 = perturb_bump(st, l, a)
    else:
        s2 = perturb(st, l, a)
    s2 = dict(s2); s2['Imin'] = min(s2['Imin'], float(s2['I'].min()))
    return evolve(s2, eps_max=eps_max, snap_eps=snap_eps)
```

A4_screen.py: paired screening runs.

```
# SCRIPT: A4-SCREEN  D4: fixed-amplitude paired runs (+f, -f, 0) at delta 0.5100 and 0.5115, per (eps_s, l);
     selection rule applied
import sys, os, json, time, numpy as np
from scipy.interpolate import CubicSpline
import restart
from restart import run_member, initial, evolve
from cpuwait import wait
F = 0.5; DELTAS = (0.5100, 0.5115)
PAIRS = [(2.0, 3), (0.5, 3), (4.0, 3), (2.0, 4), (0.5, 4), (4.0, 4), (2.0, -1)]
if len(sys.argv) > 1: PAIRS = [PAIRS[int(i)] for i in sys.argv[1].split(',')]
os.makedirs('screen', exist_ok=True)
def P(*a): print(*a, flush=True)
def get(tag, fn):
    p = f'screen/{tag}.npz'
    if os.path.exists(p):
        z = np.load(p, allow_pickle=True); return dict(status=str(z['status']), eps=float(z['eps']), rec=z['
     rec'], Imin=float(z['Imin']), max2MR=float(z['max2MR']))
    w = wait(P); t1 = time.time()
    o = fn()
    np.savez(p, status=o['status'], eps=o['eps'], rec=o['rec'], Imin=o['Imin'], max2MR=o['max2MR'])
    P(f'  run {tag}: {o["status"]} at eps={o["eps"]:.3f}  Imin={o["Imin"]:.3e}  ({time.time()-t1:.0f}s, waited
      {w:.0f}s)')
    return dict(status=o['status'], eps=o['eps'], rec=o['rec'], Imin=o['Imin'], max2MR=o['max2MR'])
def spl(rec, col):
    e = rec[:, 0]; m = np.concatenate([[True], np.diff(e) > 0]); return CubicSpline(e[m], rec[m, col])
summary = json.load(open('screen_res.json')) if os.path.exists('screen_res.json') else {}
for eps_s, l in PAIRS:
    key = f'es{eps_s:g}_l{l}'; res = {}
    P(f'== pair eps_s={eps_s} l={l}  (f = +/-{F}, bump support 0.3-2 r_m)')
    for d in DELTAS:
```

```
        cont = get(f'cont_d{d}', lambda: evolve(initial(d)))
        z = get(f'zero_d{d}_es{eps_s:g}', lambda: run_member(d, eps_s, l, 0.0, 'cache'))
        p = get(f'plus_d{d}_es{eps_s:g}_l{l}', lambda: run_member(d, eps_s, l, +F, 'cache', mode='bump'))
        m = get(f'minus_d{d}_es{eps_s:g}_l{l}', lambda: run_member(d, eps_s, l, -F, 'cache', mode='bump'))
        hi = min(6.0, cont['eps'], z['eps'], p['eps'], m['eps']) - 0.02
        g = np.linspace(eps_s + 0.02, hi, 400)
        noise = max(float(np.max(np.abs(spl(z['rec'], 1)(g) - spl(cont['rec'], 1)(g)))), 1e-12)
        dC = spl(p['rec'], 1)(g) - spl(m['rec'], 1)(g)
        dPc = spl(p['rec'], 4)(g) - spl(m['rec'], 4)(g)
        dPk = spl(p['rec'], 5)(g) - spl(m['rec'], 5)(g)
        sep = float(np.max(np.abs(dC)))
        rows = [(float(E), float(spl(p['rec'], 1)(E) - spl(m['rec'], 1)(E)), float(spl(p['rec'], 4)(E) - spl(m
    ['rec'], 4)(E)))
                for E in (2.5, 3.0, 4.0, 5.0, 5.5, 6.0) if eps_s + 0.02 <= E <= hi + 1e-9]
        res[str(d)] = dict(cont=(cont['status'], cont['eps']), zero=(z['status'], z['eps']), plus=(p['status
    '], p['eps']), minus=(m['status'], m['eps']),
                           Imin=(p['Imin'], m['Imin']), noise=noise, sep=sep, sepPc=float(np.max(np.abs(dPc)))
    , sepPk=float(np.max(np.abs(dPk))),
                           rows=rows, eps_hi=float(hi), C5=(float(spl(p['rec'], 1)(min(5.0, hi))), float(spl(m
    ['rec'], 1)(min(5.0, hi))), float(spl(z['rec'], 1)(min(5.0, hi)))))
        P(f'  delta={d}: outcomes cont {cont["status"]}@{cont["eps"]:.2f}, zero {z["status"]}@{z["eps"]:.2f},
    +f {p["status"]}@{p["eps"]:.2f}, -f {m["status"]}@{m["eps"]:.2f}')
        P(f'     record eps {eps_s+0.02:.2f}-{hi:.2f}: max|dCmax(+ - -)| = {sep:.3e}; noise floor {noise:.3e};
     ratio {sep/noise:.1f};'
          f' max|d(P_r/e_nu) centre| = {res[str(d)]["sepPc"]:.3e}, at C-peak cell = {res[str(d)]["sepPk"]:.3e}
    ')
        for E, a, b in rows: P(f'       eps={E}: dCmax={a:+.3e}  d(P_r/e_nu)_centre={b:+.3e}')
    flip = any(res[str(d)]['plus'][0] != res[str(d)]['minus'][0] for d in DELTAS)
    above = any(res[str(d)]['sep'] > 10*res[str(d)]['noise'] for d in DELTAS)
    sel = flip or above
    # linearised estimate: dCmax at common eps (<=5) over dCmax/d delta from the zero runs at the two deltas
    E = min(res[str(d)]['eps_hi'] for d in DELTAS); E = min(E, 5.0)
    def at(tag): return float(spl(np.load(f'screen/{tag}.npz')['rec'], 1)(E))
    slope = (at(f'zero_d{DELTAS[1]}_es{eps_s:g}') - at(f'zero_d{DELTAS[0]}_es{eps_s:g}'))/(DELTAS[1]-DELTAS
     [0])
    dCE = np.mean([at(f'plus_d{d}_es{eps_s:g}_l{l}') - at(f'minus_d{d}_es{eps_s:g}_l{l}') for d in DELTAS])
    est = -dCE/slope
    P(f'  SELECTION: outcome flip={flip}, separation>10x noise={above} -> {"TRISECT" if sel else "NOT SELECTED
      (null)"};'
      f' linearised estimate delta_c(+)-delta_c(-) ~ {est:+.2e} (dCmax at eps={E:.2f} = {dCE:+.3e}, dCmax/d
     delta = {slope:.3f})')
    summary[key] = dict(res=res, selected=sel, flip=flip, above=above, est=est, slope=slope, dCE=dCE, E=E)
    json.dump(summary, open('screen_res.json', 'w'), indent=1)
```

A4_trisect.py: threshold search.

```
# SCRIPT: A4-TRISECT  strict-endpoint trisection, one process, for one member (eps_s, l, f) of a restart pair
# usage: python3 trisect.py TAG eps_s l f lo hi wtarget [K]
import sys, os, json, time
import restart
from restart import run_member
from cpuwait import wait
tag = sys.argv[1]; eps_s = float(sys.argv[2]); l = int(sys.argv[3]); f = float(sys.argv[4])
lo, hi, wt = float(sys.argv[5]), float(sys.argv[6]), float(sys.argv[7])
if len(sys.argv) > 8: restart.K = int(sys.argv[8])
def P(*a): print(*a, flush=True)
log = []
def trial(d):
    w = wait(P); t1 = time.time()
    o = run_member(d, eps_s, l, f, 'cache', mode='bump')
    log.append((d, o['status'], float(o['eps']), float(o['Imin']), float(o['max2MR'])))
    P(f'{tag} delta={d:.7f}: {o["status"]} eps={o["eps"]:.3f} Imin={o["Imin"]:.2e} ({time.time()-t1:.0f}s,
     waited {w:.0f}s)')
    json.dump(dict(tag=tag, eps_s=eps_s, l=l, f=f, K=restart.K, lo=lo, hi=hi, log=log), open(f'trisect_{tag}.
     json', 'w'), indent=1)
    return o['status']
t0 = time.time()
# endpoint verification: lo must disperse, hi must form a horizon; widen by the bracket width until they do
```

```
s = trial(lo)
while s != 'dispersed':
    if s == 'BH': hi = lo; lo = lo - 1.5e-3
    else: lo = lo - 2e-4                                  # undecided/crash/breakdown: does not move the bracket;
      step aside
    s = trial(lo)
s = trial(hi)
while s != 'BH':
    if s == 'dispersed': lo = hi; hi = hi + 1.5e-3
    else: hi = hi + 2e-4
    s = trial(hi)
while hi - lo > wt:
    a = lo + (hi-lo)/3; b = lo + 2*(hi-lo)/3
    sa = trial(a)
    if sa == 'BH': hi = a; continue
    sb = trial(b)
    if sa == 'dispersed': lo = a
    if sb == 'BH': hi = b
    elif sb == 'dispersed': lo = b
    if sa not in ('BH', 'dispersed') and sb not in ('BH', 'dispersed'):
        P(f'{tag}: both trial points undecisive; bracket not moved; stopping'); break
dc = 0.5*(lo+hi)
P(f'RESULT {tag}: delta_c = {dc:.7f}  bracket [{lo:.7f}, {hi:.7f}]  width {hi-lo:.2e}  ({(time.time()-t0)
    /60:.1f} min)')
json.dump(dict(tag=tag, eps_s=eps_s, l=l, f=f, K=restart.K, lo=lo, hi=hi, dc=dc, w=hi-lo, log=log), open(f'
    trisect_{tag}.json', 'w'), indent=1)
```

B.24 The readout Gramian, the order swap and the closures (Section 8)

A5_PREREGISTERED.txt: predictions and results.

```
# SCRIPT: A5-PREREGISTERED (preregistration text, not code)
Written: 2026-11-11T14:10Z (UTC). Before any computation in this folder. Only reading done so far
(v5.1 Secs 2, 5, 6, App. A.5; witness_run.py, witness_kernel.py, memory_curve.py; A4's restart.py grid).
Prior-art fetch: 4 web requests spent (2 curl downloads refused by proxy 403, 1 WebFetch refused as not on
allowlist, 1 WebSearch returned titles only). The two papers were NOT read this session.

COMMON OBJECT (same as v5.1 memory_curve.py / witness_run.py)
 dv/dtau = -(i k M + D(tau)) v,  M the normalized-Legendre Jacobi matrix (c_l=(l+1)/sqrt((2l+1)(2l+3))),
 D = sigma(tau) diag(0,1,1,1,...) (single rate) or sigma(tau) diag(0,1,r,r,...) (split).
 sigma(tau) = ed^n / tau^(n+1). Start v = |0> at tau0 = 0.02. LM = 60. Profile weights w_k ~ k^2 exp(-k^2/2)
 on 36 Gauss-Legendre k in [0.25, 9]. A(tau) = sum_k w_k |v_0|^2.  A_inst = same with sigma = 0.
 Memory form g(e*) = (1-A(e*))/(1-A_inst(e*)), g2 = 0.89 g(1.7) + 0.11 g(4.0) (w frozen at v5.1's 0.890).
 Real basis used for computation: w_l = i^l v_l, generator k K - D with K real antisymmetric.

=== TEST 1: RATE-ORDER SWAP ===
Family R (time reversal inside a window W=[ta,tb]; sigma unchanged outside W; reversed sigma_R(t)=sigma(ta+tb-
    t) on W):
 R1: n=3, ed=1, W=[0.5,1.5]     R2: n=3, ed=1, W=[0.25,1.5]    R3: n=6, ed=1, W=[0.5,1.5]
 R4: n=3, ed=2, W=[1.0,3.5] (readout 1.7 lies INSIDE W; only e*=4 is after W)
 R5/R6: split versions of R1 with r = 4 and r = 1/4 (sigma1=sigma, sigma2=r sigma, both reversed together).
Family S (n=6 v5.1 histories ed6 in {1,2,3} against an n=3 history whose ed3 is rescaled so that the optical
    depth
 int_W sigma dtau is equal):  windows W in {[0.02,4] (whole run), [0.5,4], [1.0,4]}.
Readouts at e* = 1.7 and 4.0: A, the profile-weighted l<=2 moment powers |v_l|^2 (l=0,1,2), the relative
 difference of the complex l<=2 vectors R_diff = sqrt(sum_k w|dv_{0..2}|^2 / sum_k w|v_{0..2}|^2), g(1.7), g
    (4), g2.
PREDICTIONS (scalar summaries):
 P1a Optical depth int_W sigma: identical within each pair, by construction -> BLIND to order (both families).
 P1b Time-integrated connection energy int_W (1/8)||[L,L*]||^2 dtau = (k^2/3) int sigma^2 (split: k^2[s1
    ^2/3+4(s2-s1)^2/15]):
    identical under time reversal (any int f(sigma) is) -> BLIND in family R. NOT equal in family S (n=6 vs n
    =3 put
    different sigma^2 mass on W): it is not order-blind there, it is simply a different number; values
    reported.
```

```
 P1c Instantaneous operator at readout (sigma(e*), connection energy at e*, E_B=k^2/3): identical in family R
     whenever
     W closes before e* -> BLIND (R1,R2,R3,R5,R6 at both e*; R4 at e*=4). Not equal in family S.
 P1d The exact propagated readout is NOT blind: for every pair, the l<=2 readout differs.
 KILL for P1d (per pair): R_diff < 1e-3 AND |dg(e*)| < 1e-3 at both readout epochs -> order is invisible to
     the linear
     readout for that pair; P1d killed for that pair.
 Lean only (no kill): in family R the reversed (rising-rate) member has lost more kept population by e* (g
     larger),
     because it streams early in W and the late recoupling destroys flux that would otherwise partly return.
 Lean only: in family S, at equal optical depth on W the n=6 member keeps more (g smaller) when W starts late.

=== TEST 2: READOUT GRAMIAN OF THE HIDDEN SECTOR ===
 G(s,T) = int_s^T Q^T Phi(t,s)^T R^T R Phi(t,s) Q dt, R = rows l=0,1,2, Q = embedding of l>=3, T = 4.
 Plant epochs: ed=0.5 (witness history): s = 0.125, 0.5, 1.0 (coupled / transition / free, tau_s/ed = 0.25, 1,
      2);
 also ed=2, n=3 (v5.1 reference): s = 0.5, 2.0, 3.0. n=3 main; n=6 at ed=0.5 as a second history.
 G is computed per k, parity-symmetrized over k and -k (G(-k) = P G(k) P, P = diag((-1)^l)), and profile-
     summed.
 Also the terminal (snapshot) Gramian Phi(T,s)^T R^T R Phi(T,s) restricted to Q (the witness's R_ret object).
 P2a (algebraic): for k != 0 and every c_l != 0 the truncated chain is observable from l<=2: N_inf = {0} and G
      is
     positive definite. There is no exact predictive quotient smaller than the full hidden space; the quotient
      that
     matters is numerical (eigenvalues of G above a tolerance). Checked by rank of the Kalman matrix [R; RB;
     RB^2..]
     in exact-size floating point for the time-frozen generator (reported with its singular values).
 P2b Coupled planting suppresses the whole spectrum of G relative to the transition planting by roughly the
     square
     of the hidden survival factor, exp[-2(ed^n/n)(1/s_c^n - 1/s_t^n)] (n=3, ed=0.5: s=0.125 vs 0.5 -> ~ e^{
     -42.5}).
     Kill: ratio of top eigenvalues off from that factor by more than 10^{+-3}.
 P2c The top eigenvector at the transition and free epochs lies in the odd-parity sector and has l=3 weight >
     0.5.
     Kill: top eigenvector even, or l=3 weight <= 0.5.
 P2d Effective rank (eigenvalues > 1e-3 of the top) at the free epoch is larger than at the transition epoch.
     Lean, kill if smaller.
 Output for the restart agent: the direction d(mu) on McCormick's 32 bin midpoints, built from discrete
     orthogonal
     polynomials p_l (A4's construction, extended in l), with discrete l<=2 components exactly zero, the
     positivity bound |a| <= 1/max_k|d_k|, and the readout per unit admissible amplitude compared with pure
     p_3, p_4.
     The JSON states plainly that the mapping chain -> radially uniform angular perturbation is not exact.

=== TEST 3: CLOSURE COMPARISON (linear level) ===
 Histories: n=3 ed in {0.25,0.5,1,2,4,8}; n=6 ed in {1,2,3}; start |0> at tau0=0.02; horizon tau in [tau0,4].
 Comparators (fixed now):
  EX   exact chain, LM=60 (reference).
  P1   truncation at l=1 (v_2 := 0).  This is ALSO the linearized M1 closure: Levermore / Minerbo / maximum-
     entropy
       Eddington factors are chi(f)=1/3+O(f^2), so about isotropy the closed l=2 moment vanishes at linear
     order.
  M2   maximum-entropy (Levermore-style) closure for the l=3 moment with l<=2 evolved: f=exp(a0+a1 mu+a2 P2)
     linearized
       about isotropy has no l=3 part, so linearized M2 == P2. Stated as algebra; compared numerically as a
     check.
  P2, P3, P5   truncations at l=2, 3, 5.
  CE2  P1 plus the quasi-static (Chapman-Enskog / viscous) l=2: v_2 = -i k c_1 v_1 / sigma (sigma>0), the
       linear shear-viscosity closure.
  OP2  P2 plus a short-memory (Markovian) approximation of the exact elimination kernel of the hidden block l
     >=3:
       d v_2/dt += -k^2 c_2^2 theta(t) v_2, theta(t)=int_tau0^t exp(-int_r^t sigma) dr  (streaming inside the
     hidden
       block neglected; the "crescendo" form I recall from optimal-prediction closures -- recalled, not read).
  MFe  his memory form as an exact block elimination: l<=2 kept, hidden block l>=3 eliminated with its closed-
     form
       propagator exp(-[Sigma(t)-Sigma(r)]) exp(-ik(t-r)M_QQ) (A.5 with the cut moved from l=0|1 to l=2|3).
       Hidden moments start empty, so the source is zero. Predicted error: numerical only (< 1e-5 relative).
```

```
  MFf  "frozen" memory form: the same kernel with Sigma frozen to the reference history (n=3, ed=2) for every
     history,
       while l<=2 use the actual sigma. Defined here as the analogue of fitting once and freezing.
 Error metrics (fixed now): E_traj = sqrt( sum_k w int_tau0^4 |v^c_{0..2}-v^ex_{0..2}|^2 / sum_k w int |v^ex_{
     0..2}|^2 );
  E_A(e*) = |A^c-A^ex| at e*=1.7, 4; E_g = |g^c - g^ex| at 1.7, 4 and for g2.
 PREDICTIONS:
  P3a P1 vs M1-linearized identical and P2 vs M2-linearized identical (algebra; numerical difference < 1e-10).
  P3b Truncations converge: E_traj(P5) < E_traj(P3) < E_traj(P2) < E_traj(P1) for every history.
      Kill (per history): any inversion.
  P3c Worst case for every truncation is the earliest decoupling, n=3 ed=0.25 (longest free streaming).
      Kill: worst case elsewhere.
  P3d MFe error < 1e-5 everywhere (it is exact in the linear model). Kill: > 1e-4 (would be a setup error).
  P3e No instantaneous closure on l<=2 (P1/M1, P2/M2, CE2) gets E_traj < 0.05 for the early histories (n=3 ed
     <=0.5).
      Kill: any of them below 0.05 there.
  Lean only: OP2 beats P2 for late decoupling (ed>=2) and loses for ed<=0.5 (theta grows without bound when
     free).
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=================== RESULTS (appended 2026-11-11, after the runs; nothing above edited) ===================
Setup check (t0_reproduce): real-basis chain gives g(1.7)=0.3432, g2=0.4110 (n=3 ed=2) vs v5.1 0.344/0.411; n
     =3 ed=0.25 0.9404/0.9467 vs
 0.941/0.947; n=6 ed=3 0.0256/0.1175 vs 0.026/0.118. (v5.1's 1.7 index is the first grid point >= 1.7, hence
     the 0.001.)
TEST 1  P1a, P1b, P1c: confirmed (optical depths equal to all digits printed; int sigma^2 equal under reversal
     ; sigma(e*) equal
 wherever W closes before e*). Family S: int sigma^2 of the n=6 member is 2.09-2.15x the n=3 member's at equal
      optical depth.
 P1d: HOLDS for every pair. R_diff 0.33-0.72; |dg| 0.009-0.184; dg2 R1 -0.064, R2 -0.107, R3 -0.118, R4
     +0.161, R5 -0.063, R6 -0.067.
 Lean (reversed member loses more): FAILED - the reversed member keeps MORE (g smaller) in R1,R2,R3,R5,R6 at
     both epochs and R4 at e*=4.
 Lean (family S, n=6 keeps more when W starts late): FAILED in 5 of 6 late-window cases (holds only W=[1,4],
     ed6=1).
TEST 2  P2a: confirmed. (R B^j)_{2,2+j} = k^j c_2...c_{j+1} != 0 for all j: Kalman matrix full rank, N_inf={0}
     , G positive definite.
  Floating-point: smallest singular values 1e-29..1e-11; G eigenvalues below ~1e-19 are round-off.
 P2b: KILLED for the integral Gramian (coupled/transition top ratio 2.5e-9 for n=3 ed=0.5 vs predicted ~e^{-42
     }=6e-19; n=6: 5e-27 vs
  e^{-1365}). Cause: the prediction left out the leakage (k/sigma)^l through damped rungs into the conserved l
     =0 (v5.1's witness
  predictions had it). Terminal Gramian ratio 4.8e-17 (n=3) is within 10^1.1 of the survival factor.
 P2c: HOLDS (top eigenvector odd, l=3 weight 0.919, 0.923, 0.918, 0.922, 0.953, 0.990).
 P2d: KILLED (effective rank free < transition: 8 vs 9 at ed=0.5, 3 vs 5 at ed=2; confounded with the shorter
     remaining horizon).
TEST 3  P3a: HOLDS (entropy closures: closed moment = 0.3333 eps^2 (M1) and 0.300 eps^2 (M2) as eps->0;
     linearized M1==P1, M2==P2).
 P3b: HOLDS in all 9 histories.  P3c: HOLDS (worst for P1,P2,P3,P5 is n=3 ed=0.25).  P3d: HOLDS (MFe max
     E_traj 4.6e-6).
 P3e: HOLDS (smallest instantaneous-closure E_traj at ed<=0.5 is P2 0.433).
 Lean (OP2 loses for early decoupling): FAILED - OP2 beats P2 in all 9 histories (ed=0.25: 0.247 vs 0.496).
```

A5_chain_lib.py: the chain.

```
# SCRIPT: A5-CHAIN-LIB  linear angular chain (v5.1 memory_curve / witness object) in the real basis w_l = i^l
     v_l
import os
os.environ.setdefault('OMP_NUM_THREADS', '1')
import numpy as np, scipy.sparse as sp
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre

kn, kw = roots_legendre(36); KMIN, KMAX = 0.25, 9.0
KS = 0.5*(KMAX-KMIN)*(kn+1)+KMIN; WK = 0.5*(KMAX-KMIN)*kw
WE = WK*KS**2*np.exp(-KS**2/2); WE /= WE.sum()
TAU0 = 0.02
```

```
def cl(l):
    return (l+1)/np.sqrt((2*l+1)*(2*l+3))

def Kreal(LM):
    """real generator of streaming in w basis: dw/dt = k K w;  K[l,l-1]=c_{l-1}, K[l-1,l]=-c_{l-1}"""
    N = LM+1; K = np.zeros((N, N))
    for l in range(1, N):
        K[l, l-1] = cl(l-1); K[l-1, l] = -cl(l-1)
    return K

def sigma_powerlaw(ed, n):
    return lambda t: ed**n/t**(n+1)

def Sigma_powerlaw(ed, n):   # antiderivative, Sigma' = sigma
    return lambda t: -ed**n/(n*t**n)

def reversed_in_window(sig, ta, tb):
    def s(t):
        return sig(ta+tb-t) if ta <= t <= tb else sig(t)
    return s

def damp_vec(LM, r=1.0, shear_only=False):
    d = np.ones(LM+1); d[0] = 0.0
    if shear_only:
        d[2] = r
    else:
        d[2:] = r
    return d

def evolve(k, sig, LM, tgrid, w0=None, breaks=(), dvec=None, rtol=1e-9, atol=1e-12, t0=TAU0):
    """exact (truncated at LM) chain; returns w[l, t] on tgrid (real basis). sig(t) scalar rate; dvec diag
     pattern."""
    N = LM+1; K = sp.csr_matrix(Kreal(LM)*k)
    d = np.r_[0.0, np.ones(LM)] if dvec is None else dvec
    Dm = sp.diags(d, format='csr')
    if w0 is None:
        w0 = np.zeros(N); w0[0] = 1.0
    jac = lambda t, y: (K - sig(t)*Dm)
    rhs = lambda t, y: K@y - sig(t)*(d*y)
    edges = [t0]+[b for b in breaks if t0 < b < tgrid[-1]]+[tgrid[-1]]
    out = np.zeros((N, len(tgrid))); y = np.array(w0, float)
    for a, b in zip(edges[:-1], edges[1:]):
        m = (tgrid >= a) & (tgrid <= b) if b == edges[-1] else (tgrid >= a) & (tgrid < b)
        te = np.unique(np.r_[tgrid[m], b])
        s = solve_ivp(rhs, (a, b), y, method='BDF', jac=jac, t_eval=te, rtol=rtol, atol=atol)
        assert s.success, s.message
        out[:, m] = s.y[:, np.searchsorted(te, tgrid[m])]
        y = s.y[:, -1]
    return out

def to_complex(w):
    """v_l = (-i)^l w_l"""
    l = np.arange(w.shape[0])
    return ((-1j)**l)[:, None]*w if w.ndim == 2 else ((-1j)**l)*w
```

A5_t1_order.py: order swap.

```
# SCRIPT: A5-T1-ORDER  rate-order swap: same int sigma over a window, different order; readouts at e*=1.7, 4
import os; os.environ['OMP_NUM_THREADS']='1'
import numpy as np, json, time
from scipy.integrate import quad
from chain_lib import *
LM=60; EP=np.array([1.7,4.0]); W2=0.89
t1=time.time()
def readout(sig,breaks=(),dvec=None):
    A=np.zeros(2); P=np.zeros((3,2)); V=[]
    for k,w in zip(KS,WE):
```

```
        wl=evolve(k,sig,LM,EP,breaks=breaks,dvec=dvec)
        v=to_complex(wl[:3]); V.append(v)
        A+=w*wl[0]**2; P+=w*np.abs(v)**2
    return A,P,np.array(V)   # V[k, l, epoch]
Ai,_,_=readout(lambda t:0.0)
def g_of(A): return (1-A)/(1-Ai)
def cmp(name,sigF,sigR,ta,tb,brF=(),brR=(),dvec=None,r=1.0):
    AF,PF,VF=readout(sigF,brF,dvec); AR,PR,VR=readout(sigR,brR,dvec)
    dv2=np.einsum('k,kle->e',WE,np.abs(VF-VR)**2); n2=np.einsum('k,kle->e',WE,np.abs(VF)**2)
    gF,gR=g_of(AF),g_of(AR)
    s1=lambda f: quad(f,ta,tb,limit=400,points=[ta,tb])[0]
    ce=lambda f: s1(lambda t: f(t)**2/3+4*((r-1)*f(t))**2/15)   # per k^2
    row=dict(name=name,W=[ta,tb],r=r,
        optdepth=[s1(sigF),s1(sigR)], conn_energy_int_per_k2=[ce(sigF),ce(sigR)],
        sigma_at_readout=[[sigF(e) for e in EP],[sigR(e) for e in EP]],
        A=[AF.tolist(),AR.tolist()], P_l=[PF.tolist(),PR.tolist()],
        R_diff=np.sqrt(dv2/n2).tolist(), g=[gF.tolist(),gR.tolist()], dg=(gR-gF).tolist(),
        g2=[float(W2*gF[0]+(1-W2)*gF[1]),float(W2*gR[0]+(1-W2)*gR[1])])
    print(name,'W',[ta,tb],'r',r,'| optdepth %.6g %.6g | connE/k2 %.6g %.6g'%(*row['optdepth'],*row['
     conn_energy_int_per_k2']),flush=True)
    for i,e in enumerate(EP):
        print('   e*=%.1f  A %.5f -> %.5f | P0 %.5f %.5f P1 %.5f %.5f P2 %.5f %.5f | R_diff %.3e | g %.4f ->
     %.4f (dg %+.4f) | sigma(e*) %.3g %.3g'%(
            e,AF[i],AR[i],PF[0,i],PR[0,i],PF[1,i],PR[1,i],PF[2,i],PR[2,i],row['R_diff'][i],gF[i],gR[i],gR[i]-
     gF[i],sigF(e),sigR(e)),flush=True)
    print('   g2 %.4f -> %.4f'%tuple(row['g2']),flush=True)
    return row
out=dict(A_inst=Ai.tolist(),family_R=[],family_S=[])
# family R: time reversal inside W
for name,n,ed,ta,tb,r in [('R1',3,1,0.5,1.5,1),('R2',3,1,0.25,1.5,1),('R3',6,1,0.5,1.5,1),('R4',3,2,1.0,3.5,1)
     ,
                          ('R5',3,1,0.5,1.5,4.0),('R6',3,1,0.5,1.5,0.25)]:
    s=sigma_powerlaw(ed,n); sR=reversed_in_window(s,ta,tb)
    dv=None if r==1 else damp_vec(LM,r)
    out['family_R'].append(dict(n=n,ed=ed,**cmp(name,s,sR,ta,tb,(ta,tb),(ta,tb),dv,r)))
# family S: n=6 v5.1 histories against n=3 rescaled to equal optical depth on W
for (ta,tb) in [(TAU0,4.0),(0.5,4.0),(1.0,4.0)]:
    for ed6 in (1,2,3):
        I6=ed6**6/6*(ta**-6-tb**-6); ed3=(3*I6/(ta**-3-tb**-3))**(1/3)
        row=cmp('S_W[%g,%g]_ed6=%g'%(ta,tb,ed6),sigma_powerlaw(ed3,3),sigma_powerlaw(ed6,6),ta,tb)
        row.update(ed3=ed3,ed6=ed6); print('   (n=3 member ed3 = %.5f; first member n=3, second n=6)'%ed3,
     flush=True)
        out['family_S'].append(row)
json.dump(out,open('t1_order.json','w'),indent=1)
print('time %.0fs'%(time.time()-t1))
```

A5_t2_gramian.py: Gramian.

```
# SCRIPT: A5-T2-GRAMIAN  readout Gramian of the hidden sector l>=3 (R = l<=2), parity-symmetrized, profile-
    summed
import os; os.environ['OMP_NUM_THREADS']='1'
import numpy as np, json, time, sys
from scipy.linalg import expm
from chain_lib import *

def gramian(ed, n, s, T=4.0, LM=60, dt=2e-3, ks=KS, ws=WE):
    N=LM+1; K=Kreal(LM); d=np.r_[0.0,np.ones(LM)]; Sg=Sigma_powerlaw(ed,n)
    nst=int(np.ceil((T-s)/dt)); tg=np.linspace(s,T,nst+1); h=tg[1]-tg[0]
    G=np.zeros((N-3,N-3)); GT=np.zeros((N-3,N-3))
    for k,w in zip(ks,ws):
        X=np.zeros((N,N-3)); X[3:,:]=np.eye(N-3)
        Gk=0.5*h*X[:3].T@X[:3]
        for i in range(nst):
            seff=(Sg(tg[i+1])-Sg(tg[i]))/h          # exact step-average rate
            X=expm(h*(k*K-seff*np.diag(d)))@X
            Gk+=(h if i<nst-1 else 0.5*h)*X[:3].T@X[:3]
        GTk=X[:3].T@X[:3]
        P=np.diag((-1.0)**np.arange(3,N))           # k -> -k
```

```
        G+=w*0.5*(Gk+P@Gk@P); GT+=w*0.5*(GTk+P@GTk@P)
    return G,GT

def summarize(G,label,ntop=3):
    lam,V=np.linalg.eigh(G); idx=np.argsort(lam)[::-1]; lam=lam[idx]; V=V[:,idx]
    ls=np.arange(3,3+G.shape[0])
    rows=[]
    for j in range(ntop):
        v=V[:,j]; v=v*np.sign(v[np.argmax(abs(v))]); wts=v**2
        par='odd' if wts[ls%2==1].sum()>0.999 else ('even' if wts[ls%2==0].sum()>0.999 else 'mixed')
        rows.append(dict(eig=float(lam[j]),parity=par,l_weights={int(l):float(x) for l,x in zip(ls[:10],wts
     [:10])},vec=v.tolist()))
    effrank=int(np.sum(lam>1e-3*lam[0]))
    print('  %s: top eigs %s | eff.rank(>1e-3 top) %d | min eig %.2e'%(label,' '.join('%.3e'%x for x in lam
     [:6]),effrank,lam[-1]),flush=True)
    for j,r in enumerate(rows):
        print('     v%d [%s] l-weights: %s'%(j,r['parity'],' '.join('%d:%.3f'%(l,x) for l,x in list(r['
     l_weights'].items())[:7])),flush=True)
    return dict(eigs=lam.tolist(),effrank=effrank,top=rows)

if __name__=='__main__':
    t1=time.time(); out={}
    confs=[('ed0.5_n3',0.5,3,[0.125,0.5,1.0]),('ed0.5_n6',0.5,6,[0.125,0.5,1.0]),('ed2_n3
     ',2.0,3,[0.5,2.0,3.0])]
    for name,ed,n,ss in confs:
        for s,lab in zip(ss,['coupled','transition','free']):
            G,GT=gramian(ed,n,s)
            print('%s plant s=%g (%s)  [%.0fs]'%(name,s,lab,time.time()-t1),flush=True)
            out['%s_s%g'%(name,s)]=dict(ed=ed,n=n,s=s,epoch=lab,integral=summarize(G,'integral G'),terminal=
     summarize(GT,'terminal G_T'))
            np.save('G_%s_s%g.npy'%(name,s),G)
    json.dump(out,open('t2_gramian.json','w'),indent=1)
    print('time %.0fs'%(time.time()-t1))
```

A5_t3_closures.py: closures.

```
# SCRIPT: A5-T3-CLOSURES  linear-level closure comparison against the exact chain, v5.1 histories (comparators
      fixed in PREREGISTERED.txt)
import os; os.environ['OMP_NUM_THREADS']='1'
import numpy as np, json, time, sys
from scipy.integrate import solve_ivp
from scipy.linalg import expm
from chain_lib import *
H=float(sys.argv[1]) if len(sys.argv)>1 else 1e-3
HIST=[(3,0.25),(3,0.5),(3,1),(3,2),(3,4),(3,8),(6,1),(6,2),(6,3)]
if len(sys.argv)>2: HIST=[HIST[int(i)] for i in sys.argv[2].split(',')]
TG=TAU0+np.arange(int(round((4.0-TAU0)/H))+1)*H; TG[-1]=4.0
i17=int(np.argmin(abs(TG-1.7))); i40=len(TG)-1
c0,c1,c2=cl(0),cl(1),cl(2)
LMX=60; KQQ=Kreal(LMX)[3:,3:]
def ode(rhs,jac,y0,sig):
    s=solve_ivp(rhs,(TAU0,4.0),y0,method='BDF',jac=jac,t_eval=TG,rtol=1e-10,atol=1e-13); assert s.success,s.
     message; return s.y
def run_CE2(k,sig):
    rhs=lambda t,y: np.array([-k*c0*y[1], k*c0*y[0]-(sig(t)+k*k*c1*c1/sig(t))*y[1]])
    jac=lambda t,y: np.array([[0,-k*c0],[k*c0,-(sig(t)+k*k*c1*c1/sig(t))]])
    y=ode(rhs,jac,np.array([1.0,0.0]),sig); w2=k*c1*y[1]/np.array([sig(t) for t in TG]); return np.vstack([y,
     w2])
def run_OP2(k,sig):
    def rhs(t,y):
        s=sig(t); return np.array([-k*c0*y[1], k*c0*y[0]-k*c1*y[2]-s*y[1], k*c1*y[1]-s*y[2]-k*k*c2*c2*y[3]*y
     [2], 1-s*y[3]])
    def jac(t,y):
        s=sig(t); return np.array([[0,-k*c0,0,0],[k*c0,-s,-k*c1,0],[0,k*c1,-s-k*k*c2*c2*y[3],-k*k*c2*c2*y
     [2]],[0,0,0,-s]])
    return ode(rhs,jac,np.array([1.0,0,0,0]),sig)[:3]
def run_MF(k,sig,Sg_kernel):
    """l<=2 kept; hidden l>=3 eliminated: dw2 += -k^2 c2^2 int exp(-(S(t)-S(r))) g(t-r) w2(r) dr; implicit
```

```
    trapezoid."""
    Nt=len(TG); E=expm(H*k*KQQ); e=np.zeros(KQQ.shape[0]); e[0]=1; g=np.zeros(Nt)
    for j in range(Nt): g[j]=e[0]; e=E@e
    Sk=Sg_kernel(TG); sv=np.array([sig(t) for t in TG])
    Y=np.zeros((3,Nt)); Y[0,0]=1.0; C=k*k*c2*c2
    def Aloc(s): return np.array([[0,-k*c0,0],[k*c0,-s,-k*c1],[0,k*c1,-s]])
    mem=np.zeros(Nt)   # m(t_i)
    I3=np.eye(3)
    for i in range(Nt-1):
        j=np.arange(i+1); Kr=np.exp(-(Sk[i+1]-Sk[j]))*g[i+1-j]; wq=np.full(i+1,H); wq[0]=H/2
        known=-C*np.sum(wq*Kr*Y[2,j])          # memory at t_{i+1} from r<t_{i+1} (trapezoid, last node weight
     H/2 implicit)
        Mimp=np.zeros((3,3)); Mimp[2,2]=-C*H/2  # r=t_{i+1}: exp(0) g(0)=1
        lhs=I3-H/2*(Aloc(sv[i+1])+Mimp)
        rhsv=Y[:,i]+H/2*(Aloc(sv[i])@Y[:,i]+np.array([0,0,mem[i]])+np.array([0,0,known]))
        Y[:,i+1]=np.linalg.solve(lhs,rhsv); mem[i+1]=known-C*H/2*Y[2,i+1]
    return Y
def trap(y): return np.sum(0.5*(y[1:]+y[:-1])*np.diff(TG))
Sref=Sigma_powerlaw(2.0,3)
t1=time.time()
Ainst=sum(w*evolve(k,lambda t:0.0,LMX,TG[[i17,i40]])[0]**2 for k,w in zip(KS,WE))
COMPS=['P1','M1lin','P2','M2lin','P3','P5','CE2','OP2','MFe','MFf']
res={}
for (n,ed) in HIST:
    sig=sigma_powerlaw(ed,n); Sg=Sigma_powerlaw(ed,n)
    num={c:0.0 for c in COMPS}; den=0.0; Ac={c:np.zeros(2) for c in COMPS+['EX']}
    for k,w in zip(KS,WE):
        ex=evolve(k,sig,LMX,TG,rtol=1e-10,atol=1e-13)[:3]
        tr={}
        for N in (1,2,3,5):
            y=evolve(k,sig,N,TG,rtol=1e-10,atol=1e-13); z=np.zeros((3,len(TG))); m=min(3,N+1); z[:m]=y[:m]; tr
     [N]=z
        Y={'P1':tr[1],'M1lin':tr[1],'P2':tr[2],'M2lin':tr[2],'P3':tr[3],'P5':tr[5],'CE2':run_CE2(k,sig),'OP2':
     run_OP2(k,sig),
           'MFe':run_MF(k,sig,Sg),'MFf':run_MF(k,sig,Sref)}
        den+=w*trap(np.sum(ex**2,0)); Ac['EX']+=w*ex[0,[i17,i40]]**2
        for c in COMPS:
            num[c]+=w*trap(np.sum((Y[c]-ex)**2,0)); Ac[c]+=w*Y[c][0,[i17,i40]]**2
    gex=(1-Ac['EX'])/(1-Ainst); row={}
    for c in COMPS:
        gc=(1-Ac[c])/(1-Ainst)
        row[c]=dict(E_traj=float(np.sqrt(num[c]/den)),E_A=np.abs(Ac[c]-Ac['EX']).tolist(),E_g=np.abs(gc-gex).
     tolist(),
                    E_g2=float(abs(0.89*(gc[0]-gex[0])+0.11*(gc[1]-gex[1]))),g=gc.tolist())
    row['EX']=dict(g=gex.tolist(),g2=float(0.89*gex[0]+0.11*gex[1]))
    res['n%d_ed%g'%(n,ed)]=row
    print('n=%d ed=%g  exact g(1.7)=%.4f g(4)=%.4f g2=%.4f   [%.0fs]'%(n,ed,gex[0],gex[1],row['EX']['g2'],time
     .time()-t1),flush=True)
    for c in COMPS:
        r=row[c]; print('   %-6s E_traj %.3e | E_A(1.7) %.2e E_A(4) %.2e | E_g(1.7) %.2e E_g(4) %.2e E_g2 %.2e
     '%(c,r['E_traj'],*r['E_A'],*r['E_g'],r['E_g2']),flush=True)
json.dump(dict(h=H,res=res),open('t3_closures_h%g_%s.json'%(H,sys.argv[2].replace(',','-') if len(sys.argv)>2
     else 'all'),'w'),indent=1)
print('time %.0fs'%(time.time()-t1))
```

B.25 Figures 8–10

`make_figs_v6.py`: every number copied from the outputs above.

```
# SCRIPT: NU-V6-FIGS  figures 8-10 for v6; every number is copied from the printed outputs named beside it
import numpy as np, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
D = __file__.rsplit('/', 1)[0] + '/figs/'
INK, MUT, RUST, TEAL, GOLD, VIOL = '#1d1a16', '#8a8478', '#b5451b', '#0f7fa8', '#c8900a', '#5b3fa8'
plt.rcParams.update({'font.family': 'serif', 'font.size': 10, 'axes.spines.top': False, 'axes.spines.right':
     False, 'savefig.dpi': 200})
```

```
# ---- Fig 8: the door in the full collapse (A4_restart/PREREGISTERED.txt, trisection results; widths = full
    bracket widths)
base = 0.5107500   # f = 0 restart at eps_s = 2
pairs = {  # eps_s: (dplus, dminus, wplus, wminus)
 ('l3', 0.5): (0.5107685, 0.5107510, 1.85e-5, 8.2e-6),
 ('l3', 2.0): (0.5125833, 0.5088395, 1.85e-5, 1.65e-5),
 ('l3', 4.0): (0.5108796, 0.5106358, 1.85e-5, 1.23e-5),
 ('l4', 0.5): (0.5107685, 0.5107500, 1.85e-5, 1.85e-5),
 ('l4', 2.0): (0.5096749, 0.5117726, 8.2e-6, 1.03e-5),
 ('l4', 4.0): (0.5106759, 0.5108241, 1.85e-5, 1.85e-5),
 ('d', 2.0): (0.5134352, 0.5079300, 1.85e-5, 8.2e-6)}
dK = 1.65e-4  # drift allowance, Nmu 32 -> 48 on the (2, l3) pair
fig, axs = plt.subplots(1, 2, figsize=(10.5, 3.7))
ax = axs[0]
for key, col, mk, off, lab in [('l3', TEAL, 'o', -0.06, r'$\ell=3$'), ('l4', RUST, 's', 0.06, r'$\ell=4$'), ('
    d', VIOL, 'D', 0.0, 'readout-Gramian direction')]:
    xs, ys = [], []
    for (k, es), (p, m, wp, wm) in pairs.items():
        if k != key: continue
        diff = p - m; err = 0.5 * (wp + wm) + 2 * dK
        ax.errorbar(es + off, diff * 1e3, yerr=err * 1e3, fmt=mk, color=col, ms=6, capsize=3, label=lab if es
    == 2.0 else None)
        xs.append(es); ys.append(diff)
ax.axhline(0, color=MUT, lw=0.8); ax.axvspan(1.6, 2.4, color=GOLD, alpha=0.12)
ax.set_xticks([0.5, 2, 4]); ax.set_xticklabels(['0.5\ncoupled', '2\ndecoupling', '4\nfree streaming'])
ax.set_xlabel(r'restart surface $\epsilon_s$ where the hidden structure is planted'); ax.set_ylabel(r'$\
    delta_c(+f)-\delta_c(-f)$  [$10^{-3}$]')
ax.set_title('Same density, flux and stress; different hidden angular structure', fontsize=9.5); ax.legend(
    frameon=False, fontsize=8)
ax.text(2.45, 4.6, 'error bars: bracket + drift allowance', fontsize=7.5, color=MUT)
ax = axs[1]
labs = [r'$+p_3$', r'$-p_3$', r'$+p_4$', r'$-p_4$', '+d', '-d']
vals = [0.5125833, 0.5088395, 0.5096749, 0.5117726, 0.5134352, 0.5079300]
cols = [TEAL, TEAL, RUST, RUST, VIOL, VIOL]
ax.barh(range(6), [(v - base) * 1e3 for v in vals], color=cols, height=0.6)
ax.axvline(0, color=INK, lw=0.8); ax.set_yticks(range(6)); ax.set_yticklabels(labs); ax.invert_yaxis()
ax.set_xlabel(r'threshold shift from the unperturbed restart, $\epsilon_s=2$  [$10^{-3}$]')
ax.set_title('Odd structure raises the threshold, even lowers it', fontsize=9.5)
fig.tight_layout(); fig.savefig(D + 'fig8_door_nonlinear.png'); plt.close(fig)

# ---- Fig 9: self-scattering in McCormick's physical model (A2_selfscatter/REPORT.md, A2b_decisive/
    PREREGISTERED.txt)
fig, ax = plt.subplots(figsize=(9.6, 3.2))
rows = [('perfect fluid (his Table VII)', 0.49742, 0, MUT), ('with self-scattering (his run)', 0.519406,
    0.000156, INK),
        ('r = 1: same isotropy history, no flux-shear split', 0.51972, 0.00009, TEAL), ('without self-
    scattering (his run)', 0.526391, 0.000109, INK),
        ('instantaneous free streaming (this paper)', 0.533125, 0.00012, MUT)]
for i, (l, v, e, c) in enumerate(rows):
    ax.errorbar(v, i, xerr=e, fmt='o', color=c, ms=6, capsize=3)
ax.set_yticks(range(5)); ax.set_yticklabels([r[0] for r in rows], fontsize=8.5)
ax.vlines(0.5194, 1.6, 2.4, color=GOLD, lw=2); ax.vlines(0.5204, 1.6, 2.4, color=RUST, lw=2)
ax.annotate('timing reading 0.5194', (0.5194, 2.4), (0.5056, 2.9), fontsize=7.5, color=GOLD, arrowprops=dict(
    arrowstyle='-', color=GOLD, lw=0.6))
ax.annotate('frozen link law 0.5204', (0.5204, 1.6), (0.5225, 1.25), fontsize=7.5, color=RUST, arrowprops=dict
    (arrowstyle='-', color=RUST, lw=0.6))
ax.invert_yaxis(); ax.set_xlabel(r'$\delta_c$ at $T_H=1$ MeV, McCormick multigroup model'); ax.set_xlim(0.495,
     0.536)
ax.set_title('Self-scattering lowers the threshold mostly through timing; the flux-shear split carries
    +0.0003', fontsize=9.5)
fig.tight_layout(); fig.savefig(D + 'fig9_selfscatter.png'); plt.close(fig)

# ---- Fig 10: which hidden directions the future reads (A5_kinetic/REPORT.md)
fig, axs = plt.subplots(1, 2, figsize=(10.5, 3.5))
ax = axs[0]
ep = {'coupled\n(s=0.5)': [1.1e-7, 9e-12, 1e-15], 'decoupling\n(s=2)': [0.367, 0.148, 0.040], 'free\n(s=3)':
    [0.147, 0.024, 0.0016]}
```

```
for j, (k, v) in enumerate(ep.items()):
    for i, x in enumerate(v): ax.scatter([j + (i - 1) * 0.15], [x], color=[TEAL, RUST, MUT][i], s=30, zorder
    =3)
ax.set_yscale('log'); ax.set_xticks(range(3)); ax.set_xticklabels(list(ep)); ax.set_ylabel('readout Gramian
    eigenvalues')
ax.set_title(r'History $n=3,\ \epsilon_{\rm dec}=2$: top three directions' + '\n(blue odd, l=3-led; red even,
    l=4-led)', fontsize=9)
ax = axs[1]
pairs = ['R1', 'R2', 'R3', 'R4', 'R5\n(r=4)', 'R6\n(r=1/4)']; fw = [0.705, 0.705, 0.662, 0.411, 0.712, 0.700];
     rv = [0.641, 0.598, 0.544, 0.572, 0.649, 0.633]
x = np.arange(6); ax.bar(x - 0.18, fw, 0.36, color=TEAL, label='rate falling (forward)'); ax.bar(x + 0.18, rv,
     0.36, color=GOLD, label='same opacity, reversed order')
ax.set_xticks(x); ax.set_xticklabels(pairs, fontsize=8); ax.set_ylabel(r'memory form $g_2$'); ax.set_ylim(0,
    0.95); ax.legend(frameon=False, fontsize=8, loc='upper left', ncol=2)
ax.set_title('Equal optical depth and equal connection energy,\ndifferent order: the memory form tells them
    apart', fontsize=9)
fig.tight_layout(); fig.savefig(D + 'fig10_gramian_order.png'); plt.close(fig)
print('ok')
```

B.26 The onset of the kept-sector pole (Section 2)

The closed pole and non-orthogonality factor of Appendix A.3a against the moment chain, and the truncation onset against $2/\pi$.

```
# SCRIPT: NU-V7-ONSET  exact onset kappa_c = 2k/pi of the kept-sector pole and its non-orthogonality K = 4 phi
    ^2/sin^2(2 phi)
import numpy as np
from mpmath import mp, mpf, cot, atan, pi, sin
mp.dps = 30
def chain(N, kap, k=1.0):
    l = np.arange(N-1); c = (l+1)/np.sqrt((2*l+1)*(2*l+3))
    M = np.diag(c,1)+np.diag(c,-1)
    D = np.full(N, -kap); D[0] = 0.0
    return -1j*k*M + np.diag(D)
def least(N, kap):
    w, V = np.linalg.eig(chain(N, kap)); i = np.argmax(w.real); return w[i], V[:, i]
print('1) exact pole from I4: s = k cot(k/kappa) - kappa, kappa_c = 2k/pi =', float(2/pi))
for kap in [1.0, 2.0, 0.8]:
    phi = 1/mpf(kap); s = cot(phi) - kap
    g = atan(1/(s+kap)); print('  kappa=%.2f  s=%.12f  1-kappa*g=%.2e' % (kap, float(s), float(1-kap*g)))
print('2) chain eigenvalue vs exact pole and K vs 4phi^2/sin^2(2phi)')
for kap in [2.0, 1.0, 0.8, 0.7]:
    w, v = least(400, kap); phi = 1/kap
    Kc = (np.vdot(v, v).real)**2/abs(v @ v)**2; Ke = 4*phi**2/np.sin(2*phi)**2
    print('  kappa=%.2f  chain s=%.10f%+.1ej  exact s=%.10f  K chain=%.6f exact=%.6f' % (kap, w.real, w.imag,
     np.cos(phi)/np.sin(phi)-kap, Kc, Ke))
print('3) truncation onset (least-damped mode real) vs N')
def isreal(N, kap): w, _ = least(N, kap); return abs(w.imag) < 1e-9
for N in [50, 100, 200, 400, 800]:
    lo, hi = 0.55, 0.75
    for _ in range(40):
        m = 0.5*(lo+hi)
        if isreal(N, m): hi = m
        else: lo = m
    print('  N=%4d  onset kappa/k = %.5f   minus 2/pi = %+.5f' % (N, hi, hi-2/np.pi))
```

```
1) exact pole from I4: s = k cot(k/kappa) - kappa, kappa_c = 2k/pi = 0.6366197723675814
  kappa=1.00  s=-0.357907384066  1-kappa*g=0.00e+00
  kappa=2.00  s=-0.169512278288  1-kappa*g=0.00e+00
  kappa=0.80  s=-0.467726582745  1-kappa*g=0.00e+00
2) chain eigenvalue vs exact pole and K vs 4phi^2/sin^2(2phi)
  kappa=2.00  chain s=-0.1695122783+6.9e-17j  exact s=-0.1695122783  K chain=1.412283 exact=1.412283
  kappa=1.00  chain s=-0.3579073841+1.1e-16j  exact s=-0.3579073841  K chain=4.837802 exact=4.837802
  kappa=0.80  chain s=-0.4677265827+0.0e+00j  exact s=-0.4677265827  K chain=17.449868 exact=17.449868
  kappa=0.70  chain s=-0.5568083072-4.4e-15j  exact s=-0.5568083072  K chain=103.656747 exact=103.656747
```

```
3) truncation onset (least-damped mode real) vs N
  N=  50  onset kappa/k = 0.66492   minus 2/pi = +0.02830
  N= 100  onset kappa/k = 0.65199   minus 2/pi = +0.01537
  N= 200  onset kappa/k = 0.64495   minus 2/pi = +0.00833
  N= 400  onset kappa/k = 0.64112   minus 2/pi = +0.00450
  N= 800  onset kappa/k = 0.63904   minus 2/pi = +0.00242
```

```
# SCRIPT: NU-V7-FIG-ONSET  figure for the exact onset kappa_c = 2k/pi
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
import importlib.util, sys, os
plt.rcParams.update({'font.size': 11, 'font.family': 'serif'})
def chain(N, kap):
    l = np.arange(N-1); c = (l+1)/np.sqrt((2*l+1)*(2*l+3))
    D = np.full(N, -kap); D[0] = 0.0
    return -1j*(np.diag(c,1)+np.diag(c,-1)) + np.diag(D)
kc = 2/np.pi
ka = np.linspace(kc+1e-4, 3.0, 600); s = 1/np.tan(1/ka) - ka
kd = np.array([0.66, 0.7, 0.8, 1.0, 1.3, 1.7, 2.2, 3.0])
sd = [max(np.linalg.eigvals(chain(400, x)), key=lambda z: z.real).real for x in kd]
fig, ax = plt.subplots(1, 2, figsize=(10, 3.8))
ax[0].plot(ka, s, color='#1f4e79', lw=2, label='pole $k\\cot(k/\\kappa)-\\kappa$')
ax[0].plot(ka, -ka, '--', color='#a33', lw=1.2, label='branch-cut centre $-\\kappa$')
ax[0].plot(kd, sd, 'o', color='#e08a00', ms=6, label='400-moment chain')
ax[0].axvline(kc, color='0.4', lw=0.8, ls=':'); ax[0].text(kc+0.04, -0.95, '$2k/\\pi$', color='0.3')
ax[0].set_xlabel('$\\kappa/k$'); ax[0].set_ylabel('$s/k$'); ax[0].set_ylim(-3, 0.05); ax[0].legend(frameon=
    False, fontsize=9, loc='center right')
ph = np.linspace(0.01, np.pi/2-0.02, 500); K = 4*ph**2/np.sin(2*ph)**2
ax[1].semilogy(ph, K, color='#1f4e79', lw=2)
ax[1].axvline(np.pi/2, color='0.4', lw=0.8, ls=':'); ax[1].text(np.pi/2-0.3, 2e3, '$\\phi=\\pi/2$', color
    ='0.3')
ax[1].set_xlabel('$\\phi=k/\\kappa$'); ax[1].set_ylabel('$K=4\\phi^2/\\sin^2 2\\phi$')
ax[1].text(0.05, 300, 'chain onset $\\kappa/k$:\n50: 0.665\n200: 0.645\n800: 0.639\nexact: $2/\\pi$ = 0.637',
    fontsize=9)
fig.tight_layout(); fig.savefig(os.path.join(os.path.dirname(__file__), '../../figs/fig_v7_onset.png'), dpi
    =200)
```

B.27 The third resolution level (Section 7)

Predictions and kill line written before the runs, with the appended timing and the change of tolerance; the queue scripts; the run log; the $r = 1.25$ result.

```
PREREGISTERED.txt -- v6 task A6: a third resolution level for the r = 1.25 / 0.8 odd rate
Written 2026-11-11T20:44Z, before any run in this folder.
Results are appended below after the runs; nothing above an appended result is edited.

CODE (byte copies of A1's files, nu6/A1_stage/, nothing changed): run_split_stage.py (stage-correct kappa),
    ms_nuc_split.py,
  ms_fluid.py, search.py (sequential bisection, A1's endpoint rule), oddrate.py (four-corner rates), run_point
    .py (A1's
  single-point helper; its hard-coded N=600 grid is NOT used here -- a level-3 copy is written only if a
    single point is needed).
  Only parameters change, passed on the command line.

LEVELS (read from A1's res_*.json kw and RUNLOG; physics fixed: fnu 0.409, eps0 0.1, eps_dec 2, n 3, eps_max
    15, Gaussian q 1)
  level 1 (A1 S1): N = 400, Nmu = 32, Courant 0.4/0.4, cH = 0.005,  retry Courant 0.3
  level 2 (A1 S3): N = 600, Nmu = 48, Courant 0.2/0.2, cH = 0.0025, retry Courant 0.15
  level 3 (here) : N = 900, Nmu = 72, Courant 0.1/0.1, cH = 0.00125, retry Courant 0.075
  Stiff collision cap 0.5/(kappa max(1,r) max A) left as it is (as A1). Grid ratio per level q = 1.5 in r and
    mu; the
  CFL-limited dt shrinks by 1/3 per level (dx x 2/3, Courant x 1/2), the H-limited dt by 1/2.

LEVEL 1 AND 2 VALUES (recomputed here with oddrate.py from A1's stored brackets, v5.1 references df 0.49773,
    di 0.52832)
  L1: dc(1.25) = 0.50984158, dc(0.8) = 0.51169233, odd rate R1 = 1.11021 [1.10983, 1.11060]
```

```
  L2: dc(1.25) = 0.50991687, dc(0.8) = 0.51175643, odd rate R2 = 1.10198 [1.10159, 1.10237]
  s12 = R2 - R1 = -0.00823.  Threshold shifts L1->L2: +7.53e-5 (r = 1.25), +6.41e-5 (r = 0.8).

PREDICTIONS (a): if the scheme converges at order p in the grid spacing (ratio q = 1.5), s23 = s12 (2/3)^p:
  p = 1: s23 = -0.00549, R3 = 1.0965;  thresholds +5.02e-5 -> 0.50996706 (1.25), +4.27e-5 -> 0.51179917 (0.8)
  p = 2: s23 = -0.00366, R3 = 1.0983;  thresholds +3.35e-5 -> 0.50995033 (1.25), +2.85e-5 -> 0.51178492 (0.8)
  (p = 3 for comparison: s23 = -0.00244, R3 = 1.0995.)  Limits implied: p = 1 -> R_inf = 1.0855, p = 2 ->
    R_inf = 1.0954.
  If the time-step error dominated instead (dt ratio 1/3 per level), s23 would be s12 (1/3)^p_t: -0.0027 (p_t
    = 1).

KILL LINE (b): if |R3 - 1.10198| > |R2 - R1| = 0.00823 (the shift does not shrink), the odd rate is NOT
    converging at these
  resolutions and the paper must say so. Also flagged (not a kill): a sign change of s23 relative to s12 (
    oscillatory, then
  Richardson is not applied in the monotone form).

RICHARDSON (fixed before the runs): with three levels at constant ratio q = 1.5,
  p_obs = ln(s12/s23)/ln q  (only if s12/s23 > 1),  R_inf = R3 + s23/(q^p_obs - 1).
  Error on R_inf, reported as separate parts:
   (i) bracket part: R_inf recomputed over the four-corner extremes of each level's rate range (all 8
     combinations of
       range ends), the half-spread of the resulting R_inf values;
   (ii) order part: the spread between R_inf(p_obs) and the two-level extrapolations from L2, L3 with p = 1
     and p = 2;
   the quoted error is (i) + (ii) linear.  The fluid-reference shift +-7.8e-5 is common to every level and
     quoted separately.
  If p_obs falls outside [0.5, 4] the Richardson value is reported but marked untrustworthy.

SEARCH: search.py as A1 (both ends first; an end with the wrong outcome extends that side by 2 x half = 4e-4;
    bisection to
  width <= 1e-6; undecided midpoint -> points at 0.4 then 0.6; crash/BH_breakdown retried once at Courant
    0.075, then
  reported, never moves a bracket).  Starting brackets: L2 values +- 2e-4: 0.50991687 +- 2e-4 (r = 1.25),
    0.51175643 +- 2e-4
  (r = 0.8).  One simulation process at a time, OMP/NUMBA threads 1.  A run that runs far past the typical
    level-3 time
  without ending is handled as A1 handled its stalled point: stopped by hand, reported undecided, the 0.4 (
    then 0.6) point used.
TIMING RULE: the first run is timed; if one run takes > 3 h, r = 1.25 is run alone first and reported, then r
    = 0.8.

TIMING (appended 2026-11-11T21:54Z): first level-3 run 4168 s (69.5 min), 1153907 steps; < 3 h, so both
    thresholds queued back to back.

DEVIATION (appended 2026-11-13T07:07Z): r = 1.25 bisection stopped at width 1.25e-5 instead of 1e-6, because
    runs this close
to threshold no longer finish inside one 2 h compute window. Level-3 bracket [0.50994187, 0.50995437],
    midpoint 0.50994812.
Shift L2->L3 = +3.125e-5 (range +2.50e-5 .. +3.75e-5); against L1->L2 +7.53e-5 this gives a per-threshold
    observed order
p = 2.17 (range 1.72 .. 2.72). p = 2 prediction 0.50995033 inside the bracket; p = 1 prediction 0.50996706
    outside.
r = 0.8 run with tolerance 1.25e-5 (queue2.sh).
```

```
# SCRIPT: NU-V6-QUEUE-A6  level 3 (N=900, Nmu=72, Courant 0.1, cH 0.00125) thresholds r=1.25 then r=0.8; one
    process at a time
cd "$(dirname "$0")"
export OMP_NUM_THREADS=1 NUMBA_NUM_THREADS=1
F='"fnu":0.409,"N":900,"K":72,"eps0":0.1,"eps_dec":2.0,"eps_max":15.0,"nexp":3,"cfl_r":0.1,"cfl_mu":0.1,"cH
    ":0.00125'
go() { [ -f res_$1.json ] || python3 search.py "$@"; }
go L3_r125 run_split_stage 0.50991687 2e-4 1e-6 "{$F,\"rr\":1.25}" 0.075
go L3_r08  run_split_stage 0.51175643 2e-4 1e-6 "{$F,\"rr\":0.8}" 0.075
echo "$(date +%H:%M:%S) QUEUE DONE" >> RUNLOG.txt
```

```
# SCRIPT: NU-V6-QUEUE-A6-V2 (r=0.8 tol 1.25e-5)  level 3 (N=900, Nmu=72, Courant 0.1, cH 0.00125) thresholds r
    =1.25 then r=0.8; one process at a time
cd "$(dirname "$0")"
export OMP_NUM_THREADS=1 NUMBA_NUM_THREADS=1
F='"fnu":0.409,"N":900,"K":72,"eps0":0.1,"eps_dec":2.0,"eps_max":15.0,"nexp":3,"cfl_r":0.1,"cfl_mu":0.1,"cH
    ":0.00125'
go() { [ -f res_$1.json ] || python3 search.py "$@"; }
go L3_r125 run_split_stage 0.50991687 2e-4 1e-6 "{$F,\"rr\":1.25}" 0.075
go L3_r08  run_split_stage 0.51175643 2e-4 1.25e-5 "{$F,\"rr\":0.8}" 0.075
echo "$(date +%H:%M:%S) QUEUE DONE" >> RUNLOG.txt
```

```
14:44:59 [start] level-3 queue launched (queue.sh)
[L3_r125] 15:54:27 L3_r125 delta=0.50971687: dispersed eps=6.5193 minI=1.043e-08 steps=1153907 (4168s)
15:54:34 [note] timing: first level-3 run (delta=0.50971687, r=1.25) took 4168 s = 69.5 min, 1153907 steps (
    level 2 lower end: 433 s, 251302 steps; ratio 9.6x time, 4.6x steps). Under the 3 h rule, so r=1.25 and r
    =0.8 run back to back. Estimated ~12 runs x ~75 min = ~15 h per threshold.
[L3_r125] 17:02:20 L3_r125 delta=0.51011687: BH eps=6.4993 minI=1.052e-08 steps=1150339 (4072s)
[L3_r125] 18:16:22 L3_r125 delta=0.50991687: dispersed eps=7.1891 minI=4.308e-09 steps=1275088 (4442s)
[L3_r125] 19:27:29 L3_r125 delta=0.51001687: BH eps=6.8404 minI=7.754e-09 steps=1211984 (4267s)
[L3_r125] 20:43:35 L3_r125 delta=0.50996687: BH eps=7.2466 minI=9.622e-10 steps=1285705 (4567s)
21:42:28 [restart] container restarted; queue relaunched, search resumes from prog_L3_r125.json
[L3_r125] 23:06:59 L3_r125 delta=0.50994187: dispersed eps=7.3545 minI=7.340e-10 steps=1327361 (5057s)
02:53:09 [restart] 2h limit; queue relaunched
[L3_r125] 04:00:04 L3_r125 delta=0.50995437: BH eps=7.3625 minI=4.353e-10 steps=1320474 (4000s)
20:04:27 [restart] relaunched
22:04:33 [restart] 2h limit; queue relaunched
23:07:22 [restart] container restart; queue relaunched
01:07:46 r125 closed at width 1.25e-5 (dc=0.50994812); r0.8 started with tol 1.25e-5 via queue2.sh
[L3_r08] 02:20:25 L3_r08 delta=0.51155643: dispersed eps=6.5011 minI=9.923e-09 steps=1149928 (4357s)
03:07:52 [restart] 2h limit; queue2 relaunched
```


```
{
 "tag": "L3_r125",
 "runner": "run_split_stage",
 "kw": {
  "fnu": 0.409,
  "N": 900,
  "K": 72,
  "eps0": 0.1,
  "eps_dec": 2.0,
  "eps_max": 15.0,
  "nexp": 3,
  "cfl_r": 0.1,
  "cfl_mu": 0.1,
  "cH": 0.00125,
  "rr": 1.25
 },
 "dc": 0.50994812,
 "w": 1.2499999999970868e-05,
 "lo": 0.50994187,
 "hi": 0.50995437,
 "log": [
  [
   0.50971687,
   "dispersed",
   6.519310442867602,
   1.0428159983762577e-08,
   8099.99999999999,
   "",
   4167.6
  ],
  [
   0.51011687,
   "BH",
   6.499296632428603,
   1.052477881818825e-08,
   8099.999999999992,
```

```
   "",
   4072.4
  ],
  [
   0.50991687,
   "dispersed",
   7.189060162113385,
   4.308247981720386e-09,
   8099.99999999999,
   "",
   4442.0
  ],
  [
   0.51001687,
   "BH",
   6.84036316489117,
   7.753634635898005e-09,
   8099.999999999992,
   "",
   4266.9
  ],
  [
   0.50996687,
   "BH",
   7.246648222913391,
   9.621710774582058e-10,
   8099.99999999999,
   "",
   4566.8
  ],
  [
   0.50994187,
   "dispersed",
   7.354515357645314,
   7.340224805851244e-10,
   8099.99999999999,
   "",
   5056.6
  ],
  [
   0.50995437,
   "BH",
   7.362499600802576,
   4.3531776130686064e-10,
   8099.99999999999,
   "",
   4000.5
  ]
 ],
 "note": "stopped at width 1.25e-5: near-critical runs exceed the 2h window"
}
```


C Lean statements

The statements below are proved in Lean 4 with Mathlib (v4.34.1) in the repository `https://github.com/dicipler-pixel/neutrinos-leave-lean`. A full build of the version-7 statements at commit `d181d6d` (12 November 2026, 8934 build jobs) completes with no `sorry`, and the three self-scattering theorems use only the standard axioms. The earlier build at commit `c372d3f` (4 November 2026) compiles every module, replays every proof in an independent kernel checker, finds that all 100 theorems of the repository use only the standard axioms (`propext`, `Quot.sound`, `Classical.choice`), and confirms that all twelve false controls are rejected. Lean checks the finite moment model and the verdict arithmetic. It does not check McCormick's code, the collapse runs,

the classification of runs near threshold, the bracket edges (inputs copied from the run logs) or the numerical witness.

1. `crossLink`: $|2\,\mathrm{Re}(\bar{a}(-ih)b)| \le 2|h||a||b|$; `crossLink_sq`: with $A = |a|^2$, $|a|^2 + |b|^2 \le 1$, the square is at most $4|h|^2 A(1-A)$ (Appendix A.6).

2. `twoMomentConnectionEnergy`: $\frac{1}{8}\|[L, L^*]\|_F^2 = \kappa^2 k^2/3$ for the two-moment L.

3. `splitRateEnergy`: $\frac{1}{8}\|[L, L^*]\|_F^2 = k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$ for the three-moment L.

4. `shearCounterexample`: $R_{00}(1) = 32/37$, $R'_{00}(1) = -5/1369$, $\partial_{\kappa_2} E = 0$ at $\kappa_2 = \kappa_1$; with the complex 3×3 resolvent entry equal to the closed form $2(15\kappa_2 + 17)/(35\kappa_2 + 39)$ for $\kappa_2 > -1$ (Appendix A.7).

5. `scalarCollisionPropagator`: $e^{t(X-\sigma I)} = e^{-\sigma t} e^{tX}$, with generators $X - \sigma(t)I$ commuting pairwise (Appendix A.5).

6. `testB/C/D_verdict`: $G_B = 970/3059$, $G_C = 1983/3059$, $G_D = 472/3059$ against the pre-registered kill lines.

7. `gam_inv`, `odd_part_free_of_anchor`: the link coordinate $(r-1)/(r+1)$ is odd under $r \mapsto 1/r$, and the anchor cancels from the odd part (Section 7).

8. `cSq_zero`, `cSq_one`, `cSq_two`, `weight_ratio_01`, `weight_ratio_12`: $c_\ell^2 = (\ell+1)^2/((2\ell+1)(2\ell+3))$ gives $1/3$, $4/15$, $9/35$, and the ratios $5/4$ and $28/27$.

9. `testK_clean_range`, `testK_clean_below_sqrt`, `testK_clean_above_trend`: at $r = 1.25$ against 0.8 the odd rate lies in $[1.112, 1.114]$ over the whole bracket pair; `testK_crash_edge_false_kill`, `testK_first_sqrt_window`, `testK_first_trend_not_killed`: what the same pair would have decided had a failed run been used as a bracket edge (it would have decided wrongly), which is why the search rule of Section 7 never lets one.

10. `testJ_straddles`: over its full bracket the $r = 2$ odd rate straddles 1.108 and stays at most 1.128.

11. `selfScatterEnergy`, `selfScatterEnergy_ab`, `selfScatterEnergy_no_nn` (new in version 6, module `Neutrinos.SelfScatter`): with $\kappa_1 = \kappa_e$ and $\kappa_2 = \kappa_e + \kappa_n$, $\frac{1}{8}\|[L, L^*]\|_F^2 = a^2\kappa_e^2 + b^2\kappa_n^2$ for any streaming weights, $k^2\kappa_e^2/3 + \frac{4k^2}{15}\kappa_n^2$ in the continuum weights, and $a^2\kappa_e^2$ without self-scattering (Appendix A.13). The discrete weights $c_0^2 = \frac{1}{3} - \Delta\mu^2/12$, $c_1^2 = \frac{4}{15}(1 - \Delta\mu^2)$ and the diagonalisation of McCormick's operator are not checked in Lean.

12. `testL_rate_ge_six_fifths`, `testL_rate_le`: the shear-only odd rate lies in $[1.2, 1.21]$.

Each verdict is a whole-number inequality: with $O(\delta) = (\delta^{\mathrm{inst}} - \delta)/(\delta - \delta^{\mathrm{fluid}})$ and $Q = O(\delta_r)/O(\delta_{1/r})$, the odd rate is at least p/q exactly when $r^p \le Q^q$; O decreases in δ, so the two corners of the bracket pair that make Q smallest and largest are the ones checked, and the kernel decides each inequality exactly.